Linear Algebra for Engineering

Tutorial notes, explanations and solutions



Editing and teaching: Chen Gadi

Course website, materials and questions

`https://chengadi.science`

These notes may contain errors, inaccuracies, omissions or imperfect wording.
Use the website linked above for comments, corrections and questions.
The lectures and official course announcements determine the required material.

My sincere thanks to everyone who contributed in any way through comments, corrections, explanations, ideas and support along the way, and to everyone who taught me, offered advice, sent templates, shared exercises or helped in any other way.

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About this handbook

Linear Algebra for Engineering

Written and edited by Chen Gadi.

English translation prepared with assistance from ChatGPT.

NOTE

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This handbook develops tutorial notes written across previous academic years into a continuous study resource for engineering students. Questions from examinations written by Chen Gadi are explicitly labelled.

The English edition also supports the international engineering program. Its topic order may differ from the lecture sequence. The current syllabus and course announcements determine examinable material, submission rules and assessment.

Thank you to everyone who contributed comments, corrections, explanations, templates, exercises and ideas over the years. The cover illustration was created with an AI tool; the mathematical diagrams in the text use specified calculations.

Use the website system for corrections, questions and messages:

chengadi.science

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Study-book edition | Hebrew and English | 13 tutorials and a logic appendix

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How to Use These Notes

MOTIVATION AND GUIDING IDEA

Linear algebra provides a language for describing systems: constraints, signals, images, computational models, and maps between vector spaces. These notes are intended for engineering students across disciplines. The engineering examples explain the meaning of the calculations; the definitions and proofs state the conditions under which those calculations are valid.

NOTE

This English edition adapts Chen Gadi's tutorial notes from previous years for supplementary use in the International Engineering Program. The order of the tutorials may differ from the current lecture sequence. Follow the current syllabus, lectures, and official course announcements for the required material and course policies.

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A Study Routine

1. **Before the tutorial:** Review the definitions and basic examples from the lectures. State the given data, the space in which you are working, and the underlying field.
2. **While solving a problem:** Attempt the problem before reading its solution. Justify every step: a row operation, an application of a theorem, or a choice of basis.
3. **After the calculation:** Check the result by substitution. Check dimensions, boundary cases, and the geometric interpretation. A diagram can support understanding, but it does not replace a proof.
4. **At the end of a topic:** Summarize a definition, a central theorem, an example, and a counterexample. Distinguish necessary conditions from sufficient conditions.

The Boxes in These Notes

DEFINITION

Formal definition: Concepts, notation, and conditions for their use. Read the assumptions as carefully as the conclusion.

RECALL AND TOOLS

Lecture reminder: Facts and tools used in the solutions.

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NOTE

Mathematical note: A clarification of an assumption, a warning about a common error, or an explanation of a boundary case.

EXAMPLE

Example: A short illustration of a concept. A solved exercise appears in a separate numbered box, without an individual descriptive title, with the question and solution clearly separated.

NOTATION AND CONVENTIONS

An exercise number gives its tutorial and its position within that tutorial. For example, 3.7 is the seventh exercise in Tutorial 3. Exercise numbering is independent of definition and example numbering. The **FOUNDATION** tag marks practice with definitions and calculations. The **CHALLENGE** tag marks a problem that combines ideas or calls for a proof. The **EXAM QUESTION** tag identifies a question from an examination written by Chen Gadi. Each tutorial includes a conceptual introduction, notation and tools, followed at its end by two additional exercises at different difficulty levels, with complete solutions.

The Website and Course Announcements

Study materials and updates are available through chengadi.science. Send messages, logistics requests, and questions about the material through the website's system. The detailed communication guidelines and group logistics appear at the beginning of Tutorial 1, in the section entitled ``Course Logistics.'' The website is also available at:

`https://chengadi.science`

The official syllabus and course announcements determine the required material, submission deadlines, assessment rules, and office hours. The accompanying homework booklet retains the original distinction between questions marked for submission and questions for independent practice. Check the website to find out which questions are actually assigned for submission in the current course.

Notation and Conventions

- $\mathbb{F}$ denotes a field; $\mathbb{R}$ and $\mathbb{C}$ denote the real and complex fields. State any field assumption required by a claim.
- A basis is an ordered list. Column j of a representing matrix contains the coordinates of the image of the jth basis vector.
- In a complex inner product space, the inner product is **linear in the first argument** and conjugate-linear in the second. In particular, the standard inner product is

$$\langle x, y \rangle = \sum_j x_j \overline{y_j}.$$

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- For $A \in M_n(\mathbb{F})$, our characteristic polynomial convention is

$$p_A(t) = \det(tI - A).$$

  The alternative $\det(A - tI)$ may also be used, provided that the factor $(-1)^n$ is tracked consistently.

NOTE

These notes include solutions and supplementary explanations for learning. Errors may remain despite review. Use the website to report corrections or make suggestions. This booklet does not establish new course policies.

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TUTORIAL 1

Complex Numbers and Induction

1.1 Course Logistics

NOTE

These logistics are adapted from the teaching arrangements accompanying the original tutorial notes. They provide context and study guidance; the current International Engineering syllabus and official announcements determine the actual class arrangements, submission deadlines, office hours, and assessment rules. In particular, WeBWorK is optional practice and contributes neither to the grade nor to a protective component in the international course.

1.1.1 Access to Tutorials

The original tutorial arrangements provide for teaching over Zoom and for recordings, to accommodate the constraints of the period. The way links and recordings are made available may differ between staff members; they will usually be found on Moodle. Technical problems may prevent a tutorial from being made available. The course team will try to avoid such problems, but students remain responsible for catching up on any material missed because of a technical failure. Follow the current course announcements for the arrangements that apply to your group.

1.1.2 Homework and Assessment

- In the original homework format, each weekly sheet has two parts. The first contains two questions marked for submission, to be solved independently and submitted for grading. The second contains additional questions for independent practice, which are not submitted. Full solutions to some of these questions are provided so that you can assess your own work. Check the current assignment instructions for the questions you must actually submit.
- The homework is based on material from the lectures and/or tutorials.
- Submitting work gives you professional feedback. The additional questions help you consolidate the material and practise solving relevant problems. You are strongly encouraged to attempt as many questions as possible.
- The prior-year publication routine was Tuesday morning, with submissions accepted until Saturday evening of the following week (described in the original arrangements as a 12-day period). The current course announcements determine the deadlines for the international program.
- The original course also issued a WeBWorK sheet approximately every week or at a longer interval, open for two weeks. International Engineering students may use available WeBWorK questions as optional practice. The material may be in Hebrew; WeBWorK scores do not contribute to the international course grade or to any protective component.

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- The final assessment rules are stated in the current syllabus. Homework instructions appear in the relevant course section. Direct questions about grades and examinations to the course lecturer.

1.1.3 Office Hours

- Office hours with the teaching assistant are intended for questions arising from the tutorial material and homework.
- The teaching assistant sets the time, location, and format of office hours. Office hours are not an additional tutorial and do not replace independent study.

Procedures may change; updates will be announced when necessary.

1.1.4 Study Recommendations

Mathematics is a language, and learning it requires practice. This course is an integral (and differential) part of the mathematical foundations that support your later academic studies. The following suggestions are intended to help you learn the material and manage the course.
Scope of this advice: The recommendations in this section do not make the author responsible for readers' study decisions, understanding, or application of the material. Each reader remains responsible for their own learning and use of the information.

- Attempt all questions marked for submission and all practice questions; they have been selected carefully.
- If you fall slightly behind, use the recordings and tutorials to catch up, and check that you can solve the homework questions on the same topic.
- If you fall substantially behind, prioritize the theoretical foundations and basic practice questions, and try to rejoin the class's pace as soon as possible.
- Consider making a ``three magic sheets'' summary: one sheet for definitions, one for propositions and theorems, and one for useful techniques.
- A competitive atmosphere does not help the group or the individual. Help one another. Group learning and support for classmates can strengthen both learning and resilience.

1.1.5 Clarifications, Personal Recommendations, and Expectations

- Send messages and questions through the website system at chengadi.science.

1. Logistics requests should include your full name, student ID, and course name. Requests lacking these details will not be processed under the original communication guidelines. If you have received no response within three days, I may have missed your request; send a reminder through the system and include or quote the original request.
2. Questions about the material: in previous years and semesters, I received a very large number of questions (at least 20 requests per day). Questions may be sent through the website system, subject to the following guidelines:
   (a) First make an independent effort to understand the issue, using notes, books, the internet, and tools such as ChatGPT. Check mathematical claims independently.

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(b) Keep your message short and specific. Include the course name and a question that can be answered in one or two lines. Attach an image of the relevant material where possible, rather than referring only to a particular theorem.

(c) Do not send requests such as ``Is my method correct?'' or ``Check this part.'' I will not grade homework solutions through messages. You can discuss an approach during office hours or obtain feedback through the homework grader.

(d) You may ask for clarification of the wording of a homework question.

(e) I will try to answer within a few days. Messages that are too long, or that require more than a few minutes to read and answer, may not receive a response. Thank you for your understanding.

- I may upload videos of myself solving problems on YouTube. These are an additional resource for students, rather than part of my formal responsibilities; please do not send requests about their production or availability.
- During tutorials, I will not summarize all the theorems and conclusions covered in lectures. I recommend preparing your own summary. Tutorials are intended to practise the lecture material and work on somewhat more demanding questions that encourage thought and independent learning.
- I will also make sure to include basic exercises.
- Time will be set aside for questions during class. I will not repeatedly answer questions that have already been addressed.
- **The lectures determine the required material.**
  It is not possible to cover every topic in the tutorials or practise every type of question on each topic.
- I strongly recommend exploring the material through the resources available to you: books, online sources, earlier summaries (for example, materials from the students' association), office hours, and similar resources. **Each student is responsible for their own learning.**
- Tutorial pace and presentation: I will try to work at a reasonable pace, write clearly, and use headings. Please let me know if the presentation needs adjustment.
- WeBWorK is optional and recommended as additional practice with feedback. These questions were not written by me. **For the International Engineering Program, WeBWorK does not count toward the final grade and does not provide protective credit.**
- I will try to provide examples, proofs, and counterexamples for each topic.
- I will conduct an anonymous mid-semester teaching survey, and I welcome feedback.
- I will take questions at the designated times rather than in the middle of solving an exercise.
- I will try to upload typed tutorial sheets before the tutorials, but they may contain errors or imprecise statements.
  The material presented in the lectures, both in writing and orally, remains authoritative.
- Come prepared: review the definitions and material covered in the lectures before each tutorial.

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MOTIVATION AND GUIDING IDEA

Cartesian form is well suited to addition and comparison of components; polar form is well suited to multiplication, powers, and roots. Before calculating, choose a representation that fits the requested operation. For induction, first choose the precise statement and the first value for which it is supposed to hold.

NOTATION AND CONVENTIONS

For $z = x + iy \in \mathbb{C}$, the symbols $\operatorname{Re} z = x$ and $\operatorname{Im} z = y$ denote real numbers; in particular, the imaginary part is y, not iy. Both z^* and $\overline{z}$ denote the conjugate $x - iy$. The modulus $|z|$ is a nonnegative real number. When $z \neq 0$, $\arg z$ refers to an argument modulo integer multiples of 2π, whereas $\operatorname{Arg} z \in [0, 2\pi)$ is the fixed choice in these notes. The notation $a \mid b$ means that $b = am$ for some $m \in \mathbb{Z}$. The expression $P(n) \Rightarrow P(n+1)$ is a logical implication; it does not by itself establish the base case.

DEFINITION 1.1

Existence, uniqueness, and conditions of use. Every $z \in \mathbb{C}$ has a unique representation $z = x + iy$ with $x, y \in \mathbb{R}$. Thus two complex numbers are equal if and only if their real parts are equal and their imaginary parts are equal. For $z \neq 0$, its inverse exists and is unique, since $zz^* = |z|^2 > 0$; check this condition before dividing by z. If $r > 0$ and $\theta \in \mathbb{R}$, de Moivre's formula gives

$$(re^{i\theta})^n = r^n e^{in\theta} \qquad (n \in \mathbb{Z}).$$

Negative powers require $r > 0$. For nonnegative integer powers, the formula also holds when $r = 0$, except for 0^0, which is not defined here. When finding nth roots of a nonzero number, for $n \geq 1$, list all n roots rather than selecting a single angle.

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How to use these notes

These notes are intended for students in all engineering disciplines. They combine definitions, guiding ideas, examples, and detailed solutions.

MOTIVATION AND GUIDING IDEA

Objectives. We will convert between Cartesian and polar forms, calculate powers and roots, and interpret operations in the complex plane. We will then formulate an induction proof, stating the domain on which its claim holds.

MOTIVATION AND GUIDING IDEA

A working habit. In every solution, state what is given, what must be established, which definition or result you use, and why every operation is permitted. Correct calculations and clear explanations complement one another.

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1.2 Complex numbers: Cartesian form

DEFINITION 1.2

The imaginary unit i satisfies $i^2 = -1$.
A complex number has the form $z = x + iy$, where $x, y \in \mathbb{R}$.

1. If $\mathrm{Re}(z) = x = 0$, then z is purely imaginary.
2. If $\mathrm{Im}(z) = y = 0$, then z is real.

EXERCISE 1.1

Solve $z^2 - 4z + 13 = 0$ over $\mathbb{C}$. Factor the polynomial into linear factors over $\mathbb{C}$, and determine whether such a factorization exists over $\mathbb{R}$.

WORKED SOLUTION

The discriminant is $\Delta = (-4)^2 - 4 \cdot 1 \cdot 13 = 16 - 52 = -36$. Therefore

$$z_{1,2} = \frac{4 \pm \sqrt{-36}}{2} = \frac{4 \pm 6i}{2} = 2 \pm 3i.$$

Consequently,

$$z^2 - 4z + 13 = (z - (2 + 3i))(z - (2 - 3i)).$$

This is a factorization over $\mathbb{C}$. Over $\mathbb{R}$, the discriminant is negative, so there are no real roots and no factorization into two real linear factors.

DEFINITION 1.3

- **Complex conjugate:** for $z = x + iy$, write $z^* := x - iy$.
- **Modulus:** $|z| = \sqrt{x^2 + y^2} \in \mathbb{R}$.
- **Multiplicative inverse:** for $z \neq 0$, $z^{-1} = \dfrac{z^*}{|z|^2}$.

EXERCISE 1.2

Prove the following statements. **(a)** $\mathrm{Re}\, z = \dfrac{z + z^*}{2}$ and $\mathrm{Im}\, z = \dfrac{z - z^*}{2i}$. **(b)** $|z| \geq \mathrm{Im}\, z$ and $|z| \geq \mathrm{Re}\, z$ for every $z \in \mathbb{C}$. **(c)** (At home) $(z^*)^* = z$. **(d)** (At home) If z is real, then $z^* = z$.

WORKED SOLUTION

Write $z = x + iy$. Then

$$\frac{z + z^*}{2} = x = \mathrm{Re}\, z, \qquad \frac{z - z^*}{2i} = y = \mathrm{Im}\, z,$$

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EXERCISE 1.2 CONTINUED

and

$$|z| = \sqrt{x^2 + y^2} \geq |x| \geq x = \operatorname{Re} z, \qquad |z| \geq |y| \geq y = \operatorname{Im} z.$$

(c) $(z^*)^* = (x - iy)^* = x + iy = z$.
(d) If z is real, then $y = 0$, and hence $z^* = x - 0i = x = z$.

1.2.1 Useful properties

1. $|z| = |z^*|$.
2. $z \cdot z^* = |z|^2$ (multiplication by the conjugate gives a real number).
3. $|zw| = |z|\,|w|$.
4. $(zw)^* = z^* w^*$.

EXERCISE 1.3

For $z \neq 0$, prove that $(z^{-1})^* = (z^*)^{-1}$.

WORKED SOLUTION

Since $z \neq 0$, we also have $z^* \neq 0$ and $|z|^2 > 0$. The inverse formula gives

$$(z^{-1})^* = \left(\frac{z^*}{|z|^2}\right)^* = \frac{z}{|z|^2}, \qquad (z^*)^{-1} = \frac{(z^*)^*}{|z^*|^2} = \frac{z}{|z|^2}.$$

Both sides equal the same expression, so $(z^{-1})^* = (z^*)^{-1}$.

1.3 Polar form

1.3.1 Converting between Cartesian and polar forms

DEFINITION 1.4

For every $z = x + iy \neq 0$, there are $r = |z| > 0$ and an angle θ such that

$$z = r(\cos\theta + i\sin\theta) = re^{i\theta}, \qquad x = r\cos\theta, \quad y = r\sin\theta.$$

Conversely, $r = \sqrt{x^2 + y^2}$. If $x \neq 0$, then $\tan\theta = y/x$, but this ratio alone does not determine the quadrant. **Remarks:**

- The angle is determined up to adding $2\pi k$, where $k \in \mathbb{Z}$. For $z = 0$, we have $r = 0$, and no argument is defined.
- All angles in exponents $e^{i\theta}$ are measured in radians. Degrees are used only for numerical illustration.

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EXERCISE 1.4

Convert $z = 3 + 3i$ and $w = -4 + 4i$ to polar form. Then write $a = 6e^{i\pi/3}$ in Cartesian form. For each polar representation, choose an angle in $[0, 2\pi)$.

WORKED SOLUTION

For z, we have $|z| = \sqrt{3^2 + 3^2} = 3\sqrt{2}$. Its components are equal and positive, so the angle is $\pi/4$:

$$z = 3\sqrt{2}e^{i\pi/4}.$$

For w, we have $|w| = \sqrt{(-4)^2 + 4^2} = 4\sqrt{2}$. The point lies in the second quadrant, and the absolute values of its components have ratio 1. Therefore

$$w = 4\sqrt{2}e^{3i\pi/4}.$$

Finally, use Euler's identity and the trigonometric values at $\pi/3$:

$$a = 6\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right) = 6\left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right) = 3 + 3\sqrt{3}i.$$

The angles are in radians. Both given complex numbers have positive modulus, so the undefined argument at zero causes no difficulty.

1.3.2 Arithmetic operations

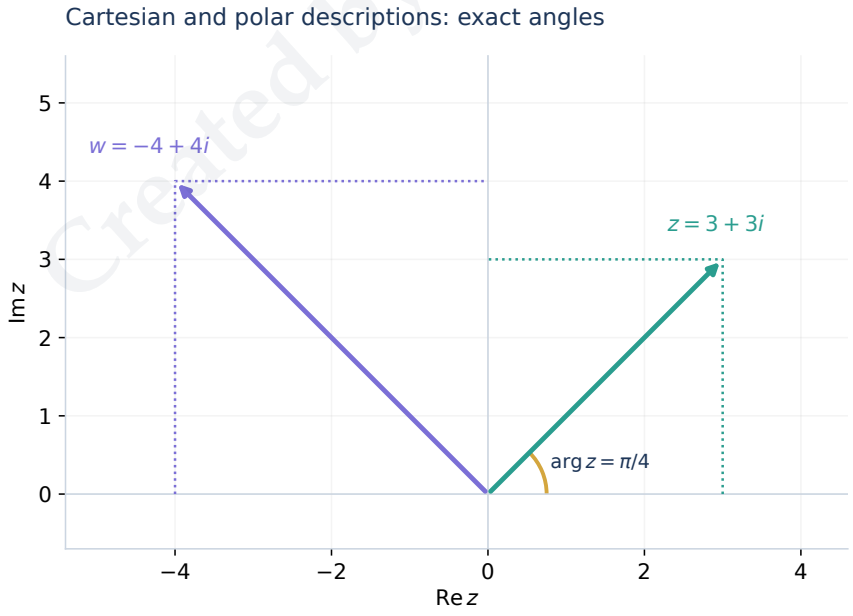


Figure 1.1. Cartesian and polar forms in the same plane: the modulus is the distance from the origin, and the argument is the direction angle.

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DEFINITION 1.5

Multiplication: $(r_1 e^{i\theta_1})(r_2 e^{i\theta_2}) = r_1 r_2\, e^{i(\theta_1+\theta_2)}$.

Division (when the denominator is nonzero): $\dfrac{r_1 e^{i\theta_1}}{r_2 e^{i\theta_2}} = \dfrac{r_1}{r_2}\, e^{i(\theta_1-\theta_2)}$.

Conjugation: $(re^{i\theta})^* = r\, e^{-i\theta}$.

Periodicity: $re^{i(\theta+2\pi k)} = re^{i\theta}$, $re^{i(\theta+\pi)} = -re^{i\theta}$.

EXERCISE 1.5

Let $z = 1 + i$ and $w = -4 + 4i$. Calculate zw and w/z in two ways: using polar form and using Cartesian arithmetic. Check that the resulting moduli agree with the multiplication and division formulas.

WORKED SOLUTION

Write $z = \sqrt{2}e^{i\pi/4}$ and $w = 4\sqrt{2}e^{3i\pi/4}$. Then

$$zw = 8e^{i\pi} = -8, \qquad \frac{w}{z} = 4e^{i\pi/2} = 4i.$$

Cartesian arithmetic gives

$$(1+i)(-4+4i) = -4 + 4i - 4i + 4i^2 = -8,$$

$$\frac{-4+4i}{1+i} = \frac{(-4+4i)(1-i)}{(1+i)(1-i)} = \frac{-4+8i+4}{2} = 4i.$$

Checking the moduli, $|z||w| = \sqrt{2}\cdot 4\sqrt{2} = 8 = |-8|$, and $|w|/|z| = 4 = |4i|$. The two calculations agree.

1.3.3 Powers

EXERCISE 1.6

Calculate $(1+i)^{12}$ using polar form. Verify your answer using $(1+i)^4$.

WORKED SOLUTION

Since $1 + i = \sqrt{2}e^{i\pi/4}$,

$$(1+i)^{12} = (\sqrt{2})^{12} e^{12i\pi/4} = 2^6 e^{3i\pi} = -64.$$

For verification, $(1+i)^2 = 2i$, so $(1+i)^4 = (2i)^2 = -4$. Thus $(1+i)^{12} = ((1+i)^4)^3 = (-4)^3 = -64$.

1.3.4 Complex roots

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DEFINITION 1.6

Let $0 \neq z \in \mathbb{C}$, and let $n \geq 1$ be an integer. The equation $w^n = z = re^{i\theta}$ has n distinct roots:

$$w_j = \sqrt[n]{r}\, e^{i\frac{\theta+2\pi j}{n}}, \quad j = 0, 1, \ldots, n-1.$$

Geometric meaning: the roots lie on the circle of radius $\sqrt[n]{r}$ at equal angular intervals of $2\pi/n$. For $n \geq 3$, they are the vertices of a regular polygon; for $n = 2$, they are two opposite points, and for $n = 1$, there is one point.

EXERCISE 1.7

Find all solutions of $z^4 = -i$.

WORKED SOLUTION

Write $-i = e^{-i\pi/2}$. Therefore

$$z_j = e^{i\frac{-\pi/2+2\pi j}{4}} = e^{i\frac{\pi(4j-1)}{8}}, \quad j = 0, 1, 2, 3,$$

so the angles are $-\pi/8,\ 3\pi/8,\ 7\pi/8,\ 11\pi/8$, all at radius 1.

EXERCISE 1.8

Let $n \geq 2$. Find all nth roots of unity and prove that their sum is zero. What happens when $n = 1$?

WORKED SOLUTION

The roots are $w_j = e^{i\frac{2\pi j}{n}}$, $j = 0, \ldots, n-1$. Put $w = e^{i\frac{2\pi}{n}}$. Then $w_j = w^j$, and therefore

$$\sum_{j=0}^{n-1} w^j = \frac{w^n - 1}{w - 1} = 0,$$

because $w^n = 1$ and $w \neq 1$ when $n \geq 2$. For $n = 1$, the only root is 1, so the sum is 1, rather than zero.

EXERCISE 1.9

(At home) Consider $a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0 = 0$, with real coefficients. Prove that if z is a solution, then z^* is also a solution.

WORKED SOLUTION

Use the identities $(z_1 + \cdots + z_m)^* = z_1^* + \cdots + z_m^*$ and $(z_1 \cdots z_m)^* = z_1^* \cdots z_m^*$, together with $a^* = a$ for real a. If $a_0 + a_1 z + \cdots + a_n z^n = 0$, conjugating gives $a_0 + a_1 z^* + \cdots + a_n (z^*)^n = 0$. Thus z^* is a solution.

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1.4 Further properties of complex numbers

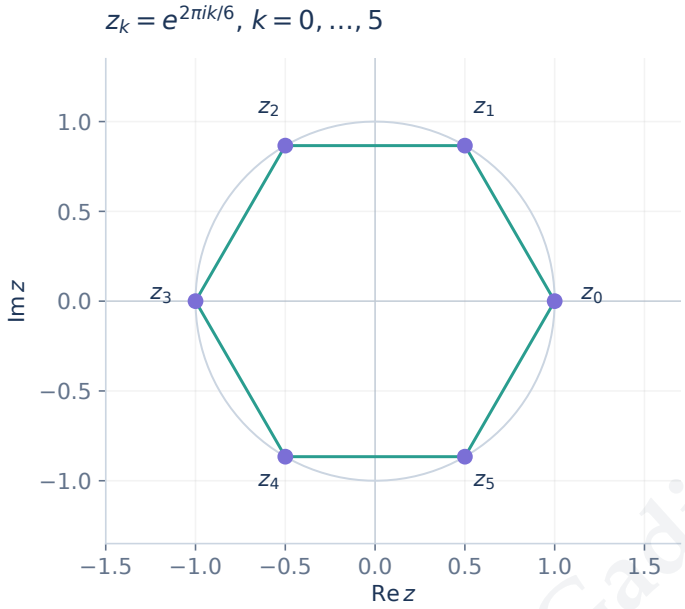


Figure 1.2. The roots of unity are equally spaced around the unit circle; their sum is zero whenever their order is greater than one.

1.4.1 Multiplication by i

Multiplication by i rotates a complex number through $\frac{\pi}{2}$ counterclockwise.

EXERCISE 1.10

Prove that multiplication by i gives a rotation through $\frac{\pi}{2}$.

WORKED SOLUTION

Cartesian representation:
Let $z = a + bi$. Multiplication by i gives

$$z \cdot i = (a + bi)i = -b + ai.$$

The vector (a, b) becomes $(-b, a)$. This is exactly a counterclockwise rotation through $\pi/2$ about the origin: the new x coordinate is $-b$, and the new y coordinate is a.
Polar representation:
For $z \neq 0$, write $z = re^{i\theta}$, where $r = |z|$ and θ is an angle from the positive x axis to z. Then

$$z \cdot i = re^{i\theta} \cdot i = re^{i\theta} \cdot e^{i\frac{\pi}{2}} = re^{i(\theta+\frac{\pi}{2})}.$$

Multiplication by i adds $\frac{\pi}{2}$ to θ, so z rotates through $\frac{\pi}{2}$ about the origin, as required. The origin is fixed, since $0 \cdot i = 0$.

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1.4.2 Division in polar form

DEFINITION 1.7

If $z_1 = r_1 e^{i\theta_1}$ and $z_2 = r_2 e^{i\theta_2}$, with $r_2 \neq 0$, then

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} e^{i(\theta_1 - \theta_2)}.$$

EXERCISE 1.11

Calculate $\frac{z_1}{z_2}$ for $z_1 = 4e^{i\frac{\pi}{3}}$ and $z_2 = 2e^{i\frac{\pi}{6}}$.

WORKED SOLUTION

We obtain

$$\frac{z_1}{z_2} = \frac{4}{2} e^{i\left(\frac{\pi}{3} - \frac{\pi}{6}\right)} = 2e^{i\frac{\pi}{6}}.$$

EXERCISE 1.12

Let $z_1, z_2 \neq 0$. Prove that an argument of z_1/z_2 is the difference of arguments of z_1 and z_2, up to an integer multiple of 2π.

WORKED SOLUTION

Write $z_1 = r_1 e^{i\theta_1}$ and $z_2 = r_2 e^{i\theta_2}$, where $r_1, r_2 > 0$. Division gives

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} e^{i(\theta_1 - \theta_2)}.$$

Thus $\theta_1 - \theta_2$ is an argument of the quotient. If an argument in a prescribed interval is required, add a suitable integer multiple of 2π.

1.4.3 Choosing an argument in a fixed interval

In these notes, $\mathrm{Arg}(z)$ denotes the unique argument in $[0, 2\pi)$, for $z \neq 0$.

RECALL AND TOOLS

The equalities $\mathrm{Arg}(z_1 \cdot z_2) = \mathrm{Arg}(z_1) + \mathrm{Arg}(z_2)$ and $\mathrm{Arg}\left(\frac{z_1}{z_2}\right) = \mathrm{Arg}(z_1) - \mathrm{Arg}(z_2)$ do not always hold, because the expression on the right may lie outside $[0, 2\pi)$.

EXERCISE 1.13

Calculate $\mathrm{Arg}(z_1 \cdot z_2)$ for $z_1 = e^{i\frac{3\pi}{4}}$ and $z_2 = e^{i\frac{5\pi}{4}}$.

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EXERCISE 1.13 CONTINUED

WORKED SOLUTION

First calculate the product itself:

$$z_1 z_2 = e^{i3\pi/4} e^{i5\pi/4} = e^{i2\pi} = 1.$$

The angle 2π is an argument of 1, but it does not belong to $[0, 2\pi)$. Hence

$$\text{Arg}(z_1 z_2) = \text{Arg}(1) = 0, \qquad \text{Arg}(z_1) + \text{Arg}(z_2) = 2\pi.$$

These two values are unequal, but congruent modulo 2π.

1.4.4 Calculating arguments and identifying quadrants

DEFINITION 1.8

For a complex number $z = a + bi \neq 0$ with $a \neq 0$, every argument θ satisfies

$$\tan(\theta) = \frac{b}{a}.$$

To determine θ precisely, take the quadrant into account:

- If $a > 0, b > 0$, then $\theta = \arctan\left(\frac{b}{a}\right)$.
- If $a < 0, b > 0$, add π to obtain an angle in the second quadrant: $\theta = \pi + \arctan\left(\frac{b}{a}\right)$.
- If $a < 0, b < 0$, add π for the third quadrant: $\theta = \pi + \arctan\left(\frac{b}{a}\right)$.
- If $a > 0, b < 0$, add 2π to obtain a positive angle in the fourth quadrant: $\theta = 2\pi + \arctan\left(\frac{b}{a}\right)$.

If $b = 0, a > 0$, then $\text{Arg}(z) = 0$; if $b = 0, a < 0$, then $\text{Arg}(z) = \pi$. If $a = 0$, the angle depends only on b:

- If $b > 0$, then $\theta = \frac{\pi}{2}$.
- If $b < 0$, then $\theta = \frac{3\pi}{2}$.

EXERCISE 1.14

Find the angle θ for $z = -1 + \sqrt{3}i$.

WORKED SOLUTION

Here $a = -1, b = \sqrt{3}$. First calculate

$$\tan(\theta) = \frac{\sqrt{3}}{-1} = -\sqrt{3}.$$

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EXERCISE 1.14 CONTINUED

Since $a < 0$ and $b > 0$, the number lies in the second quadrant. Therefore

$$\theta = \pi - \frac{\pi}{3} = \frac{2\pi}{3}.$$

EXERCISE 1.15

Find the angle θ for $z = -5 - 5i$.

WORKED SOLUTION

Here $a = -5, b = -5$. First calculate

$$\tan(\theta) = \frac{b}{a} = \frac{-5}{-5} = 1.$$

Since $a < 0, b < 0$, the number lies in the third quadrant. Therefore

$$\theta = \pi + \arctan(1) = \pi + \frac{\pi}{4} = \frac{5\pi}{4}.$$

1.5 The induction principle

DEFINITION 1.9

The principle of mathematical (weak) induction:
Let n_0 be a nonnegative integer, and let $P(n)$ be a statement defined for every integer $n \geq n_0$. Suppose that:

1. **Base case:** $P(n_0)$ is true.
2. **Inductive step:** for every integer $n \geq n_0$, the assumption $P(n)$ implies $P(n+1)$.

Then $P(n)$ is true for every integer $n \geq n_0$.

EXERCISE 1.16

Prove by induction that, for every $n \geq 1$,

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

WORKED SOLUTION

The statement: $P(n) : \sum_{i=1}^{n} i = \frac{n(n+1)}{2}$.
Base case: for $n = 1$, the left side is 1, and the right side is $\frac{1 \cdot 2}{2} = 1$.
Induction hypothesis: suppose $\sum_{i=1}^{k} i = \frac{k(k+1)}{2}$ for some $k \geq 1$.

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EXERCISE 1.16 CONTINUED

Inductive step:

$$\sum_{i=1}^{k+1} i = \left(\sum_{i=1}^{k} i\right) + (k+1) = \frac{k(k+1)}{2} + (k+1) = \frac{(k+1)(k+2)}{2}.$$

This is the formula for $k+1$.
Conclusion: the statement holds for every $n \geq 1$.

EXERCISE 1.17

Prove that $7 \mid (8^n - 1)$ for every $n \geq 1$.

WORKED SOLUTION

The statement: $P(n):\ 7 \mid (8^n - 1)$.
Base case: for $n = 1$, $8^1 - 1 = 7$, and indeed $7 \mid 7$.
Induction hypothesis: suppose $7 \mid (8^k - 1)$ for some $k \geq 1$. Thus there is $m \in \mathbb{Z}$ such that $8^k - 1 = 7m$.
Inductive step:

$$8^{k+1} - 1 = 8 \cdot 8^k - 1 = 8(8^k - 1) + 7.$$

Using $8^k - 1 = 7m$, we get

$$8^{k+1} - 1 = 8 \cdot 7m + 7 = 7(8m + 1),$$

so $7 \mid (8^{k+1} - 1)$.
Conclusion: $7 \mid (8^n - 1)$ for every $n \geq 1$.
A shorter observation: $8 \equiv 1 \pmod 7 \Rightarrow 8^n \equiv 1 \pmod 7$.

EXERCISE 1.18

Prove that, for every $n \geq 4$,

$$2^n \geq n^2.$$

WORKED SOLUTION

We use induction on n.
Base case: for $n = 4$, $2^4 = 16 = 4^2$.
Induction hypothesis: suppose $2^k \geq k^2$ for some $k \geq 4$.
Inductive step:

$$2^{k+1} = 2 \cdot 2^k \ \geq\ 2k^2.$$

It suffices to show $2k^2 \geq (k+1)^2 \iff k^2 - 2k - 1 \geq 0$. For $k \geq 4$, $k^2 - 2k - 1 > 0$, so $2^{k+1} \geq (k+1)^2$.
By induction, $2^n \geq n^2$ for every $n \geq 4$.

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1.6 Supplementary exercises: geometry and signal calibration

EXERCISE 1.19

Find and describe geometrically all $z \in \mathbb{C}$ satisfying $|z-1| = |z+1|$. Which of them also satisfy $|z| = 2$?

WORKED SOLUTION

Write $z = x + iy$. Both sides are nonnegative, so squaring preserves the equality:

$$|z-1|^2 = (x-1)^2 + y^2, \qquad |z+1|^2 = (x+1)^2 + y^2.$$

Therefore

$$(x-1)^2 + y^2 = (x+1)^2 + y^2 \iff x^2 - 2x + 1 = x^2 + 2x + 1 \iff x = 0.$$

This is the imaginary axis: the set of points equidistant from 1 and -1. The additional condition gives $y^2 = 4$, so the intersection consists of $z = 2i$ and $z = -2i$.

EXERCISE 1.20

In a simple model of a signal with two components, a calibration system applies $T(z) = (1+i)z$. How does this change the modulus and angle? Calculate $T(1-i)$, and find an input whose output is $2i$.

WORKED SOLUTION

Since $1 + i = \sqrt{2}e^{i\pi/4}$, every $z = re^{i\theta} \neq 0$ satisfies $T(z) = \sqrt{2}\, r\, e^{i(\theta+\pi/4)}$. Thus the modulus is multiplied by $\sqrt{2}$, and the direction rotates counterclockwise through $\pi/4$. We also have $T(0) = 0$. Direct calculation gives $T(1-i) = (1+i)(1-i) = 1 - i^2 = 2$. To obtain $2i$, solve $(1+i)z = 2i$:

$$z = \frac{2i}{1+i} = \frac{2i(1-i)}{(1+i)(1-i)} = \frac{2i+2}{2} = 1 + i.$$

Check: $(1+i)^2 = 1 + 2i + i^2 = 2i$.

Additional practice

EXERCISE 1.21 FOUNDATION

Let $z = -3 + 4i$. Calculate $|z|$, z^*, and z^{-1}. Solve $zw = 5 + 10i$, and verify the solution by direct multiplication.

WORKED SOLUTION

We have $|z| = \sqrt{9+16} = 5$, so $z^* = -3 - 4i$ and $z^{-1} = (-3-4i)/25$. Since $z \neq 0$, division by it is permitted and gives a unique solution:

$$w = \frac{5+10i}{-3+4i} = \frac{(5+10i)(-3-4i)}{25} = \frac{25-50i}{25} = 1 - 2i.$$

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EXERCISE 1.21 FOUNDATION CONTINUED

For verification, $(-3+4i)(1-2i) = -3+6i+4i-8i^2 = 5+10i$. Uniqueness follows because multiplication by a nonzero number is invertible.

EXERCISE 1.22 CHALLENGE

Let $z \in \mathbb{C} \setminus \{-1\}$. Prove that $(z-1)/(z+1)$ is purely imaginary if and only if $|z| = 1$. Then solve $(z-1)/(z+1) = 2i$, and check that the solution satisfies the geometric condition.

WORKED SOLUTION

Multiply the numerator and denominator by the conjugate of the denominator. Since $z \neq -1$, the denominator $|z+1|^2$ is positive, and

$$\frac{z-1}{z+1} = \frac{(z-1)(z^*+1)}{|z+1|^2} = \frac{|z|^2 - 1 + z - z^*}{|z+1|^2}.$$

The expression $z - z^*$ is purely imaginary. Hence

$$\operatorname{Re}\left(\frac{z-1}{z+1}\right) = \frac{|z|^2-1}{|z+1|^2}.$$

The real part is zero if and only if $|z|^2 = 1$, equivalent to $|z| = 1$ because the modulus is nonnegative. This proves both directions and includes $z = 1$, whose quotient is 0. In the given equation, multiplying by $z+1 \neq 0$ gives $z - 1 = 2iz + 2i$. Thus

$$(1-2i)z = 1+2i, \qquad z = \frac{1+2i}{1-2i} = \frac{(1+2i)^2}{5} = \frac{-3+4i}{5}.$$

This number differs from -1 and has modulus $\sqrt{9+16}/5 = 1$. Moreover,

$$\frac{z-1}{z+1} = \frac{-8+4i}{2+4i} = 2i.$$

It is therefore the unique solution and lies on the unit circle.

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TUTORIAL 2

Proof Methods, Sets, and Fields

MOTIVATION AND GUIDING IDEA

A proof must work within the given setting: equality of sets is proved by two inclusions, and a field identity is proved from the field axioms. In a finite field, addition, subtraction, multiplication, and division by nonzero elements are permitted; numerical notation does not imply an ordering like that of $\mathbb{R}$.

NOTATION AND CONVENTIONS

The symbol $x \in A$ expresses membership of an element, whereas $A \subseteq B$ expresses inclusion of sets. The quantifiers $\forall$ and $\exists$ mean ``for every'' and ``there exists''; their order matters. The set $A \setminus B$ consists of elements of A absent from B, and $|A|$ is its cardinality when A is finite. For a field F, write $F^\times = F \setminus \{0\}$. The notation $a \equiv b \pmod{p}$ means $p \mid (a - b)$. For prime p, writing $\mathbb{F}_p = \{0, \dots, p-1\}$ uses representatives of residue classes. The symbol $\mathbb{F}_q$ denotes a field with q elements; when $q = p^n$ with $n > 1$, it must not be identified with arithmetic modulo q.

DEFINITION 2.1

Equality of sets and cancellation. For sets A, B, we have $A = B$ if and only if $A \subseteq B$ and $B \subseteq A$. To prove equality, take an arbitrary element from each side and prove that it belongs to the other side. In a field F, for $a, b, c \in F$ with $c \neq 0$,

$$ac = bc \quad \Longleftrightarrow \quad a = b,$$

because multiplication by c^{-1} is permitted. The condition $c \neq 0$ is essential. The characteristic of a field is the least positive integer p for which $p \cdot 1 = 0$, if such an integer exists; otherwise its characteristic is 0. The characteristic of a finite field is prime.

MOTIVATION AND GUIDING IDEA

Objectives. We will strengthen proofs by induction and proofs of equivalence, identify the field axioms, and calculate in finite fields. We will distinguish familiar real arithmetic from arithmetic adapted to the specified field.

The first part is a focused review of proof methods and the language of sets. The second part introduces fields and finite field arithmetic; the order used in class will follow the progress of the lectures.

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2.1 The induction principle

DEFINITION 2.2

The principle of mathematical (weak) induction:
Let n_0 be a nonnegative integer, and let $P(n)$ be a statement defined for every integer $n \geq n_0$. Suppose that:

1. **Base case:** $P(n_0)$ is true.
2. **Inductive step:** for every integer $n \geq n_0$, the assumption $P(n)$ implies $P(n+1)$.

Then $P(n)$ is true for every integer $n \geq n_0$.

EXERCISE 2.1

Prove by induction that, for every $n \geq 1$,

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

WORKED SOLUTION

The statement: $P(n)$: $\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$.
Base case: for $n = 1$, the left side is 1, and the right side is $\frac{1\cdot 2}{2} = 1$.
Induction hypothesis: suppose $\sum_{i=1}^{k} i = \frac{k(k+1)}{2}$ for some $k \geq 1$.
Inductive step:

$$\sum_{i=1}^{k+1} i = \left(\sum_{i=1}^{k} i\right) + (k+1) = \frac{k(k+1)}{2} + (k+1) = \frac{(k+1)(k+2)}{2}.$$

This is the formula for $k+1$.
Conclusion: the statement holds for every $n \geq 1$.

EXERCISE 2.2

Prove by induction that $13 \mid (14^n - 1)$ for every integer $n \geq 1$. Also explain how modular arithmetic gives a short verification of the claim.

WORKED SOLUTION

For the base case $n = 1$, $14 - 1 = 13$ is divisible by 13. Suppose that, for some $k \geq 1$, there is $m \in \mathbb{Z}$ such that $14^k - 1 = 13m$. Then

$$14^{k+1} - 1 = 14(14^k - 1) + 13 = 14 \cdot 13m + 13 = 13(14m + 1).$$

This expression is divisible by 13, completing the inductive step. Since the base case and inductive step hold, the claim is true for every $n \geq 1$. For an additional check, $14 \equiv 1 \pmod{13}$, so $14^n \equiv 1^n = 1 \pmod{13}$.

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EXERCISE 2.3

Prove that, for every $n \geq 4$,

$$2^n \geq n^2.$$

WORKED SOLUTION

We use induction on n.
Base case: for $n = 4$, $2^4 = 16 = 4^2$.
Induction hypothesis: suppose $2^k \geq k^2$ for some $k \geq 4$.
Inductive step:

$$2^{k+1} = 2 \cdot 2^k \;\geq\; 2k^2.$$

It suffices to show $2k^2 \geq (k+1)^2 \iff k^2 - 2k - 1 \geq 0$. For $k \geq 4$, $k^2 - 2k - 1 > 0$, so $2^{k+1} \geq (k+1)^2$.
By induction, $2^n \geq n^2$ for every $n \geq 4$.

2.2 Sets

DEFINITION 2.3

- **Inclusion:** $A \subseteq B$ means that every element of A also belongs to B.
- **Union:** $A \cup B = \{\, x \mid x \in A \text{ or } x \in B \,\}$.
- **Intersection:** $A \cap B = \{\, x \mid x \in A \text{ and } x \in B \,\}$.

$\Omega = \{1,2,3,4,5\}, \quad A = \{1,2,3\}, \quad B = \{2,3,4\}$

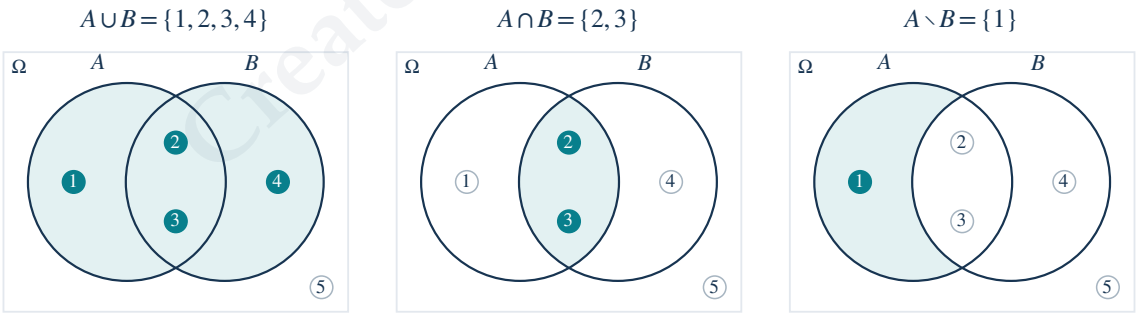


Figure 2.1. For $A = \{1,2,3\}$ and $B = \{2,3,4\}$ in $\Omega = \{1,2,3,4,5\}$, the shaded regions show $A \cup B$, $A \cap B$, and $A \setminus B$, respectively. Each numeral's position records its membership; 5 belongs to the universal set but to neither A nor B.

EXERCISE 2.4

Prove that $A \subseteq B$ if and only if $A \cup B = B$.

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EXERCISE 2.4 CONTINUED

WORKED SOLUTION

First direction: suppose $A \subseteq B$. The definition of union gives $A \cup B \subseteq B$, while $B \subseteq A \cup B$ always holds. Therefore $A \cup B = B$.
Second direction: suppose $A \cup B = B$. Let $x \in A$. Then $x \in A \cup B$, so $x \in B$. Therefore $A \subseteq B$.

EXERCISE 2.5

Prove that $A \subseteq B$ if and only if $A \cap B = A$.

WORKED SOLUTION

First direction: suppose $A \subseteq B$. If $x \in A \cap B$, then $x \in A$ and $x \in B$, so $A \cap B \subseteq A$. Conversely, if $x \in A$, then the inclusion gives $x \in B$. Thus $x \in A \cap B$, proving $A \subseteq A \cap B$. Hence $A \cap B = A$.

Second direction: suppose $A \cap B = A$. Let $x \in A$. Then $x \in A \cap B$, so $x \in B$. Therefore $A \subseteq B$.

2.3 Algebraic structure: fields

2.3.1 The field axioms

RECALL AND TOOLS

A field is a set F equipped with two operations, addition and multiplication. For all $a, b, c \in F$, the operations satisfy the following axioms. In particular, $0_{\mathbb{F}} \neq 1_{\mathbb{F}}$ is required:

- **Closure:** if $a, b \in F$, then $a +_{\mathbb{F}} b \in F$ and $a \times_{\mathbb{F}} b \in F$.
- **Commutativity:** $a +_{\mathbb{F}} b = b +_{\mathbb{F}} a$ and $a \times_{\mathbb{F}} b = b \times_{\mathbb{F}} a$.
- **Associativity:** $(a +_{\mathbb{F}} b) +_{\mathbb{F}} c = a +_{\mathbb{F}} (b +_{\mathbb{F}} c)$ and $(a \times_{\mathbb{F}} b) \times_{\mathbb{F}} c = a \times_{\mathbb{F}} (b \times_{\mathbb{F}} c)$.
- **Distributivity:** $a \times_{\mathbb{F}} (b +_{\mathbb{F}} c) = a \times_{\mathbb{F}} b +_{\mathbb{F}} a \times_{\mathbb{F}} c$ and $(a +_{\mathbb{F}} b) \times_{\mathbb{F}} c = a \times_{\mathbb{F}} c +_{\mathbb{F}} b \times_{\mathbb{F}} c$.
- **Identity elements:** there is $0_{\mathbb{F}} \in F$ such that $a +_{\mathbb{F}} 0_{\mathbb{F}} = a$ for every $a \in F$, and there is $1_{\mathbb{F}} \in F$ such that $a \times_{\mathbb{F}} 1_{\mathbb{F}} = a$ for every $a \in F$.
- **Additive and multiplicative inverses:** every $a \in F$ has an additive inverse $-a$ with $a +_{\mathbb{F}} (-a) = 0_{\mathbb{F}}$; every nonzero $a \in F$ has a multiplicative inverse a^{-1} with $a \times_{\mathbb{F}} a^{-1} = 1_{\mathbb{F}}$.

EXAMPLE 2.1

- **Examples:** $\mathbb{Q}$ (rational numbers), $\mathbb{R}$ (real numbers), and $\mathbb{C}$ (complex numbers) are fields with their usual addition and multiplication.
- **Nonexamples:** $\mathbb{Z}$ and $\mathbb{N}$ are not fields, because most of their elements have no multiplicative inverse; for example, 2 has no inverse in $\mathbb{Z}$.

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2.3.2 Notation

A general field is usually denoted by F (sometimes $\mathbb{F}$). When no particular field is intended, we write ``Let F be a field.''
Notation convention: when the field is clear from context, we omit its subscript from the operations and the elements $0, 1$.

EXERCISE 2.6

Let $a \in F$ with $a \neq 0_{\mathbb{F}}$. Prove that $(a^{-1})^{-1} = a$.

WORKED SOLUTION

Recall that every invertible field element has a unique multiplicative inverse, and in particular $a \times_{\mathbb{F}} a^{-1} = 1_{\mathbb{F}}$. Put $x = (a^{-1})^{-1}$ and calculate:

$$(a^{-1}) \times_{\mathbb{F}} x = 1_{\mathbb{F}} \quad / \times_{\mathbb{F}} a$$

$$a \times_{\mathbb{F}} (a^{-1}) \times_{\mathbb{F}} x = a \times_{\mathbb{F}} 1_{\mathbb{F}}$$

$$1_{\mathbb{F}} \times_{\mathbb{F}} x = a \qquad (*)$$

Thus $x = a$, that is, $(a^{-1})^{-1} = a$.
Remark: step $(*)$ uses $a \times_{\mathbb{F}} a^{-1} = 1_{\mathbb{F}}$.

EXERCISE 2.7

Let $a \in F$ with $a \neq 0_{\mathbb{F}}$. Prove that the multiplicative inverse of a is unique.

WORKED SOLUTION

Suppose $b \in F$ satisfies $a \times_{\mathbb{F}} b = 1_{\mathbb{F}}$. Multiply on the left by a^{-1}:

$$a \times_{\mathbb{F}} b = 1_{\mathbb{F}} \quad / \times_{\mathbb{F}} a^{-1}$$

$$(a^{-1}) \times_{\mathbb{F}} a \times_{\mathbb{F}} b = a^{-1} \times_{\mathbb{F}} 1_{\mathbb{F}}$$

$$1_{\mathbb{F}} \times_{\mathbb{F}} b = a^{-1} \qquad (*)$$

Hence $b = a^{-1}$, so the inverse of a is unique. *Remark:* again, step $(*)$ uses $(a^{-1}) \times_{\mathbb{F}} a = 1_{\mathbb{F}}$.

2.4 Further exercises on fields

EXERCISE 2.8

Let a, b be elements of a field $\mathbb{F}$. Prove that if $a \times_{\mathbb{F}} b = 0_{\mathbb{F}}$, then $a = 0_{\mathbb{F}}$ or $b = 0_{\mathbb{F}}$.

WORKED SOLUTION

We distinguish two cases according to b. **Case 1:** $b = 0_{\mathbb{F}}$, and the conclusion already holds. **Case 2:** $b \neq 0_{\mathbb{F}}$. Since $\mathbb{F}$ is a field, b has an inverse b^{-1}. Multiplying both sides by b^{-1} gives

$$a \times_{\mathbb{F}} b \times_{\mathbb{F}} b^{-1} = 0_{\mathbb{F}} \times_{\mathbb{F}} b^{-1} \Longrightarrow a \times_{\mathbb{F}} 1_{\mathbb{F}} = a = 0_{\mathbb{F}}$$

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PROPOSITION 2.1

A field has no nontrivial zero divisors.

EXERCISE 2.9

Let $\mathbb{F}$ be a field. Define a new structure $(F \times F, \oplus, \otimes)$ by

$$(a, b) \oplus (c, d) = (a +_{\mathbb{F}} c, b +_{\mathbb{F}} d),$$
$$(a, b) \otimes (c, d) = (a \times_{\mathbb{F}} c, b \times_{\mathbb{F}} d).$$

Is this new structure a field?

WORKED SOLUTION

Every element of this structure is a pair of elements of the given field, as with a point in a coordinate system.
The operation $\oplus$ adds the first and second components separately; multiplication $\otimes$ is defined in the same componentwise way.
We will show that the structure is not a field by finding two nonzero elements whose product is zero.
Suppose, for a contradiction, that $\mathbb{F} \times \mathbb{F}$ is a field. Its zero element is $(0_{\mathbb{F}}, 0_{\mathbb{F}})$, as follows from the field operations in each component. However,

$$\begin{aligned}(0_{\mathbb{F}}, 1_{\mathbb{F}}) \otimes (1_{\mathbb{F}}, 0_{\mathbb{F}}) &= (0_{\mathbb{F}} \times_{\mathbb{F}} 1_{\mathbb{F}}, 1_{\mathbb{F}} \times_{\mathbb{F}} 0_{\mathbb{F}}) \\ &= (0_{\mathbb{F}}, 0_{\mathbb{F}}).\end{aligned}$$

Neither factor is zero, contradicting the result of the preceding exercise.

2.5 Finite fields

Definition and notation

DEFINITION 2.4

- A **finite field** is a field with finitely many elements. Write $|\mathbb{F}| = q$.
- Every finite field has $q = p^n$ elements, where p is prime and $n \geq 1$ is an integer. This is a structural theorem about finite fields.
- For a prime p, the set $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$, with addition and multiplication modulo p, is a field. It is called the **prime field of characteristic** p.
- Addition and multiplication in $\mathbb{F}_p$ are defined by

$$a +_{\mathbb{F}_p} b := (a + b) \bmod p, \quad a \times_{\mathbb{F}_p} b := (a \cdot b) \bmod p.$$

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Examples

EXAMPLE 2.2

- $\mathbb{F}_2 = \{0, 1\}$, with $1 + 1 = 0$ and $1 \cdot 1 = 1$.
- $\mathbb{F}_3 = \{0, 1, 2\}$, with addition and multiplication modulo 3.
- $\mathbb{F}_5 = \{0, 1, 2, 3, 4\}$, with operations modulo 5.

Basic properties

PROPOSITION 2.2

In every finite field $\mathbb{F}_q$, the set $\mathbb{F}_q^\times$ of nonzero elements is closed under multiplication, and each of its elements has a unique inverse.

PROPOSITION 2.3

In every finite field $\mathbb{F}_q$, each $a \in \mathbb{F}_q$ satisfies

$$a^q = a.$$

PROPOSITION 2.4

Every nonzero a in a finite field $\mathbb{F}_q$ satisfies $a^{q-1} = 1$.

Explanation: Closure follows because a field has no zero divisors. The existence and uniqueness of inverses follow from the field axioms and the proof above. To prove both power identities, let $a \neq 0$ and put

$$P = \prod_{x \in \mathbb{F}_q^\times} x \neq 0.$$

Multiplication by a permutes the nonzero elements: the map $x \mapsto ax$ is invertible, with inverse $x \mapsto a^{-1}x$. Thus

$$P = \prod_{x \in \mathbb{F}_q^\times} (ax) = a^{q-1}P.$$

Since $P \neq 0$, cancellation is permitted and gives $a^{q-1} = 1$, hence $a^q = a$. For $a = 0$, the identity $0^q = 0$ is immediate.

EXAMPLE 2.3

Calculate all inverses in $\mathbb{F}_7 = \mathbb{Z}/7\mathbb{Z}$, for $1 \leq a \leq 6$:

a	1	2	3	4	5	6
a^{-1}	1	4	5	2	3	6

For example, $2 \cdot 4 = 8 \equiv 1$, $3 \cdot 5 = 15 \equiv 1$, and $4 \cdot 2 \equiv 1$, and similarly for the remaining entries.

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EXERCISE 2.10

Show that, in every finite field $\mathbb{F}_q$,

$$(a-b)^q = a^q - b^q.$$

WORKED SOLUTION

Use the identity $x^q = x$, proved for every $x \in \mathbb{F}_q$. In particular, it applies to $a-b$, so

$$(a-b)^q = a - b = a^q - b^q.$$

This proves the claim directly, without calculating binomial coefficients.

EXERCISE 2.11

Work in $\mathbb{F}_{13} = \mathbb{Z}/13\mathbb{Z}$.

1. Find the inverses of $2, 3, 4, 5, 6$.
2. Evaluate the following expression, checking that every denominator is nonzero:

$$E = (5^{-1} + 2)(4 - 3^{-1})^{-1} - 1.$$

WORKED SOLUTION

All arithmetic is modulo 13. Check the products

$$2 \cdot 7 = 14 \equiv 1, \quad 3 \cdot 9 = 27 \equiv 1, \quad 4 \cdot 10 = 40 \equiv 1,$$

$$5 \cdot 8 = 40 \equiv 1, \qquad 6 \cdot 11 = 66 \equiv 1.$$

The inverses, in the stated order, are therefore $7, 9, 10, 8, 11$. Now $5^{-1}+2 = 8+2 = 10$, whereas

$$4 - 3^{-1} = 4 - 9 = -5 \equiv 8 \neq 0.$$

The inverse of 8 is 5, because $8 \cdot 5 = 40 \equiv 1$. Consequently,

$$E \equiv 10 \cdot 5 - 1 = 49 \equiv 10 \pmod{13}.$$

Every inverse appearing in the expression is defined, and the answer is the residue class 10.

EXERCISE 2.12

Show that for every nonzero a in a finite field $\mathbb{F}_q$, there is $n \in \mathbb{N}$ with $a^n = 1$.

WORKED SOLUTION

The set $\mathbb{F}_q^\times$ contains exactly $q-1$ elements and is closed under multiplication. Put $m = q-1$ and consider the list

$$a, a^2, a^3, \ldots, a^{m+1}.$$

It has $m+1$ entries but only m possible distinct values, so two entries are equal, say $a^k = a^l$

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EXERCISE 2.12 CONTINUED

with $k > l$. Multiplying by a^{-l} gives $a^{k-l} = 1$. Thus every nonzero element has a positive integer power equal to the multiplicative identity.

2.6 Calculations and applications in finite fields

EXERCISE 2.13

A system stores two information bits $a, b \in \mathbb{F}_2$ and adds a check bit c such that $a + b + c = 0$. Find c. If one of the three bits is flipped, does the check detect it? What happens if exactly two bits are flipped?

WORKED SOLUTION

In $\mathbb{F}_2$, $-x = x$, so $c = a + b$. For example, if $a = b = 1$, then $c = 0$. Let $e_1, e_2, e_3 \in \mathbb{F}_2$ denote the errors, where $e_j = 1$ means that bit j is flipped. The sum of the received bits is

$$(a + e_1) + (b + e_2) + (c + e_3) = (a + b + c) + (e_1 + e_2 + e_3) = e_1 + e_2 + e_3.$$

For a single flip, this sum is 1, so an error is detected. For two flips, it is $1 + 1 = 0$, so that error is undetected. The check detects any single flip but does not identify which bit was flipped.

Additional practice

EXERCISE 2.14 FOUNDATION

In the field $\mathbb{F}_7$, define

$$A = \{x \in \mathbb{F}_7 : 2x + 3 = 1\}, \qquad B = \{x \in \mathbb{F}_7 : x^2 = 1\}.$$

Find $A, B, A \cup B, A \cap B$, and determine whether $A \subseteq B$. Explain why you have found every element of B.

WORKED SOLUTION

All arithmetic is modulo 7. The first equation is equivalent to $2x = 5$. The inverse of 2 is 4, so $x = 4 \cdot 5 = 20 = 6$ in the field, giving $A = \{6\}$. In the second equation, $x^2 - 1 = (x-1)(x+1) = 0$. Since a field has no nontrivial zero divisors, necessarily $x = 1$ or $x = -1 = 6$. Both values do satisfy the equation, so $B = \{1, 6\}$. Hence

$$A \subseteq B, \qquad A \cup B = \{1, 6\} = B, \qquad A \cap B = \{6\} = A.$$

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EXERCISE 2.15 **CHALLENGE**

Let F be a finite field with $q > 2$ elements. Prove that

$$\sum_{x \in F} x = 0, \qquad \sum_{x \in F^\times} x^{-1} = 0.$$

Explain why both statements fail in the field with two elements. Do not assume an ordering on F.

WORKED SOLUTION

Put $S = \sum_{x \in F} x$. Since $q > 2$, choose $c \in F \setminus \{0, 1\}$. The map $x \mapsto cx$ is bijective, with inverse $x \mapsto c^{-1}x$. It therefore only rearranges the summands, and

$$cS = \sum_{x \in F} cx = \sum_{x \in F} x = S.$$

Thus $(c - 1)S = 0$. Since $c - 1 \neq 0$, field cancellation gives $S = 0$. Removing the summand 0 does not change the sum, so $\sum_{x \in F^\times} x = 0$. The map $x \mapsto x^{-1}$ also permutes $F^\times$, because applying it twice returns x. Therefore

$$\sum_{x \in F^\times} x^{-1} = \sum_{x \in F^\times} x = 0.$$

In $\mathbb{F}_2$, we have $0 + 1 = 1 \neq 0$, and the sum of the nonzero inverses is $1^{-1} = 1 \neq 0$. Indeed, no element c distinct from $0, 1$ exists there, so the choice used in the proof is unavailable.

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TUTORIAL 3

Linear Systems and Subspaces

MOTIVATION AND GUIDING IDEA

Row reduction replaces a problem with an equivalent one that is easier to read. After calculating, return to the question: is the system consistent, which variables are free, and what is the solution set? For a nonhomogeneous system, the difference of two solutions solves the associated homogeneous system. This explains the form ``a particular solution plus a homogeneous solution.''

NOTATION AND CONVENTIONS

If $A \in M_{m\times n}(F)$, then $Ax = b$ is a system of m equations in n variables, with $x \in F^n$ and $b \in F^m$. The notation $[A \mid b]$ denotes its augmented matrix, and $A \sim B$ denotes row equivalence. The rank rank A is the number of pivots. Define $\ker A = \{x \in F^n : Ax = 0\}$. The expression $\operatorname{span}\{v_1, \dots, v_k\}$ denotes all combinations $\alpha_1 v_1 + \cdots + \alpha_k v_k$ with $\alpha_j \in F$. The notation $x_0 + U = \{x_0 + u : u \in U\}$ denotes a translate of a subspace U. Distinguish the empty set $\varnothing$, with no elements, from $\{0\}$, which has one element.

DEFINITION 3.1

Vector space axioms and solution tools. A vector space over a field F is a set V equipped with operations $V \times V \to V$ and $F \times V \to V$ satisfying, for all $u, v, w \in V$ and $\alpha, \beta \in F$,

$$u + v = v + u, \qquad (u + v) + w = u + (v + w).$$

There is $0_V \in V$ such that $v + 0_V = v$, and every $v \in V$ has an additive inverse $-v \in V$ satisfying $v + (-v) = 0_V$. In addition,

$$\alpha(u + v) = \alpha u + \alpha v, \qquad (\alpha + \beta)v = \alpha v + \beta v,$$

$$(\alpha\beta)v = \alpha(\beta v), \qquad 1_F v = v.$$

For $A \in M_{m\times n}(F)$, the system $Ax = b$ is consistent if and only if $\operatorname{rank} A = \operatorname{rank}[A \mid b]$. If x_0 is a particular solution, then its full solution set is $x_0 + \ker A$: indeed, $Ax = b$ is equivalent to $A(x - x_0) = 0$. This solution set is a subspace of F^n if and only if $b = 0$, since containing the zero vector forces $A0 = b = 0$.

MOTIVATION AND GUIDING IDEA

Objectives. We will formulate problems as linear systems, perform row reduction while preserving the solution set, and distinguish free variables from pivot variables. We will connect homogeneous systems with vector spaces and test closure systematically.

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3.1 Linear systems and row reduction

3.1.1 A system of linear equations

RECALL AND TOOLS

In general, for a positive integer n, a linear equation in n unknowns (variables) has the form

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$$

Here $a_1, a_2, ..., a_n$ and b are **given scalars** in a specified field $\mathbb{F}$, such as $\mathbb{R}$, $\mathbb{C}$, or a finite field, and $x_1, x_2, ..., x_n$ are the **unknowns**, or variables.
The scalar a_i is the **coefficient** of x_i, and b is the **constant term** of the equation.
A system of m equations in n unknowns has the general form

$$\begin{cases} a_{1,1}x_1 + a_{1,2}x_2 + \cdots + a_{1,n}x_n = b_1 \\ a_{2,1}x_1 + a_{2,2}x_2 + \cdots + a_{2,n}x_n = b_2 \\ \vdots \\ a_{m,1}x_1 + a_{m,2}x_2 + \cdots + a_{m,n}x_n = b_m \end{cases}$$

To solve the system, it is convenient to collect the **coefficients** in a matrix, preserving the variable order. To shorten the notation, we write either the matrix of variable coefficients, called the **coefficient matrix**, or the **augmented matrix**, which also includes the constant terms. Following the basic rules consistently makes it clear which coefficient belongs to each variable.

EXAMPLE 3.1

$$\begin{cases} 2x_1 - x_2 = 0 \\ -4x_1 + 3x_2 = 1 \end{cases} \xrightarrow{\text{matrix form}} \left(\begin{array}{cc|c} 2 & -1 & 0 \\ -4 & 3 & 1 \end{array}\right)$$

3.1.2 Elementary operations

DEFINITION 3.2

Elementary row operations:
An elementary row operation modifies a matrix in one of the following ways:

1. Interchange rows i and j, denoted by $R_i \longleftrightarrow R_j$.
2. Multiply row i by a **nonzero** scalar λ, denoted by $R_i \longleftarrow \lambda R_i$.
   (Dividing by a nonzero scalar a is multiplication by $1/a$.)
3. Add λ times row j to a different row i, where $i \neq j$, denoted by $R_i \longleftarrow R_i + \lambda R_j$.

- Elementary row operations preserve the solutions of a linear system.
- Every elementary row operation has an inverse that is also an elementary row operation.

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DEFINITION 3.3

Matrices A and B are **row equivalent** (or simply equivalent), denoted by $A \sim B$, if B is obtained by applying a finite sequence of elementary row operations to A.

DEFINITION 3.4

Leading entry (pivot): in every nonzero row of a matrix, the first nonzero entry is called its leading entry.

DEFINITION 3.5

Row echelon form: each leading entry is strictly to the left of the leading entry in the next nonzero row. Any zero rows appear below all nonzero rows.

DEFINITION 3.6

Reduced row echelon form: a matrix in row echelon form such that, in each nonzero row, the leading entry equals 1, and every other entry above and below it in the same column equals 0. These requirements apply to every column, including the constant column of an augmented matrix.

PROPOSITION 3.1

Every matrix is row equivalent to a unique matrix in reduced row echelon form.

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Examples:

1. $\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$ is not in row echelon form.
2. $\begin{pmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$ is in row echelon form, but not in reduced row echelon form.
3. $\begin{pmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \end{pmatrix}$ is in reduced row echelon form.

3.1.3 Gaussian elimination

RECALL AND TOOLS

1. Start with the first active row and the first column. Look in that column for a nonzero entry in the active row or below it. If none exists, move to the next column.
2. Interchange rows to bring this entry into the active row. Division by it is permitted because it has been checked to be nonzero.
3. Multiply the active row by the inverse of its leading entry to obtain a leading 1, and use it to eliminate the entries below.

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4. Move to the next active row and seek a pivot column to the right of the previous one. A pivot column need not have the same index as its row.

5. After reaching row echelon form, eliminate the entries above each pivot, working from the last pivot to the first, to obtain reduced row echelon form. This completes Gauss--Jordan elimination.

EXERCISE 3.1

Over $\mathbb{R}$, reduce the augmented matrix

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 2 & 1 & 3 & 12 \\ 1 & 2 & 2 & 6 \end{array}\right)$$

to reduced row echelon form, and read off the solution of the system.

WORKED SOLUTION

Subtract multiples of the first row:

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 2 & 1 & 3 & 12 \\ 1 & 2 & 2 & 6 \end{array}\right) \xrightarrow[R_3 \leftarrow R_3 - R_1]{R_2 \leftarrow R_2 - 2R_1} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & -1 & 1 & 4 \\ 0 & 1 & 1 & 2 \end{array}\right).$$

Multiply R_2 by -1, and subtract it from R_3:

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & 1 & -1 & -4 \\ 0 & 0 & 2 & 6 \end{array}\right) \xrightarrow{R_3 \leftarrow R_3/2} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & 1 & -1 & -4 \\ 0 & 0 & 1 & 3 \end{array}\right).$$

Apply $R_2 \leftarrow R_2 + R_3$, followed by $R_1 \leftarrow R_1 - R_3 - R_2$:

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 3 \end{array}\right).$$

The unique solution is therefore $(2, -1, 3)^T$. Substitution gives $2 - 1 + 3 = 4$, $4 - 1 + 9 = 12$, and $2 - 2 + 6 = 6$, as required.

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NOTE

To solve a linear system, reduce its **augmented matrix**, including the constant terms. You may stop at row echelon form and use back substitution, or continue to reduced row echelon form and read off the solutions.

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EXAMPLE 3.2

Consider the following system:

$$\begin{cases} 2x + 2y + 2z = 8 \\ 4x + 2z = 12 \\ 2x + 2y = 8 \end{cases}$$

Write the system in matrix form:

$$\left(\begin{array}{ccc|c} 2 & 2 & 2 & 8 \\ 4 & 0 & 2 & 12 \\ 2 & 2 & 0 & 8 \end{array}\right)$$

Applying the same row-reduction method gives

$$\left(\begin{array}{ccc|c} 2 & 2 & 2 & 8 \\ 4 & 0 & 2 & 12 \\ 2 & 2 & 0 & 8 \end{array}\right) \sim \left(\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{array}\right)$$

Returning to the equations,

$$\begin{cases} 1 \cdot x + 0 \cdot y + 0 \cdot z = 3 \\ 0 \cdot x + 1 \cdot y + 0 \cdot z = 1 \\ 0 \cdot x + 0 \cdot y + 1 \cdot z = 0 \end{cases} \longrightarrow \begin{cases} x = 3 \\ y = 1 \\ z = 0 \end{cases}$$

we find that the solution set of the system is $S = \{ \begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix} \}$.

This example shows how row reduction helps solve a linear system. We will next see that a system need not have a unique solution.

EXERCISE 3.2

Reduce the following matrix over $\mathbb{F}_7 = \mathbb{Z}/7\mathbb{Z}$ to reduced row echelon form. Perform all operations modulo 7, using the representatives $0, 1, \ldots, 6$:

$$\left(\begin{array}{cccc|c} 1 & 0 & 5 & 0 & 1 \\ 0 & 1 & 2 & 1 & 2 \\ 3 & 4 & 0 & 1 & 3 \end{array}\right)$$

WORKED SOLUTION

$$\left(\begin{array}{cccc|c} 1 & 0 & 5 & 0 & 1 \\ 0 & 1 & 2 & 1 & 2 \\ 3 & 4 & 0 & 1 & 3 \end{array}\right) \xrightarrow{R_3 \to R_3 - 3R_1} \left(\begin{array}{cccc|c} 1 & 0 & 5 & 0 & 1 \\ 0 & 1 & 2 & 1 & 2 \\ 0 & 4 & 6 & 1 & 0 \end{array}\right) \xrightarrow{R_3 \to R_3 - 4R_2}$$

$$\left(\begin{array}{cccc|c} 1 & 0 & 5 & 0 & 1 \\ 0 & 1 & 2 & 1 & 2 \\ 0 & 0 & 5 & 4 & 6 \end{array}\right)$$

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EXERCISE 3.2 CONTINUED

$$\xrightarrow{R_3 \to 5^{-1}R_3} \left(\begin{array}{cccc|c} 1 & 0 & 5 & 0 & 1 \\ 0 & 1 & 2 & 1 & 2 \\ 0 & 0 & 1 & 5 & 4 \end{array}\right)$$

$$\xrightarrow[R_2 \to R_2 - 2R_3]{R_1 \to R_1 - 5R_3} \left(\begin{array}{cccc|c} 1 & 0 & 0 & 3 & 2 \\ 0 & 1 & 0 & 5 & 1 \\ 0 & 0 & 1 & 5 & 4 \end{array}\right)$$

EXERCISE 3.3

Solve the following system over $\mathbb{C}$:

$$\begin{cases} iz + (1+i)w = 3 - i, \\ 2iz + (3+2i)w = 7 - 4i. \end{cases}$$

Verify the solution in both equations.

WORKED SOLUTION

Subtract twice the first equation from the second. The coefficient of z becomes zero, and the coefficient of w becomes $(3+2i) - 2(1+i) = 1$. Thus

$$w = (7 - 4i) - 2(3 - i) = 1 - 2i.$$

Substitute into the first equation and calculate the product:

$$(1+i)(1-2i) = 1 - 2i + i - 2i^2 = 3 - i.$$

Hence $iz = (3 - i) - (3 - i) = 0$. Since $i \neq 0$, we get $z = 0$. Checking the second equation gives

$$(3+2i)(1-2i) = 3 - 6i + 2i - 4i^2 = 7 - 4i.$$

Thus $(z, w) = (0, 1-2i)$ solves both equations, and the elimination steps show that it is unique.

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3.2 Solving linear systems

DEFINITION 3.7

After row reduction, a variable whose column in the coefficient matrix contains a leading entry is called a **pivot variable** (or dependent variable). A variable whose column contains no leading entry is called a **free variable**. In a consistent system, every free variable can be assigned an arbitrary value in the field, after which the pivot variables are determined by the equations. Over an infinite field, a free parameter gives infinitely many choices; over a finite field, the number of choices is finite. The number of free variables is the number of degrees of freedom of a consistent system.

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DEFINITION 3.8

The rank of a matrix A, denoted by $\text{rank}(A)$, is the number of nonzero rows in a row echelon form. Equivalently, it is the number of leading entries (pivots) in that form.

Remark: if a system with coefficient matrix A and n variables is consistent, its number of degrees of freedom is $n - \text{rank}(A)$. Over $\mathbb{F}_q$, it has exactly $q^{n-\text{rank}(A)}$ solutions.

EXERCISE 3.4

Solve the following system over $\mathbb{R}$: $\begin{cases} x + 2y - 3z = 6 \\ 2x - y + 4z = 2 \\ 4x + 3y - 2z = 14 \end{cases}$.

WORKED SOLUTION

Write the system in matrix form and perform row reduction:

$$\left(\begin{array}{ccc|c} 1 & 2 & -3 & 6 \\ 2 & -1 & 4 & 2 \\ 4 & 3 & -2 & 14 \end{array}\right)$$

$$\xrightarrow[R_3 \longrightarrow R_3 - 4R_1]{R_2 \longrightarrow R_2 - 2R_1} \left(\begin{array}{ccc|c} 1 & 2 & -3 & 6 \\ 0 & -5 & 10 & -10 \\ 0 & -5 & 10 & -10 \end{array}\right)$$

$$\xrightarrow{R_3 \longrightarrow R_3 - R_2} \left(\begin{array}{ccc|c} 1 & 2 & -3 & 6 \\ 0 & -5 & 10 & -10 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

$$\xrightarrow{R_2 \longrightarrow \frac{-1}{5} R_2} \left(\begin{array}{ccc|c} 1 & 2 & -3 & 6 \\ 0 & 1 & -2 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

$$\xrightarrow{R_1 \longrightarrow R_1 - 2R_2} \left(\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 1 & -2 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

The matrix is now in reduced row echelon form. The first two variable columns contain leading entries, whereas the third does not.
The matrix has rank 2, and the number of degrees of freedom is 1.
Therefore x, y are dependent variables, whereas the third variable z is free and may take any real value.
Returning to the system of equations,

$$\begin{gathered} x + z = 2 \\ y - 2z = 2 \\ 0 = 0 \end{gathered}$$

since x, y are dependent, express them in terms of the free variable: z :

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EXERCISE 3.4 CONTINUED

$$x = 2 - z$$
$$y = 2 + 2z$$

The solution set is therefore

$$S = \{ \begin{pmatrix} 2-s \\ 2+2s \\ s \end{pmatrix} \mid s \in \mathbb{R} \} = \left\{ \begin{pmatrix} 2 \\ 2 \\ 0 \end{pmatrix} + \begin{pmatrix} -s \\ 2s \\ s \end{pmatrix} \mid s \in \mathbb{R} \right\} = \left\{ \begin{pmatrix} 2 \\ 2 \\ 0 \end{pmatrix} + s \cdot \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix} \mid s \in \mathbb{R} \right\}$$

.

There are **infinitely many solutions**: for every choice of a real number s, substitution determines the other two coordinates, giving a different solution.

For example, setting $s = 0, 1$ gives the solutions $\begin{pmatrix} 2 \\ 2 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 4 \\ 1 \end{pmatrix}$. Substitution confirms that each satisfies the original system.

EXERCISE 3.5

Solve the following system:

$$\begin{cases} 3x_1 + 12x_2 - 6x_3 = 4 \\ x_1 + 4x_2 - 2x_3 = 1 \end{cases}$$

.

WORKED SOLUTION

Perform row reduction until a contradiction is found:

$$\left(\begin{array}{cccc} 3 & 12 & -6 & 4 \\ 1 & 4 & -2 & 1 \end{array} \right) \xrightarrow{R_2 \leftrightarrow R_1} \left(\begin{array}{cccc} 1 & 4 & -2 & 1 \\ 3 & 12 & -6 & 4 \end{array} \right)$$

$$\xrightarrow{R_2 \to R_2 - 3 \cdot R_1} \left(\begin{array}{cccc} 1 & 4 & -2 & 1 \\ 0 & 0 & 0 & 1 \end{array} \right)$$

A contradiction row has already appeared, so there is no need to complete the reduction. Returning to the system of equations,

$$x_1 + 4x_2 - 2x_3 = 1$$
$$0 = 1$$

we obtain a false statement, a contradiction, so the system has no solution.
In future, returning to the equations is unnecessary: recognize a **contradiction row** directly in the matrix. Such a row has zero variable coefficients and a nonzero constant term. We can immediately conclude that there is no solution.

Geometric illustration

We have encountered three possible cases:

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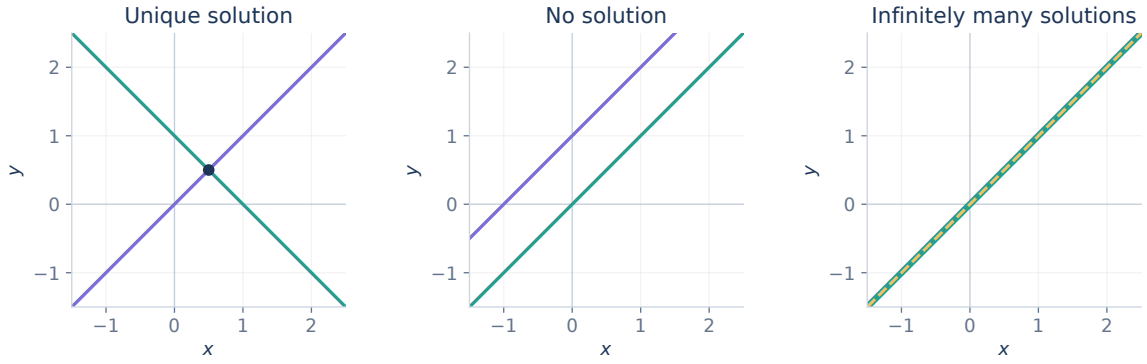


Figure 3.1. Over the real numbers, two equations in two unknowns can describe intersecting lines, distinct parallel lines, or coincident lines. These give, respectively, a unique solution, no solution, or infinitely many solutions.

EXERCISE 3.6

For which values of $a \in \mathbb{R}$ does the following system have a unique solution, infinitely many solutions, or no solution? State the rank of the coefficient matrix in each case.

$$\begin{cases} x - y = 0 \\ 2x + 2z = 1 \\ x + az = 1 \end{cases}$$

WORKED SOLUTION

Write the augmented matrix and perform row reduction:

$$\left(\begin{array}{ccc|c} 1 & -1 & 0 & 0 \\ 2 & 0 & 2 & 1 \\ 1 & 0 & a & 1 \end{array}\right)$$

$$\xrightarrow{R_2 \to R_2 - 2R_1,\ R_3 \to R_3 - R_1} \left(\begin{array}{ccc|c} 1 & -1 & 0 & 0 \\ 0 & 2 & 2 & 1 \\ 0 & 1 & a & 1 \end{array}\right)$$

$$\xrightarrow{R_2 \leftrightarrow R_3} \left(\begin{array}{ccc|c} 1 & -1 & 0 & 0 \\ 0 & 1 & a & 1 \\ 0 & 2 & 2 & 1 \end{array}\right)$$

$$\xrightarrow{R_1 \to R_1 + R_2,\ R_3 \to R_3 - 2R_2} \left(\begin{array}{ccc|c} 1 & 0 & a & 1 \\ 0 & 1 & a & 1 \\ 0 & 0 & 2 - 2a & -1 \end{array}\right)$$

We would like to make the last leading entry equal to 1, but we cannot divide by $2 - 2a$ before checking whether it vanishes. Distinguish the following cases:

Case 1: $2 - 2a = 0$, that is, $a = 1$. Substitution makes the third row a contradiction row, so there is no solution. (The coefficient matrix has rank 2.)

Case 2: $2 - 2a \neq 0$, that is, $a \neq 1$.
We can now continue the reduction by dividing by this expression to obtain a leading 1.

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EXERCISE 3.6 CONTINUED

$$\xrightarrow{R_3 \to \frac{1}{2-2a} R_3} \left(\begin{array}{ccc|c} 1 & 0 & a & 1 \\ 0 & 1 & a & 1 \\ 0 & 0 & 1 & \frac{-1}{2-2a} \end{array}\right) \xrightarrow{R_1 \to R_1 - aR_3,\, R_2 \to R_2 - aR_3} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 + \frac{a}{2-2a} \\ 0 & 1 & 0 & 1 + \frac{a}{2-2a} \\ 0 & 0 & 1 & \frac{-1}{2-2a} \end{array}\right)$$

The solution is unique: $S = \left\{ \begin{pmatrix} 1 + \frac{a}{2-2a} \\ 1 + \frac{a}{2-2a} \\ \frac{-1}{2-2a} \end{pmatrix} \right\}$. The matrix has rank 3 and there are no degrees of freedom.

Conclusion: for $a = 1$ there is no solution; for $a \neq 1$ there is a unique solution; no value of a gives infinitely many solutions.

3.2.1 Homogeneous systems

RECALL AND TOOLS

All constant terms are 0. In this case:

- We may write only the coefficient matrix rather than the augmented matrix: the zero column is unchanged by all row operations, so there is no need to keep writing it.
- A solution always exists: no contradiction row can occur, so inconsistency is impossible. We must still determine whether the zero solution is the only solution. Over an infinite field, the existence of a nonzero solution implies infinitely many solutions.
- If there is a unique solution, it is the zero solution, also called the **trivial solution**.
- The solution set of a homogeneous system in n variables is a vector subspace of $\mathbb{F}^n$ and can always be expressed as a span.

EXERCISE 3.7

For which values of $a \in \mathbb{R}$ does the following system have a unique solution, infinitely many solutions, or no solution?

$$\begin{cases} x - y = 0 \\ 2x + 2z = 0 \\ x + az = 0 \end{cases}$$

WORKED SOLUTION

Write only the coefficient matrix and perform row reduction. This is the same coefficient matrix as in the preceding exercise, so we immediately obtain

$$\begin{pmatrix} 1 & -1 & 0 \\ 2 & 0 & 2 \\ 1 & 0 & a \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & a \\ 0 & 1 & a \\ 0 & 0 & 2 - 2a \end{pmatrix}$$

As before, distinguish the following cases:

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EXERCISE 3.7 CONTINUED

Case 1: $2-2a \neq 0$, that is, $a \neq 1$.
As before, divide by this nonzero expression to obtain a leading 1.

$$\xrightarrow{R_3 \to \frac{1}{2-2a} R_3} \begin{pmatrix} 1 & 0 & a \\ 0 & 1 & a \\ 0 & 0 & 1 \end{pmatrix} \xrightarrow{R_1 \to R_1 - aR_3,\, R_2 \to R_2 - aR_3} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Here the solution is unique: $S = \{ \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \}$. This is the zero solution, or trivial solution.

Case 2: $2-2a = 0$, that is, $a = 1$. Substitution gives a zero row:

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

The first two variables are dependent and the third is free. The equations are

$$\begin{cases} x + z = 0 \\ y + z = 0 \\ 0 = 0 \end{cases} \longrightarrow \begin{cases} x = -z \\ y = -z \\ z = z \end{cases} \longrightarrow S = \{ \begin{pmatrix} -s \\ -s \\ s \end{pmatrix} \mid s \in \mathbb{R} \} = \text{span} \left\{ \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix} \right\}$$

Conclusion: for $a = 1$ there are infinitely many solutions; for $a \neq 1$ there is a unique, trivial solution; no value of a gives no solution.

EXERCISE 3.8 EXAM QUESTION

Work over the field $\mathbb{Z}_5 = \{0, 1, 2, 3, 4\}$, with all arithmetic modulo 5. Solve the following system in terms of the parameter $\omega \in \mathbb{Z}_5$:

$$\begin{cases} x + y + z = 1, \\ x + \omega y + z = 2, \\ x + y + (\omega + 1)z = 1, \end{cases} \qquad x, y, z \in \mathbb{Z}_5.$$

Determine which values of ω give a unique solution, no solution, or more than one solution. Write the solution set explicitly in every case.

WORKED SOLUTION

Subtract the first equation from each of the other two equations. This gives the equivalent system over $\mathbb{Z}_5$:

$$\begin{cases} x + y + z = 1, \\ (\omega - 1)y = 1, \\ \omega z = 0. \end{cases}$$

The subtraction is useful because the second and third equations now each contain only one

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EXERCISE 3.8 **EXAM QUESTION CONTINUED**

unknown. Separate the cases before dividing by a parameter-dependent expression.

1. If $\omega = 1$, the second equation is $0 \cdot y = 1$. This is a contradiction in the field, so
$$S_1 = \varnothing.$$

2. If $\omega = 0$, the second equation is $-y = 1$, giving $y = 4$. The third equation places no restriction on z. Put $z = t \in \mathbb{Z}_5$. Then
$$x = 1 - y - z = 1 - 4 - t = 2 - t \pmod 5.$$
There are exactly five solutions:
$$\begin{aligned} S_0 &= \{(2-t, 4, t) : t \in \mathbb{Z}_5\} \\ &= \{(2,4,0), (1,4,1), (0,4,2), (4,4,3), (3,4,4)\}. \end{aligned}$$

3. If $\omega \in \{2, 3, 4\}$, both ω and $\omega - 1$ are nonzero, so both are invertible in the field. We obtain
$$z = 0, \qquad y = (\omega - 1)^{-1}, \qquad x = 1 - (\omega - 1)^{-1}.$$
Calculate the inverses explicitly:
$$1^{-1} = 1, \qquad 2^{-1} = 3, \qquad 3^{-1} = 2 \pmod 5.$$
Thus each of these three cases has a unique solution:
$$S_2 = \{(0,1,0)\}, \qquad S_3 = \{(3,3,0)\}, \qquad S_4 = \{(4,2,0)\}.$$

As an additional check, the determinant of the coefficient matrix is $\omega(\omega - 1)$. Only $\omega = 0, 1$ therefore require a distinction between inconsistency and multiple solutions.

3.3 Vector spaces

RECALL AND TOOLS

1. To define a vector space V over a field $\mathbb{F}$, we specify an addition of vectors in V and a multiplication of a vector in V by a scalar in $\mathbb{F}$, satisfying the vector space axioms. These are **in addition** to addition and multiplication in $\mathbb{F}$. A product of two vectors is not part of the vector space axioms; if needed later, it will be defined separately.

2. Every field is a vector space over itself, with the field's addition and multiplication.

3. A vector space V is always defined over a field $\mathbb{F}$. Always specify the field of scalars. For example, as discussed in the lecture, $\mathbb{C}$ can be regarded as a vector space over $\mathbb{C}$ or over $\mathbb{R}$. These are two different vector spaces.

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EXAMPLE 3.3

Let $\mathbb{F}$ be a field.

(a) $\mathbb{F}^n$: the set of n-tuples over $\mathbb{F}$, where $n \in \mathbb{N}$.

(b) $M_{m\times n}(\mathbb{F})$: the set of $m \times n$ matrices over $\mathbb{F}$.

(c) $\mathbb{F}[X]$: the set of all polynomials with coefficients in $\mathbb{F}$.

EXERCISE 3.9

Determine whether the following sets are subspaces of $\mathbb{R}[X]$ over $\mathbb{R}$, with the usual operations:

1. $V = \{p(X) \in \mathbb{R}[X] : p(5) = 0, p(4) \neq 0\}$.
2. $V = \{p(X) \in \mathbb{R}[X] : \exists c \in \mathbb{R}, p(c) = 0\}$.

WORKED SOLUTION

1. No. The zero polynomial is absent, since its value at 4 is 0. Addition also fails to be closed: the polynomials $p(X) = (X-5)(X-3)$ and $q(X) = (X-5)^2$ belong to the set, because $p(4) = -1$ and $q(4) = 1$, but
$$(p+q)(X) = (X-5)(2X-8), \qquad (p+q)(4) = 0.$$
therefore $p+q$ does not belong to the set.
2. No. The set is not closed under addition. For example, $(X+1)^2, -2X \in V$, but $(X+1)^2 - 2X = X^2 + 1 \notin V$.

3.4 Subspaces

DEFINITION 3.9

Let V be a vector space over a field $\mathbb{F}$, and let $U \subseteq V$ be a subset that is itself a vector space over $\mathbb{F}$ under the addition and scalar multiplication inherited from V. Then U is a vector subspace of V.

PROPOSITION 3.2

Let V be a vector space over a field $\mathbb{F}$, and let $U \subseteq V$ be a subset. Then U is a vector subspace if and only if all of the following conditions hold:

1. $0 \in U$, where this is the zero vector of V.
2. U is closed under addition.
3. U is closed under scalar multiplication.

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EXAMPLE 3.4

1. $V = \{p(x) \in \mathbb{F}[X] | p(5) = 0\}$ is a subspace of $\mathbb{F}[X]$.
   Proof:
   (a) Clearly $0 \in V$.
   (b) Suppose $p(x), q(x) \in V$. Define $r(x) = p(x) + q(x)$. Then $r(5) = p(5) + q(5) = 0$, so $r(x) \in V$.
   (c) Suppose $p \in V$ and let $\alpha \in \mathbb{F}$. Define $r = \alpha p$. Then $r(5) = \alpha p(5) = 0$.

2. $V = \{(x_1, \ldots, x_n) \in \mathbb{R}^n | x_2 = 0\}$ (where $n \geq 2$). This is a subspace of $\mathbb{R}^n$.
   Proof:
   (a) Clearly $0 \in V$.
   (b) Let $x = (x_1, \ldots, x_n), y = (y_1, \ldots, y_n) \in V$. Then $x_2 = y_2 = 0$, so the second coordinate of $x + y$ is 0; hence $x + y \in V$
   (c) If $x \in V$ and $\alpha \in \mathbb{R}$, then $\alpha x_2 = 0$, so $\alpha x \in V$.

EXERCISE 3.10

Prove or disprove:

1. The set $U = \left\{ \begin{pmatrix} a \\ a^2 \end{pmatrix} \mid a \in \mathbb{R} \right\}$ is a vector subspace of $\mathbb{R}^2$.

2. The set of symmetric matrices $Sym_n(\mathbb{F}) := \left\{A \in M_{n\times n}(\mathbb{F}) \mid A = A^T\right\}$ is a vector subspace of $M_{n\times n}(\mathbb{F})$.

WORKED SOLUTION

1. This is not a vector subspace, because addition is not closed. For example, $\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1^2 \end{pmatrix} \in U$, but
$$\begin{pmatrix} 1 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \end{pmatrix} \notin U$$

2. The zero matrix is symmetric. Suppose $A, B \in \text{Sym}_n(\mathbb{F})$ and $\lambda \in \mathbb{F}$. Then
$$(A + B)^T = A^T + B^T = A + B,$$
so $A + B$ is symmetric. Also,
$$(\lambda A)^T = \lambda A^T = \lambda A,$$
so λA is symmetric. All three conditions of the subspace test hold.

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EXERCISE 3.11

An engineering sensing system models a normalized signal by a real polynomial of degree at most 3:

$$p(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3.$$

Time t is measured in normalized time units. This is a mathematical approximation of a signal, without assuming any particular sensor type.

1. Find all polynomials satisfying

$$p(0) = 0, \quad p(1) = 0.2, \quad p'(0) = 0, \quad p'(0.5) = 0.$$

   Is the solution set a subspace of $\mathbb{R}[t]$? Is $t = 0.5$ a local maximum or minimum point?

2. Repeat the question when the measurements change to

$$p(0) = 0.1, \quad p(1) = 0.3, \quad p'(0) = 0, \quad p'(0.5) = 0.$$

3. In the second part, add the measurement $p'(1) = 0.4$. Is the system still consistent? Is its solution set a subspace?

WORKED SOLUTION

Differentiate the polynomial:

$$p'(t) = a_1 + 2a_2 t + 3a_3 t^2.$$

Part 1. The conditions at $t = 0$ give $a_0 = a_1 = 0$. The remaining two conditions give

$$\begin{cases} a_2 + a_3 = 0.2, \\ a_2 + 0.75 a_3 = 0. \end{cases}$$

Subtract the second equation from the first:

$$0.25 a_3 = 0.2 \quad \Longrightarrow \quad a_3 = 0.8, \qquad a_2 = -0.75 \cdot 0.8 = -0.6.$$

Thus there is a unique solution,

$$p_1(t) = -0.6 t^2 + 0.8 t^3.$$

Check: $p_1(1) = -0.6 + 0.8 = 0.2$ and $p_1'(0.5) = -1.2 \cdot 0.5 + 2.4 \cdot 0.25 = 0$. The solution set is not a subspace: the zero polynomial does not satisfy $p(1) = 0.2$. The condition $p'(0.5) = 0$ alone gives a stationary point, which need not be a peak. Here,

$$p_1''(t) = -1.2 + 4.8t, \qquad p_1''(0.5) = 1.2 > 0,$$

so this is a local minimum.

Part 2. This time $a_0 = 0.1$ and $a_1 = 0$. The condition $p(1) = 0.3$ gives $a_2 + a_3 = 0.2$, so the two equations for a_2, a_3 are identical to those in the first part. We obtain

$$p_2(t) = 0.1 - 0.6 t^2 + 0.8 t^3 = p_1(t) + 0.1.$$

Again, the solution is unique, and its solution set is not a subspace because it does not contain

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EXERCISE 3.11 CONTINUED

the zero polynomial. The derivatives of p_1, p_2 agree, so $t = 0.5$ is again a local minimum point.
Part 3. The unique polynomial from part 2 satisfies

$$p_2'(1) = -1.2 + 2.4 = 1.2 \neq 0.4.$$

Thus the additional condition introduces a contradiction. There is no solution, and the empty solution set is not a subspace because it does not contain the zero vector. By contrast, $\{0\}$ is a subspace.

3.5 Supplementary exercises: measurements and counting solutions

EXERCISE 3.12

Three sensors measure sums of real state variables:

$$y_1 = x_1 + x_2 = 5, \qquad y_2 = x_2 + x_3 = 7, \qquad y_3 = x_1 + x_3 = 6.$$

Recover the three state variables. A control sensor measures $y_4 = x_1 + x_2 + x_3$. What is its consistent reading, and what does a reading $y_4 = 10$ imply about the exact model?

WORKED SOLUTION

Add the first two equations and subtract the third:

$$2x_2 = 5 + 7 - 6 = 6, \qquad x_2 = 3.$$

Hence $x_1 = 5 - 3 = 2$ and $x_3 = 7 - 3 = 4$. The third equation gives $2 + 4 = 6$. The sum of the three readings is twice the sum of the state variables:

$$y_1 + y_2 + y_3 = 2(x_1 + x_2 + x_3) = 2y_4.$$

The consistent reading is therefore $y_4 = (5 + 7 + 6)/2 = 9$. The reading 10 contradicts this, so the four equations are inconsistent in the exact model. This consistency check can indicate a measurement error or an unsuitable model.

EXERCISE 3.13

Solve the following system over $\mathbb{F}_2$:

$$\begin{cases} x + y + z = 0, \\ x + z = 1. \end{cases}$$

How many solutions are there? Compare with the system given by the same equations over $\mathbb{R}$.

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EXERCISE 3.13 CONTINUED

WORKED SOLUTION

Over $\mathbb{F}_2$, subtracting the second equation from the first gives $y = -1 = 1$. Put $z = t$; then $x = 1 - t = 1 + t$. Thus

$$(x, y, z) = (1 + t, 1, t), \qquad t \in \mathbb{F}_2.$$

There are exactly two solutions: $(1, 1, 0)$ and $(0, 1, 1)$. Over $\mathbb{R}$, subtraction gives $y = -1$, and we obtain

$$(x, y, z) = (1 - t, -1, t), \qquad t \in \mathbb{R}.$$

Here there are infinitely many solutions. Both systems have one free variable; the number of solutions depends on the field.

Additional practice

EXERCISE 3.14 FOUNDATION

Solve the following system over $\mathbb{R}$:

$$\begin{cases} x + y + z = 4, \\ x - y + z = 2. \end{cases}$$

Write the solution set as a particular solution plus a multiple of a fixed vector. Determine whether it is a subspace, and describe the set of differences of two solutions.

WORKED SOLUTION

Subtracting the second equation from the first gives $2y = 2$, so $y = 1$. Substitution into the second equation gives $x + z = 3$. Put $z = t \in \mathbb{R}$. Then

$$S = \{(3 - t, 1, t) : t \in \mathbb{R}\} = (3, 1, 0) + \text{span}\{(-1, 0, 1)\}.$$

These are all solutions because each subtraction and substitution is equivalent to the original system. The set is not a subspace: the zero vector does not satisfy the equations. The difference of solutions with parameters t, s is

$$(3 - t, 1, t) - (3 - s, 1, s) = (t - s)(-1, 0, 1).$$

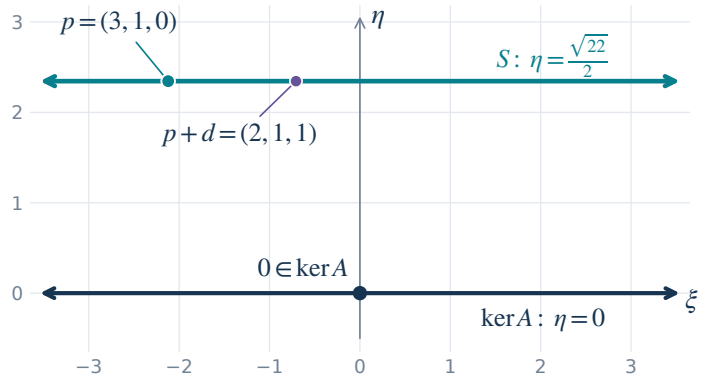
Every real multiple occurs, for example by setting $s = 0$. Thus the difference set is exactly $\text{span}\{(-1, 0, 1)\}$, which is the solution set of the associated homogeneous system.

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$$S=\{(3-t,1,t):t\in\mathbb{R}\}=p+\ker A$$

$$\Phi(\xi,\eta)=\xi u+\eta v,\qquad u=\frac{(-1,0,1)}{\sqrt{2}},\quad v=\frac{(3,2,3)}{\sqrt{22}}$$



$$\ker A=\operatorname{span}\{d\},\quad d=(-1,0,1),\qquad 0\notin S$$

Figure 3.2. The real system $x+y+z=4$, $x-y+z=2$ has solution set $S=p+\ker A$, where $p=(3,1,0)$ and $\ker A=\operatorname{span}\{(-1,0,1)\}$. The ξ,η axes describe the plane $\operatorname{span}\{u,v\}\subseteq\mathbb{R}^3$ through the map Φ displayed in the figure. The vectors u,v are perpendicular unit vectors, so the chart preserves lengths and angles. The translated line excludes the origin, whereas the kernel contains it.

EXERCISE 3.15 **CHALLENGE**

Let $a,b\in\mathbb{R}$. Classify the solutions of

$$\begin{cases}x+y+z=2,\\ x+ay+z=b,\\ x+y+az=b.\end{cases}$$

In each case, find the coefficient matrix rank, the number of free variables when consistent, and the solution set. Also find the kernel of the coefficient matrix, and determine whether the solution set is a subspace.

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WORKED SOLUTION

Subtract the first equation from both the second and the third. The equivalent system is

$$\begin{cases}x+y+z=2,\\ (a-1)y=b-2,\\ (a-1)z=b-2.\end{cases}$$

1. If $a\neq 1$, division by $a-1$ is permitted. Put $c=(b-2)/(a-1)$. The unique solution is

   $$S=\{(2-2c,c,c)\}.$$

   There are three pivots, so the rank is 3 and there are no free variables. In the homogeneous system, $y=z=0$ and then $x=0$, so $\ker A=\{0\}$.

2. If $a=1$ and $b=2$, the last two equations after subtraction are $0=0$. The rank is 1, and

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EXERCISE 3.15 **CHALLENGE CONTINUED**

there are two free variables. Choosing $y = s, z = t$ gives

$$S = \{(2 - s - t, s, t) : s, t \in \mathbb{R}\} = (2, 0, 0) + \text{span}\{(-1, 1, 0), (-1, 0, 1)\}.$$

The kernel is $\text{span}\{(-1, 1, 0), (-1, 0, 1)\}$.

3. If $a = 1$ and $b \neq 2$, the equation $0 = b - 2$ is a contradiction, so $S = \varnothing$. The coefficient matrix still has rank 1, while the augmented matrix has rank 2. The kernel depends only on A, so it is the same kernel as in the preceding case.

The solution set is never a subspace: in the consistent cases the zero vector fails to satisfy $x+y+z = 2$, and in the inconsistent case the set is empty. A number of degrees of freedom of a solution set cannot be defined through a choice of solution parameters when no solutions exist.

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TUTORIAL 4

Linear Dependence, Spanning Sets, and Sums of Subspaces

MOTIVATION AND GUIDING IDEA

Motivation. A homogeneous linear constraint allows two admissible states to be added, or an admissible state to be scaled, without leaving the admissible set. Span describes which states can be constructed; independence identifies components that are not redundant. Together, these ideas describe a system using a small collection of independent components.

NOTATION AND CONVENTIONS

Notation guide. $\mathbb{F}$ is the scalar field, V a vector space over it, 0_V the zero vector, and $\alpha \in \mathbb{F}$ a scalar. $v \in S$ denotes membership and $S \subseteq V$ inclusion. $\operatorname{span}_{\mathbb{F}} S$ is the span over the indicated field. $U \cup W$ denotes union, $U \cap W$ intersection, $U + W$ a sum of subspaces, and $U \oplus W$ a direct sum. The symbol $\setminus$ denotes set difference and $\varnothing$ the empty set.

DEFINITION 4.1

A precise formulation with hypotheses. For $S \subseteq V$, its span consists of all finite sums $\sum_{j=1}^{k} \alpha_j v_j$ with $v_j \in S$ and $\alpha_j \in \mathbb{F}$, including the empty sum, which equals 0_V. It is the smallest subspace containing S. Independence means that every finite zero linear combination of distinct members of S has only zero coefficients. In particular, the empty set is independent and its span is $\{0_V\}$.
For subspaces $U, W \leq V$ over the same field, $U + W$ is the smallest subspace containing both. The condition $U \cap W = \{0_V\}$ is equivalent to every $x \in U + W$ having a unique expression $x = u + w$ with $u \in U, w \in W$. Indeed, two expressions give $u - u' = w' - w \in U \cap W$. Conversely, if $z \in U \cap W$, then $z = z + 0_V = 0_V + z$, so uniqueness forces $z = 0_V$.

NOTE

A vector is an element of the vector space, so a matrix or polynomial can also be a vector. In an independence test, the unknowns are the *coefficients of the combination*; in a span-membership test, the target vector is given. For a nonhomogeneous condition such as $x + y + z = 1$, first check whether the zero vector belongs to the set.

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MOTIVATION AND GUIDING IDEA

Tutorial objectives. Test linear dependence and independence, find spanning sets, and distinguish the union, intersection, and sum of subspaces. State the scalar field in every test and connect the equations with their geometric interpretation.

4.1 Vector Spaces

RECALL AND TOOLS

1. To define a vector space V over a field $\mathbb{F}$, specify addition of vectors in V and multiplication of a vector in V by a scalar in $\mathbb{F}$, satisfying the vector-space axioms. These operations are specified **in addition** to the addition and multiplication in $\mathbb{F}$. Multiplication of two vectors is not a vector-space axiom; any such operation used later must be defined separately.
2. Every field is a vector space over itself, using its own addition and multiplication.
3. A vector space V is always defined over a field $\mathbb{F}$, which must be stated. For example, as discussed in the lecture, $\mathbb{C}$ can be viewed as a vector space over $\mathbb{C}$ or over $\mathbb{R}$. These are two different vector-space structures.

EXAMPLE 4.1

Let $\mathbb{F}$ be a field.

(a) $\mathbb{F}^n$: the set of n-tuples over $\mathbb{F}$ ($n \in \mathbb{N}$).

(b) $M_{m\times n}(\mathbb{F})$: the set of $m \times n$ matrices over $\mathbb{F}$.

(c) $\mathbb{F}[X]$: the set of all polynomials with coefficients in $\mathbb{F}$.

EXERCISE 4.1

Determine whether the following sets are subspaces of $\mathbb{R}[X]$ over $\mathbb{R}$, with the usual operations.

1. $V = \{p(X) \in \mathbb{R}[X] : p(5) = 0, p(4) \neq 0\}$.
2. $V = \{p(X) \in \mathbb{R}[X] : \exists c \in \mathbb{R}, p(c) = 0\}$.

WORKED SOLUTION

1. No. The zero polynomial is not in the set because its value at 4 is 0. Closure under addition also fails: $p(X) = (X-5)(X-3)$ and $q(X) = (X-5)^2$ belong to the set, since $p(4) = -1$ and $q(4) = 1$, but

$$(p+q)(X) = (X-5)(2X-8), \qquad (p+q)(4) = 0.$$

   Thus $p+q$ does not belong to the set.

2. No. The set is not closed under addition. For example, $(X+1)^2, -2X \in V$, but $(X+$

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EXERCISE 4.1 CONTINUED

$1)^2 - 2X = X^2 + 1 \notin V$.

4.1.1 Subspaces

DEFINITION 4.2

Let V be a vector space over a field $\mathbb{F}$. A subset $U \subseteq V$ is a **vector subspace** of V if it is itself a vector space over $\mathbb{F}$ under the addition and scalar multiplication inherited from V.

PROPOSITION 4.1

Let V be a vector space over a field $\mathbb{F}$ and $U \subseteq V$. Then U is a subspace if and only if all the following conditions hold:

1. $0 \in U$, where 0 is the zero vector of V.
2. U is closed under addition.
3. U is closed under scalar multiplication.

EXAMPLE 4.2

1. $V = \{p(X) \in \mathbb{F}[X] : p(5) = 0\}$ is a subspace of $\mathbb{F}[X]$. **Proof.** The zero polynomial belongs to V. If $p, q \in V$ and $r = p + q$, then $r(5) = p(5) + q(5) = 0$, so $r \in V$. If $p \in V$, $\alpha \in \mathbb{F}$, and $r = \alpha p$, then $r(5) = \alpha p(5) = 0$.
2. $V = \{(x_1, \ldots, x_n) \in \mathbb{R}^n : x_2 = 0\}$, where $n \geq 2$, is a subspace of $\mathbb{R}^n$. **Proof.** Clearly $0 \in V$. If $x = (x_1, \ldots, x_n), y = (y_1, \ldots, y_n) \in V$, then $x_2 = y_2 = 0$, so the second coordinate of $x + y$ is zero and $x + y \in V$. If $x \in V$ and $\alpha \in \mathbb{R}$, then $\alpha x_2 = 0$, hence $\alpha x \in V$.

EXERCISE 4.2

Prove or disprove.

1. The set $U = \left\{ \begin{pmatrix} a \\ a^2 \end{pmatrix} : a \in \mathbb{R} \right\}$ is a subspace of $\mathbb{R}^2$.
2. The set of symmetric matrices $\mathrm{Sym}_n(\mathbb{F}) := \{A \in M_{n\times n}(\mathbb{F}) : A = A^T\}$ is a subspace of $M_{n\times n}(\mathbb{F})$.

WORKED SOLUTION

1. This is not a subspace, since it is not closed under addition. Indeed,

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1^2 \end{pmatrix} \in U, \qquad \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \end{pmatrix} \notin U.$$

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EXERCISE 4.2 CONTINUED

2. The zero matrix is symmetric. Let $A, B \in \mathrm{Sym}_n(\mathbb{F})$ and $\lambda \in \mathbb{F}$. Then

$$(A+B)^T = A^T + B^T = A + B,$$

so $A+B$ is symmetric. Also,

$$(\lambda A)^T = \lambda A^T = \lambda A,$$

so λA is symmetric. All three conditions of the subspace test hold.

EXERCISE 4.3

An engineering sensing system models a normalized signal by a real polynomial of degree at most 3:

$$p(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3.$$

Time t is measured in normalized time units. This is a mathematical approximation to a signal, with no assumption about the sensor type.

1. Find all polynomials satisfying

$$p(0) = 0, \quad p(1) = 0.2, \quad p'(0) = 0, \quad p'(0.5) = 0.$$

Is the solution set a subspace of $\mathbb{R}[t]$? Is $t = 0.5$ a local maximum or minimum?

2. Repeat the question with the modified measurements

$$p(0) = 0.1, \quad p(1) = 0.3, \quad p'(0) = 0, \quad p'(0.5) = 0.$$

3. Add the measurement $p'(1) = 0.4$ to part 2. Is the system still consistent? Is its solution set a subspace?

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WORKED SOLUTION

Differentiate:

$$p'(t) = a_1 + 2a_2 t + 3a_3 t^2.$$

Part 1. The conditions at $t = 0$ give $a_0 = a_1 = 0$, and the remaining conditions give

$$\begin{cases} a_2 + a_3 = 0.2, \\ a_2 + 0.75 a_3 = 0. \end{cases}$$

Subtract the second equation from the first:

$$0.25 a_3 = 0.2 \quad \Longrightarrow \quad a_3 = 0.8, \qquad a_2 = -0.75 \cdot 0.8 = -0.6.$$

There is therefore a unique solution,

$$p_1(t) = -0.6 t^2 + 0.8 t^3.$$

Verification: $p_1(1) = -0.6 + 0.8 = 0.2$ and $p_1'(0.5) = -1.2 \cdot 0.5 + 2.4 \cdot 0.25 = 0$. The solution set

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EXERCISE 4.3 CONTINUED

is not a subspace: the zero polynomial does not satisfy $p(1) = 0.2$. The condition $p'(0.5) = 0$ alone identifies a stationary point, not necessarily a peak. Here

$$p_1''(t) = -1.2 + 4.8t, \qquad p_1''(0.5) = 1.2 > 0,$$

so the point is a local minimum.

Part 2. Now $a_0 = 0.1$ and $a_1 = 0$. The condition $p(1) = 0.3$ gives $a_2 + a_3 = 0.2$, so the two equations for a_2, a_3 are the same as in part 1. Thus

$$p_2(t) = 0.1 - 0.6t^2 + 0.8t^3 = p_1(t) + 0.1.$$

Again the solution is unique, and the solution set is not a subspace because it does not contain the zero polynomial. The derivatives of p_1, p_2 coincide, so $t = 0.5$ is again a local minimum.

Part 3. The unique polynomial from part 2 satisfies

$$p_2'(1) = -1.2 + 2.4 = 1.2 \neq 0.4.$$

The added condition is therefore inconsistent with the others. There is no solution, and the empty solution set is not a subspace because it does not contain the zero vector. In contrast, $\{0\}$ is a subspace.

4.2 Linear Dependence and Independence

DEFINITION 4.3

Let V be a vector space over $\mathbb{F}$ and $S \subseteq V$.

1. The set S is **linearly dependent** if there are pairwise distinct vectors $v_1, \ldots, v_k \in S$ and scalars $\alpha_1, \ldots, \alpha_k \in \mathbb{F}$, not all zero, such that
$$\alpha_1 v_1 + \cdots + \alpha_k v_k = 0.$$
2. The set S is **linearly independent** if, for every finite choice of pairwise distinct vectors $v_1, \ldots, v_k \in S$, this equality implies $\alpha_1 = \cdots = \alpha_k = 0$.

The vectors need not be tuples: they may also be polynomials, matrices, or elements of another vector space.

EXAMPLE 4.3

The set

$$\left\{ \begin{pmatrix} 0 \\ -3 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right\}$$

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is linearly dependent, since

$$1 \cdot \begin{pmatrix} 0 \\ -3 \\ 1 \end{pmatrix} - 1 \cdot \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} + 3 \cdot \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = 0.$$

Rearranging this equation expresses $\begin{pmatrix} 0 \\ -3 \\ 1 \end{pmatrix}$ in terms of the other vectors.

How Do We Test Dependence?

MOTIVATION AND GUIDING IDEA

- Write a **homogeneous** system whose unknowns are the coefficients in the linear combination.
- Form its matrix and row-reduce.
- **Linear independence:** the only solution is the trivial, or zero, solution.
- **Linear dependence:** there is a nonzero solution. Over an infinite field there are then infinitely many solutions.
- A homogeneous system always has a solution: row reduction cannot produce an inconsistent row.

EXERCISE 4.4

Let $v_1 = (1, 2, 0)^T$, $v_2 = (0, 1, 1)^T$, and $v_3 = (2, 1, -3)^T$ in $\mathbb{R}^3$. Determine whether they are linearly dependent. Find all dependence relations and express one vector in terms of the others.

WORKED SOLUTION

Write $av_1 + bv_2 + cv_3 = 0$. Comparing coordinates gives

$$a + 2c = 0, \qquad 2a + b + c = 0, \qquad b - 3c = 0.$$

The first equation gives $a = -2c$, and the third gives $b = 3c$. The second then becomes $-4c + 3c + c = 0$, imposing no further restriction. All dependence relations are

$$(a, b, c) = t(-2, 3, 1), \qquad t \in \mathbb{R}.$$

In particular, $-2v_1 + 3v_2 + v_3 = 0$, hence $v_3 = 2v_1 - 3v_2$. This is a nontrivial relation. The first two vectors are independent: the first coordinate of $av_1 + bv_2 = 0$ gives $a = 0$, and then $b = 0$.

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4.2.1 Useful Criteria for Linear Dependence

1. Every set containing the zero vector is linearly dependent.
2. A singleton consisting of a nonzero vector is linearly independent.
3. **Testing a set for linear dependence.**
   - As in the previous tutorial, set a linear combination equal to zero, obtain a homogeneous system, row-reduce, and determine its solutions.
   - Recall: only the zero solution means independence; a nonzero solution means dependence.
   - Another method from the lecture is to place the vectors as the **rows** of a matrix and row-reduce. If a zero row occurs, the original rows are dependent; if there are no zero rows, they are independent. **Explain why.**

4.3 Linear Combinations

DEFINITION 4.4

Let V be a vector space over a field $\mathbb{F}$ and let $v_1, \ldots, v_k \in V$. A vector $v \in V$ is a **linear combination** of $v_1, \ldots, v_k$ if there are $a_1, \ldots, a_k \in \mathbb{F}$ such that $v = a_1 v_1 + \cdots + a_k v_k$.

EXERCISE 4.5

Is $A = \begin{pmatrix} 2 & 5 \\ -1 & -1 \end{pmatrix}$ in the span over $\mathbb{R}$ of $B_1 = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$ and $B_2 = \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}$? Justify your answer by comparing entries.

WORKED SOLUTION

For every $a, b \in \mathbb{R}$,

$$aB_1 + bB_2 = \begin{pmatrix} a & a \\ b & b \end{pmatrix}.$$

Equality with A would require both $a = 2$ and $a = 5$, a contradiction. Thus A is not in the span. Matching the second row using $b = -1$ is insufficient: all four entries must match simultaneously.

EXERCISE 4.6

Let $A = \begin{pmatrix} 1 & 3 \\ 0 & 0 \end{pmatrix}$ over $\mathbb{R}$. Characterize all matrices B satisfying $AB = BA$, and show that they are precisely the linear combinations of A and $I_2 - A$.

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EXERCISE 4.6 CONTINUED

4

WORKED SOLUTION

Write $B = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and compute the two products separately:

$$AB = \begin{pmatrix} 1a+3c & 1b+3d \\ 0a+0c & 0b+0d \end{pmatrix} = \begin{pmatrix} a+3c & b+3d \\ 0 & 0 \end{pmatrix},$$

$$BA = \begin{pmatrix} a1+b0 & a3+b0 \\ c1+d0 & c3+d0 \end{pmatrix} = \begin{pmatrix} a & 3a \\ c & 3c \end{pmatrix}.$$

Entry $(2,1)$ gives $c = 0$, and entry $(1,2)$ gives $b = 3(a-d)$. All remaining equalities then hold automatically. Consequently,

$$B = \begin{pmatrix} a & 3(a-d) \\ 0 & d \end{pmatrix} = a\begin{pmatrix} 1 & 3 \\ 0 & 0 \end{pmatrix} + d\begin{pmatrix} 0 & -3 \\ 0 & 1 \end{pmatrix} = aA + d(I_2 - A).$$

The converse holds as well: substituting $c = 0$ and $b = 3(a-d)$ in the two products gives the same expression.

4.3.1 Spanning Sets

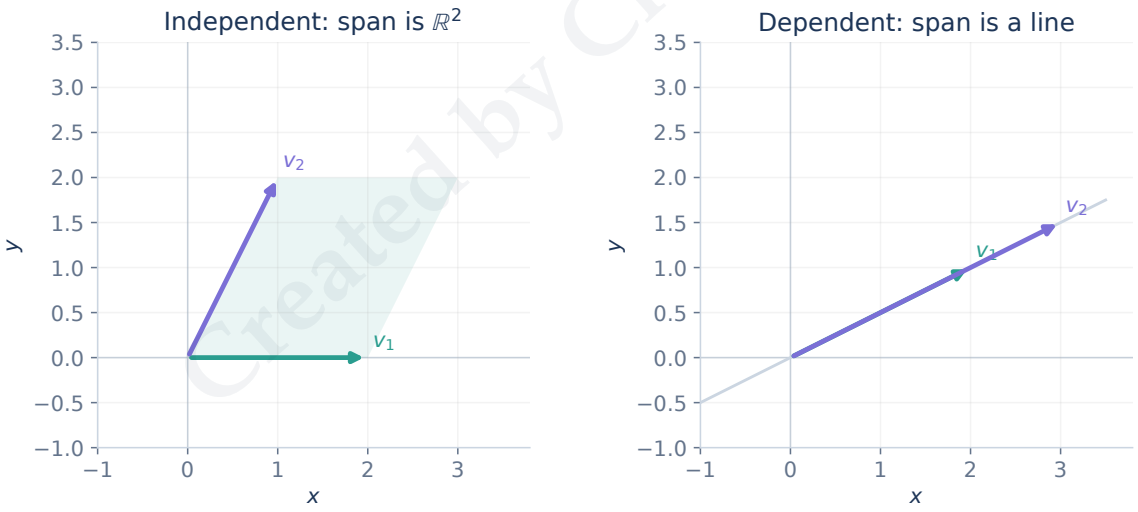


Figure 4.1. Geometric span: a nonzero vector spans a line through the origin; two independent vectors span a plane through the origin.

DEFINITION 4.5

Let V be a vector space over a field $\mathbb{F}$. For a set $S \subseteq V$, not necessarily finite, define $\operatorname{span}_{\mathbb{F}} S$ as the set of all finite linear combinations of elements of S:

$$\operatorname{span}_{\mathbb{F}} S = \left\{ \sum_{j=1}^{k} \alpha_j v_j \;\middle|\; k \geq 0,\ v_j \in S,\ \alpha_j \in \mathbb{F} \right\}.$$

The empty sum ($k = 0$) is allowed and equals the zero vector. In particular, $\operatorname{span}_{\mathbb{F}} \varnothing = \{0\}$.

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NOTE

If V is a vector space over $\mathbb{F}$ and $S \subseteq V$, then span S is a subspace of V.

DEFINITION 4.6

If V is a vector space over $\mathbb{F}$ and $S \subseteq V$ satisfies $V = \operatorname{span} S$, then S **spans** V and is called a **spanning set**. If S is finite, V is **finitely generated**.

NOTE

Pay attention to the scalar field. A set $S \subseteq V$ can span over one field and fail to span over another. When more than one field is possible, write $\operatorname{span}_{\mathbb{F}} S$ to specify the field.

EXERCISE 4.7

Let V be a vector space over $\mathbb{F}$, let $\varnothing \neq S \subseteq V$ be a finite set of size k, and let $v \in V$. Prove that

$$\operatorname{span}(S \cup \{v\}) = \operatorname{span} S \quad \Longleftrightarrow \quad v \in \operatorname{span} S.$$

WORKED SOLUTION

$\Longrightarrow$: Assume $\operatorname{span}(S \cup \{v\}) = \operatorname{span} S$. Write $v = 0 \cdot v_1 + \cdots + 0 \cdot v_k + 1 \cdot v$. Thus v is a linear combination of elements of $S \cup \{v\}$, so $v \in \operatorname{span}(S \cup \{v\})$. The assumed equality gives $v \in \operatorname{span} S$.
$\Longleftarrow$: The inclusion $\operatorname{span}(S \cup \{v\}) \supseteq \operatorname{span} S$ always holds: any linear combination of elements of S is also one of elements of $S \cup \{v\}$, with coefficient zero on v. For the opposite inclusion, write $S = \{s_1, \ldots, s_k\}$ and take $x \in \operatorname{span}(S \cup \{v\})$. There are $a_1, \ldots, a_{k+1} \in \mathbb{F}$ with

$$a_1 s_1 + \cdots + a_k s_k + a_{k+1} v = x.$$

Since $v \in \operatorname{span} S$, there are $b_1, \ldots, b_k \in \mathbb{F}$ with $b_1 s_1 + \cdots + b_k s_k = v$. Substituting gives

$$(a_1 + a_{k+1} b_1) s_1 + \cdots + (a_k + a_{k+1} b_k) s_k = x.$$

Hence $x \in \operatorname{span} S$, as required.

EXERCISE 4.8

Find a spanning set for the solution space of each homogeneous system in $\mathbb{R}^3$:

1. $\begin{cases} x + y = 0, \\ x + z = 0. \end{cases}$
2. $y + z = 0.$

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EXERCISE 4.8 CONTINUED

WORKED SOLUTION

1. The second equation gives $x = -z$, and the first gives $y = -x = z$. Set $z = t \in \mathbb{R}$. Then

$$(x, y, z) = (-t, t, t) = t(-1, 1, 1),$$

and therefore

$$S_1 = \{t(-1, 1, 1) : t \in \mathbb{R}\} = \text{span}_{\mathbb{R}}\{(-1, 1, 1)\}.$$

Every multiple of the given vector satisfies both equations, so this is the entire solution set.

2. From $y + z = 0$ we obtain $y = -z$, while x is free. Set $x = s$, $z = t$, where $s, t \in \mathbb{R}$. Then

$$(x, y, z) = (s, -t, t) = s(1, 0, 0) + t(0, -1, 1).$$

Thus

$$S_2 = \text{span}_{\mathbb{R}}\{(1, 0, 0), (0, -1, 1)\}.$$

Both vectors satisfy the equation, and every solution has the required form.

4.4 Intersections and Sums of Subspaces

MOTIVATION AND GUIDING IDEA

Distinguish union from sum. For example, the two coordinate axes in $\mathbb{R}^2$ are subspaces, but their union is not closed under addition: $(1, 0)$ and $(0, 1)$ belong to the union, whereas their sum $(1, 1)$ does not. The sum of the two axes is the whole plane.

RECALL AND TOOLS

If $U, W \subseteq V$ are subspaces of a vector space V, their **intersection** is

$$U \cap W = \{v \in V : v \in U \text{ and } v \in W\}.$$

DEFINITION 4.7

For subspaces $U, W \subseteq V$, define their **sum** by

$$U + W = \{u + w : u \in U, w \in W\}.$$

Then $U + W$ is a subspace of V containing both U and W. If also $U \cap W = \{0\}$, the sum is called a **direct sum** and is denoted $U \oplus W$.

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EXERCISE 4.9

Write

$$\mathrm{Sym}(\mathbb{F}) = \{A \in M_n(\mathbb{F}) : A = A^T\}, \qquad \mathrm{aSym}(\mathbb{F}) = \{A \in M_n(\mathbb{F}) : A = -A^T\}$$

for the symmetric and skew-symmetric matrices. Prove that both sets are subspaces of $M_n(\mathbb{F})$. In addition, prove that if $\mathrm{char}(\mathbb{F}) \neq 2$, that is, $1 + 1 \neq 0$ in the field, then

$$M_n(\mathbb{F}) = \mathrm{Sym}(\mathbb{F}) \oplus \mathrm{aSym}(\mathbb{F}).$$

WORKED SOLUTION

We verify that $\mathrm{Sym}(\mathbb{F})$ is a subspace. For $\mathrm{aSym}(\mathbb{F})$, the zero matrix belongs to the set, and if $A^T = -A$, $B^T = -B$, and $\lambda \in \mathbb{F}$, then

$$(A + B)^T = A^T + B^T = -A - B = -(A + B), \qquad (\lambda A)^T = \lambda A^T = -\lambda A.$$

Thus skew-symmetric matrices are also closed under addition and scalar multiplication and form a subspace.
Now let $A, B \in \mathrm{Sym}(\mathbb{F})$ and $\lambda \in \mathbb{F}$. First, $0 \in \mathrm{Sym}(\mathbb{F})$. For addition, $(A+B)^T = A^T+B^T = A+B$, so $A + B$ is symmetric. For scalar multiplication, $(\lambda A)^T = \lambda A^T = \lambda A$, so λA is symmetric. To prove the direct-sum assertion, first prove that the sum is the whole matrix space and then prove that the intersection is zero. For the equality $M_n(\mathbb{F}) = \mathrm{Sym}(\mathbb{F}) + \mathrm{aSym}(\mathbb{F})$, check both inclusions:

1. $M_n(\mathbb{F}) \supseteq \mathrm{Sym}(\mathbb{F}) + \mathrm{aSym}(\mathbb{F})$ holds because both summands are subspaces of $M_n(\mathbb{F})$.
2. For the reverse inclusion, take $A \in M_n(\mathbb{F})$ and write

$$A = \frac{A + A^T}{2} + \frac{A - A^T}{2}.$$

The first summand is symmetric, since

$$\left(\frac{A + A^T}{2}\right)^T = \frac{A^T + (A^T)^T}{2} = \frac{A + A^T}{2},$$

and the second is skew-symmetric, since

$$\left(\frac{A - A^T}{2}\right)^T = \frac{A^T - (A^T)^T}{2} = \frac{A^T - A}{2} = -\frac{A - A^T}{2}.$$

Hence $M_n(\mathbb{F}) \subseteq \mathrm{Sym}(\mathbb{F}) + \mathrm{aSym}(\mathbb{F})$.

Finally, suppose $A \in \mathrm{Sym}(\mathbb{F}) \cap \mathrm{aSym}(\mathbb{F})$. Symmetry gives $A = A^T$, while skew-symmetry gives $-A = A^T$. Thus $A = -A$, and $a_{ij} = -a_{ij}$ for every $1 \leq i, j \leq n$. Consequently $a_{ij} + a_{ij} = 2a_{ij} = 0$. Because $\mathrm{char}(\mathbb{F}) \neq 2$, the scalar $2 = 1 + 1$ is nonzero and invertible in the field. Therefore every $a_{ij} = 0$, so $A = 0$, as required.

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RECALL AND TOOLS

In characteristic 2, $-A = A$, so the conditions $A^T = A$ and $A^T = -A$ coincide. This is exactly why the symmetric/skew-symmetric decomposition using division by 2 requires characteristic different from 2.

4.5 Supplementary Exercises: Span, Redundancy, and the Scalar Field

EXERCISE 4.10

Three signal components satisfy the balance constraint $x + y + z = 0$. Let

$$U = \{(x, y, z) \in \mathbb{R}^3 : x + y + z = 0\},$$

$$u = (1, -1, 0), \quad v = (0, 1, -1), \quad w = (1, 0, -1).$$

Show that $U = \operatorname{span}_{\mathbb{R}}\{u, v\}$, determine whether u, v are independent, and explain what changes when w is added to the spanning set. Does $x + y + z = 1$ define a subspace?

WORKED SOLUTION

Every combination $au + bv = (a, -a + b, -b)$ has coordinate sum zero, hence belongs to U. Conversely, if $(x, y, z) \in U$, then $y = -x - z$, so

$$(x, y, z) = (x, -x - z, z) = x(1, -1, 0) - z(0, 1, -1) = xu - zv.$$

The two inclusions prove $U = \operatorname{span}\{u, v\}$. If $au + bv = 0$, the first coordinate gives $a = 0$, and the third gives $-b = 0$. Thus the pair is independent. Since $w = u + v$, adding w leaves the span unchanged and introduces the relation $u + v - w = 0$. The constraint $x + y + z = 1$ does not define a subspace because the zero vector does not satisfy it. It describes a translated plane parallel to U.

EXERCISE 4.11

Consider $S = \{1, i\} \subseteq \mathbb{C}$. Test linear dependence and spanning first over $\mathbb{R}$, then over $\mathbb{C}$.

WORKED SOLUTION

Over $\mathbb{R}$, if $a + bi = 0$ with $a, b \in \mathbb{R}$, comparing real and imaginary parts gives $a = b = 0$. Thus S is independent. Every $z = x + iy$ can be written $x \cdot 1 + y \cdot i$, so $\operatorname{span}_{\mathbb{R}} S = \mathbb{C}$. Over $\mathbb{C}$, complex coefficients are allowed, and

$$i \cdot 1 + (-1) \cdot i = 0.$$

This is a zero linear combination with nonzero coefficients, so S is dependent. It still spans: for every $z \in \mathbb{C}$, $z = z \cdot 1$. In particular, $\{1\}$ alone spans over $\mathbb{C}$.

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Additional practice

EXERCISE 4.12 FOUNDATION

Work over $\mathbb{R}$. Let $u = (1, 1, 0)^T$ and $v = (0, 1, 1)^T$.

1. Describe $U = \text{span}\{u, v\}$ by an equation.
2. Determine whether $(2, 5, 3)^T$ and $(2, 4, 3)^T$ belong to U, finding coefficients when possible.
3. Determine whether u, v are independent.

WORKED SOLUTION

For $a, b \in \mathbb{R}$,

$$au + bv = (a, a + b, b)^T.$$

Every vector in the span therefore satisfies $y = x+z$. Conversely, if $(x, y, z)^T$ satisfies $y = x+z$, then

$$(x, y, z)^T = xu + zv.$$

Thus $U = \{(x, y, z)^T : y = x + z\}$. For the first vector, $5 = 2 + 3$, and the vector is $2u + 3v$. For the second, $4 \neq 2 + 3$, so it is not in the span. Finally, if $au + bv = 0$, the first and third coordinates give $a = b = 0$. Thus u, v are independent; in particular, the coefficients found above are unique.

EXERCISE 4.13 CHALLENGE

Let U, W be subspaces of a vector space V over an arbitrary field $\mathbb{F}$. Prove that the following are equivalent:

1. $U \cup W$ is a subspace of V.
2. $U \subseteq W$ or $W \subseteq U$.
3. $U + W = U \cup W$.

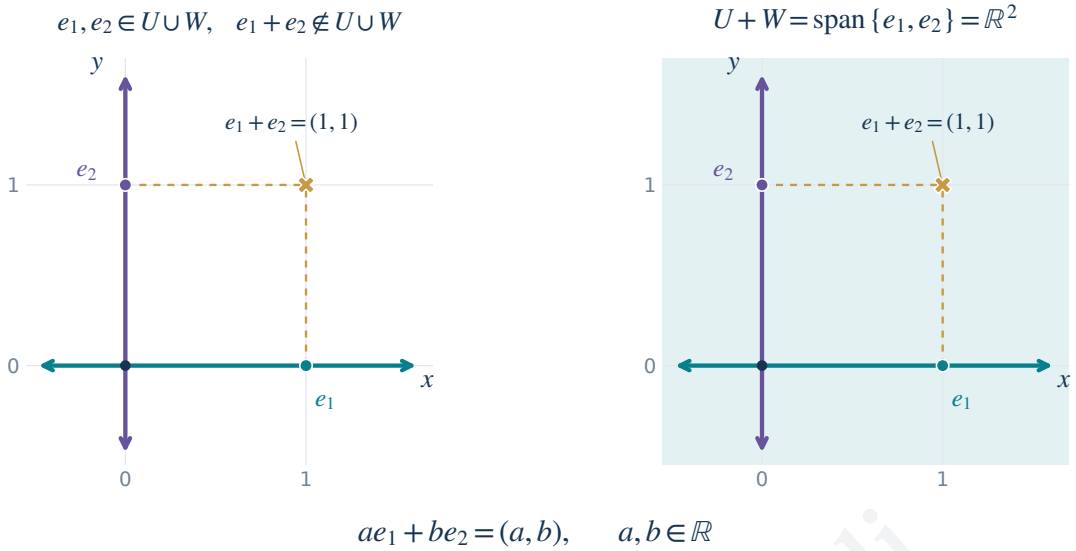
WORKED SOLUTION

(1) $\Rightarrow$ (2): Suppose neither inclusion holds. Choose $u \in U \setminus W$ and $w \in W \setminus U$. Both belong to the union, so closure under addition gives $u + w \in U \cup W$. If $u + w \in U$, then $w = (u + w) - u \in U$, a contradiction. If $u + w \in W$, then $u = (u + w) - w \in W$, again a contradiction. Hence one inclusion must hold.
(2) $\Rightarrow$ (3): If $U \subseteq W$, then $U \cup W = W$ and $U + W = W$: every such sum belongs to W, and each $w \in W$ equals $0_V + w$. The other case follows by interchanging the subspaces.
(3) $\Rightarrow$ (1): The sum $U + W$ is a subspace, so the equal set $U \cup W$ is also a subspace. The proof divides by no scalar and therefore applies to finite fields and characteristic 2 as well.

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$$U = \mathbb{R}e_1, \qquad W = \mathbb{R}e_2, \qquad e_1 = (1,0), \quad e_2 = (0,1)$$



$$ae_1 + be_2 = (a,b), \qquad a,b \in \mathbb{R}$$

Figure 4.2. An illustration over $\mathbb{R}$: the coordinate axes $U = \mathbb{R}e_1$ and $W = \mathbb{R}e_2$ are subspaces. Their union contains e_1 and e_2 but excludes their sum $(1,1)$, so it is not closed under addition. In contrast, every $(a,b) \in \mathbb{R}^2$ equals $ae_1 + be_2$, giving $U + W = \mathbb{R}^2$. Shading in the right panel represents the full plane within the viewing window.

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TUTORIAL 5

Matrix Operations, Bases, and Dimension

MOTIVATION AND GUIDING IDEA

Motivation. Matrix multiplication describes composition of linear operations: in ABx, apply B first and then A. Checking dimensions reveals in advance whether the composition is possible. A basis gives a unique description of every vector, and dimension counts the coefficients required for that description.

NOTATION AND CONVENTIONS

Notation guide. $M_{m\times n}(\mathbb{F})$ is the space of matrices with m rows and n columns; a_{ij} is the entry in row i, column j. $I_n = (\delta_{ij})$, where $\delta_{ij} = 1$ if $i = j$ and is zero otherwise. e_j is the jth standard basis vector, and Ae_j is column j of A. A^T denotes transpose, $\operatorname{diag}(d_1, \ldots, d_n)$ a diagonal matrix, $|B|$ the cardinality of a finite set, and $\dim_{\mathbb{F}} V$ the dimension over $\mathbb{F}$.

DEFINITION 5.1

Hypotheses to check. If $A \in M_{k\times m}(\mathbb{F})$ and $B \in M_{m\times n}(\mathbb{F})$, then $AB \in M_{k\times n}(\mathbb{F})$ with $(AB)_{ij} = \sum_{r=1}^{m} a_{ir}b_{rj}$. Also $(AB)^T = B^T A^T$. Over $\mathbb{C}$, the transpose A^T does not include complex conjugation.

For square matrices, define $\operatorname{Sym}_n(\mathbb{F}) = \{A : A^T = A\}$ and $\operatorname{ASym}_n(\mathbb{F}) = \{A : A^T = -A\}$. Both sets are subspaces over any field. If the characteristic is different from 2, a skew-symmetric matrix has zero diagonal and every matrix has a unique symmetric/skew-symmetric decomposition using division by 2. In characteristic 2 the two conditions coincide and do not imply a zero diagonal.

If $\dim_{\mathbb{F}} V = n < \infty$, a set of n vectors is a basis if and only if it is independent, and if and only if it spans. For $U, W \leq V$, $\dim(U + W) = \dim U + \dim W - \dim(U \cap W)$; subtracting the intersection dimension prevents shared directions from being counted twice.

MOTIVATION AND GUIDING IDEA

Tutorial objectives. Compute with matrices while checking dimensions and multiplication order, recognize symmetric structure, and choose bases. Connect algebraic calculations with sums and intersections of subspaces.

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5.1 Matrix Arithmetic

Throughout this section, let

$$A = \begin{pmatrix} 1 & 5 \\ 2 & 3 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 1 \\ 3 & 6 \end{pmatrix}.$$

5.1.1 Scalar Multiplication

Scalar multiplication is defined for every matrix. Multiply **every** entry by the scalar: $(\lambda A)_{ij} = \lambda(A)_{ij}$.

EXAMPLE 5.1

Compute

$$7A = \begin{pmatrix} 7\cdot 1 & 7\cdot 5 \\ 7\cdot 2 & 7\cdot 3 \end{pmatrix} = \begin{pmatrix} 7 & 35 \\ 14 & 21 \end{pmatrix}.$$

5.1.2 Matrix Addition and Subtraction

These are defined only for two matrices of **exactly the same shape**. Add or subtract corresponding entries: $(A \pm B)_{ij} = (A)_{ij} \pm (B)_{ij}$.

EXAMPLE 5.2

Compute

$$A + B = \begin{pmatrix} 1 & 5 \\ 2 & 3 \end{pmatrix} + \begin{pmatrix} 1 & 1 \\ 3 & 6 \end{pmatrix} = \begin{pmatrix} 1+1 & 5+1 \\ 2+3 & 3+6 \end{pmatrix} = \begin{pmatrix} 2 & 6 \\ 5 & 9 \end{pmatrix}.$$

5.1.3 Matrix Multiplication

A product is defined only when the number of columns of the first (left) matrix equals the number of rows of the second (right) matrix. If A is $k \times m$ and B is $m \times n$, then AB is $k \times n$ and is defined by

$$(AB)_{ij} = \sum_{r=1}^{m} a_{ir} b_{rj} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{im}b_{mj}.$$

This is the entry in row i, column j of the product. Follow row i of A and column j of B, each containing m entries: multiply matching entries and add. **Note:** the existence of AB does **not** ensure the existence of BA. With the dimensions above, the latter also requires $n = k$.

EXERCISE 5.1

Compute AB and BA for the matrices defined above.

WORKED SOLUTION

$$AB = \begin{pmatrix} 1\cdot 1 + 5\cdot 3 & 1\cdot 1 + 5\cdot 6 \\ 2\cdot 1 + 3\cdot 3 & 2\cdot 1 + 3\cdot 6 \end{pmatrix} = \begin{pmatrix} 16 & 31 \\ 11 & 20 \end{pmatrix},$$

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EXERCISE 5.1 CONTINUED

$$BA = \begin{pmatrix} 1\cdot 1+1\cdot 2 & 1\cdot 5+1\cdot 3 \\ 3\cdot 1+6\cdot 2 & 3\cdot 5+6\cdot 3 \end{pmatrix} = \begin{pmatrix} 3 & 8 \\ 15 & 33 \end{pmatrix}.$$

For each entry we added the products of matching row and column entries. The results differ because the rows and columns involved change with the order of the factors.

NOTE

Important conclusion: matrix multiplication is not commutative. The equality $AB = BA$ need not hold. In the example, $AB \neq BA$ even though both products are defined.

EXERCISE 5.2

Let A be a $k \times m$ matrix with a zero row, and let B be $m \times n$. Prove that the $k \times n$ matrix AB has a zero row.

WORKED SOLUTION

Let i be the index of a zero row of A, where $1 \leq i \leq k$. Thus $(A)_{ir} = 0$ for every $1 \leq r \leq m$. In row i of AB, for every $1 \leq j \leq n$,

$$(AB)_{ij} = \sum_{r=1}^{m} (A)_{ir}(B)_{rj} = \sum_{r=1}^{m} 0 = 0.$$

All entries in row i of AB are zero, as required. **The zero matrix** is a matrix of any specified shape whose entries are all zero.

5.2 Diagonal Matrices

A **diagonal matrix** is a **square** matrix whose off-diagonal entries are all zero. Write

$$D = \operatorname{diag}(d_1, \ldots, d_n) = \begin{pmatrix} d_1 & & 0 \\ & \ddots & \\ 0 & & d_n \end{pmatrix}.$$

Equivalently,

$$[D]_{ij} = \begin{cases} 0, & i \neq j, \\ d_i, & i = j. \end{cases}$$

A **scalar matrix** is a diagonal matrix with equal diagonal entries. The **identity matrix** is a diagonal matrix with all diagonal entries equal to 1, hence is a scalar matrix. A scalar matrix can therefore be written aI, where a is its common diagonal entry. Write I_n for the $n \times n$ identity, or simply I when its size is clear. For every $n \times m$ matrix A, $AI_m = I_n A = A$.

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EXAMPLE 5.3

$$IA = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 5 \\ 2 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 5 \\ 2 & 3 \end{pmatrix} = A, \qquad AI = \begin{pmatrix} 1 & 5 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 5 \\ 2 & 3 \end{pmatrix}.$$

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EXERCISE 5.3

Let A, B be diagonal matrices of order n. Prove that:

1. $A + B$ is diagonal.
2. AB is diagonal and $[AB]_{ii} = a_{ii}b_{ii}$ for every $1 \leq i \leq n$.
3. A and B **commute**: $AB = BA$.

WORKED SOLUTION

1. For $i \neq j$, $(A + B)_{ij} = a_{ij} + b_{ij} = 0 + 0 = 0$. Thus $A + B$ is diagonal, with diagonal entries $(A + B)_{ii} = a_{ii} + b_{ii}$.
2. By the definition of multiplication,

$$(AB)_{ij} = \sum_{k=1}^{n} a_{ik}b_{kj}.$$

   In row i of A, only a_{ii} can be nonzero, so the sum reduces to $a_{ii}b_{ij}$. If $i \neq j$, then $b_{ij} = 0$, giving $(AB)_{ij} = 0$. If $i = j$, then $(AB)_{ii} = a_{ii}b_{ii}$. Hence AB is diagonal.
3. Similarly BA is diagonal with diagonal entries $b_{ii}a_{ii}$. Scalars in a field commute, so

$$a_{ii}b_{ii} = b_{ii}a_{ii} \qquad (1 \leq i \leq n).$$

   All entries of AB and BA agree; therefore $AB = BA$.

NOTE

Check dimensions before computing. The product $A_{k\times m}B_{m\times n}$ has k rows and n columns. Equality of matrices requires both equal dimensions and equal corresponding entries.

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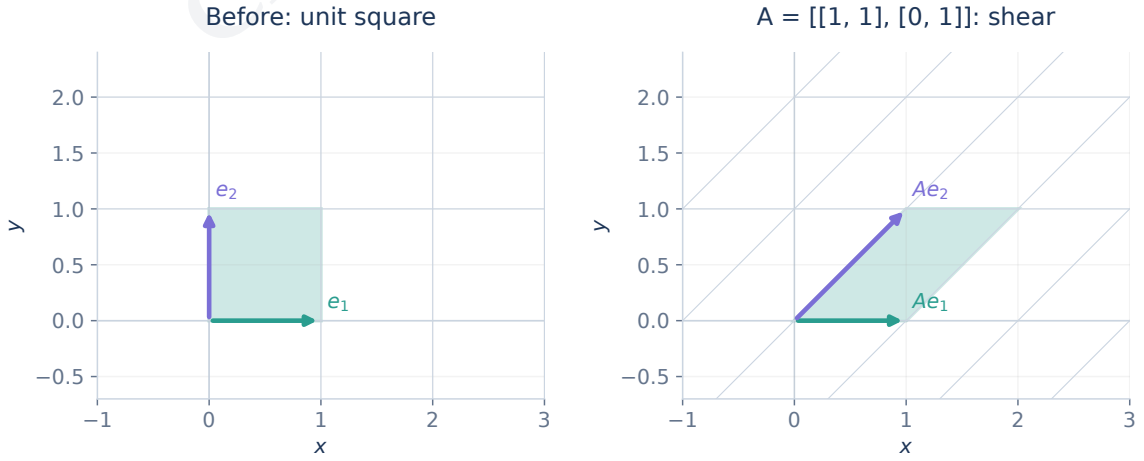


Figure 5.1. A linear map takes a straight grid to a straight grid. The columns of the matrix describe the images of the basis vectors.

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5.3 Transpose

- The transpose is defined for every matrix. The **transpose** A^T satisfies $(A^T)_{ij} = (A)_{ji}$.
- Rows become columns and columns become rows.
- If $A \in \mathbb{R}^{m\times n}$, then $A^T \in \mathbb{R}^{n\times m}$: the numbers of rows and columns are interchanged.

EXAMPLE 5.4

If

$$C = \begin{pmatrix} 1 & -2 \\ 4 & 5 \\ 7 & 3 \end{pmatrix} \in \mathbb{R}^{3\times 2}, \qquad C^T = \begin{pmatrix} 1 & 4 & 7 \\ -2 & 5 & 3 \end{pmatrix} \in \mathbb{R}^{2\times 3}.$$

The matrix C has 3 rows and 2 columns. Transposing gives a matrix with 3 columns and 2 rows.

5.4 Symmetry and Skew-Symmetry

- A **square** matrix A is **symmetric** if $A^T = A$.
- A **square** matrix A is **skew-symmetric** if $A^T = -A$.
- Every diagonal matrix is symmetric; in particular, every scalar matrix and the identity matrix are symmetric.

The square condition matters: $A \in \mathbb{F}^{m\times n}$ and $A^T \in \mathbb{F}^{n\times m}$ have the same shape only when $n = m$. Intuitively, symmetry means reflection across the main diagonal; skew-symmetry means the same reflection followed by a change of sign.

EXERCISE 5.4

Over $\mathbb{R}$, determine whether each of the following matrices is symmetric, skew-symmetric, or neither:

$$A = \begin{pmatrix} 1 & 3 \\ 3 & 2 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix}, \qquad C = \begin{pmatrix} 1 & \pi \\ -5 & 0 \end{pmatrix}.$$

WORKED SOLUTION

1. $A = \begin{pmatrix} 1 & 3 \\ 3 & 2 \end{pmatrix}$ has $A^T = \begin{pmatrix} 1 & 3 \\ 3 & 2 \end{pmatrix}$. Thus $A^T = A$ and A is symmetric.
2. $B = \begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix}$ has $B^T = \begin{pmatrix} 0 & 2 \\ -2 & 0 \end{pmatrix}$. Thus $B^T = -B$ and B is skew-symmetric.
3. $C = \begin{pmatrix} 1 & \pi \\ -5 & 0 \end{pmatrix}$ has $C^T = \begin{pmatrix} 1 & -5 \\ \pi & 0 \end{pmatrix}$. This equals neither C nor $-C$, so C is neither symmetric nor skew-symmetric.

As discussed in class, sums and scalar multiples of symmetric matrices are symmetric. This assertion also appears in the exercise later in this chapter.

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EXERCISE 5.5

Prove or disprove.

1. If A, B are symmetric, then AB is symmetric.
2. If A, B are both nonsymmetric, then AB is nonsymmetric.

WORKED SOLUTION

Part 1. Investigate the transpose:

$$(AB)^T = B^T A^T = BA.$$

The first equality is the transpose-of-a-product rule; the second uses symmetry. Since BA need not equal AB, consider the counterexample

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 2 \\ 2 & 0 \end{pmatrix}, \quad AB = \begin{pmatrix} 3 & 2 \\ 1 & 2 \end{pmatrix}.$$

Both factors are symmetric, but their product is not.

Part 2. Take $A = B = \begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix}$. Neither is symmetric, but

$$AB = \begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}.$$

The product is diagonal and hence symmetric, disproving the assertion.

EXERCISE 5.6

Let $A \in M_n(\mathbb{F})$, where $\text{char}(\mathbb{F}) \neq 2$. Prove that:

1. $A + A^T$ is symmetric.
2. $A - A^T$ is skew-symmetric.
3. A can be written as the sum of a symmetric matrix and a skew-symmetric matrix.

WORKED SOLUTION

1. $(A + A^T)^T = A^T + (A^T)^T = A^T + A = A + A^T$.
2. $(A - A^T)^T = A^T - A = -(A - A^T)$, proving skew-symmetry.
3. Add the matrices from the preceding parts:

$$2A = (A + A^T) + (A - A^T),$$
$$A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T).$$

As shown above, the first summand is symmetric and the second is skew-symmetric.

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5.5 Intersections and Sums of Subspaces

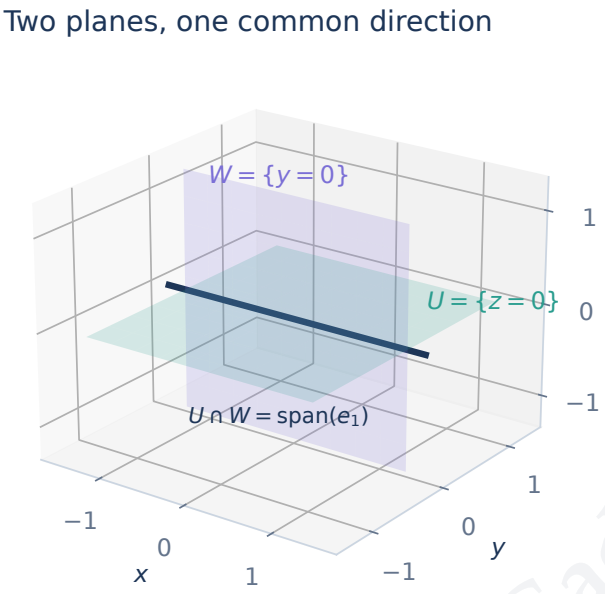


Figure 5.2. Two distinct planes through the origin in $\mathbb{R}^3$ may intersect in a line. Here $U = \{z = 0\}$ and $W = \{y = 0\}$, so $U \cap W = \operatorname{span}\{(1, 0, 0)\}$ and $U + W = \mathbb{R}^3$.

RECALL AND TOOLS

For subspaces $U, W \subseteq V$ of a vector space V, their intersection is

$$U \cap W = \{v \in V : v \in U \text{ and } v \in W\}.$$

DEFINITION 5.2

For subspaces $U, W \subseteq V$, define

$$U + W = \{u + w : u \in U, w \in W\}.$$

This is a subspace of V containing U and W. If also $U \cap W = \{0\}$, the sum is a **direct sum**, denoted $U \oplus W$.

EXERCISE 5.7

Let

$$\operatorname{Sym}(\mathbb{F}) = \{A \in M_n(\mathbb{F}) : A = A^T\}, \qquad \operatorname{aSym}(\mathbb{F}) = \{A \in M_n(\mathbb{F}) : A = -A^T\}.$$

Prove that the symmetric and skew-symmetric matrices are subspaces of $M_n(\mathbb{F})$, and that whenever $\operatorname{char}(\mathbb{F}) \neq 2$,

$$M_n(\mathbb{F}) = \operatorname{Sym}(\mathbb{F}) \oplus \operatorname{aSym}(\mathbb{F}).$$

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EXERCISE 5.7 CONTINUED

WORKED SOLUTION

We show that $\mathrm{Sym}(\mathbb{F})$ is a subspace. For the skew-symmetric set, the corresponding identities are

$$0^T = -0, \quad (A+B)^T = -A - B = -(A+B), \quad (\lambda A)^T = -\lambda A,$$

so this set is also a subspace. For the symmetric set, take $A, B \in \mathrm{Sym}(\mathbb{F})$ and $\lambda \in \mathbb{F}$. Clearly $0 \in \mathrm{Sym}(\mathbb{F})$. Closure under addition follows from $(A+B)^T = A^T + B^T = A + B$. Closure under scalar multiplication follows from $(\lambda A)^T = \lambda A^T = \lambda A$. To establish the direct sum, first prove equality with the sum and then show that the intersection is zero. Check both inclusions for $M_n(\mathbb{F}) = \mathrm{Sym}(\mathbb{F}) + \mathrm{aSym}(\mathbb{F})$:

1. $M_n(\mathbb{F}) \supseteq \mathrm{Sym}(\mathbb{F}) + \mathrm{aSym}(\mathbb{F})$ follows because both summands are subspaces of $M_n(\mathbb{F})$.
2. Take $A \in M_n(\mathbb{F})$. Then
$$A = \frac{A + A^T}{2} + \frac{A - A^T}{2}.$$
The first summand is symmetric because
$$\left(\frac{A + A^T}{2}\right)^T = \frac{A^T + (A^T)^T}{2} = \frac{A + A^T}{2}.$$
The second is skew-symmetric because
$$\left(\frac{A - A^T}{2}\right)^T = \frac{A^T - (A^T)^T}{2} = \frac{A^T - A}{2} = -\frac{A - A^T}{2}.$$
This proves the reverse inclusion.

Now take $A \in \mathrm{Sym}(\mathbb{F}) \cap \mathrm{aSym}(\mathbb{F})$. Symmetry gives $A = A^T$, and skew-symmetry gives $-A = A^T$. Therefore $A = -A$, so $a_{ij} = -a_{ij}$ for every $1 \le i, j \le n$. Thus $a_{ij} + a_{ij} = 2a_{ij} = 0$. As the field has characteristic different from 2, the scalar 2 is nonzero, so every $a_{ij} = 0$. Hence $A = 0$, as required.

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5.6 Bases and Dimension

DEFINITION 5.3

Let V be a vector space over a field $\mathbb{F}$. A set $\{u_1, \ldots, u_n\} \subseteq V$ is a **basis** of V if it is linearly independent and spans V.

A finite-dimensional vector space can have many bases, but all its bases have the same number of elements. This number is the **dimension** of the space.

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EXAMPLE 5.5

$$\dim(\mathbb{F}^n) = n, \qquad \dim(M_{n\times m}(\mathbb{F})) = nm, \qquad \dim(\mathbb{F}_n[x]) = n+1.$$

Familiar spaces have **standard bases**, denoted $e_1, e_2, \ldots, e_{\dim V}$. For instance, in $\mathbb{R}^2$, $e_1 = \begin{pmatrix}1\\0\end{pmatrix}$ and $e_2 = \begin{pmatrix}0\\1\end{pmatrix}$; in $\mathbb{R}_2[x]$, $e_1 = 1, e_2 = x, e_3 = x^2$.

RECALL AND TOOLS

In a vector space of known finite dimension n, any two of the following three properties suffice to establish that a set is a basis:

1. Linear independence.
2. Spanning the space.
3. Having exactly n elements, the dimension of the space.

EXERCISE 5.8

Determine whether

$$v_1 = (1, 0, i)^T, \qquad v_2 = (1+i, 1, 0)^T, \qquad v_3 = (2+i, 1+i, 1+2i)^T$$

form a basis of $\mathbb{C}^3$. If not, give an explicit dependence relation.

WORKED SOLUTION

For three vectors in $\mathbb{C}^3$, it suffices to test independence. Place them in columns and row-reduce:

$$\begin{pmatrix}1 & 1+i & 2+i\\ 0 & 1 & 1+i\\ i & 0 & 1+2i\end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 - iR_1} \begin{pmatrix}1 & 1+i & 2+i\\ 0 & 1 & 1+i\\ 0 & 1-i & 2\end{pmatrix}$$

$$\xrightarrow{R_3 \leftarrow R_3 - (1-i)R_2} \begin{pmatrix}1 & 1+i & 2+i\\ 0 & 1 & 1+i\\ 0 & 0 & 0\end{pmatrix}.$$

The last step uses $(1-i)(1+i) = 2$. The rank is 2, so the set is not a basis. To identify the relation, verify directly:

$$(2-i)v_1 + (1+i)v_2 = \begin{pmatrix}(2-i) + (1+i)^2\\ 1+i\\ (2-i)i\end{pmatrix} = \begin{pmatrix}2+i\\ 1+i\\ 1+2i\end{pmatrix} = v_3.$$

Thus $(2-i)v_1 + (1+i)v_2 - v_3 = 0$ is a nontrivial relation confirming the row reduction.

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EXERCISE 5.9

Determine whether $B = (x^2 - 1,\ 2x + 3,\ 1 + x + x^2)$ is a basis of $\mathbb{R}_2[x]$, and then whether it is a basis of $\mathbb{R}_3[x]$. If it is a basis in the first case, find the coordinates of $p(x) = ax^2 + bx + c$.

WORKED SOLUTION

Our convention is that $\mathbb{R}_j[x]$ consists of polynomials of degree at most j. Seek α, β, γ such that

$$ax^2 + bx + c = \alpha(x^2 - 1) + \beta(2x + 3) + \gamma(1 + x + x^2).$$

Comparing coefficients gives

$$\alpha + \gamma = a, \qquad 2\beta + \gamma = b, \qquad -\alpha + 3\beta + \gamma = c.$$

The second equation gives $\gamma = b - 2\beta$, and the first gives $\alpha = a - b + 2\beta$. Substitute in the third:

$$-(a - b + 2\beta) + 3\beta + b - 2\beta = c \iff -a + 2b - \beta = c.$$

Consequently,

$$\beta = -a + 2b - c, \quad \gamma = 2a - 3b + 2c, \quad \alpha = -a + 3b - 2c.$$

For every a, b, c there is a unique solution, so B is a basis of $\mathbb{R}_2[x]$, with

$$[p]_B = (-a + 3b - 2c,\ -a + 2b - c,\ 2a - 3b + 2c)^T.$$

However, B does not span $\mathbb{R}_3[x]$: every linear combination of its members has degree at most 2, and therefore cannot equal x^3.

EXERCISE 5.10

Let $\{u, v, w\}$ be a basis of $\mathbb{R}^3$. Are the following sets also bases of $\mathbb{R}^3$?

1. $\{u, u + v, u - v\}$.
2. $\{\frac{1}{2}u, \frac{1}{3}v, \frac{1}{4}w\}$.
3. $\{u + v + w, u + 2v, v - w\}$.

WORKED SOLUTION

Each displayed collection has three vectors in a space of dimension 3, so testing independence determines whether it is a basis.

1. Suppose $\alpha_1 u + \alpha_2(u + v) + \alpha_3(u - v) = 0$. Rearrange:

$$(\alpha_1 + \alpha_2 + \alpha_3)u + (\alpha_2 - \alpha_3)v + 0 \cdot w = 0.$$

Since $\{u, v, w\}$ is a basis, independence forces all the rearranged coefficients to be zero:

$$\begin{cases} \alpha_1 + \alpha_2 + \alpha_3 = 0, \\ \alpha_2 - \alpha_3 = 0. \end{cases}$$

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EXERCISE 5.10 CONTINUED

There are nontrivial solutions: $\alpha_2 = \alpha_3$ and $\alpha_1 = -2\alpha_3$. Thus the set is dependent and is not a basis.

2. Suppose $\alpha_1 \frac{1}{2}u + \alpha_2 \frac{1}{3}v + \alpha_3 \frac{1}{4}w = 0$. Since $\{u, v, w\}$ is a basis, $\frac{\alpha_1}{2} = 0$, $\frac{\alpha_2}{3} = 0$, and $\frac{\alpha_3}{4} = 0$. Hence the set is independent and is a basis.

3. Suppose $\alpha_1(u + v + w) + \alpha_2(u + 2v) + \alpha_3(v - w) = 0$. Rearrange:

$$(\alpha_1 + \alpha_2)u + (\alpha_1 + 2\alpha_2 + \alpha_3)v + (\alpha_1 - \alpha_3)w = 0.$$

The basis property gives

$$\begin{cases} \alpha_1 + \alpha_2 = 0, \\ \alpha_1 + 2\alpha_2 + \alpha_3 = 0, \\ \alpha_1 - \alpha_3 = 0. \end{cases}$$

In matrix form,

$$\begin{pmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 1 & 0 & -1 \end{pmatrix} \dashrightarrow \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$

There are solutions other than the trivial solution, so the set is dependent and is not a basis.

5.7 The Dimension Formula

PROPOSITION 5.1

Let V be a finitely generated vector space over a field $\mathbb{F}$, and let U be a subspace of V.

1. U is finitely generated and $\dim U \leq \dim V$.
2. $\dim U = \dim V$ if and only if $U = V$.

The dimension formula:

PROPOSITION 5.2

If V is finite-dimensional over $\mathbb{F}$ and $U, W \subseteq V$ are subspaces, then

$$\dim(U + W) = \dim U + \dim W - \dim(U \cap W).$$

EXERCISE 5.11

1. Let V have dimension n over $\mathbb{F}$, and let $U, W \subseteq V$ be subspaces such that $\dim U = n - 1$ and $W \nsubseteq U$. Prove that:
   (a) $V = U + W$.
   (b) $\dim(U \cap W) = \dim W - 1$.

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EXERCISE 5.11 CONTINUED

2. Let $V = \mathbb{R}_2[X]$, $U = \operatorname{span}\{X, X^2+1\}$, and $W = \{P \in V : P'(1) = 0\}$.
   (a) Compute $\dim(U+W)$ and $\dim(U \cap W)$.
   (b) Is $U + W = V$? If so, is the sum direct, that is, is $U \oplus W = V$?

WORKED SOLUTION

1. (a) By assumption, there is $x \in W \setminus U$. Hence $U \subsetneq U + W \subseteq V$, and the preceding dimension lemma gives

$$n - 1 = \dim U < \dim(U+W) \leq \dim V = n.$$

   Thus $\dim(U+W) = n$, and the equality case in the lemma gives $V = U + W$.

   (b) Apply the dimension formula:

$$\begin{aligned} \dim(U+W) &= \dim U + \dim W - \dim(U \cap W), \\ n &= n - 1 + \dim W - \dim(U \cap W). \end{aligned}$$

   Therefore $\dim(U \cap W) = \dim W - 1$, as required.

2. (a) We have $\dim V = 3$ and $\dim U = 2$, since its displayed spanning vectors are independent. Also $W \not\subseteq U$: for instance, $Q(X) = 1$ belongs to $W \setminus U$. All hypotheses of part 1 hold, giving $V = U + W$ and hence $\dim(U+W) = \dim V = 3$. The second conclusion of part 1 gives $\dim(U \cap W) = \dim W - 1$, so it remains to find $\dim W$. For $P(X) = aX^2 + bX + c \in W$, with $a, b, c \in \mathbb{R}$, the condition is $P'(1) = 2a + b = 0$, that is, $b = -2a$. Thus

$$\begin{aligned} W &= \{aX^2 - 2aX + c : a, c \in \mathbb{R}\} \\ &= \{a(X^2 - 2X) + c \cdot 1 : a, c \in \mathbb{R}\} \\ &= \operatorname{span}\{1, X^2 - 2X\}. \end{aligned}$$

   Consequently $\dim W = 2$, and $\dim(U \cap W) = 1$.

   (b) This is indeed a sum equal to V: $U + W \subseteq V$, and $\dim(U+W) = \dim V$ implies $U + W = V$. The sum is not direct, since

$$(X-1)^2 = (X^2+1) - 2X \in U, \qquad \frac{d}{dX}(X-1)^2\Big|_{X=1} = 0,$$

   so $(X-1)^2 \in W$. In fact $U \cap W = \operatorname{span}\{(X-1)^2\}$, consistent with the computed dimension 1.

EXERCISE 5.12

A drafting program applies a stretch and a shear in the plane:

$$D = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}, \qquad H = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}.$$

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EXERCISE 5.12 CONTINUED

Compute DH and HD entry by entry, compare their action on e_2, and find all v such that $DHv = HDv$.

WORKED SOLUTION

By definition, each entry is a sum of row-by-column products:

$$DH = \begin{pmatrix} 3\cdot 1+0\cdot 0 & 3\cdot 2+0\cdot 1 \\ 0\cdot 1+1\cdot 0 & 0\cdot 2+1\cdot 1 \end{pmatrix} = \begin{pmatrix} 3 & 6 \\ 0 & 1 \end{pmatrix},$$

$$HD = \begin{pmatrix} 1\cdot 3+2\cdot 0 & 1\cdot 0+2\cdot 1 \\ 0\cdot 3+1\cdot 0 & 0\cdot 0+1\cdot 1 \end{pmatrix} = \begin{pmatrix} 3 & 2 \\ 0 & 1 \end{pmatrix}.$$

Therefore $DHe_2 = (6,1)^T$, whereas $HDe_2 = (2,1)^T$. The rightmost operation is applied first, and changing the order changes the result. For $v = (x,y)^T$, the difference between the results is $((3x+6y)-(3x+2y),\ y-y)^T = (4y,0)^T$. The results agree if and only if $y = 0$, precisely on $\mathrm{span}\{e_1\}$.

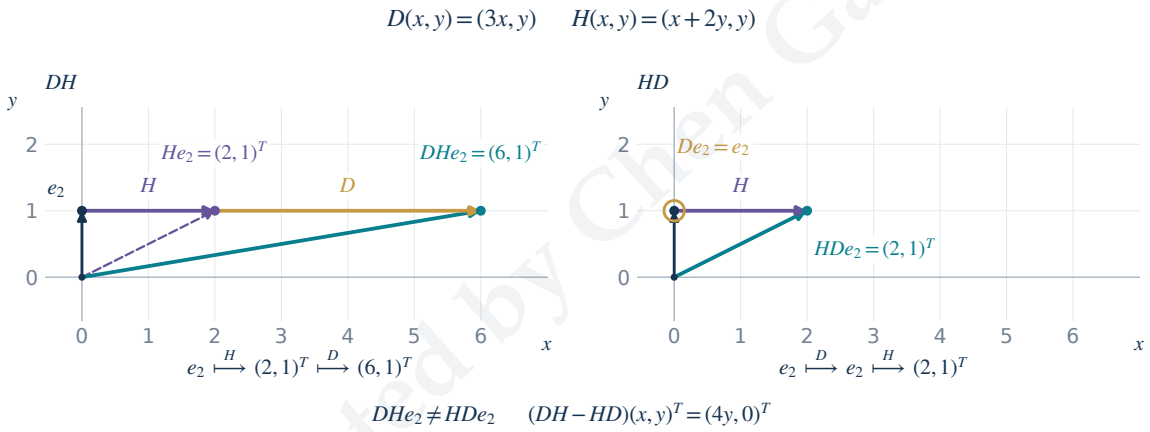


Figure 5.3. Operation order on the same vector: DH applies H first, whereas HD applies D first. The ring marks the fixed point $De_2 = e_2$, but the final results differ: $DHe_2 = (6,1)^T \neq (2,1)^T = HDe_2$.

Additional practice

EXERCISE 5.13 FOUNDATION

Over $\mathbb{R}$, let

$$A = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 2 \\ 3 & 0 \\ 0 & 1 \end{pmatrix}.$$

Determine the dimensions of AB, BA, compute both products, and use $e_2 \in \mathbb{R}^2$ to verify $(AB)e_2 = A(Be_2)$. Can AB and BA be equal as matrices?

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EXERCISE 5.13 FOUNDATION CONTINUED

5

WORKED SOLUTION

The shapes of A, B are 2×3 and 3×2, so AB has order 2 and BA order 3. Entry-by-entry calculation gives

$$AB = \begin{pmatrix} 1\cdot 1+0\cdot 3+2\cdot 0 & 1\cdot 2+0\cdot 0+2\cdot 1 \\ 0\cdot 1+1\cdot 3-1\cdot 0 & 0\cdot 2+1\cdot 0-1\cdot 1 \end{pmatrix} = \begin{pmatrix} 1 & 4 \\ 3 & -1 \end{pmatrix},$$

$$BA = \begin{pmatrix} 1\cdot 1+2\cdot 0 & 1\cdot 0+2\cdot 1 & 1\cdot 2-2\cdot 1 \\ 3\cdot 1+0\cdot 0 & 3\cdot 0+0\cdot 1 & 3\cdot 2-0\cdot 1 \\ 0\cdot 1+1\cdot 0 & 0\cdot 0+1\cdot 1 & 0\cdot 2-1\cdot 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 0 \\ 3 & 0 & 6 \\ 0 & 1 & -1 \end{pmatrix}.$$

Now $Be_2 = (2, 0, 1)^T$, so $A(Be_2) = (4, -1)^T$. Also $(AB)e_2$ is the second column of AB, namely $(4, -1)^T$. Both products are defined, but they have different dimensions; they act on different spaces and cannot be equal as matrices.

EXERCISE 5.14 CHALLENGE

Let $n \geq 2$ and $A \in M_n(\mathbb{F})$ over an arbitrary field. Prove:

1. A commutes with every diagonal matrix if and only if A is diagonal.
2. A commutes with every matrix in $M_n(\mathbb{F})$ if and only if $A = \lambda I_n$ for some $\lambda \in \mathbb{F}$.

You may use the matrix units E_{ij}, whose entry (i, j) is 1 and all other entries are zero.

WORKED SOLUTION

Suppose A commutes with every diagonal matrix. In particular, $AE_{kk} = E_{kk}A$ for every k. In entry (i, k) with $i \neq k$, the left side has value a_{ik} and the right side has value zero. Hence $a_{ik} = 0$ for every $i \neq k$, so A is diagonal. Conversely, two diagonal matrices commute because their product is diagonal and its diagonal entries are products of commuting scalars.
If A commutes with every matrix, part 1 gives $A = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$. For $i \neq j$,

$$AE_{ij} = \lambda_i E_{ij}, \qquad E_{ij}A = \lambda_j E_{ij}.$$

Comparing entry (i, j) gives $\lambda_i = \lambda_j$. All diagonal entries are therefore equal, so $A = \lambda I_n$. Conversely, for every B, $(\lambda I_n)B = \lambda B = B(\lambda I_n)$. The proof uses only the scalars $0, 1$ and field multiplication, so it applies to finite fields as well.

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TUTORIAL 6

Bases, Invertibility, and Elementary Matrices

MOTIVATION AND GUIDING IDEA

Motivation. A basis turns the construction of a vector into the determination of unique coefficients. An invertible matrix guarantees that every input can be reconstructed from its output. Elementary row operations are reversible steps, so row reduction provides both a reconstruction algorithm and an algebraic explanation of why it works.

NOTATION AND CONVENTIONS

Notation guide. For an ordered basis $B = (b_1, \dots, b_n)$, the notation $[v]_B = (c_1, \dots, c_n)^T$ means $v = \sum_{j=1}^{n} c_j b_j$. A^{-1} denotes the inverse, A^T the transpose, and $(A \mid I_n)$ an augmented matrix. $R_i \leftrightarrow R_j$ interchanges rows; $R_i \leftarrow \lambda R_i$ requires $\lambda \neq 0$; $R_i \leftarrow R_i + \lambda R_j$ requires $i \neq j$. rank A is the number of pivots in a row-echelon form. Distinguish a matrix unit E_{ij} from an entire elementary matrix E.

DEFINITION 6.1

Hypotheses and tools. Let V have dimension $n < \infty$ over $\mathbb{F}$. Every independent set in V can be extended to a basis; relative to an ordered basis, every $v \in V$ has a unique expression. If $A \in M_n(\mathbb{F})$, $n \geq 1$, is invertible, then $A^{-1} \in M_n(\mathbb{F})$ satisfies $AA^{-1} = A^{-1}A = I_n$. The inverse is unique: if B, C are inverses, then $B = B(AC) = (BA)C = C$.

The following are equivalent: A is invertible; its reduced row-echelon form is I_n; rank $A = n$; $Ax = b$ has a unique solution for every $b \in \mathbb{F}^n$; $Ax = 0$ has only the zero solution; the columns of A form a basis of $\mathbb{F}^n$. For square matrices of the same order, $(AB)^{-1} = B^{-1}A^{-1}$ when both factors are invertible.

If row operations are applied in the order $E_1, E_2, \dots, E_k$, the result is $E_k \cdots E_2 E_1 A$. If this equals I_n, then $A^{-1} = E_k \cdots E_1$, whereas $A = E_1^{-1} \cdots E_k^{-1}$. The factor order is part of the assertion.

MOTIVATION AND GUIDING IDEA

Tutorial objectives. Develop methods for choosing and extending bases, recognize invertible matrices and calculate their inverses, and connect row reduction with products of elementary matrices. Check each solution through its meaning for a system of equations.

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6.1 Bases and Dimension

DEFINITION 6.2

Let V be a vector space over a field $\mathbb{F}$. A set $\{u_1, \ldots, u_n\} \subseteq V$ is a **basis** of V if it is linearly independent and spans V.

A finite-dimensional vector space can have many bases, but all its bases have the same number of elements. This number is the **dimension** of the space.

EXAMPLE 6.1

$$\dim(\mathbb{F}^n) = n, \qquad \dim(M_{n\times m}(\mathbb{F})) = nm, \qquad \dim(\mathbb{F}_n[x]) = n+1.$$

Familiar spaces have **standard bases**, denoted $e_1, e_2, \ldots, e_{\dim V}$. For example, in $\mathbb{R}^2$, $e_1 = \begin{pmatrix}1\\0\end{pmatrix}$ and $e_2 = \begin{pmatrix}0\\1\end{pmatrix}$; in $\mathbb{R}_2[x]$, $e_1 = 1, e_2 = x, e_3 = x^2$.

RECALL AND TOOLS

Let V be a vector space over a field $\mathbb{F}$ with finite dimension, $\dim V < \infty$.

1. If B_1, B_2 are bases of V, then $|B_1| = |B_2|$. Thus dimension is well-defined: $\dim V$ is the cardinality of any basis of V.
2. If $W = \{w_1, \ldots, w_m\} \subseteq V$ is linearly independent, then
$$|W| = m \leq \dim V.$$
One cannot choose more independent vectors than the number of vectors in a basis.
3. If $W = \{w_1, \ldots, w_m\} \subseteq V$ spans V, that is, $V = \operatorname{span}(W)$, then
$$|W| = m \geq \dim V.$$
Every spanning set therefore has at least $\dim V$ vectors.
4. Suppose $W = \{w_1, \ldots, w_m\} \subseteq V$ satisfies:
   - $\dim V = |W| = m$;
   - W is either linearly independent or spans V.

   Then W is a basis of V. If W is independent and has exactly $\dim V$ vectors, no further vector can be added while preserving independence, so W already spans. If W spans and has exactly $\dim V$ vectors, there can be no nontrivial zero linear combination of its elements, so they are independent.

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PROPOSITION 6.1

An additional important fact is the following. Let V have dimension n, and let $W = \{w_1, \ldots, w_k\} \subseteq V$ be linearly independent, with $k \leq n$. There is a set

$$U = \{u_{k+1}, \ldots, u_n\} \subseteq V$$

such that

$$B = W \cup U = \{w_1, \ldots, w_k, u_{k+1}, \ldots, u_n\}$$

is a basis of V. Thus **every linearly independent set in a finite-dimensional space can be extended to a basis**. This is a central tool for constructing bases and calculating dimensions.

RECALL AND TOOLS

To establish that a set is a basis of a space of known finite dimension n, it suffices to verify any two of:

1. Linear independence.
2. Spanning the space.
3. Having exactly n elements, the dimension of the space.

EXERCISE 6.1

Determine whether

$$v_1 = (1, 0, i)^T, \qquad v_2 = (1 + i, 1, 0)^T, \qquad v_3 = (2 + i, 1 + i, 1 + 2i)^T$$

form a basis of $\mathbb{C}^3$. If not, give an explicit dependence relation.

WORKED SOLUTION

For three vectors in $\mathbb{C}^3$, it suffices to test independence. Place them in columns and row-reduce:

$$\begin{pmatrix} 1 & 1+i & 2+i \\ 0 & 1 & 1+i \\ i & 0 & 1+2i \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 - iR_1} \begin{pmatrix} 1 & 1+i & 2+i \\ 0 & 1 & 1+i \\ 0 & 1-i & 2 \end{pmatrix}$$

$$\xrightarrow{R_3 \leftarrow R_3 - (1-i)R_2} \begin{pmatrix} 1 & 1+i & 2+i \\ 0 & 1 & 1+i \\ 0 & 0 & 0 \end{pmatrix}.$$

The last step uses $(1 - i)(1 + i) = 2$. The rank is 2, so the set is not a basis. To identify the relation, verify directly:

$$(2 - i)v_1 + (1 + i)v_2 = \begin{pmatrix} (2 - i) + (1 + i)^2 \\ 1 + i \\ (2 - i)i \end{pmatrix} = \begin{pmatrix} 2 + i \\ 1 + i \\ 1 + 2i \end{pmatrix} = v_3.$$

Thus $(2 - i)v_1 + (1 + i)v_2 - v_3 = 0$ is a nontrivial relation confirming the row reduction.

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EXERCISE 6.2

Determine whether $B = (x^2 - 1,\ 2x + 3,\ 1 + x + x^2)$ is a basis of $\mathbb{R}_2[x]$, and then whether it is a basis of $\mathbb{R}_3[x]$. If it is a basis in the first case, find the coordinates of $p(x) = ax^2 + bx + c$.

WORKED SOLUTION

Our convention is that $\mathbb{R}_j[x]$ consists of polynomials of degree at most j. Seek α, β, γ such that

$$ax^2 + bx + c = \alpha(x^2 - 1) + \beta(2x + 3) + \gamma(1 + x + x^2).$$

Comparing coefficients gives

$$\alpha + \gamma = a, \qquad 2\beta + \gamma = b, \qquad -\alpha + 3\beta + \gamma = c.$$

The second equation gives $\gamma = b - 2\beta$, and the first gives $\alpha = a - b + 2\beta$. Substitute in the third:

$$-(a - b + 2\beta) + 3\beta + b - 2\beta = c \iff -a + 2b - \beta = c.$$

Consequently,

$$\beta = -a + 2b - c, \quad \gamma = 2a - 3b + 2c, \quad \alpha = -a + 3b - 2c.$$

For every a, b, c there is a unique solution, so B is a basis of $\mathbb{R}_2[x]$, with

$$[p]_B = (-a + 3b - 2c,\ -a + 2b - c,\ 2a - 3b + 2c)^T.$$

However, B does not span $\mathbb{R}_3[x]$: every linear combination of its members has degree at most 2, and therefore cannot equal x^3.

EXERCISE 6.3

Work over $\mathbb{R}$. The same proof holds over any field $\mathbb{F}$ of characteristic different from 2, after changing the scalar field in all definitions. Let

$$M_4 := M_{4\times 4}(\mathbb{R})$$

be the space of all real 4×4 matrices. Let E_{ij} have 1 in entry (i, j) and zero elsewhere. Define two subspaces of M_4:

$$\mathrm{Sym}_4(\mathbb{R}) := \{A \in M_4 : A^T = A\} \quad \text{(symmetric matrices)},$$

$$\mathrm{ASym}_4(\mathbb{R}) := \{A \in M_4 : A^T = -A\} \quad \text{(skew-symmetric matrices)}.$$

Define

$$D_i := E_{ii} \quad (1 \le i \le 4),$$

$$S_{ij} := E_{ij} + E_{ji} \quad (1 \le i < j \le 4),$$

$$K_{ij} := E_{ij} - E_{ji} \quad (1 \le i < j \le 4).$$

1. Show that

$$\mathcal{B}_{\mathrm{sym}} := \{D_1, D_2, D_3, D_4,\ S_{12}, S_{13}, S_{14}, S_{23}, S_{24}, S_{34}\}$$

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EXERCISE 6.3 **CONTINUED** 6

is a basis of $\mathrm{Sym}_4(\mathbb{R})$.

2. Show that
$$\mathcal{B}_{\text{skew}} := \{K_{12}, K_{13}, K_{14}, K_{23}, K_{24}, K_{34}\}$$
is a basis of $\mathrm{ASym}_4(\mathbb{R})$.

3. Show that the union
$$\mathcal{B} := \mathcal{B}_{\text{sym}} \cup \mathcal{B}_{\text{skew}}$$
is a basis of M_4. Thus a basis of the subspace $\mathrm{Sym}_4(\mathbb{R})$ is extended to a basis of all of M_4 by adding vectors from $\mathrm{ASym}_4(\mathbb{R})$.

4. Compute
$$\dim \mathrm{Sym}_4(\mathbb{R}), \quad \dim \mathrm{ASym}_4(\mathbb{R}), \quad \dim M_4,$$
and explicitly verify
$$\dim \mathrm{Sym}_4(\mathbb{R}) + \dim \mathrm{ASym}_4(\mathbb{R}) = \dim M_4.$$
Briefly explain how the decomposition
$$M_4 = \mathrm{Sym}_4(\mathbb{R}) \oplus \mathrm{ASym}_4(\mathbb{R})$$
appears in the bases and the dimension calculation.

WORKED SOLUTION

1. $\mathcal{B}_{\text{sym}}$ **is a basis of** $\mathrm{Sym}_4(\mathbb{R})$**.** First, every member of the set is symmetric:
$$D_i^T = D_i, \qquad S_{ij}^T = (E_{ij} + E_{ji})^T = E_{ji} + E_{ij} = S_{ij}.$$
Thus $\mathcal{B}_{\text{sym}} \subseteq \mathrm{Sym}_4(\mathbb{R})$.

Spanning. Let $A = (a_{kl}) \in \mathrm{Sym}_4(\mathbb{R})$. For $1 \le i \le 4$, the diagonal entry a_{ii} is supplied by $a_{ii}D_i$. For $1 \le i < j \le 4$, $a_{ij} = a_{ji}$, and this pair of entries is supplied by $a_{ij}S_{ij}$. Therefore
$$A = \sum_{i=1}^{4} a_{ii} D_i + \sum_{1 \le i < j \le 4} a_{ij} S_{ij}.$$
Every A is a linear combination of members of $\mathcal{B}_{\text{sym}}$, so $\mathrm{Sym}_4(\mathbb{R}) \subseteq \mathrm{span}(\mathcal{B}_{\text{sym}})$.

Independence. Suppose
$$\sum_{i=1}^{4} \alpha_i D_i + \sum_{1 \le i < j \le 4} \beta_{ij} S_{ij} = 0.$$
Only D_i contributes to entry (i,i), giving $\alpha_i = 0$ for every i. For $i < j$, only S_{ij} contributes to entry (i,j), whose value is β_{ij}. Equality with the zero matrix therefore gives $\beta_{ij} = 0$ for every $i < j$. All coefficients vanish, so $\mathcal{B}_{\text{sym}}$ is independent. Since it also spans, it is a basis. Moreover,
$$|\mathcal{B}_{\text{sym}}| = 4 + 6 = 10, \qquad \dim \mathrm{Sym}_4(\mathbb{R}) = 10.$$

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EXERCISE 6.3 CONTINUED

2. **$\mathcal{B}_{\text{skew}}$ is a basis of** $\text{ASym}_4(\mathbb{R})$**.** Each member is skew-symmetric:

$$K_{ij}^T = (E_{ij} - E_{ji})^T = E_{ji} - E_{ij} = -K_{ij},$$

so $K_{ij} \in \text{ASym}_4(\mathbb{R})$.

Spanning. If $A = (a_{kl}) \in \text{ASym}_4(\mathbb{R})$, then

$$a_{ii} = 0 \quad \text{for every } i, \qquad a_{ji} = -a_{ij} \quad \text{for } i < j.$$

It is therefore enough to specify the entries a_{ij} with $1 \le i < j \le 4$; the entries below the diagonal are then determined. We can write

$$A = \sum_{1 \le i < j \le 4} a_{ij} K_{ij},$$

because $K_{ij} = E_{ij} - E_{ji}$ supplies exactly the entries (i, j) and (j, i) with opposite signs. Thus every $A \in \text{ASym}_4(\mathbb{R})$ lies in $\text{span}(\mathcal{B}_{\text{skew}})$.

Independence. Suppose

$$\sum_{1 \le i < j \le 4} \gamma_{ij} K_{ij} = 0.$$

For $i < j$, entry (i, j) is exactly γ_{ij}. Equality with the zero matrix forces $\gamma_{ij} = 0$ for every $i < j$. All coefficients vanish, so the set is independent. Its size is

$$|\mathcal{B}_{\text{skew}}| = 6, \qquad \dim \text{ASym}_4(\mathbb{R}) = 6.$$

3. **The union $\mathcal{B}$ is a basis of M_4 and extends the first basis.** The conceptual facts are:

- $\mathcal{B}_{\text{sym}}$ is a basis of $\text{Sym}_4(\mathbb{R})$;
- $\mathcal{B}_{\text{skew}}$ is a basis of $\text{ASym}_4(\mathbb{R})$;
- the intersection of these subspaces is exactly $\{0\}$.

Recall the standard decomposition. For each $A \in M_4$, define

$$S := \frac{1}{2}(A + A^T), \qquad K := \frac{1}{2}(A - A^T).$$

Then

$$S^T = \frac{1}{2}(A^T + A) = S, \qquad K^T = \frac{1}{2}(A^T - A) = -K.$$

Thus $S \in \text{Sym}_4(\mathbb{R})$ and $K \in \text{ASym}_4(\mathbb{R})$, with $A = S + K$. Consequently,

$$M_4 = \text{Sym}_4(\mathbb{R}) + \text{ASym}_4(\mathbb{R}).$$

For the intersection, if A is in both summands, then $A^T = A$ and $A^T = -A$, giving

$$A = -A \implies 2A = 0 \implies A = 0,$$

since we work over $\mathbb{R}$, where $2 \neq 0$. Hence

$$\text{Sym}_4(\mathbb{R}) \cap \text{ASym}_4(\mathbb{R}) = \{0\},$$

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EXERCISE 6.3 CONTINUED

and

$$M_4 = \mathrm{Sym}_4(\mathbb{R}) \oplus \mathrm{ASym}_4(\mathbb{R}).$$

For the basis conclusion, use the three established facts:

- $\mathcal{B}_{\text{sym}}$ is a basis of $\mathrm{Sym}_4(\mathbb{R})$;
- $\mathcal{B}_{\text{skew}}$ is a basis of $\mathrm{ASym}_4(\mathbb{R})$;
- $\mathrm{Sym}_4(\mathbb{R}) \cap \mathrm{ASym}_4(\mathbb{R}) = \{0\}$.

Their union $\mathcal{B} = \mathcal{B}_{\text{sym}} \cup \mathcal{B}_{\text{skew}}$ is therefore a basis of the direct sum, hence a basis of all of M_4. In other words:

- We started with a basis $\mathcal{B}_{\text{sym}}$ of the subspace of symmetric matrices.
- We extended it to a basis of all of M_4 by adding the members of $\mathcal{B}_{\text{skew}}$.
- This concretely illustrates the theorem that every basis of a subspace can be extended to a basis of the whole space.

4. We already found

$$\dim \mathrm{Sym}_4(\mathbb{R}) = 10, \qquad \dim \mathrm{ASym}_4(\mathbb{R}) = 6.$$

Since M_4 contains all 4×4 matrices, $\dim M_4 = 16$. Indeed,

$$\dim \mathrm{Sym}_4(\mathbb{R}) + \dim \mathrm{ASym}_4(\mathbb{R}) = 10 + 6 = 16 = \dim M_4.$$

This is exactly the dimension relation for the direct decomposition

$$M_4 = \mathrm{Sym}_4(\mathbb{R}) \oplus \mathrm{ASym}_4(\mathbb{R}).$$

Interpretation.

- A symmetric matrix is specified by its 4 diagonal entries and its 6 entries above the diagonal, giving 10 free parameters.
- A skew-symmetric matrix has zero diagonal, and the entries below the diagonal are the negatives of those above, giving 6 free parameters.
- Together there are $10+6$ parameters, exactly the number of entries in a 4×4 matrix.
- The combined basis $\mathcal{B}$ corresponds to decomposing each matrix unit E_{ij} into symmetric and skew-symmetric parts.

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6.2 The Dimension Formula

PROPOSITION 6.2

Let V be a finitely generated vector space over a field $\mathbb{F}$, and let U be a subspace of V.

1. U is finitely generated and $\dim U \leq \dim V$.
2. $\dim U = \dim V$ if and only if $U = V$.

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The dimension formula:

PROPOSITION 6.3

If V is finite-dimensional over $\mathbb{F}$ and $U, W \subseteq V$ are subspaces, then

$$\dim(U+W) = \dim U + \dim W - \dim(U \cap W).$$

EXERCISE 6.4

1. Let V have dimension n over $\mathbb{F}$, and let $U, W \subseteq V$ be subspaces such that $\dim U = n-1$ and $W \nsubseteq U$. Prove that:
   (a) $V = U + W$.
   (b) $\dim(U \cap W) = \dim W - 1$.
2. Let $V = \mathbb{R}_2[X]$, $U = \operatorname{span}\{X, X^2+1\}$, and $W = \{P \in V : P'(1) = 0\}$.
   (a) Compute $\dim(U+W)$ and $\dim(U \cap W)$.
   (b) Is $U + W = V$? If so, is the sum direct, that is, is $U \oplus W = V$?

WORKED SOLUTION

1. (a) By assumption, there is $x \in W \setminus U$. Hence $U \subsetneq U + W \subseteq V$, and the preceding dimension lemma gives

$$n - 1 = \dim U < \dim(U+W) \leq \dim V = n.$$

   Thus $\dim(U+W) = n$, and the equality case in the lemma gives $V = U + W$.

   (b) Apply the dimension formula:

$$\begin{aligned} \dim(U+W) &= \dim U + \dim W - \dim(U \cap W), \\ n &= n - 1 + \dim W - \dim(U \cap W). \end{aligned}$$

   Therefore $\dim(U \cap W) = \dim W - 1$, as required.

2. (a) We have $\dim V = 3$ and $\dim U = 2$, since its displayed spanning vectors are independent. Also $W \nsubseteq U$: for instance, $Q(X) = 1$ belongs to $W \setminus U$. All hypotheses of part 1 hold, giving $V = U + W$ and hence $\dim(U+W) = \dim V = 3$. The second conclusion of part 1 gives $\dim(U \cap W) = \dim W - 1$, so it remains to find $\dim W$. For $P(X) = aX^2 + bX + c \in W$, with $a, b, c \in \mathbb{R}$, the condition is $P'(1) = 2a + b = 0$, that is, $b = -2a$. Thus

$$\begin{aligned} W &= \{aX^2 - 2aX + c : a, c \in \mathbb{R}\} \\ &= \{a(X^2 - 2X) + c \cdot 1 : a, c \in \mathbb{R}\} \\ &= \operatorname{span}\{1, X^2 - 2X\}. \end{aligned}$$

   Consequently $\dim W = 2$, and $\dim(U \cap W) = 1$.

   (b) This is indeed a sum equal to V: $U + W \subseteq V$, and $\dim(U+W) = \dim V$ implies

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EXERCISE 6.4 CONTINUED

$U + W = V$. The sum is not direct, since

$$(X-1)^2 = (X^2+1) - 2X \in U, \qquad \frac{d}{dX}(X-1)^2\Big|_{X=1} = 0,$$

so $(X-1)^2 \in W$. In fact $U \cap W = \text{span}\{(X-1)^2\}$, consistent with the computed dimension 1.

MOTIVATION AND GUIDING IDEA

From a basis to unique reconstruction. The columns of a square matrix form a basis if and only if every right-hand-side vector can be expressed uniquely using those columns. This is the meaning of invertibility and explains why an inverse solves a system for every right-hand side.

6.3 Invertible Matrices

DEFINITION 6.3

A matrix $A \in M_n(\mathbb{F})$, with $n \geq 1$, is **invertible** if there is $B \in M_n(\mathbb{F})$ satisfying $AB = BA = I_n$.

The unique inverse B is denoted A^{-1}.

Basic Properties

RECALL AND TOOLS

- If A is invertible, so is A^{-1}.
- If A, B are invertible, so is AB, and $(AB)^{-1} = B^{-1}A^{-1}$.
- If A is invertible, so is A^T, and $(A^T)^{-1} = (A^{-1})^T$.
- The identity is invertible, with $I^{-1} = I$.
- A square zero matrix of positive order is not invertible.

6.3.1 Exercises on the Algebra of Invertible Matrices

EXERCISE 6.5

Let A, B be square matrices, with A invertible, such that $A = B^2$ and $B = A^2$. Prove that B is the inverse of A, that is, $B = A^{-1}$.

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EXERCISE 6.5 CONTINUED

WORKED SOLUTION

We have

$$\begin{aligned} AB = AA^2 = A^3, \\ A^3 = A^3(AA^{-1}) = A^4A^{-1} \\ = (A^2A^2)A^{-1} = B^2A^{-1} = AA^{-1} = I. \end{aligned}$$

Since $B = A^2$, also $BA = A^2A = A^3 = AB = I$. Therefore $AB = BA = I$, and $B = A^{-1}$.

EXERCISE 6.6

Let A, B be square matrices such that AB is invertible. Prove that both A and B are invertible. **This is the converse of the product assertion from the lecture.**

WORKED SOLUTION

Use the fact that for square matrices of the same order, a one-sided inverse is also a two-sided inverse. Indeed, $AX = I$ implies $n = \text{rank}(AX) \leq \text{rank}(A) \leq n$, so A is invertible. From $(AB)(AB)^{-1} = I$, obtain $A[B(AB)^{-1}] = I$. Thus A is invertible, with inverse $B(AB)^{-1}$. Similarly, $(AB)^{-1}(AB) = I$ gives $[(AB)^{-1}A]B = I$. Thus B is invertible, with inverse $(AB)^{-1}A$.

6.3.2 An Algorithm for Finding the Inverse

PROPOSITION 6.4

A square matrix A is invertible if and only if it is row-equivalent to the identity matrix.

Algorithm

1. Form $(A \mid I)$ and row-reduce, aiming to put the left block into **reduced row-echelon form**.
2. Inspect the resulting left block:
   - If it is I, then A is invertible, and the right block is its inverse: $(I \mid A^{-1})$.
   - If it is not I, then A is not invertible.

EXERCISE 6.7

Determine whether the following matrices are invertible. For each invertible matrix, find its inverse by row reduction and verify the result:

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 3 \\ 0 & 1 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 2 & -4 \\ 3 & -6 \end{pmatrix}.$$

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EXERCISE 6.7 CONTINUED

WORKED SOLUTION

Row-reduce $(A \mid I_3)$:

$$\left(\begin{array}{ccc|ccc} 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right) \xrightarrow{R_1 \leftrightarrow R_2} \left(\begin{array}{ccc|ccc} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right).$$

After $R_3 \leftarrow R_3 - R_2$, this becomes

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{array}\right).$$

Finally, $R_1 \leftarrow R_1 - 3R_3$ gives

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 3 & 1 & -3 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{array}\right).$$

Hence $A^{-1} = \begin{pmatrix} 3 & 1 & -3 \\ 1 & 0 & 0 \\ -1 & 0 & 1 \end{pmatrix}$. To verify, left multiplication by A forms, respectively, the second row of the inverse, its first row plus three times its third row, and the sum of its second and third rows:

$$AA^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 3-3 & 1 & -3+3 \\ 1-1 & 0 & 1 \end{pmatrix} = I_3.$$

For B, the second row is $3/2$ times the first. Thus $R_2 \leftarrow R_2 - (3/2)R_1$ creates a zero row, so B is not invertible.

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6.4 Elementary Matrices

DEFINITION 6.4

- Elementary matrices are square matrices representing the elementary row operations used in row reduction.
- Applying an elementary row operation to a matrix is equivalent to multiplying it **on the left** by the corresponding elementary matrix.
- Each elementary matrix is obtained by applying its row operation to the identity matrix.

Consequence. Matrices A, B are row-equivalent if and only if there is a finite sequence of elementary matrices such that $B = E_1 E_2 \cdots E_k A$.

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1. **Interchanging rows** i and j is represented by $E^{(1)}_{i,j}$:

$$E^{(1)}_{i,j} = \begin{pmatrix} 1 & & & & & \\ & 0 & & 1 & & \\ & & 1 & & & \\ & 1 & & 0 & & \\ & & & & 1 & \\ & & & & & \end{pmatrix}.$$

   - This matrix is its own inverse: interchange the same two rows again.

2. **Multiplying row** i by a nonzero scalar λ is represented by $E^{(2)}_{i,\lambda}$:

$$E^{(2)}_{i,\lambda} = \begin{pmatrix} 1 & & & & \\ & \ddots & & & \\ & & \lambda & & \\ & & & \ddots & \\ & & & & 1 \end{pmatrix}.$$

   - Its inverse is $E^{(2)}_{i,\lambda^{-1}}$: multiply the same row by the reciprocal scalar.

3. **Adding** λ **times row** j to row i, with $i \neq j$, is represented by $E^{(3)}_{i,j,\lambda}$. The entry λ is in row i, column j:

$$E^{(3)}_{i,j,\lambda} = \begin{pmatrix} 1 & & & & & \\ & \ddots & & & & \\ & & 1 & & \lambda & \\ & & & \ddots & & \\ & & & & 1 & \\ & & & & & 1 \end{pmatrix}.$$

   - Its inverse is $E^{(3)}_{i,j,-\lambda}$: add $-\lambda$ times row j to row i.

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EXERCISE 6.8

Write $A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 3 \\ 0 & 1 & 1 \end{pmatrix}$ as a product of elementary matrices. Use the row reduction from the previous exercise and check the order of the factors.

WORKED SOLUTION

The three row operations interchange R_1, R_2, subtract R_2 from R_3, and subtract $3R_3$ from R_1. Their matrices are

$$E_1 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad E_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix}, \quad E_3 = \begin{pmatrix} 1 & 0 & -3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Multiplication is on the left, so $E_3E_2E_1A = I_3$. Multiplying by the inverses in the reverse

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EXERCISE 6.8 CONTINUED

order yields

$$A = E_1^{-1}E_2^{-1}E_3^{-1} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Indeed, the last two factors multiply to $\begin{pmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}$, and the first interchanges the first two rows to recover A. Each of the three factors represents one elementary row operation.

EXERCISE 6.9

Let $A, B, C \in \mathbb{R}^{n\times n}$ be invertible, and let $\alpha \in \mathbb{R}$ with $\alpha \neq 0$. Define

$$D := \alpha A^2 B^T C^3 A.$$

1. Prove that D is invertible.
2. Find an explicit expression for D^{-1} in terms of A, B, C, α.

WORKED SOLUTION

1. Each of $\alpha I_n, A, A^2, B^T, C^3$ is invertible:
   - $\alpha \neq 0$, so αI_n is invertible.
   - A is invertible by assumption, so A^2 is invertible.
   - Since B is invertible, B^T is invertible with $(B^T)^{-1} = (B^{-1})^T$.
   - Since C is invertible, C^3 is invertible with $(C^3)^{-1} = (C^{-1})^3$.

   A product of invertible matrices is invertible, so D is invertible.

2. Write the ordered product $D = \alpha A^2 B^T C^3 A$. The inverse of a product is the product of the inverses in reverse order:

   $$(XYZ)^{-1} = Z^{-1}Y^{-1}X^{-1}.$$

   Thus

   $$D^{-1} = A^{-1}(C^3)^{-1}(B^T)^{-1}(A^2)^{-1}(\alpha I_n)^{-1}.$$

   Compute each factor:

   $$(C^3)^{-1} = (C^{-1})^3, \quad (B^T)^{-1} = (B^{-1})^T, \quad (A^2)^{-1} = (A^{-1})^2, \quad (\alpha I_n)^{-1} = \frac{1}{\alpha} I_n.$$

   Substitution gives

   $$D^{-1} = A^{-1}(C^{-1})^3(B^{-1})^T(A^{-1})^2 \cdot \frac{1}{\alpha} I_n = \frac{1}{\alpha} A^{-1}(C^{-1})^3(B^{-1})^T(A^{-1})^2.$$

   This is the required expression for D^{-1}.

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EXERCISE 6.10

Let

$$A = \operatorname{diag}(\lambda_1, \ldots, \lambda_n) = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix} \in \mathbb{R}^{n \times n}.$$

Prove that A is invertible if and only if every $\lambda_i \neq 0$, and find an explicit formula for A^{-1}.

WORKED SOLUTION

If A is invertible, then every $\lambda_i \neq 0$. Suppose A is invertible and, for a contradiction, that $\lambda_k = 0$ for some index k. Consider the standard vector

$$\vec{e}_k = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix},$$

with 1 in position k and zero in every other coordinate. Then

$$A\vec{e}_k = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix} \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ \lambda_k \\ 0 \\ \vdots \\ 0 \end{pmatrix} = \vec{0},$$

since $\lambda_k = 0$. Thus $A\vec{e}_k = \vec{0}$ with $\vec{e}_k \neq \vec{0}$, a nontrivial solution of $Ax = 0$. If A were invertible, left multiplication by A^{-1} would give $e_k = A^{-1}0 = 0$, a contradiction. Therefore every $\lambda_i \neq 0$.

If every $\lambda_i \neq 0$, then A is invertible; we find its inverse. Assume all these scalars are nonzero. Define

$$B := \operatorname{diag}\left(\frac{1}{\lambda_1}, \ldots, \frac{1}{\lambda_n}\right) = \begin{pmatrix} \frac{1}{\lambda_1} & 0 & \cdots & 0 \\ 0 & \frac{1}{\lambda_2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{1}{\lambda_n} \end{pmatrix}.$$

Calculate

$$\begin{aligned} AB &= \operatorname{diag}(\lambda_1, \ldots, \lambda_n) \operatorname{diag}\left(\frac{1}{\lambda_1}, \ldots, \frac{1}{\lambda_n}\right) \\ &= \operatorname{diag}\left(\lambda_1 \frac{1}{\lambda_1}, \ldots, \lambda_n \frac{1}{\lambda_n}\right) \\ &= \operatorname{diag}(1, \ldots, 1) = I_n. \end{aligned}$$

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EXERCISE 6.10 CONTINUED

The same calculation in reverse order gives $BA = I_n$. Hence B is the inverse of A, so

$$A^{-1} = B = \operatorname{diag}\left(\frac{1}{\lambda_1}, \ldots, \frac{1}{\lambda_n}\right).$$

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6.4.1 Equivalent Conditions

PROPOSITION 6.5

For a square matrix $A \in M_n(\mathbb{F})$, the following conditions are equivalent:

1. A is invertible.
2. A is row-equivalent to I_n.
3. $\text{rank}(A) = n$.
4. For every column vector $\vec{b} \in \mathbb{F}^n$, the system $A\vec{x} = \vec{b}$ has a solution.
5. For every column vector $\vec{b} \in \mathbb{F}^n$, the system $A\vec{x} = \vec{b}$ has a unique solution $\vec{x} \in \mathbb{F}^n$.

6.4.2 The Connection with the Homogeneous System

In the preceding assertions, take $\vec{b} = \vec{0}$. There is then a unique solution, the trivial solution. Thus:

PROPOSITION 6.6

Another equivalent condition: $A \in M_n(\mathbb{R})$ is invertible if and only if its homogeneous system has only the trivial solution. In other words, the only solution of $A\vec{x} = \vec{0}$ is $\vec{x} = \vec{0}$.

EXERCISE 6.11

For $\alpha, \gamma, \phi \in \mathbb{R}$, consider $Ax = b$, where

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 3 \\ 0 & 1 & 1 \end{pmatrix}, \qquad b = (\alpha, 4\gamma, 5\phi)^T.$$

1. Express the solution in terms of the parameters.
2. Prove that the system is homogeneous if and only if its unique solution is zero.

WORKED SOLUTION

Using the inverse calculated earlier,

$$x = A^{-1}b = \begin{pmatrix} 3 & 1 & -3 \\ 1 & 0 & 0 \\ -1 & 0 & 1 \end{pmatrix} \begin{pmatrix} \alpha \\ 4\gamma \\ 5\phi \end{pmatrix} = \begin{pmatrix} 3\alpha + 4\gamma - 15\phi \\ \alpha \\ 5\phi - \alpha \end{pmatrix}.$$

The coordinates are obtained from the sums $3\alpha + 1 \cdot 4\gamma - 3 \cdot 5\phi$, α, and $-\alpha + 5\phi$. Direct substitution in the equations gives $x_2 = \alpha$, $x_1 + 3x_3 = 4\gamma$, and $x_2 + x_3 = 5\phi$. Because A is invertible, this solution is unique. The system is homogeneous exactly when $b = 0$, that is, when $\alpha = \gamma = \phi = 0$; the formula then gives $x = 0$. Conversely, if $x = 0$ is a solution, then $b = A0 = 0$, so the system is homogeneous. Invertibility ensures uniqueness in this case as well.

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EXERCISE 6.12

Three channels are coupled by $A = I_3 + N$, where

$$N = \begin{pmatrix} 0 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$

Compute N^2, N^3, find A^{-1} without row reduction, and reconstruct the state from $y = Ax = (7, 4, 2)^T$.

WORKED SOLUTION

The only nonzero entry of N^2 is $(N^2)_{13} = N_{12}N_{23} = 2 \cdot 1 = 2$. Hence

$$N^2 = \begin{pmatrix} 0 & 0 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad N^3 = 0.$$

Compute the products in both orders:

$$(I + N)(I - N + N^2) = I - N + N^2 + N - N^2 + N^3 = I,$$

$$(I - N + N^2)(I + N) = I + N - N - N^2 + N^2 + N^3 = I.$$

Therefore

$$A^{-1} = I - N + N^2 = \begin{pmatrix} 1 & -2 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}.$$

Now

$$x = \begin{pmatrix} 7 - 2 \cdot 4 + 2 \cdot 2 \\ 4 - 2 \\ 2 \end{pmatrix} = (3, 2, 2)^T.$$

Verification: $Ax = (3 + 2 \cdot 2, 2 + 2, 2)^T = (7, 4, 2)^T$. Three applications of the coupling give zero, so the inverse series terminates.

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Additional practice

EXERCISE 6.13 FOUNDATION

Over $\mathbb{R}$, let $w_1 = (1, 1, 0)^T$ and $w_2 = (0, 1, 1)^T$.

1. Show that adding $e_3 = (0, 0, 1)^T$ extends the pair to an ordered basis $B = (w_1, w_2, e_3)$ of $\mathbb{R}^3$.
2. Use row reduction to find the inverse of the matrix A whose columns are the members of B.
3. Compute $[b]_B$ for $b = (2, 5, 7)^T$ and verify the representation.

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EXERCISE 6.13 FOUNDATION CONTINUED

WORKED SOLUTION

If $aw_1 + bw_2 + ce_3 = 0$, comparison of coordinates gives $a = 0$, $a + b = 0$, and $b + c = 0$, so $a = b = c = 0$. Three independent vectors in $\mathbb{R}^3$ form a basis; in particular, the first two are independent as well.

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}.$$

Row-reduce the augmented matrix:

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right) \xrightarrow{R_2 \leftarrow R_2 - R_1} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right)$$

$$\xrightarrow{R_3 \leftarrow R_3 - R_2} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & 1 & -1 & 1 \end{array}\right).$$

Thus

$$A^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 1 & -1 & 1 \end{pmatrix}, \qquad [b]_B = A^{-1}b = \begin{pmatrix} 2 \\ -2+5 \\ 2-5+7 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix}.$$

Indeed, $2w_1 + 3w_2 + 4e_3 = (2, 2+3, 3+4)^T = (2, 5, 7)^T$. Both products $AA^{-1}, A^{-1}A$ equal I_3, and the representation is unique because B is a basis.

EXERCISE 6.14 CHALLENGE

Let $u, v \in \mathbb{R}^n$, $n \geq 1$, and define $A = I_n + uv^T$.

1. Prove that A is invertible if and only if $1 + v^T u \neq 0$, and find its inverse in this case without using a determinant.
2. Apply the formula to $u = (1, 1, 0)^T$, $v = (1, 0, 1)^T$, and solve $Ax = (5, 7, 1)^T$.

WORKED SOLUTION

Write $P = uv^T$ and $c = v^T u \in \mathbb{R}$. Associativity gives

$$P^2 = (uv^T)(uv^T) = u(v^T u)v^T = cP.$$

If $1 + c \neq 0$, define $B = I_n - P/(1 + c)$. Then

$$AB = (I_n + P)\left(I_n - \frac{P}{1+c}\right) = I_n + \left(1 - \frac{1}{1+c} - \frac{c}{1+c}\right)P = I_n.$$

Multiplication in the opposite order gives exactly the same expansion, so $BA = I_n$. Therefore

$$A^{-1} = I_n - \frac{uv^T}{1 + v^T u}.$$

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EXERCISE 6.14 **CHALLENGE CONTINUED**

If $1 + c = 0$, then $c = -1$, so $u \neq 0$, whereas

$$Au = u + u(v^T u) = (1 + c)u = 0.$$

An inverse would imply $u = 0$, a contradiction. This proves both directions.
In the example, $v^T u = 1$, so

$$A = \begin{pmatrix} 2 & 0 & 1 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}, \qquad A^{-1} = \begin{pmatrix} \frac{1}{2} & 0 & -\frac{1}{2} \\ -\frac{1}{2} & 1 & -\frac{1}{2} \\ 0 & 0 & 1 \end{pmatrix}.$$

Hence

$$x = A^{-1} \begin{pmatrix} 5 \\ 7 \\ 1 \end{pmatrix} = \begin{pmatrix} (5-1)/2 \\ -5/2 + 7 - 1/2 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \\ 1 \end{pmatrix}.$$

Direct verification: $Ax = (2 \cdot 2 + 1, 2 + 4 + 1, 1)^T = (5, 7, 1)^T$.

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TUTORIAL 7

Rank, Row and Column Spaces, and Measurement Systems

MOTIVATION AND GUIDING IDEA

Motivation. In $Ax = b$, the column space determines attainable measurements, while the kernel determines undetectable changes in the state. Distinguishing these roles allows existence and uniqueness to be tested before solving explicitly.

NOTATION AND CONVENTIONS

- $\mathbb{F}^{m\times n}$ is the space of matrices with m rows and n columns over the field $\mathbb{F}$; A^T denotes the transpose.
- $\mathcal{R}(A) = \text{Row}(A) \subseteq \mathbb{F}^n$ is the row space; $\mathcal{C}(A) = \text{Col}(A) \subseteq \mathbb{F}^m$ is the column space. Identify a row with its transposed column when matrix multiplication requires it.
- $N(A) = \ker A = \{x \in \mathbb{F}^n : Ax = 0\}$ is the kernel; $\text{nullity}(A) = \dim \ker A$ is the nullity; $\text{rank}(A) = \dim \text{Col}(A) = \dim \text{Row}(A)$ is the rank.
- span S consists of all finite linear combinations of vectors in S; e_j is the standard basis vector with entry 1 at index j and 0 elsewhere.
- $x_0 + \ker A = \{x_0 + v : v \in \ker A\}$ is an affine space, a translate of the kernel. Its affine dimension is $\dim \ker A$; it need not be a vector subspace.

DEFINITION 7.1

Let $A \in \mathbb{F}^{m\times n}$, and define $T_A : \mathbb{F}^n \to \mathbb{F}^m$ by $T_A(x) = Ax$. Then $\text{Im}\, T_A = \text{Col}(A)$ and $\ker T_A = N(A)$, so

$$\text{rank}(A) + \dim \ker A = n.$$

The right-hand side is the dimension of the **domain**. The system $Ax = b$ is solvable if and only if $b \in \text{Col}(A)$. If x_0 is one solution, **all** solutions form $x_0 + \ker A$. An existing solution is therefore unique if and only if $\ker A = \{0\}$. The nonzero echelon rows form a basis of the row space, but a column-space basis must use the **original** columns at the pivot indices.

MOTIVATION AND GUIDING IDEA

Tutorial objectives. Distinguish the row space, column space and null space, and use rank and nullity to quantify information and freedom in a system. The measurement examples apply to engineering systems in many disciplines.

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7.1 Row and column spaces and the rank theorem

RECALL AND TOOLS

The **row space** $\mathcal{R}(A)$ of $A \in \mathbb{F}^{m\times n}$ is the subspace of $\mathbb{F}^n$ spanned by its rows. .

- Elementary row operations preserve the row space of a matrix.

PROPOSITION 7.1

- **Claim 1.** If $A, B \in M_{m\times n}(\mathbb{F})$ are row equivalent, then $\mathcal{R}(A) = \mathcal{R}(B)$.
- **Claim 2:** If A is in row echelon form, its nonzero rows form a basis of $\mathcal{R}(A)$, (and are therefore linearly independent in $\mathbb{F}^n$).

RECALL AND TOOLS

The **column space** $\mathcal{C}(A)$ of $A \in \mathbb{F}^{m\times n}$ is the subspace of $\mathbb{F}^m$ spanned by its columns. .

- Row reduction **does not in general preserve the column space**; only its **dimension** is preserved.
- $\mathcal{C}(A) = \mathcal{R}(A^T)$ when a row vector is identified with its transposed column vector.

PROPOSITION 7.2

The rank theorem: Every matrix A satisfies $\dim \mathcal{C}(A) = \dim \mathcal{R}(A)$. This common dimension is called the **rank of the matrix**.

RECALL AND TOOLS

- The row and column spaces need not be the same subspace. For a rectangular matrix they even lie in different ambient spaces, $\mathbb{F}^n$ and $\mathbb{F}^m$, respectively; their dimensions are nevertheless equal.
- Equivalently, rank$A$ is the number of pivots in the reduced row echelon form.
- For $A \in \mathbb{R}^{m\times n}$, $\text{rank}(A) \leq \min(m, n)$.
- A square matrix $A \in \mathbb{R}^{n\times n}$ is invertible if and only if $rank(A) = n$
- $rank(A \cdot B) \leq min\{rank(A), rank(B)\}$
- $rank(A + B) \leq rank(A) + rank(B)$

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EXERCISE 7.1

Find a basis and the dimension of the row space and column space of

$$A = \begin{pmatrix} 3 & -6 & 9 \\ 1 & -2 & 3 \end{pmatrix}.$$

Explain which matrix supplies the vectors in each basis.

WORKED SOLUTION

Interchange the rows, then subtract three times the first row from the second:

$$\begin{pmatrix} 3 & -6 & 9 \\ 1 & -2 & 3 \end{pmatrix} \sim \begin{pmatrix} 1 & -2 & 3 \\ 3 & -6 & 9 \end{pmatrix} \sim \begin{pmatrix} 1 & -2 & 3 \\ 0 & 0 & 0 \end{pmatrix}.$$

Row operations preserve the row space, so $\{(1, -2, 3)\}$ is a basis for it. For the column space, choose the original column at the pivot index: $\{(3, 1)^T\}$. Indeed, the second and third columns are -2 and 3 times the first. Both spaces have dimension 1, but the rows lie in $\mathbb{R}^3$ and the columns in $\mathbb{R}^2$.

EXERCISE 7.2

Let $A \in \mathbb{R}^{m\times n}, B \in \mathbb{R}^{n\times m}$ satisfy $AB = I_m$. Find the ranks of A and B.
What can be concluded about m and n?

WORKED SOLUTION

The rank inequalities give $\text{rank}(AB) = \text{rank}(I_m) = m \leq \text{rank}(A) \leq m$.
Thus $\text{rank}(A) = m$. For B, $m = \text{rank}(AB) \leq \text{rank}(B) \leq m$, so $\text{rank}(B) = m$ as well.
Since $\text{rank}(A) = m \leq \min(m, n)$, we obtain $m \leq n$.
The matrices are not assumed square, so one cannot conclude that B is an inverse of A.

7.1.1 Reducing spanning sets and extending independent sets to bases in $\mathbb{F}^n$

Every spanning set contains a basis, and every linearly independent set extends to a basis.

Reducing a spanning set to a basis

MOTIVATION AND GUIDING IDEA

1. **Method A:** We want a basis consisting of vectors from the original set (as in the preceding exercise).
   - Form a matrix whose **columns** are the vectors of S and reduce it to row echelon form.
   - For each pivot column, choose the corresponding **original column** as a basis vector. These columns are linearly independent.
   - The column space is not preserved by row operations; therefore the columns must be taken from the original matrix.
2. **Method B:** Find a **new** linearly independent set spanning the same space.

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- Form a matrix whose **rows** are the vectors of S, and reduce it to row echelon form.
- Its nonzero echelon rows form a basis.
- Row operations preserve the row space, so the basis may be taken after reduction.

Extending a linearly independent set to a basis

MOTIVATION AND GUIDING IDEA

- Form a matrix whose **rows** are the vectors of S, and reduce it to row echelon form.
- For each nonpivot column j, adjoin the standard basis vector e_j to S. To justify this construction, first replace the original rows by the echelon rows, preserving their span. At the pivot indices, the added standard basis vectors are zero, whereas the echelon rows give a triangular matrix with nonzero diagonal. Hence a vanishing linear combination first forces the coefficients of the echelon rows to be zero, and then those of the added standard basis vectors. Since the original rows are independent, replacing the echelon rows back by the originals preserves the basis property.

EXERCISE 7.3

Let $S = \left\{ \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{pmatrix}, \begin{pmatrix} 2 \\ 4 \\ 2 \\ 4 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 0 \\ 1 \\ 2 \end{pmatrix} \right\} \subseteq \mathbb{R}^5$

1. Show that S is linearly independent.
2. Extend S to a basis of $\mathbb{R}^5$.

WORKED SOLUTION

1. Form a matrix with these vectors as rows, and row reduce:

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 4 & 2 & 4 & 2 \\ 1 & 2 & 0 & 1 & 2 \end{pmatrix} \xrightarrow{\text{row operations}} \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 0 & 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 & 3 \end{pmatrix}$$

   The echelon form has no zero row, so S is linearly independent.

2. Two more vectors are needed for a basis of $\mathbb{R}^5$. Form a matrix with the given vectors as **rows**, and reduce:

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 4 & 2 & 4 & 2 \\ 1 & 2 & 0 & 1 & 2 \end{pmatrix} \xrightarrow{\text{row operations}} \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 0 & 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 & 3 \end{pmatrix}$$

   The nonpivot columns are 2 and 4. Therefore adjoin to the **original** set S the corre-

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EXERCISE 7.3 CONTINUED

sponding standard basis vectors: e_2, e_4

$$B = S \cup \{e_2, e_4\} = \left\{ \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{pmatrix}, \begin{pmatrix} 2 \\ 4 \\ 2 \\ 4 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 0 \\ 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \right\}$$

EXERCISE 7.4

1. Find a basis B of $V = \text{span}\left\{ \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix}, \begin{pmatrix} 2 \\ 3 \\ 4 \\ 5 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} \right\}$ over $\mathbb{R}$. What is its dimension?
2. Extend your basis B to a basis D of $\mathbb{R}^4$.

WORKED SOLUTION

1. Form a matrix whose rows are the given vectors, and row reduce:

$$\begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 5 \\ 1 & 1 & 1 & 1 \end{pmatrix} \xrightarrow{\text{row operations}} \begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & -1 & -2 & -3 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Since the vectors were entered as rows, row operations preserve their span. We may therefore choose the **nonzero rows** of the **echelon matrix**; they are linearly independent.

Thus the required basis is $B = \left\{ \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix}, \begin{pmatrix} 0 \\ -1 \\ -2 \\ -3 \end{pmatrix} \right\}$. It contains two vectors, so the dimension is 2.

2. Two further vectors are needed to obtain a basis of $\mathbb{R}^4$.

Write the vectors of B as rows. This matrix is already in echelon form, with pivot columns $1, 2$. Adjoin the standard basis vectors at the nonpivot indices $3, 4$, namely e_3, e_4:

$$\begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & -1 & -2 & -3 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & -1 & -2 & -3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

The enlarged matrix has four independent rows, so the required basis is

$$D = B \cup \{e_3, e_4\} = \left\{ \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix}, \begin{pmatrix} 0 \\ -1 \\ -2 \\ -3 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \right\}.$$

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EXERCISE 7.5 **EXAM QUESTION**

Work over $\mathbb{Z}_3 = \{0, 1, 2\}$, with all arithmetic modulo 3. For each $\omega \in \mathbb{Z}_3$, define

$$A(\omega) = \begin{pmatrix} 1 & \omega & 1 \\ \omega & 1 & 1 \\ 1 & 1 & \omega \end{pmatrix}, \qquad b = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}, \qquad c = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}.$$

1. Determine the values of ω for which $b \in \mathrm{Col}(A(\omega))$, and find all solutions of $A(\omega)x = b$ in those cases.
2. Determine the values of ω for which both $b \in \mathrm{Col}(A(\omega))$ and $c \in \mathrm{Col}(A(\omega))$ hold.

WORKED SOLUTION

A vector belongs to a matrix's column space if and only if the corresponding linear system is solvable. There are only three possible parameter values, so examine all three.

1. For $\omega = 0$, the system is

$$x_1 + x_3 = 1, \qquad x_2 + x_3 = 2, \qquad x_1 + x_2 = 0.$$

The first two equations give $x_1 = 1 - x_3$ and $x_2 = 2 - x_3$. Substitution into the third yields

$$(1 - x_3) + (2 - x_3) = 0 \iff -2x_3 = 0 \iff x_3 = 0,$$

because $3 = 0$ and $-2 = 1$ in $\mathbb{Z}_3$. The unique solution is therefore

$$x = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}.$$

For $\omega = 1$, all rows agree, so every vector in the column space has the form $(a, a, a)^T$. Since $b = (1, 2, 0)^T$ does not, there is no solution.

For $\omega = 2$, reduce the augmented matrix using arithmetic modulo 3:

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 2 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 \end{array}\right) \xrightarrow[R_3 \leftarrow R_3 - R_1]{R_2 \leftarrow R_2 - 2R_1} \left(\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & 0 & 2 & 0 \\ 0 & 2 & 1 & 2 \end{array}\right).$$

The second row gives $2x_3 = 0$, hence $x_3 = 0$. The third gives $2x_2 = 2$, hence $x_2 = 1$. Finally, $x_1 + 2 = 1$, so $x_1 = 2$. The unique solution is

$$x = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}.$$

Thus $b \in \mathrm{Col}(A(\omega))$ exactly when $\omega \in \{0, 2\}$.

2. Expansion along the first row gives

$$\begin{aligned} \det A(\omega) &= (\omega - 1) - \omega(\omega^2 - 1) + (\omega - 1) \\ &= -\omega^3 + 3\omega - 2. \end{aligned}$$

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EXERCISE 7.5 **EXAM QUESTION CONTINUED**

Consequently $\det A(0) = 1$ and $\det A(2) = 2$ in $\mathbb{Z}_3$. Both matrices are invertible, so their column spaces are all of $\mathbb{Z}_3^3$. In particular, both b and c lie in them. For a direct check,

$$A(0)\begin{pmatrix}1\\0\\1\end{pmatrix} = c, \qquad A(2)\begin{pmatrix}0\\1\\0\end{pmatrix} = c.$$

For $\omega = 1$, we already know that b does not belong to the column space. The answer is therefore exactly $\omega \in \{0, 2\}$.

7.2 The null space and nullity of a matrix

DEFINITION 7.2

Let $\mathbb{F}$ be a field and $A \in \mathbb{F}^{m\times n}$.
The **null space of** A, also called its *kernel*, is the set of column vectors

$$N(A) := \{x \in \mathbb{F}^n \mid Ax = 0\},$$

that solve the homogeneous system $Ax = 0$.

- $N(A)$ is a vector subspace of $\mathbb{F}^n$.
- Its dimension is the **nullity** of A, denoted

$$\dim N(A) = \text{null}(A).$$

PROPOSITION 7.3

For $A \in \mathbb{F}^{m\times n}$,

$$\text{rank}(A) + \text{null}(A) = n.$$

- A has n columns, each a vector in $\mathbb{F}^m$.
- Reduce A to reduced row echelon form.
- The number of pivot columns equals $\text{rank}(A)$. The columns of the *original matrix* at these pivot indices form a basis of $\text{Col}(A)$; row reduction itself need not preserve the column space.
- There are $n - \text{rank}(A)$ nonpivot columns. These correspond to free variables in $Ax = 0$, each contributing one independent parameter to its solution space. Consequently,

$$\dim N(A) = \text{null}(A) = n - \text{rank}(A).$$

- Therefore,

$$\text{rank}(A) + \text{null}(A) = \text{rank}(A) + \big(n - \text{rank}(A)\big) = n.$$

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EXERCISE 7.6

Find a basis and the dimension of the kernel of

$$A = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \end{pmatrix}.$$

Describe all solutions of $Ax = 0$ and verify the rank-nullity theorem.

WORKED SOLUTION

The equations are $x_1 + 2x_2 = 0$, $x_2 + x_3 = 0$. Set $x_3 = t$; then

$$x_2 = -t, \qquad x_1 = 2t,$$

and hence

$$\ker A = \{t(2, -1, 1)^T : t \in \mathbb{R}\} = \text{span}\{(2, -1, 1)^T\}.$$

The spanning vector is nonzero, so it is a basis and the nullity is 1. The matrix has two pivot columns, so its rank is 2. Indeed, $\text{rank } A + \dim \ker A = 2 + 1 = 3$, the number of columns.

MOTIVATION AND GUIDING IDEA

Different questions require different spaces. The column space describes attainable right-hand sides. The null space describes changes in the input that leave the output unchanged. Rank and nullity measure these two kinds of freedom.

7.3 Interpreting matrix-vector multiplication

DEFINITION 7.3

Let $\mathbb{F}$ be a field and $A \in \mathbb{F}^{m \times n}$. Write its columns as

$$A = \begin{pmatrix} | & & | \\ a_1 & \dots & a_n \\ | & & | \end{pmatrix}, \qquad a_j \in \mathbb{F}^m.$$

Let $v = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} \in \mathbb{F}^n$.

1. **A linear combination of columns.** Matrix-vector multiplication is defined by

$$Av = v_1 a_1 + \cdots + v_n a_n.$$

Thus Av is the linear combination of the columns of A with coefficients $v_1, \dots, v_n$. It follows that

$$\{Av \mid v \in \mathbb{F}^n\} = \text{span}\{a_1, \dots, a_n\} = \text{Col}(A).$$

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2. **A linear map.** The function

$$T_A : \mathbb{F}^n \to \mathbb{F}^m, \qquad T_A(v) = Av,$$

is linear: for all $u, v \in \mathbb{F}^n$ and $\lambda \in \mathbb{F}$,

$$T_A(u+v) = A(u+v) = Au + Av,$$

$$T_A(\lambda v) = A(\lambda v) = \lambda(Av).$$

Thus multiplication by A is a linear map, and its columns describe its action on the standard basis of $\mathbb{F}^n$:

$$T_A(e_j) = a_j.$$

EXERCISE 7.7

Let $\mathbb{F}$ be a field and $B \in \mathbb{F}^{n\times m}$. Define

$$J := \{Bv \mid v \in \mathbb{F}^m\} \subseteq \mathbb{F}^n.$$

Show that

$$J = \mathrm{Col}(B).$$

WORKED SOLUTION

Denote the columns of B by

$$B = \begin{pmatrix} | & & | \\ b_1 & \dots & b_m \\ | & & | \end{pmatrix},$$

so that $b_1, \dots, b_m \in \mathbb{F}^n$ are its column vectors.
Now let

$$v = \begin{pmatrix} \lambda_1 \\ \vdots \\ \lambda_m \end{pmatrix} \in \mathbb{F}^m.$$

By the definition of matrix-vector multiplication,

$$Bv = \lambda_1 b_1 + \cdots + \lambda_m b_m.$$

We prove both inclusions.

1. First, $J \subseteq \mathrm{Col}(B)$.

   Let $y \in J$. Then $y = Bv$ for some $v \in \mathbb{F}^m$.

   Every such vector is a linear combination of $b_1, \dots, b_m$:

$$y = Bv = \lambda_1 b_1 + \cdots + \lambda_m b_m$$

   for suitable scalars $\lambda_1, \dots, \lambda_m \in \mathbb{F}$.

   Thus y belongs to their span, namely

$$y \in \mathrm{span}\{b_1, \dots, b_m\} = \mathrm{Col}(B).$$

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EXERCISE 7.7 CONTINUED

This proves $J \subseteq \mathrm{Col}(B)$.

2. Conversely, $\mathrm{Col}(B) \subseteq J$.

   Let $w \in \mathrm{Col}(B)$. There are scalars $\eta_1, \dots, \eta_m \in \mathbb{F}$ such that

   $$w = \eta_1 b_1 + \cdots + \eta_m b_m.$$

   Define

   $$v := \begin{pmatrix} \eta_1 \\ \vdots \\ \eta_m \end{pmatrix} \in \mathbb{F}^m.$$

   The preceding calculation gives

   $$Bv = \eta_1 b_1 + \cdots + \eta_m b_m = w.$$

   Hence $w = Bv$ for some $v \in \mathbb{F}^m$, so

   $$w \in J.$$

   This proves $\mathrm{Col}(B) \subseteq J$.

The two inclusions yield

$$J = \mathrm{Col}(B),$$

as required.

EXERCISE 7.8

Let $\mathbb{F}$ be a field and let $A, B \in \mathbb{F}^{n \times n}$ satisfy

$$AB = 0.$$

Show that

$$\mathrm{Col}(B) \subseteq N(A),$$

Every column of B, and every linear combination of its columns, is therefore annihilated by A.

WORKED SOLUTION

Use the definitions of column space and null space.
It suffices to prove that $Av = 0$ for every $v \in \mathrm{Col}(B)$.
By the definition of column space, $v \in \mathrm{Col}(B)$ implies that some $w \in \mathbb{F}^n$ satisfies

$$v = Bw.$$

Compute

$$Av = A(Bw) = (AB)w.$$

Since $AB = 0$,

$$(AB)w = 0w = 0,$$

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EXERCISE 7.8 CONTINUED

and therefore

$$Av = 0.$$

We have proved that $Av = 0$ for every $v \in \mathrm{Col}(B)$, or equivalently,

$$v \in N(A).$$

Consequently,

$$\mathrm{Col}(B) \subseteq N(A),$$

as required.

7.4 Exercises

EXERCISE 7.9

Let V be a vector space over a field $\mathbb{F}$, and let $S, R \subseteq V$ be arbitrary subsets, not necessarily finite.
Write

$$\mathrm{span}(S) = \left\{ \sum_{i=1}^{k} \alpha_i s_i \;\middle|\; k \in \mathbb{Z}_{\geq 0},\ \alpha_i \in \mathbb{F},\ s_i \in S \right\},$$

The empty sum is the zero vector, so $\mathrm{span}(\varnothing) = \{0\}$. Define $\mathrm{span}(R)$ and $\mathrm{span}(S \cup R)$ similarly.
Show that, for all $S, R \subseteq V$,

$$\mathrm{span}(S) + \mathrm{span}(R) = \mathrm{span}(S \cup R).$$

WORKED SOLUTION

Write

$$U := \mathrm{span}(S), \qquad W := \mathrm{span}(R).$$

Recall that the sum $U + W$ is defined by

$$U + W := \{u + w \mid u \in U,\ w \in W\}.$$

We prove

$$\mathrm{span}(S) + \mathrm{span}(R) = \mathrm{span}(S \cup R)$$

by establishing both inclusions:

$$\mathrm{span}(S) + \mathrm{span}(R) \subseteq \mathrm{span}(S \cup R) \quad \text{and} \quad \mathrm{span}(S \cup R) \subseteq \mathrm{span}(S) + \mathrm{span}(R).$$

1. First prove

$$\mathrm{span}(S) + \mathrm{span}(R) \subseteq \mathrm{span}(S \cup R).$$

Let $v \in \mathrm{span}(S) + \mathrm{span}(R)$. Then $v = u + w$ for some $u \in \mathrm{span}(S)$ and $w \in \mathrm{span}(R)$:

$$v = u + w.$$

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EXERCISE 7.9 **CONTINUED** 7

By the definition of span(S), there are vectors $s_1, \ldots, s_k \in S$ and scalars $\alpha_1, \ldots, \alpha_k \in \mathbb{F}$ such that

$$u = \sum_{i=1}^{k} \alpha_i s_i.$$

Similarly, there are vectors $r_1, \ldots, r_\ell \in R$ and scalars $\beta_1, \ldots, \beta_\ell \in \mathbb{F}$ such that

$$w = \sum_{j=1}^{\ell} \beta_j r_j.$$

Therefore,

$$v = u + w = \sum_{i=1}^{k} \alpha_i s_i + \sum_{j=1}^{\ell} \beta_j r_j.$$

Every s_i and every r_j lies in $S \cup R$. Thus v is a finite linear combination of vectors in $S \cup R$.

By the definition of span,

$$v \in \text{span}(S \cup R).$$

Hence

$$\text{span}(S) + \text{span}(R) \subseteq \text{span}(S \cup R).$$

2. For the reverse inclusion, prove

$$\text{span}(S \cup R) \subseteq \text{span}(S) + \text{span}(R).$$

Let $v \in \text{span}(S \cup R)$. There are $x_1, \ldots, x_t \in S \cup R$ and $\gamma_1, \ldots, \gamma_t \in \mathbb{F}$ such that

$$v = \sum_{i=1}^{t} \gamma_i x_i.$$

Partition the indices into disjoint sets: assign to S every vector belonging to S, and assign the rest to R. A vector in the intersection is counted only in the first set.

Define

$$I_S := \{i \in \{1, \ldots, t\} \mid x_i \in S\}, \qquad I_R := \{1, \ldots, t\} \setminus I_S.$$

Then v can be written as

$$v = \sum_{i \in I_S} \gamma_i x_i + \sum_{i \in I_R} \gamma_i x_i.$$

Set

$$u := \sum_{i \in I_S} \gamma_i x_i, \qquad w := \sum_{i \in I_R} \gamma_i x_i.$$

Observe that

- for $i \in I_S$, $x_i \in S$, so u is a linear combination of vectors in S; thus

$$u \in \text{span}(S).$$

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EXERCISE 7.9 CONTINUED

- for $i \in I_R$, $x_i \in R$, so w is a linear combination of vectors in R; thus

$$w \in \text{span}(R).$$

We have obtained

$$v = u + w$$

with $u \in \text{span}(S)$ and $w \in \text{span}(R)$, hence

$$v \in \text{span}(S) + \text{span}(R).$$

This proves

$$\text{span}(S \cup R) \subseteq \text{span}(S) + \text{span}(R).$$

The two inclusions give

$$\text{span}(S) + \text{span}(R) = \text{span}(S \cup R),$$

as required.

EXERCISE 7.10

Let $\mathbb{R}_3[x]$ denote the real vector space of polynomials of degree at most 3.
Define the following subspaces of $\mathbb{R}_3[x]$:

$$U = \text{span}\{1,\ x,\ 1+x\},$$

$$W = \{\, p(x) \in \mathbb{R}_3[x] \mid p(0) = 0,\ p(1) = 0 \,\}.$$

Find bases and dimensions of $U + W$ and $U \cap W$.

WORKED SOLUTION

We work in $\mathbb{R}_3[x]$.

1. First consider U.

   We have

$$U = \text{span}\{1, x, 1+x\}.$$

   The linear relation

$$(1+x) - 1 - x = 0,$$

   expresses $1+x$ as a linear combination of 1 and x. Thus $\{1, x, 1+x\}$ is linearly dependent.
   Conversely, $\{1, x\}$ is linearly independent: if

$$a \cdot 1 + b \cdot x = 0$$

   as a polynomial identity, then comparison of coefficients gives

$$a = 0, \quad b = 0.$$

   Thus $\{1, x\}$ is a basis of U, and

$$\dim U = 2.$$

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EXERCISE 7.10 CONTINUED

2. Now consider W.

   By definition,

$$W = \{p(x) \in \mathbb{R}_3[x] \mid p(0) = 0,\ p(1) = 0\}.$$

   Write a general polynomial in $\mathbb{R}_3[x]$ as

$$p(x) = a_0 + a_1x + a_2x^2 + a_3x^3.$$

   The condition $p(0) = 0$ gives

$$p(0) = a_0 = 0,$$

   and hence

$$p(x) = a_1x + a_2x^2 + a_3x^3.$$

   The condition $p(1) = 0$ gives

$$p(1) = a_1 + a_2 + a_3 = 0.$$

   Introduce parameters by setting

$$a_2 = s, \quad a_3 = t,$$

   so that

$$a_1 = -s - t.$$

   Every polynomial in W therefore has the form

$$p(x) = (-s - t)x + sx^2 + tx^3.$$

   Equivalently,

$$p(x) = s(x^2 - x) + t(x^3 - x).$$

   Hence

$$W = \text{span}\{x^2 - x,\ x^3 - x\}.$$

   To show that this is a basis, suppose

$$\alpha(x^2 - x) + \beta(x^3 - x) = 0$$

   as a polynomial identity. Collecting coefficients gives

$$\beta x^3 + \alpha x^2 + (-\alpha - \beta)x = 0.$$

   A polynomial is zero if and only if all of its coefficients are zero:

$$\beta = 0, \quad \alpha = 0, \quad -\alpha - \beta = 0.$$

   Thus $\alpha = \beta = 0$, so $x^2 - x$ and $x^3 - x$ are linearly independent.

   Therefore,

$$\{x^2 - x,\ x^3 - x\}$$

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EXERCISE 7.10 CONTINUED

is a basis of W, and

$$\dim W = 2.$$

3. Next find a basis and the dimension of $U + W$.

   The calculations above show that

$$U = \text{span}\{1, x\}, \quad W = \text{span}\{x^2 - x,\ x^3 - x\},$$

   and hence

$$U + W = \text{span}\{1,\ x,\ x^2 - x,\ x^3 - x\}.$$

   Set

$$S = \{1,\ x,\ x^2 - x,\ x^3 - x\}.$$

   To test whether S is linearly independent, suppose

$$c_1 \cdot 1 + c_2 x + c_3(x^2 - x) + c_4(x^3 - x) = 0$$

   as a polynomial identity. Collecting coefficients,

$$c_1 + (c_2 - c_3 - c_4)x + c_3 x^2 + c_4 x^3 = 0.$$

   all coefficients must vanish:

$$\begin{cases} c_1 = 0, \\ c_2 - c_3 - c_4 = 0, \\ c_3 = 0, \\ c_4 = 0. \end{cases}$$

   The third and fourth equations give $c_3 = c_4 = 0$, then the second gives $c_2 = 0$ and the first gives $c_1 = 0$. Only the trivial combination vanishes, so S is linearly independent.

   There are four polynomials in S and $\dim \mathbb{R}_3[x] = 4$. Thus S is a basis of $\mathbb{R}_3[x]$, and

$$U + W = \mathbb{R}_3[x], \quad \dim(U + W) = 4.$$

4. Finally find a basis and the dimension of $U \cap W$.

   The dimension formula for a sum gives

$$\dim(U + W) = \dim U + \dim W - \dim(U \cap W).$$

   Substituting,

$$4 = 2 + 2 - \dim(U \cap W),$$

   Therefore,

$$\dim(U \cap W) = 0.$$

   For a direct check, a polynomial in U has the form

$$p(x) = a + bx,$$

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EXERCISE 7.10 CONTINUED

because $U = \text{span}\{1, x\}$.

To belong to W as well, it must satisfy

$$p(0) = 0, \quad p(1) = 0.$$

From $p(0) = 0$,

$$a = 0.$$

and from $p(1) = 0$,

$$a + b = 0.$$

Substituting $a = 0$ into the second equation gives $b = 0$. The only polynomial in the intersection is therefore

$$p(x) \equiv 0.$$

Hence

$$U \cap W = \{0\}, \quad \dim(U \cap W) = 0.$$

The empty set is a basis of this intersection, since it contains no nonzero vector.

To summarize,

$$\dim(U + W) = 4, \quad U + W = \mathbb{R}_3[x],$$
$$\dim(U \cap W) = 0, \quad U \cap W = \{0\}.$$

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EXERCISE 7.11

An engineering system seeks to recover three real state variables x_1, x_2, x_3 from four sensor readings. They may represent temperature deviations, loads or electrical signals in normalized units. Assume an exact, noise-free linear model:

$$M = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \\ 2 & 1 & 3 \end{pmatrix}, \quad \begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{pmatrix} = M \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}.$$

1. Find bases of the row and column spaces of M, and their dimensions.
2. How many readings are linearly dependent on the others? Explain why they add no information in this exact model.
3. Can all three state variables x_1, x_2, x_3 be recovered uniquely from the readings?
4. Suppose the readings are

$$\mathbf{y} = \begin{pmatrix} 2 \\ 3 \\ 5 \\ 8 \end{pmatrix}.$$

First test whether $M\mathbf{x} = \mathbf{y}$ is consistent. Then replace only the fourth reading by 7 and find all states compatible with the corrected readings.

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EXERCISE 7.11 CONTINUED

WORKED SOLUTION

1. Row reduce M:

$$M = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \\ 2 & 1 & 3 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

   Thus its rank is

$$\operatorname{rank} M = 2.$$

   The first two original rows are independent, and the third and fourth are linear combinations of them:

$$(1,1,2) = (1,0,1) + (0,1,1),$$

$$(2,1,3) = (1,0,1) + (1,1,2) = 2(1,0,1) + (0,1,1).$$

   A basis of the row space is therefore

$$\{(1,0,1),(0,1,1)\}, \quad \dim(\text{Row}(M)) = 2.$$

   The columns are

$$\mathbf{c}_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \\ 2 \end{pmatrix}, \quad \mathbf{c}_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \\ 1 \end{pmatrix}, \quad \mathbf{c}_3 = \begin{pmatrix} 1 \\ 1 \\ 2 \\ 3 \end{pmatrix}.$$

   Observe that

$$\mathbf{c}_3 = \mathbf{c}_1 + \mathbf{c}_2.$$

   Since $\mathbf{c}_1, \mathbf{c}_2$ are independent,

$$\{\mathbf{c}_1, \mathbf{c}_2\}$$

   is a basis of the column space. Its dimension is also 2, in agreement with equality of row rank and column rank.

2. The third and fourth rows are combinations of the first two, so the last two readings are linearly redundant. There are only two independent sensor readings.

3. The matrix is 4×3 and has rank 2. There are three unknowns, so

$$\dim(\ker M) = 3 - \operatorname{rank} M = 3 - 2 = 1.$$

   The homogeneous solution space has dimension 1 and contains nonzero vectors. For each attainable $\mathbf{y}$, the solutions of $M\mathbf{x} = \mathbf{y}$ form an affine line, rather than a single point. Consequently the readings do not determine the state uniquely.

4. First test consistency. The row relations imply that every attainable reading vector satisfies

$$y_3 = y_1 + y_2, \qquad y_4 = 2y_1 + y_2.$$

   For the original data $(2,3,5,8)^T$, the first relation holds, but the second requires $y_4 = 2 \cdot 2 + 3 = 7 \neq 8$. Thus the original system has **no solution**.

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EXERCISE 7.11 CONTINUED

Replacing the fourth reading by 7 gives the consistent system

$$M\mathbf{x} = \begin{pmatrix} 2 \\ 3 \\ 5 \\ 7 \end{pmatrix}, \qquad \begin{cases} x_1 + x_3 = 2, \\ x_2 + x_3 = 3, \\ x_1 + x_2 + 2x_3 = 5, \\ 2x_1 + x_2 + 3x_3 = 7. \end{cases}$$

The last two equations now follow from the first two, including their right-hand sides. Set $t = x_3$ to obtain

$$x_1 = 2 - t, \qquad x_2 = 3 - t, \qquad x_3 = t, \qquad t \in \mathbb{R}.$$

The solution set is the affine line

$$\left\{ \begin{pmatrix} 2 \\ 3 \\ 0 \end{pmatrix} + t \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix} \,\middle|\, t \in \mathbb{R} \right\}.$$

Its direction vector $(-1, -1, 1)^T$ spans $N(M)$, as predicted by rank-nullity.

EXERCISE 7.12

In a calibration system, three measured signals (m_1, m_2, m_3) depend on three state variables (q_1, q_2, q_3). This abstract model applies whenever measurements are cumulative sums of the required quantities. Their linear relationship is

$$\begin{pmatrix} m_1 \\ m_2 \\ m_3 \end{pmatrix} = C \begin{pmatrix} q_1 \\ q_2 \\ q_3 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix}.$$

1. Find a basis of the column space of C and show that its columns form a basis of $\mathbb{R}^3$.
2. Prove that C is invertible.
3. Find C^{-1} explicitly by solving $C\mathbf{q} = \mathbf{m}$ for $\mathbf{q}$ in terms of $\mathbf{m}$.
4. Explain why invertibility means that the state variables (q_1, q_2, q_3) can be recovered uniquely from (m_1, m_2, m_3).

WORKED SOLUTION

1. The columns of C are

$$\mathbf{c}_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad \mathbf{c}_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \quad \mathbf{c}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

To test linear independence, suppose

$$a\mathbf{c}_1 + b\mathbf{c}_2 + c\mathbf{c}_3 = \mathbf{0}.$$

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EXERCISE 7.12 CONTINUED

Equivalently,

$$a\begin{pmatrix}1\\1\\1\end{pmatrix}+b\begin{pmatrix}0\\1\\1\end{pmatrix}+c\begin{pmatrix}0\\0\\1\end{pmatrix}=\begin{pmatrix}0\\0\\0\end{pmatrix},$$

giving the system

$$\begin{cases}a=0,\\a+b=0,\\a+b+c=0.\end{cases}$$

The first equation gives $a=0$, the second gives $b=0$ and the third gives $c=0$. Thus the columns are independent.

Three independent vectors in $\mathbb{R}^3$ form a basis, so

$$\operatorname{Col}(C)=\mathbb{R}^3,\quad \operatorname{rank}C=3.$$

2. C is a 3×3 matrix, and the preceding part gives

$$\operatorname{rank}C=3.$$

Thus C has full rank and is invertible. In terms of linear systems, the homogeneous system $C\mathbf{q}=\mathbf{0}$ has

$$\dim(\ker C)=3-\operatorname{rank}C=3-3=0.$$

so its unique solution is $\mathbf{q}=\mathbf{0}$.

Since $\operatorname{Col}(C)=\mathbb{R}^3$, the system $C\mathbf{q}=\mathbf{m}$ is solvable for every $\mathbf{m}\in\mathbb{R}^3$.

Thus there is a *unique solution* $\mathbf{q}$ for every $\mathbf{m}$. Multiplication by C is a bijection from $\mathbb{R}^3$ to itself, and there is a matrix C^{-1} satisfying

$$C^{-1}C=CC^{-1}=I_3.$$

Therefore C is invertible. This relates matrix invertibility to injectivity and surjectivity of the corresponding linear map.

3. To compute C^{-1} explicitly, start with

$$\begin{pmatrix}1&0&0\\1&1&0\\1&1&1\end{pmatrix}\begin{pmatrix}q_1\\q_2\\q_3\end{pmatrix}=\begin{pmatrix}m_1\\m_2\\m_3\end{pmatrix}.$$

The resulting equations are

$$\begin{cases}q_1=m_1,\\q_1+q_2=m_2,\\q_1+q_2+q_3=m_3.\end{cases}$$

From the first equation,

$$q_1=m_1.$$

from the second,

$$m_1+q_2=m_2\quad\Longrightarrow\quad q_2=m_2-m_1.$$

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EXERCISE 7.12 CONTINUED

and from the third,

$$m_1 + (m_2 - m_1) + q_3 = m_3 \quad \Longrightarrow \quad m_2 + q_3 = m_3 \quad \Longrightarrow \quad q_3 = m_3 - m_2.$$

Therefore,

$$\begin{pmatrix} q_1 \\ q_2 \\ q_3 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix} \begin{pmatrix} m_1 \\ m_2 \\ m_3 \end{pmatrix}.$$

Hence

$$C^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix}.$$

4. Since C is invertible, its columns form a basis of $\mathbb{R}^3$. Every measurement vector $\mathbf{m}$ is therefore a unique linear combination of those columns:

$$\mathbf{m} = C\mathbf{q}$$

with a unique $\mathbf{q}$. In this exact model, each measurement vector (m_1, m_2, m_3) determines a *unique state* (q_1, q_2, q_3), computed using C^{-1}:

$$\begin{pmatrix} q_1 \\ q_2 \\ q_3 \end{pmatrix} = C^{-1} \begin{pmatrix} m_1 \\ m_2 \\ m_3 \end{pmatrix}.$$

This calibration matrix recovers the underlying quantities from their measurements.

EXERCISE 7.13

Let $\mathbb{F}$ be a field, and let $P_{\leq 3}$ be the space of polynomials over $\mathbb{F}$ of degree at most 3. Write a polynomial in the standard basis as

$$p(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3,$$

with coefficient vector

$$\begin{pmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{pmatrix} \in \mathbb{F}^4.$$

Define three polynomial measurements:

$$\ell_1(p) = p(0), \quad \ell_2(p) = p(1), \quad \ell_3(p) = p'(0).$$

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EXERCISE 7.13 CONTINUED

Represent them by a matrix $A \in \mathbb{F}^{3\times 4}$ such that, for every p,

$$A\begin{pmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{pmatrix} = \begin{pmatrix} \ell_1(p) \\ \ell_2(p) \\ \ell_3(p) \end{pmatrix}.$$

1. Find A explicitly.
2. Find bases and dimensions of its row space and column space.
3. Determine whether there are polynomials $p \in P_{\leq 3}$ satisfying

   $$p(0) = 1, \quad p(1) = 1, \quad p'(0) = 0.$$

   and, if so, find all of them.
4. Determine the affine dimension of the solution set. How many independent parameters remain?

WORKED SOLUTION

Write

$$p(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3.$$

1. Compute the three measurements:

   Evaluation at 0 gives

   $$p(0) = a_0.$$

   and evaluation at 1 gives

   $$p(1) = a_0 + a_1 + a_2 + a_3.$$

   The formal derivative is

   $$p'(x) = a_1 + 2a_2 x + 3a_3 x^2,$$

   so

   $$p'(0) = a_1.$$

   Therefore,

   $$A\begin{pmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{pmatrix} = \begin{pmatrix} a_0 \\ a_0 + a_1 + a_2 + a_3 \\ a_1 \end{pmatrix}.$$

   and hence

   $$A = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix}.$$

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EXERCISE 7.13 CONTINUED

2. Begin the row reduction of A:

$$A = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix}.$$

Subtract the first row from the second:

$$\sim \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix}.$$

Subtract the third row from the second:

$$\sim \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix}.$$

Reorder the rows to obtain echelon form:

$$\sim \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix}.$$

There are three nonzero echelon rows, so

$$\text{rank}(A) = 3.$$

These rows span the row space. A convenient basis is

$$\{(1,0,0,0), (0,1,0,0), (0,0,1,1)\}.$$

The pivot indices are $1, 2, 3$. The corresponding columns of the original matrix therefore form a basis of its column space. The columns are

$$c_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad c_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \quad c_3 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad c_4 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}.$$

Immediately,

$$c_4 = c_3.$$

Thus one may choose the column-space basis

$$\{c_1, c_2, c_3\},$$

whose dimension is

$$\dim(\text{Col}(A)) = 3.$$

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EXERCISE 7.13 CONTINUED

3. We seek p satisfying

$$p(0) = 1, \quad p(1) = 1, \quad p'(0) = 0.$$

In matrix form this is the system

$$A \begin{pmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}.$$

Explicitly,

$$\begin{cases} a_0 = 1, \\ a_0 + a_1 + a_2 + a_3 = 1, \\ a_1 = 0. \end{cases}$$

The first equation gives

$$a_0 = 1.$$

the third gives

$$a_1 = 0.$$

and substitution into the second gives

$$1 + 0 + a_2 + a_3 = 1.$$

Hence

$$a_2 + a_3 = 0.$$

This is one equation in a_2, a_3, leaving one free parameter. An infinite field gives infinitely many solutions; a finite field $\mathbb{F}$ gives exactly $|\mathbb{F}|$ solutions. Set

$$t = a_2,$$

so that

$$a_3 = -t.$$

Therefore,

$$a_0 = 1, \quad a_1 = 0, \quad a_2 = t, \quad a_3 = -t,$$

and all the required polynomials have the form

$$p(x) = 1 + tx^2 - tx^3, \quad t \in \mathbb{F}.$$

4. There are four unknowns and the rank is 3. By rank-nullity,

$$\text{null}(A) = 4 - \text{rank}(A) = 4 - 3 = 1.$$

The null space has dimension 1. Whenever $Ax = b$ is consistent, its solution set is a translate of the kernel, an affine line of dimension 1; it need not be a vector subspace.

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EXERCISE 7.13 CONTINUED

Here the system is consistent, and the free parameter t parametrizes that line. Thus the affine dimension of the polynomial solution set is 1, in agreement with rank-nullity.

EXERCISE 7.14

Two sensors measure Ax for $x \in \mathbb{R}^3$, where

$$A = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \end{pmatrix}.$$

An additional sensor measures $z = ax_1 + bx_2 + cx_3$. Find a necessary and sufficient condition for unique recovery, and give one useful sensor and one whose reading follows from the existing readings.

WORKED SOLUTION

First find the changes that are invisible to the two original sensors:

$$Av = 0 \iff v_1 + 2v_3 = 0,\ v_2 - v_3 = 0 \iff v = t(-2, 1, 1)^T.$$

On such a change, the new sensor returns

$$a(-2t) + bt + ct = (c - 2a + b)t.$$

The three readings determine the state uniquely exactly when this expression vanishes only for $t = 0$, namely when

$$c - 2a + b \neq 0.$$

The determinant gives the same condition: $\det \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \\ a & b & c \end{pmatrix} = c + b - 2a$. The sensor $z = x_3$ is useful. In contrast, $z = x_1 + x_2 + x_3$ is the sum of the two existing readings and adds no information; for $(a, b, c) = (1, 1, 1)$, the condition indeed fails.

EXERCISE 7.15 EXAM QUESTION

1. Let $n \geq 2$ and $A, B \in M_n(\mathbb{R})$ satisfy $B \neq 0$ and $\operatorname{rank}(A) = n - 1$. Prove that $AB = 0$ implies $\operatorname{rank}(B) = 1$.
2. Let $A, B \in M_{2026}(\mathbb{R})$ satisfy $AB = 0$. Prove that
$$\operatorname{rank}(A) + \operatorname{rank}(B) \leq 2026.$$

WORKED SOLUTION

The key implication in both parts is

$$AB = 0 \quad \Longrightarrow \quad \operatorname{Im}(B) \subseteq \ker A.$$

Indeed, every $y \in \operatorname{Im} B$ has the form $y = Bx$. Thus $Ay = A(Bx) = (AB)x = 0$, so $y \in \ker A$.

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EXERCISE 7.15 EXAM QUESTION CONTINUED

1. The inclusion and rank-nullity give

$$\begin{aligned}\operatorname{rank}(B) &= \dim \operatorname{Im}(B)\\ &\le \dim \ker A\\ &= n - \operatorname{rank}(A)\\ &= n - (n-1) = 1.\end{aligned}$$

   Since $B \neq 0$, at least one column of B is nonzero, so $\operatorname{rank}(B) \ge 1$. The two bounds yield $\operatorname{rank}(B) = 1$.

2. Apply the same inclusion in $\mathbb{R}^{2026}$:

$$\begin{aligned}\operatorname{rank}(B) &= \dim \operatorname{Im}(B)\\ &\le \dim \ker A\\ &= 2026 - \operatorname{rank}(A).\end{aligned}$$

   Rearranging gives

$$\operatorname{rank}(A) + \operatorname{rank}(B) \le 2026.$$

If every output of B is annihilated by A, the entire image of B must lie in the kernel of A. The dimension of that kernel therefore bounds the rank of B.

Additional practice

EXERCISE 7.16 FOUNDATION

Over $\mathbb{R}$, consider

$$A = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & -1 \\ 1 & 2 & 1 & 0 \end{pmatrix}.$$

Find bases of its row space, column space and kernel, and verify rank-nullity. Give a condition on $b = (b_1, b_2, b_3)^T$ for $Ax = b$ to be solvable, and find all solutions for $b = (3, 2, 5)^T$.

WORKED SOLUTION

The third row is the sum of the first two, which are independent. Thus a row-space basis is

$$\{(1, 2, 0, 1), (0, 0, 1, -1)\}, \qquad \operatorname{rank} A = 2.$$

Original columns $1, 3$ are independent and form a column-space basis:

$$\left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right\}.$$

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EXERCISE 7.16 FOUNDATION CONTINUED

In the homogeneous system, set $x_2 = s, x_4 = t$. Then $x_1 = -2s - t$ and $x_3 = t$, so

$$\ker A = \operatorname{span}\left\{ \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \\ 1 \end{pmatrix} \right\}.$$

These vectors are independent by their second and fourth entries. The nullity is 2, and $2+2 = 4$. The row relation requires $b_3 = b_1 + b_2$. This is also sufficient, since the first two equations are solvable for all b_1, b_2. For the given right-hand side, all solutions are

$$x = \begin{pmatrix} 3 \\ 0 \\ 2 \\ 0 \end{pmatrix} + s \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + t \begin{pmatrix} -1 \\ 0 \\ 1 \\ 1 \end{pmatrix}, \qquad s, t \in \mathbb{R}.$$

This is an affine plane, not a vector subspace, since $b \neq 0$.

EXERCISE 7.17 CHALLENGE

Let $A \in \mathbb{F}^{m \times n}$ and $B \in \mathbb{F}^{n \times p}$. Prove

$$\operatorname{rank}(AB) = \operatorname{rank}(B) - \dim\big(\operatorname{Col}(B) \cap \ker A\big),$$

and deduce $\operatorname{rank}(AB) \geq \operatorname{rank}(A) + \operatorname{rank}(B) - n$. Verify the identity for

$$A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ -1 & 0 \end{pmatrix}$$

over $\mathbb{R}$.

WORKED SOLUTION

Restrict multiplication by A to the linear map $L : \operatorname{Col}(B) \to \mathbb{F}^m$, $L(y) = Ay$. Its domain has dimension $\operatorname{rank}(B)$, its image is $\operatorname{Col}(AB)$, and

$$\ker L = \operatorname{Col}(B) \cap \ker A.$$

Rank-nullity for L gives the identity. Since the intersection has dimension at most $\dim \ker A = n - \operatorname{rank}(A)$, the required inequality follows. In the example, the columns of B are independent, so its rank is 2. Moreover,

$$\ker A = \operatorname{span}\{(-1, -1, 1)^T\} \subseteq \operatorname{Col}(B), \qquad AB = \begin{pmatrix} 0 & 0 \\ -1 & 1 \end{pmatrix}.$$

The inclusion follows from $(-1, -1, 1)^T = -Be_1 - Be_2$. The intersection has dimension 1, giving $\operatorname{rank}(AB) = 2 - 1 = 1$, as the product confirms. Exactly those output directions of B that lie in the kernel of A are lost.

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TUTORIAL 8

Linear Maps and Coordinates

MOTIVATION AND GUIDING IDEA

Motivation. The same information may appear as a polynomial, a matrix or a coordinate vector. A linear map compares such descriptions while preserving linear combinations. The kernel asks what is lost; the image asks what can be attained.

NOTATION AND CONVENTIONS

- $T : V \to W$ denotes a map with domain V and codomain W; $T(v) \in W$ is the image of v. The vectors $0_V, 0_W$ are the zero vectors in the corresponding spaces.
- $\ker T = \{v \in V : T(v) = 0_W\}$ is the kernel; $\operatorname{Im} T = T(V) = \{T(v) : v \in V\}$ is the image. Do not automatically identify the image with the codomain.
- $\operatorname{rank} T = \dim \operatorname{Im} T$ is the rank; $\dim \ker T$ is the nullity. The notation $V \cong W$ means that a linear isomorphism between them exists.
- $B = (v_1, \dots, v_n)$ is an **ordered** basis. If $v = \sum_{j=1}^{n} a_j v_j$, then $[v]_B = (a_1, \dots, a_n)^T$. The order of basis vectors determines the order of entries.
- $\mathbb{F}_{\leq n}[X]$ is the space of polynomials of degree at most n, including the zero polynomial. For $T : V \to V$, $T^2 = T \circ T$ denotes composition, not entrywise squaring.

DEFINITION 8.1

Let V, W be vector spaces over the same field $\mathbb{F}$. A map $T : V \to W$ is linear if it is well defined and satisfies

$$T(av + bw) = aT(v) + bT(w) \qquad (v, w \in V,\ a, b \in \mathbb{F}).$$

It is injective if $T(v) = T(w)$ implies $v = w$, and surjective if every $y \in W$ equals $T(v)$ for some $v \in V$. For a linear map, injectivity is equivalent to $\ker T = \{0_V\}$, and surjectivity to $\operatorname{Im} T = W$. An isomorphism is a bijective linear map; its inverse is linear. If $\dim V < \infty$, then

$$\dim V = \dim \ker T + \dim \operatorname{Im} T.$$

The coordinate map $v \mapsto [v]_B$ is an isomorphism $V \to \mathbb{F}^n$. A vector and its coordinate column are related by a choice of basis; they should not be equated without an appropriate identification.

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MOTIVATION AND GUIDING IDEA

Tutorial objectives. Recognize linear maps, compute kernels and images, and use coordinates to pass between polynomials, matrices and vectors. Distinguish a mathematical object from its representation in a chosen basis.

8.1 Linear maps

DEFINITION 8.2

Let V, W be vector spaces over the same field $\mathbb{F}$.
A map $T : V \to W$ is **linear** if, for all $v_1, v_2 \in V$ and $\lambda \in \mathbb{F}$,

1. $T(v_1 + v_2) = T(v_1) + T(v_2)$.
2. $T(\lambda v_1) = \lambda T(v_1)$.

Interpretation A linear map sends vectors from one vector space to another while preserving their linear structure. It respects vector addition and scalar multiplication.

The first condition says that adding and then applying the map gives the same result as applying the map and then adding. Thus addition commutes with applying the map.

The second condition says that multiplying by a scalar before applying the map gives the same result as multiplying the image by that scalar. In a real geometric setting, scaling an input by λ scales its image by the same factor.

NOTE

Together these properties imply that, for all $\alpha, \beta \in \mathbb{F}$ and $v_1, v_2 \in V$,

$$T(\alpha v_1 + \beta v_2) = \alpha T(v_1) + \beta T(v_2),$$

so a linear map preserves every linear combination.
Geometrically, real linear maps include scalings, rotations and reflections. They send a line to a line or a point and fix the zero vector. In particular, $T(0_V) = 0_W$ always holds.
The main consequences are:

- More generally, $T(\sum_{i=1}^{n} \lambda_i v_i) = \sum_{i=1}^{n} \lambda_i T(v_i)$.
- A necessary condition for $T : V \to W$ to be linear is $T(0_V) = 0_W$. (This condition alone does not establish linearity, but failure of it disproves linearity.)

EXERCISE 8.1

Let $n \geq 1$, and define

$$V = \mathbb{R}_{\leq n}[X].$$

Determine which of the following formulas define linear maps (first verify that each proposed map is **well defined** as a map $V \to V$):

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EXERCISE 8.1 CONTINUED

1. $T_1 : V \to V$, given by
$$T_1(P(X)) := P(X) + P(0).$$
2. $T_2 : V \to V$, given by
$$T_2(P(X)) := X \cdot P(X).$$
3. $T_3 : V \to V$, given by
$$T_3(P(X)) := P(X+1) - P(1).$$

WORKED SOLUTION

Recall that $T : V \to V$ is linear if, for all $P, Q \in V$ and $\alpha, \beta \in \mathbb{R}$,

$$T\big(\alpha P + \beta Q\big) = \alpha T(P) + \beta T(Q).$$

1. **T_1 is linear.**

   (a) Well-definedness. If $P \in V$, then $P(0) \in \mathbb{R}$ is constant, so $P(X) + P(0)$ has degree at most n. Hence $T_1(P) \in V$.

   (b) Linearity. Let $P, Q \in V$ and $\alpha, \beta \in \mathbb{R}$. Then
   $$T_1\big(\alpha P(X) + \beta Q(X)\big) = (\alpha P + \beta Q)(X) + (\alpha P + \beta Q)(0).$$
   Since
   $$(\alpha P + \beta Q)(X) = \alpha P(X) + \beta Q(X), \qquad (\alpha P + \beta Q)(0) = \alpha P(0) + \beta Q(0),$$
   we obtain
   $$T_1\big(\alpha P(X) + \beta Q(X)\big) = \alpha\big(P(X) + P(0)\big) + \beta\big(Q(X) + Q(0)\big) = \alpha T_1(P(X)) + \beta T_1(Q(X)).$$
   Thus T_1 is linear.

2. **T_2 does not define a map $V \to V$, so it is not a linear operator on V.**

   A linear map must first be well defined with its stated domain and codomain. We show that this requirement fails for T_2.

   Take $P(X) = X^n \in V$. Then
   $$T_2(P(X)) = X \cdot X^n = X^{n+1} \notin \mathbb{R}_{\leq n}[X] = V.$$
   Thus T_2 does not define a map $V \to V$, and cannot be a linear operator on V.

   Remark (generalization). If instead we define
   $$\widetilde{T}_2 : \mathbb{R}[X] \to \mathbb{R}[X], \qquad \widetilde{T}_2(P(X)) := X \cdot P(X),$$
   then $\widetilde{T}_2$ is well defined and linear: for all $P, Q \in \mathbb{R}[X]$ and $\alpha, \beta \in \mathbb{R}$,
   $$\widetilde{T}_2(\alpha P + \beta Q) = X(\alpha P + \beta Q) = \alpha XP + \beta XQ = \alpha \widetilde{T}_2(P) + \beta \widetilde{T}_2(Q).$$

3. **T_3 is linear.**

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EXERCISE 8.1 CONTINUED

(a) Well-definedness. If $P \in V$, then $\deg P \le n$, so $\deg P(X+1) \le n$ as well. Also, $P(1) \in \mathbb{R}$ is constant, so $P(X+1) - P(1) \in V$. Hence $T_3(P) \in V$.

(b) Linearity. Let $P, Q \in V$ and $\alpha, \beta \in \mathbb{R}$. Then

$$T_3\big(\alpha P(X) + \beta Q(X)\big) = (\alpha P + \beta Q)(X+1) - (\alpha P + \beta Q)(1).$$

Since

$$(\alpha P + \beta Q)(X+1) = \alpha P(X+1) + \beta Q(X+1), \qquad (\alpha P + \beta Q)(1) = \alpha P(1) + \beta Q(1),$$

we obtain

$$\begin{aligned} T_3(\alpha P + \beta Q) &= \alpha\big(P(X+1) - P(1)\big) \\ &\quad + \beta\big(Q(X+1) - Q(1)\big) \\ &= \alpha T_3(P) + \beta T_3(Q). \end{aligned}$$

Thus T_3 is linear.

MOTIVATION AND GUIDING IDEA

Two guiding questions. The kernel records information lost by a map; the image records attainable outputs. A trivial kernel means that distinct inputs remain distinguishable. An image equal to the codomain means that every output in that codomain is attainable.

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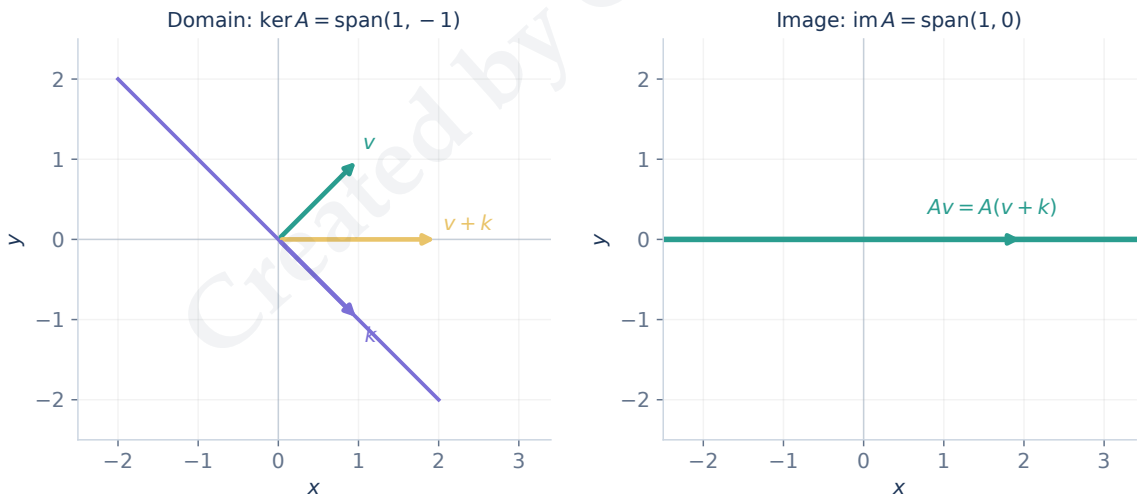


Figure 8.1. The kernel lies in the domain and the image in the codomain. Several inputs can have the same output when the kernel is nontrivial.

8.2 The kernel and image of a linear map

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RECALL AND TOOLS

Let $T : V \to W$ be linear.
The **kernel of** T consists of all vectors sent to 0_W:

$$\ker(T) = \{v \in V | T(v) = 0\}$$

The kernel is a subspace of V.
The **image of** T is the set of all outputs attained by the map:

$$\mathrm{Im}(T) = \{T(v) | v \in V\} = \{w \in W | \exists v \in V, T(v) = w\}$$

The image is a subspace of W. It consists of outputs actually attained, whereas the codomain W is specified as part of the definition of the map. These need not be equal.
The sets $\ker T$ and $\mathrm{Im}\, T$ are nonempty, since $T(0_V) = 0_W$.

- Always $\mathrm{Im}\, T \subseteq W$. If $\mathrm{Im}\, T = W$, the map is **surjective**, or **onto**.
- T is **injective**, or **one-to-one**, if, for all $v_1, v_2 \in V$,

  $$T(v_1) = T(v_2) \implies v_1 = v_2$$

  Equivalently, no two distinct inputs have the same image.

  $$v_1 \neq v_2 \implies T(v_1) \neq T(v_2)$$

- A linear map T is an **isomorphism** if it is injective and surjective.
- Vector spaces V, W are **isomorphic** if there **exists** an isomorphism $T : V \to W$. We then write $V \cong W$.

PROPOSITION 8.1

A linear map $T : V \to W$ is **injective** if and only if $\ker T = \{0_V\}$.
For a homogeneous linear system, this means that the trivial solution is the only solution.

PROPOSITION 8.2

Let $T : V \to W$ be a linear map between finite-dimensional vector spaces. Then

1. If T is injective, $\dim V \leq \dim W$.
2. If T is surjective, $\dim W \leq \dim V$.
3. If $\dim V = \dim W$, then T is injective **if and only if** it is surjective.

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EXERCISE 8.2

Let $n \geq 2$ and $V = M_n(\mathbb{F})$. Define $T : V \to \mathbb{F}$ by

$$T(A) = \mathrm{Tr}(A).$$

1. Prove that T is linear.
2. Find $\ker T$.
3. Find $\mathrm{Im}\, T$.
4. Compute $\dim \ker T$ and $\dim \mathrm{Im}\, T$, and verify rank-nullity.

WORKED SOLUTION

Write $A = (a_{ij}), B = (b_{ij})$.

1. Linearity:

$$T(A+B) = \mathrm{Tr}(A+B) = \sum_{i=1}^{n}(a_{ii}+b_{ii}) = \sum_{i=1}^{n} a_{ii} + \sum_{i=1}^{n} b_{ii} = T(A) + T(B),$$

For every $\lambda \in \mathbb{F}$,

$$T(\lambda A) = \mathrm{Tr}(\lambda A) = \sum_{i=1}^{n} \lambda a_{ii} = \lambda \sum_{i=1}^{n} a_{ii} = \lambda T(A).$$

Thus T is linear.

2. Kernel:

$$\ker(T) = \{A \in M_n(\mathbb{F}) \ : \ \mathrm{Tr}(A) = 0\}.$$

This is the subspace of matrices with trace zero.

3. Image: for any $\alpha \in \mathbb{F}$, take

$$A = \mathrm{diag}(\alpha, 0, \ldots, 0) \in M_n(\mathbb{F}),$$

Then $\mathrm{Tr}(A) = \alpha$, so

$$\mathrm{Im}(T) = \mathbb{F}.$$

4. Dimensions: $\dim M_n(\mathbb{F}) = n^2$. Since $\mathrm{Im}\, T = \mathbb{F}$, its dimension is 1. Rank-nullity therefore gives

$$\dim \ker(T) = n^2 - 1.$$

Alternatively, $n^2 - 1$ entries may be chosen freely, and the last diagonal entry is fixed by requiring trace zero.

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EXERCISE 8.3

Let $V = C[0,1]$, the real vector space of continuous real-valued functions on $[0,1]$. Define

$$S : V \to \mathbb{R}, \qquad S(f) = \int_0^1 f(x)\,dx.$$

1. Prove that S is linear.
2. Find $\ker S$.
3. Find $\operatorname{Im} S$.
4. Give an explicit nonzero $f \in \ker S$.

WORKED SOLUTION

1. Linearity: for $f, g \in V$ and $\lambda \in \mathbb{R}$,

$$S(f+g) = \int_0^1 (f+g) = \int_0^1 f + \int_0^1 g = S(f) + S(g),$$

$$S(\lambda f) = \int_0^1 \lambda f = \lambda \int_0^1 f = \lambda S(f).$$

so S is linear.

2. Kernel:

$$\ker(S) = \left\{ f \in C[0,1] \ : \ \int_0^1 f(x)\,dx = 0 \right\}.$$

Since $C[0,1]$ is infinite-dimensional, the finite-dimensional rank-nullity formula cannot here be treated as an equality of finite integers. Nevertheless,

$$C[0,1] = \ker(S) \oplus \operatorname{span}\{1\}, \qquad f = \big(f - S(f)\cdot 1\big) + S(f)\cdot 1.$$

The first summand has integral zero, and the only constant function with integral zero is the zero function.

3. Image: given $c \in \mathbb{R}$, take $f(x) \equiv c$. Then

$$S(f) = \int_0^1 c\,dx = c,$$

so $\operatorname{Im} S = \mathbb{R}$.

4. Example: $f(x) = x - \frac{1}{2}$ is continuous and nonzero, but

$$\int_0^1 \left(x - \tfrac{1}{2}\right) dx = \left[\tfrac{x^2}{2} - \tfrac{x}{2}\right]_0^1 = \tfrac{1}{2} - \tfrac{1}{2} = 0,$$

so $f \in \ker S$.

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EXERCISE 8.4

Let V be a vector space over a field F, and let $T : V \to V$ be linear. Prove:
(a) $\ker T \subseteq \ker T^2$; (b) $\operatorname{Im} T^2 \subseteq \operatorname{Im} T$.

WORKED SOLUTION

(a) We must show that every vector in the set on the left also belongs to the set on the right.
Let $v \in \ker T$. Then $T^2(v) = T(T(v)) = T(0) = 0$, so $v \in \ker T^2$.
(b) Let $w \in \operatorname{Im} T^2$. Some $v \in V$ satisfies $w = T^2(v) = T(T(v))$.
Set $u = T(v)$. Then $w = T(u)$, so $w \in \operatorname{Im} T$.

DEFINITION 8.3

If $T : V \to W$ is an **isomorphism**, each $w \in W$ has a unique $v \in V$ with $T(v) = w$.
Thus the **inverse map** $T^{-1} : W \to V$ is defined by $T^{-1}(w) = v$ whenever $T(v) = w$.
Applying T to v gives w; applying T^{-1} to w recovers v.

EXERCISE 8.5

Let $V = M_3(\mathbb{R})$.

1. Find a linear isomorphism $\Phi : V \to \mathbb{R}^9$.
2. In this part use the explicit row-by-row flattening isomorphism

$$\Phi((a_{ij})) = (a_{11}, a_{12}, a_{13}, a_{21}, a_{22}, a_{23}, a_{31}, a_{32}, a_{33}).$$

Define two subsets of $\mathbb{R}^9$:

$$U = \left\{ x \in \mathbb{R}^9 \;\middle|\; x_2 = x_4,\ x_3 = x_7,\ x_6 = x_8 \right\},$$

$$W = \left\{ x \in \mathbb{R}^9 \;\middle|\; x_1 = x_5 = x_9 = 0,\ x_2 = -x_4,\ x_3 = -x_7,\ x_6 = -x_8 \right\}.$$

Find $\dim U$, $\dim W$ and identify $A = \Phi^{-1}(U)$ and $B = \Phi^{-1}(W)$ as subspaces of $M_3(\mathbb{R})$.

WORKED SOLUTION

1. Define $\Phi : M_3(\mathbb{R}) \to \mathbb{R}^9$ by the following formula for $A = (a_{ij})$:

$$\Phi(A) = (a_{11}, a_{12}, a_{13}, a_{21}, a_{22}, a_{23}, a_{31}, a_{32}, a_{33}).$$

We verify that Φ is an isomorphism.

Linearity. Let $A = (a_{ij}), B = (b_{ij})$ and $\alpha, \beta \in \mathbb{R}$. The entries of $\alpha A + \beta B$ are $\alpha a_{ij} + \beta b_{ij}$, so

$$\Phi(\alpha A + \beta B) = (\alpha a_{11} + \beta b_{11}, \ldots, \alpha a_{33} + \beta b_{33}) = \alpha \Phi(A) + \beta \Phi(B).$$

Injectivity. If $\Phi(A) = \Phi(B)$, equality of all nine coordinates gives $a_{ij} = b_{ij}$ for all i, j, hence $A = B$.

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EXERCISE 8.5 CONTINUED

Surjectivity. Given $x = (x_1, \dots, x_9) \in \mathbb{R}^9$, define

$$A = \begin{pmatrix} x_1 & x_2 & x_3 \\ x_4 & x_5 & x_6 \\ x_7 & x_8 & x_9 \end{pmatrix}.$$

Then $\Phi(A) = x$ by definition. Thus Φ is surjective.

Consequently Φ is an isomorphism.

2. Compute the dimensions and identify the corresponding matrix subspaces.

Computing $\dim U$**.** A vector $x \in U$ is determined by freely choosing

$$x_1,\ x_2,\ x_3,\ x_5,\ x_6,\ x_9,$$

The constraints then force

$$x_4 = x_2, \qquad x_7 = x_3, \qquad x_8 = x_6.$$

There are six free parameters, so

$$\dim U = 6.$$

Computing $\dim W$**.** A vector $x \in W$ satisfies

$$x_1 = x_5 = x_9 = 0,$$

together with

$$x_4 = -x_2, \qquad x_7 = -x_3, \qquad x_8 = -x_6.$$

Exactly the following entries may be chosen freely:

$$x_2,\ x_3,\ x_6,$$

All remaining entries are then fixed. Hence

$$\dim W = 3.$$

Now identify $A = \Phi^{-1}(U)$. Let $A = (a_{ij}) \in M_3(\mathbb{R})$ and $\Phi(A) = (x_1, \dots, x_9)$, so

$$\begin{aligned} x_1 &= a_{11}, \quad x_2 = a_{12}, \quad x_3 = a_{13}, \\ x_4 &= a_{21}, \quad x_5 = a_{22}, \quad x_6 = a_{23}, \\ x_7 &= a_{31}, \quad x_8 = a_{32}, \quad x_9 = a_{33}. \end{aligned}$$

The conditions defining U are

$$x_2 = x_4 \Rightarrow a_{12} = a_{21}, \qquad x_3 = x_7 \Rightarrow a_{13} = a_{31}, \qquad x_6 = x_8 \Rightarrow a_{23} = a_{32}.$$

These are precisely $A^T = A$, so

$$\Phi^{-1}(U) = \mathrm{Sym}_3 = \{A \in M_3(\mathbb{R}) \mid A^T = A\}.$$

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EXERCISE 8.5 CONTINUED

Similarly, identify $B = \Phi^{-1}(W)$. The conditions defining W give

$$x_1 = x_5 = x_9 = 0 \Rightarrow a_{11} = a_{22} = a_{33} = 0,$$

and

$$x_2 = -x_4 \Rightarrow a_{12} = -a_{21}, \qquad x_3 = -x_7 \Rightarrow a_{13} = -a_{31}, \qquad x_6 = -x_8 \Rightarrow a_{23} = -a_{32}.$$

These are precisely $A^T = -A$, so

$$\Phi^{-1}(W) = \text{Skew}_3 = \{A \in M_3(\mathbb{R}) \mid A^T = -A\}.$$

Thus:

$$\dim U = 6, \qquad \dim W = 3, \qquad \Phi^{-1}(U) = \text{Sym}_3, \qquad \Phi^{-1}(W) = \text{Skew}_3.$$

8.3 Rank-nullity for linear maps

PROPOSITION 8.3

Let $T : V \to W$ be linear, with V finite-dimensional.
Then

$$\dim V = \dim \ker T + \dim \text{Im}\, T.$$

EXERCISE 8.6

Determine whether a linear map with the following properties exists. If it does, give it explicitly.

1. $T : \mathbb{R}^3 \to \mathbb{R}^3$, with $\ker T = \text{span}\{(1,2,1)\}$, $\text{Im}T = \text{Sp}\{(1,1,1)\}$.
2. $T : M_{2\times 2}(\mathbb{R}) \to \mathbb{R}^2$, with $\ker T = \text{span}\left\{\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}\right\}$.

WORKED SOLUTION

1. Both the image and kernel are spanned by one nonzero vector, so each has dimension 1. Rank-nullity would give

$$3 = \dim(R^3) = \dim ImT + \dim kerT = 1 + 1 = 2$$

a contradiction. Thus no such linear map exists.

2. No such map exists. The stated kernel is spanned by one nonzero vector (a matrix), so $\dim \ker T = 1$. By rank-nullity,

$$\dim ImT = \dim(M_{2\times 2}(R)) - \dim \ker T = 4 - 1 = 3$$

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EXERCISE 8.6 CONTINUED

But $\operatorname{Im} T \subseteq \mathbb{R}^2$, so $\dim \operatorname{Im} T \leq 2$, a contradiction. Hence the required map does not exist.

8.4 Coordinates

DEFINITION 8.4

Let V be a vector space over a field $\mathbb{F}$ with ordered basis $B = (v_1, \ldots, v_n)$.
Every $v \in V$ has a unique representation as a linear combination of the basis vectors. Thus there are unique $\alpha_1, \ldots, \alpha_n \in \mathbb{F}$ such that

$$v = \alpha_1 v_1 + \cdots + \alpha_n v_n$$

The coefficients $\alpha_1, \ldots, \alpha_n$ are the **coordinates of v relative to B**.

The column vector $\begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix} \in \mathbb{F}^n$ is the **coordinate vector of v relative to B**, denoted $[v]_B$.

EXERCISE 8.7

Let $B = ((1,0,1)^T, (0,1,1)^T, (1,1,1)^T)$ be an ordered basis of $\mathbb{R}^3$. Find the coordinates of $v = (5,2,4)^T$ relative to B, and verify them by reconstructing v.

WORKED SOLUTION

Seek a, b, c such that $v = a(1,0,1)^T + b(0,1,1)^T + c(1,1,1)^T$. Comparing entries gives

$$a + c = 5, \qquad b + c = 2, \qquad a + b + c = 4.$$

Adding the first two equations and subtracting the third yields $c = 5 + 2 - 4 = 3$. Thus $a = 2$ and $b = -1$, so

$$[v]_B = (2, -1, 3)^T.$$

Check: $2(1,0,1)^T - (0,1,1)^T + 3(1,1,1)^T = (5,2,4)^T$. Uniqueness follows since the matrix of basis columns has determinant $-1 \neq 0$.

Conversely, knowing a vector's coordinates in a chosen basis allows us to reconstruct the vector.

With the notation above, $v = \alpha_1 v_1 + \cdots + \alpha_n v_n$ **if and only if** $[v]_B = \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix}$.

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EXERCISE 8.8

For the ordered basis $B = ((1,2)^T, (2,3)^T)$, suppose $[v]_B = (-2,3)^T$. Find v in standard coordinates. Explain why the two coordinate vectors need not have the same numerical entries.

WORKED SOLUTION

The coordinate entries are the coefficients of the basis vectors:

$$v = -2\begin{pmatrix}1\\2\end{pmatrix} + 3\begin{pmatrix}2\\3\end{pmatrix} = \begin{pmatrix}-2+6\\-4+9\end{pmatrix} = \begin{pmatrix}4\\5\end{pmatrix}.$$

This is a basis since $\det\begin{pmatrix}1 & 2\\2 & 3\end{pmatrix} = 3-4 = -1$. The column $(-2,3)^T$ gives coefficients relative to B, while $(4,5)^T$ gives coefficients relative to the standard basis. They represent the same vector.

PROPOSITION 8.4

Let V have dimension n over a field $\mathbb{F}$, and let B be an ordered basis of V.
The coordinate map $T_B : V \to \mathbb{F}^n$, defined by $T_B(v) = [v]_B$, is an isomorphism. In particular, $V \cong \mathbb{F}^n$.

RECALL AND TOOLS

Let V be a vector space over $\mathbb{F}$, let B be an ordered basis with n vectors, and let $u_1, \dots, u_m$ be vectors in V. Then:

1. $\{u_1, \dots, u_m\}$ is linearly independent (in V) **if and only if** $\{[u_1]_B, \dots, [u_m]_B\}$ is linearly independent (in $\mathbb{F}^n$).
2. $\{u_1, \dots, u_m\}$ spans V **if and only if** $\{[u_1]_B, \dots, [u_m]_B\}$ spans $\mathbb{F}^n$.
3. $\{u_1, \dots, u_m\}$ is a basis of V **if and only if** $\{[u_1]_B, \dots, [u_m]_B\}$ is a basis of $\mathbb{F}^n$.

EXERCISE 8.9

Using coordinates in the ordered basis $(1, X, X^2)$, show that the polynomials

$$p_1 = 1 + X, \qquad p_2 = X + X^2, \qquad p_3 = 1 + X + X^2$$

are linearly independent in $\mathbb{R}_2[X]$.

WORKED SOLUTION

Their coordinate columns are

$$[p_1] = (1,1,0)^T, \qquad [p_2] = (0,1,1)^T, \qquad [p_3] = (1,1,1)^T.$$

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EXERCISE 8.9 CONTINUED

If $ap_1 + bp_2 + cp_3 = 0$, comparison of the coefficients of $1, X, X^2$ gives

$$a + c = 0, \qquad a + b + c = 0, \qquad b + c = 0.$$

Subtracting the first equation from the second gives $b = 0$. The third then gives $c = 0$, and the first gives $a = 0$. Thus only the trivial combination vanishes, proving independence. Since the space has dimension 3, they also form a basis; the coordinate matrix has determinant 1.

EXERCISE 8.10

Consider the real vector space $M_{2\times 2}(\mathbb{R})$.
Define two subspaces

$$U := \operatorname{span}\{U_1, U_2, U_3\}, \qquad V := \operatorname{span}\{V_1, V_2\},$$

where

$$U_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad U_2 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad U_3 = \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix},$$

$$V_1 = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \quad V_2 = \begin{pmatrix} 0 & 1 \\ 2 & 2 \end{pmatrix}.$$

1. Find a basis and the dimension of U and of V.
2. Find a basis and the dimension of $U \cap V$.

WORKED SOLUTION

Use the linear map

$$\Phi : M_{2\times 2}(\mathbb{R}) \to \mathbb{R}^4, \qquad \Phi\left(\begin{pmatrix} a & b \\ c & d \end{pmatrix}\right) = (a, b, c, d),$$

which is bijective and hence preserves linear independence and dimensions.

1. Find bases and dimensions of U and V.

   Their coordinates are

   $$\Phi(U_1) = (1, 0, 0, 0), \qquad \Phi(U_2) = (0, 1, 0, 0), \qquad \Phi(U_3) = (0, 0, 1, 1).$$

   If

   $$\alpha\Phi(U_1) + \beta\Phi(U_2) + \gamma\Phi(U_3) = (0, 0, 0, 0),$$

   comparison of entries gives

   $$(\alpha, \beta, \gamma, \gamma) = (0, 0, 0, 0) \Rightarrow \alpha = \beta = \gamma = 0.$$

   Thus $\{\Phi(U_1), \Phi(U_2), \Phi(U_3)\}$ is independent. Therefore $\{U_1, U_2, U_3\}$ is a basis of U, and

   $$\dim U = 3.$$

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EXERCISE 8.10 CONTINUED

For V,

$$\Phi(V_1) = (1,0,1,1), \qquad \Phi(V_2) = (0,1,2,2).$$

if

$$\alpha\Phi(V_1) + \beta\Phi(V_2) = (0,0,0,0),$$

we obtain

$$(\alpha, \beta, \alpha+2\beta, \alpha+2\beta) = (0,0,0,0) \Rightarrow \alpha = 0,\ \beta = 0.$$

Thus $\{\Phi(V_1), \Phi(V_2)\}$ is independent. Therefore $\{V_1, V_2\}$ is a basis of V, and

$$\dim V = 2.$$

2. Find a basis and the dimension of $U \cap V$.

   First show that $V \subseteq U$ by expressing its generators in terms of the generators of U:

$$V_1 = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix} = U_1 + U_3,$$

$$V_2 = \begin{pmatrix} 0 & 1 \\ 2 & 2 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} + 2\begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix} = U_2 + 2U_3.$$

   Hence $V_1, V_2 \in U$, and $V \subseteq U$.

   Whenever $V \subseteq U$, we have $U \cap V = V$. Therefore,

$$U \cap V = V = \text{span}\{V_1, V_2\}, \qquad \dim(U \cap V) = \dim V = 2.$$

8.5 Five descriptions of a linear map

Define $T : \mathbb{R}^2 \to \mathbb{R}^2$ so that, for every

$$(x, y) \in \mathbb{R}^2$$

we have

$$T(x, y) = (x + y,\ x - y).$$

We now describe this same map in several ways.

Method 1: A formula for a general vector

This is a **general formula**: it specifies $T(v)$ for arbitrary $v = (x, y) \in \mathbb{R}^2$, rather than just particular vectors:

$$T(x, y) = (x + y,\ x - y).$$

It defines the map directly on its entire domain.

Method 2: Images of a basis

Choose the standard basis:

$$e_1 = (1, 0), \qquad e_2 = (0, 1).$$

Then

$$T(e_1) = T(1, 0) = (1, 1), \qquad T(e_2) = T(0, 1) = (1, -1).$$

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For $(x, y) = xe_1 + ye_2$, extend linearly:

$$T(x, y) = xT(e_1) + yT(e_2) = x(1, 1) + y(1, -1) = (x + y,\ x - y).$$

Thus specifying the images of a basis defines the same map.

Method 3: A matrix

Write the input as a column:

$$\begin{pmatrix} x \\ y \end{pmatrix}.$$

Define

$$A = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Then

$$T(x, y) = A \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x + y \\ x - y \end{pmatrix}.$$

This matrix represents the same map in the standard basis.

Method 4: Familiar linear operations

View T as performing the following operations:

- First component: sum of the coordinates.
- Second component: difference of the coordinates.

Thus

$$T(x, y) = \big(\text{sum}(x, y), \text{diff}(x, y)\big),$$

These operations are linear, so T is linear.

Method 5: A structural decomposition

Specify that T satisfies

$$T(e_1 + e_2) = (2, 0), \qquad T(e_1 - e_2) = (0, 2),$$

and that T is linear.

Since $\{e_1 + e_2, e_1 - e_2\}$ is a basis of $\mathbb{R}^2$, these requirements determine T uniquely. Calculation gives the same formula:

$$T(x, y) = (x + y,\ x - y).$$

Remarks

- A **general formula** defines $T(v)$ for an arbitrary vector v.
- A linear map may be specified by a formula, basis images, a matrix or a structural description.
- The descriptions differ, but the map itself is the same.

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8.5.1 Summary of the methods

RECALL AND TOOLS

1. **An explicit formula for a general vector**
   Specify $T(v)$ for every $v \in V$.
   Check: the formula is well defined and satisfies both linearity conditions.

2. **Basis images and linear extension**
   If $B = \{v_1, \dots, v_n\}$ is a basis of V, freely choose

   $$T(v_1), \dots, T(v_n) \in W$$

   and define, for $v = \sum_{i=1}^{n} \alpha_i v_i$,

   $$T(v) := \sum_{i=1}^{n} \alpha_i \, T(v_i).$$

   Each choice of basis images determines exactly one linear map.

3. **A matrix**
   For $V = \mathbb{F}^n$ and $W = \mathbb{F}^m$, define

   $$T(x) = Ax$$

   where A is an $m \times n$ matrix.
   Every matrix defines a linear map. After choosing ordered bases, every linear map between finite-dimensional spaces has a matrix representation.
   Remark: the matrix depends on the chosen bases; the map itself does not.

4. **A familiar linear operation**
   Use an operation such as differentiation, integration or multiplication by a fixed scalar.
   Examples:

   $$T(P) = P', \qquad T(f) = \int_0^1 f(x)\, dx.$$

   Specify a domain and codomain, check well-definedness and verify linearity in each case.

5. **A direct-sum decomposition**
   A map can be specified by its action on the components of a direct sum.
   For example, if $V = U \oplus W$, define T by

   $$T(u) = 0 \ \ (\forall u \in U), \qquad T(w) = w \ \ (\forall w \in W).$$

   Linearity follows from the direct-sum decomposition and the linear action on each component.

Common errors

- Adding a nonzero constant vector destroys linearity: $T(v) = v + b$ with $b \neq 0$ is not linear.
- Always specify the domain and codomain: $T : V \to W$. The image $\operatorname{Im} T$ may be a proper subspace of the codomain.
- Distinguish a vector, its coordinate vector in an ordered basis, and the matrix representing a

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map.

EXERCISE 8.11

A system receives three samples $x = (x_1, x_2, x_3)^T$ and subtracts their mean. Define

$$P(x) = x - \frac{x_1 + x_2 + x_3}{3} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

Prove that P is linear, find its kernel and image, and show that a second application changes nothing: $P^2 = P$.

WORKED SOLUTION

Write $\mu(x) = (x_1 + x_2 + x_3)/3$ and $\mathbf{1} = (1, 1, 1)^T$. For vectors x, y and scalars a, b,

$$\mu(ax + by) = a\mu(x) + b\mu(y),$$

so

$$P(ax + by) = ax + by - \big(a\mu(x) + b\mu(y)\big)\mathbf{1} = aP(x) + bP(y).$$

Thus P is linear. If $P(x) = 0$, then $x = \mu(x)\mathbf{1}$; conversely, every constant vector is sent to zero. Therefore,

$$\ker P = \operatorname{span}\{\mathbf{1}\}, \qquad \dim \ker P = 1.$$

For every x, the sum of the entries of $P(x)$ is $x_1 + x_2 + x_3 - 3\mu(x) = 0$. Hence its image lies in the plane

$$H = \{y \in \mathbb{R}^3 : y_1 + y_2 + y_3 = 0\}.$$

If $y \in H$, then $\mu(y) = 0$ and $P(y) = y$. Thus $H \subseteq \operatorname{Im} P$, giving

$$\operatorname{Im} P = H = \operatorname{span}\{(1, -1, 0)^T, (1, 0, -1)^T\}, \qquad \dim \operatorname{Im} P = 2.$$

Finally, $P(x) \in H$, so $P(P(x)) = P(x)$. The dimensions satisfy $1 + 2 = 3$. Removing the mean eliminates the constant component and retains the differences between samples. After one application the mean is already zero.

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Additional practice

EXERCISE 8.12 FOUNDATION

Define $T : \mathbb{R}_{\leq 2}[X] \to \mathbb{R}^2$ by

$$T(p) = \begin{pmatrix} p(1) \\ p(-1) \end{pmatrix}.$$

Prove linearity, find the kernel and image, and verify rank-nullity. Construct a linear map $S : \mathbb{R}^2 \to \mathbb{R}_{\leq 2}[X]$ with $T \circ S = I_{\mathbb{R}^2}$. Is $S \circ T$ also the identity?

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EXERCISE 8.12 FOUNDATION CONTINUED

WORKED SOLUTION

Evaluation preserves addition and scalar multiplication, so T is linear. For $p = a + bX + cX^2$, membership in the kernel requires

$$a + b + c = 0, \qquad a - b + c = 0.$$

Subtracting gives $b = 0$, hence $a = -c$. Thus $\ker T = \operatorname{span}\{X^2 - 1\}$ has dimension 1. For every $(u, v)^T \in \mathbb{R}^2$, define

$$S((u, v)^T) = \frac{u + v}{2} + \frac{u - v}{2} X.$$

This polynomial has values u, v at $1, -1$, respectively, and its coefficients are linear in u, v. Hence S is linear, $T \circ S = I_{\mathbb{R}^2}$ and $\operatorname{Im} T = \mathbb{R}^2$. Indeed, $1 + 2 = 3 = \dim \mathbb{R}_{\leq 2}[X]$. However,

$$(S \circ T)(X^2) = S((1, 1)^T) = 1 \neq X^2,$$

so $S \circ T$ is not the identity. Explicitly, $(S \circ T)(a + bX + cX^2) = a + c + bX$: the measurements do not determine the quadratic coefficient.

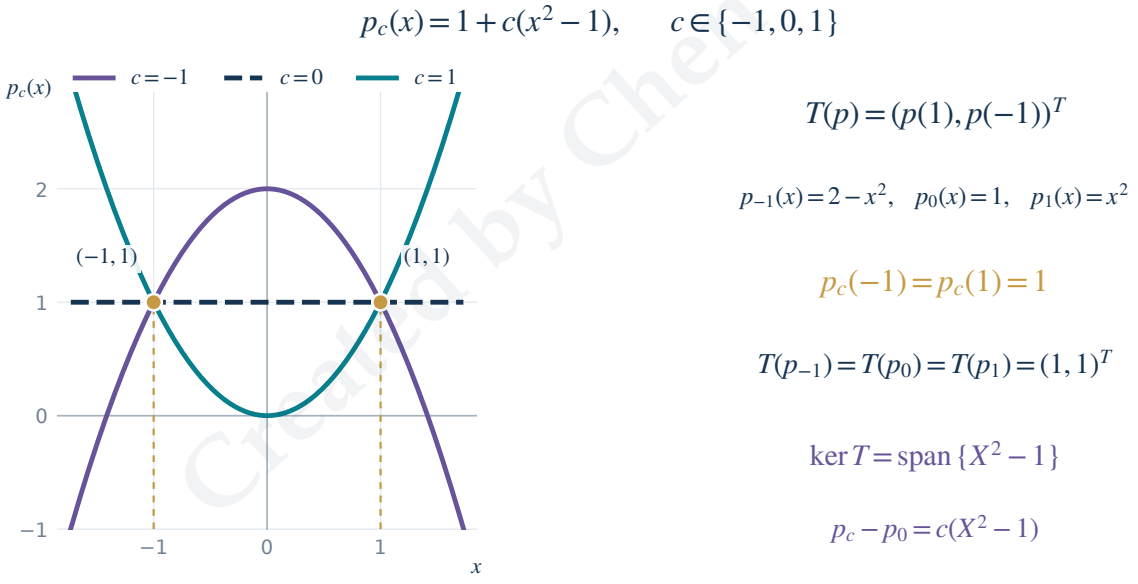


Figure 8.2. Three distinct polynomials give the same two measurements: $p_c(1) = p_c(-1) = 1$ for $c = -1, 0, 1$. Their differences $p_c - p_0 = c(X^2 - 1)$ lie in the kernel of $T(p) = (p(1), p(-1))^T$, so these measurements do not distinguish the polynomials.

EXERCISE 8.13 CHALLENGE

Let $P : V \to V$ be linear and satisfy $P^2 = P$. Prove

$$V = \ker P \oplus \operatorname{Im} P, \qquad \ker(I_V - P) = \operatorname{Im} P, \qquad \operatorname{Im}(I_V - P) = \ker P.$$

Explain why every specified decomposition $V = U \oplus W$ determines a unique such map with

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EXERCISE 8.13 **CHALLENGE CONTINUED**

kernel U and image W. Find both components of the decomposition for

$$P(x, y, z) = (x - z, y + 2z, 0) \quad \text{on } \mathbb{R}^3.$$

WORKED SOLUTION

For every $v \in V$, write $v = (v - Pv) + Pv$. The second term is in the image, and the first is in the kernel because $P(v - Pv) = Pv - P^2v = 0$. If $w = Pv \in \text{Im } P$, then $Pw = P^2v = Pv = w$. Thus a vector in both kernel and image must be zero, proving the direct sum. Also, $(I_V - P)w = 0$ is equivalent to $Pw = w$, and hence to $w \in \text{Im } P$. The decomposition proves $\text{Im}(I_V - P) \subseteq \ker P$. Conversely, if $u \in \ker P$, then $(I_V - P)u = u$. For $V = U \oplus W$, define $P(u + w) = w$ using the unique decomposition. Uniqueness makes the map well defined and linear. Immediately $P^2 = P$, its kernel is U and its image is W. Every map with these properties must annihilate U and act identically on W, proving uniqueness. In the example,

$$\ker P = \text{span}\{(1, -2, 1)^T\}, \qquad \text{Im } P = \{(a, b, 0)^T : a, b \in \mathbb{R}\},$$

and the explicit decomposition is

$$(x, y, z)^T = z(1, -2, 1)^T + (x - z, y + 2z, 0)^T.$$

Applying P again fixes the second component, so $P^2 = P$. This is projection along a direct-sum complement; orthogonality is not assumed.

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TUTORIAL 9

Linear Maps and Matrix Representations

MOTIVATION AND GUIDING IDEA

Motivation. A representing matrix is a computational tool that depends on the ordered bases of the domain and codomain. Changing coordinates leaves the map unchanged but can greatly simplify repeated calculations. Always identify the input and output coordinate systems to avoid reversing a change of basis.

NOTATION AND CONVENTIONS

- For ordered bases $B = (u_1, \dots, u_n)$ of U and $C = (v_1, \dots, v_m)$ of V, the notation $[T]_C^B$ denotes an $m \times n$ matrix taking B-coordinates as input and returning C-coordinates.
- $S \circ T$ applies T first and S second; I_V is the identity map on V.
- $P_{B \leftarrow B'} = [I_V]_B^{B'}$ takes $[v]_{B'}$ to $[v]_B$. The arrow indicates the direction of coordinate conversion.
- If B is an ordered basis of $\mathbb{F}^n$ and E is the standard basis, $S_B = [\, b_1 \ \cdots \ b_n \,] = [I]_E^B$ is the basis-column matrix. Hence $[v]_E = S_B[v]_B$.
- $[T]_B$ abbreviates $[T]_B^B$ for an operator $T : V \to V$. Square matrices A, D are similar if $D = P^{-1}AP$ for an invertible P.

DEFINITION 9.1

Let U, V be finite-dimensional vector spaces over the same field, with ordered bases B, C, and let $T : U \to V$ be linear. Its representing matrix is defined by its columns:

$$[T]_C^B = \big[\, [T(u_1)]_C \ \cdots \ [T(u_n)]_C \,\big], \qquad [T(u)]_C = [T]_C^B [u]_B.$$

For another ordered basis B' of the domain, the columns of $P_{B \leftarrow B'}$ are the B-coordinate vectors of the vectors of B'. Its inverse is $P_{B' \leftarrow B}$. If $R : V \to W$ is linear and D is an ordered basis of W, then

$$[R \circ T]_D^B = [R]_D^C [T]_C^B.$$

If T is an isomorphism, $[T^{-1}]_B^C = ([T]_C^B)^{-1}$. For an operator with standard matrix A, the matrix in the ordered basis B is $S_B^{-1} A S_B$. Domain and codomain bases must match at every matrix product.

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MOTIVATION AND GUIDING IDEA

Construct a map from prescribed vector images, check compatibility with linear dependence relations, compute a representing matrix and change bases. The emphasis is on the connection between an abstract map, coordinates and matrix calculations.

9.1 Determining a map from vector images

PROPOSITION 9.1

Let V, W be vector spaces over the same field $\mathbb{F}$, let $B = (v_1, \dots, v_n)$ be an ordered basis of V, and let $w_1, \dots, w_n \in W$ be arbitrary vectors, not necessarily independent. There is a unique linear map $T : V \to W$ with $T(v_i) = w_i$ for every i. Explicitly, if $v = \sum_{i=1}^{n} a_i v_i$, then

$$T(v) = \sum_{i=1}^{n} a_i w_i.$$

The formula is well defined because basis expansions are unique. It is linear because coordinates of sums and scalar multiples are computed entry by entry. Any linear map with the prescribed images must obey this formula, proving uniqueness. In particular, between any two vector spaces over the same field there is at least one linear map: the zero map.

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EXERCISE 9.1

Determine whether a linear map T with the following properties exists. If it does, give an explicit formula.

1. $T : \mathbb{R}^3 \to \mathbb{R}^3$ satisfying

$$T\overbrace{\begin{pmatrix}1\\1\\1\end{pmatrix}}^{u_1} = \begin{pmatrix}3\\0\\0\end{pmatrix}, \quad T\overbrace{\begin{pmatrix}2\\-1\\2\end{pmatrix}}^{u_2} = \begin{pmatrix}0\\0\\0\end{pmatrix}, \quad T\overbrace{\begin{pmatrix}1\\0\\0\end{pmatrix}}^{u_3} = \begin{pmatrix}1\\0\\0\end{pmatrix}$$

2. Does there exist a linear map $T : \mathbb{R}^3 \to \mathbb{R}^3$ satisfying

$$T\begin{pmatrix}3\\2\\6\end{pmatrix} = \begin{pmatrix}-3\\0\\2\end{pmatrix}, \qquad T\begin{pmatrix}1\\-1\\2\end{pmatrix} = \begin{pmatrix}2\\4\\-5\end{pmatrix}, \qquad T\begin{pmatrix}0\\1\\0\end{pmatrix} = \begin{pmatrix}2\\0\\5\end{pmatrix}?$$

WORKED SOLUTION

1. Test independence without determinants. If $au_1 + bu_2 + cu_3 = 0$, comparison of entries gives

$$a + 2b + c = 0, \qquad a - b = 0, \qquad a + 2b = 0.$$

The second equation gives $a = b$, and the third gives $3b = 0$. Thus $a = b = 0$, and then the first gives $c = 0$. Hence these vectors form a basis of $\mathbb{R}^3$. The theorem guarantees a unique linear map with the prescribed images.

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EXERCISE 9.1 CONTINUED

To find it, first determine the action of T on the standard basis vectors e_1, e_2, e_3. Express each as a linear combination of the given vectors.

We have $e_1 = u_3$. For e_2, write $e_2 = au_1 + bu_2 + cu_3$. This gives a system for a, b, c.

Row reduction yields $a = \frac{2}{3}, b = -\frac{1}{3}, c = 0$.

Treating e_3 similarly gives

$$e_1 = u_3 \;\; e_2 = \frac{2}{3}u_1 - \frac{1}{3}u_2, \;\; e_3 = u_1 - e_1 - e_2$$

Apply linearity and the prescribed images:

$$T(e_1) = T(u_3) = e_1,$$

$$T(e_2) = \frac{2}{3}T(u_1) - \frac{1}{3}T(u_2) = \frac{2}{3}\begin{pmatrix}3\\0\\0\end{pmatrix} - \frac{1}{3}\begin{pmatrix}0\\0\\0\end{pmatrix} = \begin{pmatrix}2\\0\\0\end{pmatrix} = 2e_1,$$

$$T(e_3) = T(u_1) - T(e_1) - T(e_2) = 3e_1 - e_1 - 2e_1 = 0.$$

For every $v = (x, y, z)^T = xe_1 + ye_2 + ze_3$,

$$T(v) = xT(e_1) + yT(e_2) + zT(e_3) = xe_1 + 2ye_1 = \begin{pmatrix}x+2y\\0\\0\end{pmatrix}.$$

Substituting the three original inputs verifies all required images.

2. Now set $u_1 = (3, 2, 6)^T$, $u_2 = (1, -1, 2)^T$, $u_3 = (0, 1, 0)^T$. These vectors are dependent, since

$$u_1 - 3u_2 - 5u_3 = (3-3,\, 2+3-5,\, 6-6)^T = 0.$$

Every linear map satisfies $T(0) = 0$. However, the prescribed images would give

$$T\begin{pmatrix}0\\0\\0\end{pmatrix} = T\left(\begin{pmatrix}3\\2\\6\end{pmatrix} - 3\begin{pmatrix}1\\-1\\2\end{pmatrix} - 5\begin{pmatrix}0\\1\\0\end{pmatrix}\right) =$$

$$T\begin{pmatrix}3\\2\\6\end{pmatrix} - 3T\begin{pmatrix}1\\-1\\2\end{pmatrix} - 5T\begin{pmatrix}0\\1\\0\end{pmatrix} =$$

$$\begin{pmatrix}-3\\0\\2\end{pmatrix} - 3\begin{pmatrix}2\\4\\-5\end{pmatrix} - 5\begin{pmatrix}2\\0\\5\end{pmatrix} = \begin{pmatrix}-19\\-12\\-8\end{pmatrix} \neq \begin{pmatrix}0\\0\\0\end{pmatrix}$$

so no such linear map exists.

A necessary distinction: One violated dependence relation suffices to disprove existence. To prove existence, all dependence relations among the inputs must be preserved by their prescribed images.

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MOTIVATION AND GUIDING IDEA

Existence criterion for prescribed images. The requirements $T(u_i) = w_i$ can be satisfied by a linear map if and only if, for every choice of coefficients a_i,

$$\sum_i a_i u_i = 0 \quad \Longrightarrow \quad \sum_i a_i w_i = 0.$$

Under this condition, define a map on $U = \operatorname{span}\{u_i\}$ by $T(\sum_i a_i u_i) = \sum_i a_i w_i$. The difference of two representations is a dependence relation, so both yield the same image. Linearity follows from the definition. If the domain is finite-dimensional, extend a basis of U to a basis of V, and freely prescribe images of the extra basis vectors. The map is unique if $U = V$. If $U \neq V$ and $W \neq \{0\}$, two different images may be chosen for an extra basis vector, so the extension is not unique. If the u_i already form a basis, there are no nontrivial dependence relations, and every choice of images defines a unique linear map.

EXERCISE 9.2

Let $V = \mathbb{R}^2$ and $W = \mathbb{R}_3[X]$. Find the unique linear map $T : V \to W$ satisfying

$$T((1,0)^T) = 2 + X^2, \qquad T((1,1)^T) = 1 + X + X^2.$$

Give an explicit formula for $T((x,y)^T)$.

WORKED SOLUTION

The vectors $(1,0)^T, (1,1)^T$ form a basis, since their column determinant is 1. Every vector has the unique expansion

$$(x,y)^T = (x-y)(1,0)^T + y(1,1)^T.$$

Linearity therefore requires

$$\begin{aligned} T((x,y)^T) &= (x-y)(2+X^2) + y(1+X+X^2) \\ &= 2x - 2y + (x-y)X^2 + y + yX + yX^2 \\ &= (2x-y) + yX + xX^2. \end{aligned}$$

The formula is linear because each polynomial coefficient is linear in x, y. The two given inputs produce $2 + X^2$ and $1 + X + X^2$, respectively. Existence and uniqueness follow from specifying the images of a basis.

9.2 Composition of maps

RECALL AND TOOLS

- Let U, V, W be vector spaces over a field $\mathbb{F}$.
- Let $T : U \to V$ and $S : V \to W$ be linear maps.
- The map $S \circ T : U \to W$, defined for $u \in U$ by $(S \circ T)(u) = S(T(u))$, is their **composition**.
- $S \circ T$ is linear over $\mathbb{F}$.

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- If both maps are surjective, their composition is surjective. If both are injective, their composition is injective. In particular, a composition of isomorphisms is an isomorphism.

EXERCISE 9.3

Compute $S \circ T$ for the following linear maps:
$T\colon \mathbb{R}^2 \longrightarrow \mathrm{M}_2(\mathbb{R})$ given by $T\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} a & a+b \\ a-b & b \end{pmatrix}$
and $S : M_2(\mathbb{R}) \to \mathbb{R}_3[X]$, given by $S\begin{pmatrix} q & r \\ s & t \end{pmatrix} = tX^2 + (r+s)X + q$

WORKED SOLUTION

Compute $S \circ T : \mathbb{R}^2 \to \mathbb{R}_3[X]$. Let $\begin{pmatrix} a \\ b \end{pmatrix} \in \mathbb{R}^2$. Then

$$S \circ T \begin{pmatrix} a \\ b \end{pmatrix} = S(T\begin{pmatrix} a \\ b \end{pmatrix}) = S\begin{pmatrix} a & a+b \\ a-b & b \end{pmatrix} = bX^2 + 2aX + a$$

9.3 Representing a linear map by a matrix

DEFINITION 9.2

Let U be an n-dimensional vector space over $\mathbb{F}$ with ordered basis $B = (u_1, \ldots, u_n)$.
Let V be an m-dimensional vector space over the same field with ordered basis $C = (v_1, \ldots, v_m)$.
For a linear map $T : U \to V$, define the matrix

$$[T]_C^B = \begin{pmatrix} | & | & | & | \\ [T(u_1)]_C & [T(u_2)]_C & \cdots & [T(u_n)]_C \\ | & | & | & | \end{pmatrix} \in \mathbb{F}^{m \times n}$$

$[T]_C^B$ is the **matrix of T relative to the ordered bases B and C.**

Construction

First compute $T(u_i)$, the images of the basis vectors of the domain.

Then express each image in coordinates relative to C, and use these coordinate columns in the given order.

PROPOSITION 9.2

Let U, V be vector spaces over $\mathbb{F}$, with ordered bases B of U and C of V. For a linear map $T : U \to V$ and every $u \in U$,

$$[T(u)]_C = [T]_C^B \cdot [u]_B$$

Compute this matrix once for the chosen ordered bases. It then converts the coordinate vector

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of any new input into the coordinate vector of its image relative to C.

EXERCISE 9.4

Consider the vector space

$$V = M_3(\mathbb{R})$$

of all real 3×3 matrices.
Define a linear map $T : V \to V$ by

$$T(A) = A - A^T.$$

Write

$$U = \{A \in V \mid A = A^T\}, \qquad W = \{A \in V \mid A = -A^T\}.$$

1. Prove that T is linear.
2. Find $\ker T$ and $\operatorname{Im} T$.
3. Show that $\ker T = U$ and $\operatorname{Im} T = W$.
4. Use rank-nullity to compute $\dim U$ and $\dim W$.
5. Find the matrix of T in an ordered basis obtained by concatenating a basis of U with a basis of W.

WORKED SOLUTION

1. For $A, B \in V$ and $\lambda \in \mathbb{R}$,

$$T(A+B) = (A+B) - (A+B)^T = (A - A^T) + (B - B^T) = T(A) + T(B),$$

and

$$T(\lambda A) = \lambda A - (\lambda A)^T = \lambda(A - A^T) = \lambda T(A),$$

so T is linear.

2. Begin with the kernel:

$$A \in \ker T \iff T(A) = 0 \iff A - A^T = 0 \iff A = A^T.$$

Therefore,

$$\ker T = U.$$

For the image, every $A \in V$ satisfies

$$T(A) = A - A^T,$$

and

$$(T(A))^T = (A - A^T)^T = A^T - A = -(A - A^T) = -T(A).$$

Thus $T(A) \in W$, so

$$\operatorname{Im} T \subseteq W.$$

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EXERCISE 9.4 CONTINUED

Conversely, if $B \in W$, then $B = -B^T$. Set $A = \frac{1}{2}B$. Then

$$T(A) = \tfrac{1}{2}(B - B^T) = \tfrac{1}{2}(B - (-B)) = B,$$

so $B \in \operatorname{Im} T$, proving

$$\operatorname{Im} T = W.$$

3. We have already shown that

$$\ker T = U, \qquad \operatorname{Im} T = W.$$

4. Apply rank-nullity. We know

$$\dim V = \dim M_3(\mathbb{R}) = 9.$$

and hence

$$\dim V = \dim \ker T + \dim \operatorname{Im} T = \dim U + \dim W.$$

A symmetric 3×3 matrix is determined by three diagonal entries and three entries above the diagonal. Thus

$$\dim U = 6.$$

Consequently,

$$\dim W = 9 - 6 = 3.$$

5. Choose the following basis of U:

$$E_{11},\ E_{22},\ E_{33},\ E_{12} + E_{21},\ E_{13} + E_{31},\ E_{23} + E_{32},$$

and the following basis of W:

$$E_{12} - E_{21},\ E_{13} - E_{31},\ E_{23} - E_{32}.$$

Form the ordered basis

$$B = (\text{basis of } U,\ \text{basis of } W).$$

Since $T(A) = 0$ for $A \in U$ and $T(B) = 2B$ for $B \in W$, the matrix of T in this ordered basis is

$$[T]_B = \begin{pmatrix} 0_{6\times6} & 0 \\ 0 & 2I_3 \end{pmatrix}.$$

Thus T annihilates the U component and multiplies the W component by 2, in accordance with

$$V = U \oplus W.$$

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EXERCISE 9.5

Let $V = \mathbb{R}^3$ with standard ordered basis $E = (e_1, e_2, e_3)$. Let $T : V \to V$ have the following matrix in E:

$$[T]_E = \begin{pmatrix} 0 & 1 & 1 \\ 0 & 2 & -1 \\ 0 & 0 & 3 \end{pmatrix}.$$

1. Show that some nonzero $v \in V$ satisfies $T(v) = 0$.
2. Show that some nonzero $u \in V$ satisfies $T(u) = 3u$.
3. Find $w \in V$ such that $T(w) = w + u$, using the u from part 2.

WORKED SOLUTION

Write $A = [T]_E$.

1. The first column of A is the coordinate vector of $T(e_1)$ relative to E. Here that column is zero, so

   $$T(e_1) = 0.$$

   Choose $v = e_1 \neq 0$; then $T(v) = 0$.

2. The third column of A is the coordinate vector of $T(e_3)$ relative to E. Here

   $$T(e_3) = \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix} = e_1 - e_2 + 3e_3.$$

   Seek $u = (x, y, z)$ with $T(u) = 3u$. Using A,

   $$T(x, y, z) = A \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} y + z \\ 2y - z \\ 3z \end{pmatrix}.$$

   we require

   $$\begin{pmatrix} y + z \\ 2y - z \\ 3z \end{pmatrix} = \begin{pmatrix} 3x \\ 3y \\ 3z \end{pmatrix}.$$

   The third equation, $3z = 3z$, is automatic. The second gives $2y - z = 3y$, or $y = -z$. The first gives $y + z = 3x$, so $x = 0$. Take $z = 1$, hence $y = -1$:

   $$u = (0, -1, 1) \neq 0, \qquad T(u) = 3u.$$

3. From part 2, $u = (0, -1, 1)$. Seek $w = (x, y, z)$ satisfying $T(w) = w + u$. As above,

   $$T(w) = \begin{pmatrix} y + z \\ 2y - z \\ 3z \end{pmatrix}, \qquad w + u = \begin{pmatrix} x \\ y - 1 \\ z + 1 \end{pmatrix}.$$

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EXERCISE 9.5 CONTINUED

solve the system

$$\begin{cases} y+z=x, \\ 2y-z=y-1, \\ 3z=z+1. \end{cases}$$

The third equation gives $3z=z+1$, hence $z=\frac{1}{2}$. The second gives $2y-z=y-1$, hence $y=z-1=-\frac{1}{2}$. The first gives $x=y+z=0$. Thus we may take

$$w=\left(0,-\tfrac{1}{2},\tfrac{1}{2}\right), \qquad \text{and then} \qquad T(w)=w+u.$$

EXERCISE 9.6

Let $V=\mathbb{C}[x]$, the vector space of all complex polynomials.
Define the linear operator $N:V\to V$ by

$$(Np)(x):=-(58-x)\,p'(x).$$

For each $\theta\in\mathbb{R}$, define $U_\theta:V\to V$ by

$$U_\theta:=\exp(-i\theta N^2)=\sum_{k=0}^{\infty}\frac{(-i\theta)^k}{k!}N^{2k}.$$

Use the following two ordered bases:

$$E=(1,x,x^2,x^3,\dots), \qquad B=(58,\ (58-x),\ (58-x)^2,\ (58-x)^3,\dots).$$

1. Prove that N and U_θ are linear.
2. For $n\geq 1$, set $b_n(x)=(58-x)^n$, and set $b_0(x)=58$. Compute $N(b_n)$ and $N^2(b_n)$.
3. Compute $U_\theta(b_n)$ for all $n\geq 0$ and deduce the form of $[U_\theta]_B$.
4. Find the matrix of U_θ restricted to $P_3=\operatorname{span}\{1,x,x^2,x^3\}$ in the ordered basis $E_3=(1,x,x^2,x^3)$.

WORKED SOLUTION

1. Differentiation is linear, so

$$N(\alpha p+\beta r)=-(58-x)(\alpha p'+\beta r')=\alpha N(p)+\beta N(r).$$

To justify the series, restrict to P_d, the polynomials of degree at most d. This space is invariant under N, so on P_d the series is the exponential series of a finite matrix, converging for every θ. These definitions agree as d increases and therefore define a map on $\mathbb{C}[x]$. Linearity follows by taking limits of the partial sums in each coordinate.

2. For $n\geq 1$,

$$b_n'(x)=\frac{d}{dx}(58-x)^n=n(58-x)^{n-1}\cdot(-1)=-n(58-x)^{n-1}.$$

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EXERCISE 9.6 CONTINUED

so

$$N(b_n) = -(58-x)\,b_n'(x) = -(58-x)\big(-n(58-x)^{n-1}\big) = n(58-x)^n = nb_n.$$

and hence

$$N^2(b_n) = N(nb_n) = nN(b_n) = n\cdot nb_n = n^2 b_n.$$

Also $N(b_0) = N(58) = 0$, so $N^2(b_0) = 0$.

3. Since $N^{2k}(b_n) = n^{2k}b_n$ for $n \geq 1$, the series gives

$$U_\theta(b_n) = \sum_{k=0}^{\infty} \frac{(-i\theta)^k}{k!}\, n^{2k}\, b_n = \exp(-i\theta n^2)\, b_n.$$

For $n = 0$, $U_\theta(b_0) = b_0$.

Thus the matrix in B is diagonal:

$$[U_\theta]_B = \operatorname{diag}\big(1,\ e^{-i\theta},\ e^{-i4\theta},\ e^{-i9\theta}, \dots\big).$$

4. Write $q_1 = e^{-i\theta}$, $q_4 = e^{-i4\theta}$ and $q_9 = e^{-i9\theta}$. On P_3, use b_0, b_1, b_2, b_3. Their images are $b_0, q_1 b_1, q_4 b_2, q_9 b_3$, respectively.

Expand these four polynomials in E_3:

$$b_0 = 58, \qquad b_1 = 58 - x, \qquad b_2 = (58-x)^2 = x^2 - 116x + 3364,$$

$$b_3 = (58-x)^3 = -x^3 + 174x^2 - 10092x + 195112.$$

Let C be the matrix with coordinate columns $[b_0]_{E_3}, [b_1]_{E_3}, [b_2]_{E_3}, [b_3]_{E_3}$:

$$C = \begin{pmatrix} 58 & 58 & 3364 & 195112 \\ 0 & -1 & -116 & -10092 \\ 0 & 0 & 1 & 174 \\ 0 & 0 & 0 & -1 \end{pmatrix}.$$

This matrix is invertible. The formula $[U_\theta]_{E_3} = C\,\operatorname{diag}(1, q_1, q_4, q_9)C^{-1}$ gives the following matrix:

$$[U_\theta]_{E_3} = \begin{pmatrix} 1 & 58(1-q_1) & 3364(1-2q_1+q_4) & 195112(1-3q_1+3q_4-q_9) \\ 0 & q_1 & 116(q_1-q_4) & 10092(q_1-2q_4+q_9) \\ 0 & 0 & q_4 & 174(q_4-q_9) \\ 0 & 0 & 0 & q_9 \end{pmatrix}.$$

EXERCISE 9.7

Let $T : \mathbb{R}^3 \to \mathbb{R}^2$ be given by

$$T((x,y,z)^T) = (2x - y + z,\ x + 3z)^T.$$

Find its matrix in the standard ordered bases by computing the images of the three basis

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EXERCISE 9.7 CONTINUED

vectors.

WORKED SOLUTION

Substitute each basis vector:

$$T(e_1) = (2\cdot 1 - 0 + 0,\ 1 + 3\cdot 0)^T = (2,1)^T,$$

$$T(e_2) = (2\cdot 0 - 1 + 0,\ 0 + 3\cdot 0)^T = (-1,0)^T,$$

$$T(e_3) = (2\cdot 0 - 0 + 1,\ 0 + 3\cdot 1)^T = (1,3)^T.$$

These images, in this order, are the columns of the matrix:

$$[T]_{E_2}^{E_3} = \begin{pmatrix} 2 & -1 & 1 \\ 1 & 0 & 3 \end{pmatrix}.$$

As a check, multiplying by a general vector yields $(2x - y + z, x + 3z)^T$, the stated formula.

EXERCISE 9.8

Let $T : \mathbb{R}_2[X] \to \mathbb{R}^2$ be the linear map given by $T(P(X)) = \begin{pmatrix} P(1) \\ P(2) \end{pmatrix}$ for every $P(X) \in \mathbb{R}_2[X]$.

Let $B = \left(1, 1+X, 1+X+X^2\right)$ and $C = \left(\begin{pmatrix}1\\1\end{pmatrix}, \begin{pmatrix}0\\1\end{pmatrix}\right)$ be ordered bases of $\mathbb{R}_2[X]$ and $\mathbb{R}^2$, respectively.

(a) Find $[T]_C^B$. (b) **(Independent practice)** Verify the identity from the preceding theorem, namely $[T(u)]_C = [T]_C^B[u]_B$

WORKED SOLUTION

By definition, compute $[T(1)]_C, [T(1+X)]_C, [T(1+X+X^2)]_C$:

$$[T(1)]_C = [\begin{pmatrix} P(1)=1 \\ P(2)=1 \end{pmatrix}]_C = [\begin{pmatrix}1\\1\end{pmatrix}]_C = [1\cdot\begin{pmatrix}1\\1\end{pmatrix} + 0\cdot\begin{pmatrix}0\\1\end{pmatrix}]_C = \begin{pmatrix}1\\0\end{pmatrix}$$

$$[T(1+X)]_C = [\begin{pmatrix} P(1)=2 \\ P(2)=3 \end{pmatrix}]_C = [\begin{pmatrix}2\\3\end{pmatrix}]_C = [2\cdot\begin{pmatrix}1\\1\end{pmatrix} + 1\cdot\begin{pmatrix}0\\1\end{pmatrix}]_C = \begin{pmatrix}2\\1\end{pmatrix}$$

$$[T(1+X+X^2)]_C = [\begin{pmatrix} P(1)=3 \\ P(2)=7 \end{pmatrix}]_C = [\begin{pmatrix}3\\7\end{pmatrix}]_C = [3\cdot\begin{pmatrix}1\\1\end{pmatrix} + 4\cdot\begin{pmatrix}0\\1\end{pmatrix}]_C = \begin{pmatrix}3\\4\end{pmatrix}$$

We obtain $[T]_C^B = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \end{pmatrix}$.

(b) To verify $[T(u)]_C = [T]_C^B[u]_B$, take a general polynomial $P(X) = aX^2 + bX + c \in \mathbb{R}_2[X]$. We check the equality for this polynomial.

First compute the left-hand side:

$$[T(P(X))]_C = [\begin{pmatrix} P(1)=a+b+c \\ P(2)=4a+2b+c \end{pmatrix}]_C = [(a+b+c)\begin{pmatrix}1\\1\end{pmatrix} + (3a+b)\begin{pmatrix}0\\1\end{pmatrix}]_C = \begin{pmatrix} a+b+c \\ 3a+b \end{pmatrix}$$

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For the right-hand side, find the coordinates of $P(X)$ relative to B: solve for $\alpha_1, \alpha_2, \alpha_3$ such that

$$aX^2 + bX + c = \alpha_1 \cdot 1 + \alpha_2 \cdot (1 + X) + \alpha_3 \cdot (1 + X + X^2)$$

Comparison of coefficients, or row reduction, gives $\alpha_1 = c - b, \alpha_2 = b - a, \alpha_3 = a$.

Thus $[P(X)]_B = \begin{pmatrix} c-b \\ b-a \\ a \end{pmatrix}$.

Using the result of part (a), the product is equal to the previously computed left-hand side:

$$[T]_C^B [P(X)]_B = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \end{pmatrix} \begin{pmatrix} c-b \\ b-a \\ a \end{pmatrix} = \begin{pmatrix} c-b+2(b-a)+3a \\ b-a+4a \end{pmatrix} = \begin{pmatrix} a+b+c \\ 3a+b \end{pmatrix}$$

9.3.1 Change-of-coordinate matrices

RECALL AND TOOLS

Definition: Let V be a finite-dimensional vector space over $\mathbb{F}$, with ordered bases B, B'. The **change-of-coordinate matrix** $[I_V]_B^{B'}$ from B' to B is the representing matrix $[T]_B^{B'}$ for the **identity map** $T : V \to V, T(v) = v$ for every $v \in V$.

- The fundamental identity is $[v]_B = [I_V]_B^{B'} [v]_{B'}$.
- The identity map is denoted I_V, with $I_V(v) = v$.
- Its columns are the coordinate vectors of the vectors of B' relative to the ordered basis B.

EXERCISE 9.9

In $V = \mathbb{R}_1[X]$, consider the ordered bases

$$B = (1 + X,\ 1 + 2X), \qquad B' = (2 + 3X,\ 3 + 5X).$$

Compute $[I_V]_B^{B'}$, converting coordinates from B' to B. For $p(X) = a + bX$, explicitly verify $[p]_B = [I_V]_B^{B'} [p]_{B'}$.

WORKED SOLUTION

The basis-column matrices in the standard ordered basis $(1, X)$ are

$$M_B = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}, \qquad M_{B'} = \begin{pmatrix} 2 & 3 \\ 3 & 5 \end{pmatrix}.$$

Both have determinant 1. For the first change-of-coordinate column, use $2 + 3X = (1 + X) + (1 + 2X)$; for the second, use $3 + 5X = (1 + X) + 2(1 + 2X)$. Thus

$$P = [I_V]_B^{B'} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

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EXERCISE 9.9 CONTINUED

For a general polynomial, solve for its coordinates in each basis:

$$\alpha+\beta=a,\ \alpha+2\beta=b \quad\Longrightarrow\quad [p]_B=(2a-b,b-a)^T,$$

$$2\gamma+3\delta=a,\ 3\gamma+5\delta=b \quad\Longrightarrow\quad [p]_{B'}=(5a-3b,-3a+2b)^T.$$

Now compute both entries of the product:

$$P[p]_{B'}=\begin{pmatrix}(5a-3b)+(-3a+2b)\\(5a-3b)+2(-3a+2b)\end{pmatrix}=\begin{pmatrix}2a-b\\b-a\end{pmatrix}=[p]_B.$$

This also verifies the direction: B'-coordinates are the input, and B-coordinates are the output.

9.3.2 Matrices of compositions and inverse maps

PROPOSITION 9.3

Let U,V,W be finite-dimensional vector spaces over $\mathbb{F}$, with ordered bases A,B,C, respectively.
Let $S:U\to V$ and $T:V\to W$ be linear. Then

$$[T\circ S]_C^A=[T]_C^B[S]_B^A$$

Corollary

If $T:V\to W$ is an **isomorphism** and B,C are ordered bases of V,W, respectively, then

$$[T^{-1}]_B^C=\left([T]_C^B\right)^{-1}$$

9.4 Similar matrices

RECALL AND TOOLS

Matrices $A,B\in\mathbb{F}^{n\times n}$ are similar if there is an invertible $P\in\mathbb{F}^{n\times n}$ such that

$$A=P^{-1}BP$$

If A,B are similar, then

1. $$\det(A)=\det(B)$$
2. $$\operatorname{trace}(A)=\operatorname{trace}(B)$$
3. $$\operatorname{rank}(A)=\operatorname{rank}(B)$$

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EXERCISE 9.10

Let $A, B \in \mathbb{F}^{n\times n}$. Prove or disprove:

1. If A, B are similar, then A^k, B^k are similar for every integer $k \geq 1$.
2. If A^k, B^k are similar for some integer $k \geq 1$, then A, B are similar.

WORKED SOLUTION

1. If $B = P^{-1}AP$ with P invertible, cancellation of $PP^{-1} = I$ gives

$$B^k = (P^{-1}AP)^k = P^{-1}A^kP.$$

   Thus the powers are similar via the same P.

2. The claim is false. Over any field, take

$$A = 0_{2\times 2}, \qquad B = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

   Then $A^2 = B^2 = 0$, but A, B are not similar: $P^{-1}AP = 0$ for every invertible P, whereas $B \neq 0$. Over $\mathbb{R}$ or $\mathbb{C}$, one may also take $A = -I_2, B = I_2$: their squares agree, and a scalar matrix is similar only to itself. This example does not disprove the statement in characteristic 2, where $-I_2 = I_2$.

Common error: The identity $A^k = (Q^{-1}BQ)^k$ does not imply $A = Q^{-1}BQ$. Taking matrix powers is not an injective operation, as these examples show.

EXERCISE 9.11

A planar system represents a force vector $v = (x, y)^T$ by two sensor readings:

$$T(v) = \begin{pmatrix} x + y \\ x - y \end{pmatrix}.$$

Use $B = ((1, 1)^T, (1, -1)^T)$ for input coordinates and the standard basis E for output coordinates. Find $[T]_E^B$, recover v from $T(v) = (6, 2)^T$, and check the answer in both coordinate systems.

WORKED SOLUTION

The columns are the images of the input basis vectors:

$$T(1, 1)^T = (2, 0)^T, \qquad T(1, -1)^T = (0, 2)^T, \qquad [T]_E^B = 2I_2.$$

Thus if $[v]_B = (a, b)^T$, then $(6, 2)^T = 2(a, b)^T$, giving $a = 3, b = 1$. Hence

$$v = 3\begin{pmatrix} 1 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \end{pmatrix}.$$

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EXERCISE 9.11 CONTINUED

In standard coordinates, $T((4,2)^T) = (4+2, 4-2)^T = (6,2)^T$. Using B for the input and E for the output gives $[T]_E^B = 2I_2$, so the two coordinate channels are uncoupled. This uses different bases for the domain and codomain; it does not assert that $[T]_B^B = 2I_2$.

EXERCISE 9.12 EXAM QUESTION

Prove or disprove the following statements.

1. There do not exist fields $\mathbb{F}_1 \subseteq \mathbb{F}_2$ and a vector space V over $\mathbb{F}_2$, also regarded as an $\mathbb{F}_1$-vector space by restricting scalars, such that
$$\dim_{\mathbb{F}_1} V = 2 \dim_{\mathbb{F}_2} V.$$
2. If $A \in M_n(\mathbb{F})$ is invertible, then every $B \in M_{n\times m}(\mathbb{F})$ satisfies
$$N(AB) = N(B).$$
3. Let $n \geq 1$. For every integer k with $0 \leq k \leq n^2$, there exists $A \in M_n(\mathbb{R})$ with $\operatorname{tr}(A) = k$ that is similar only to itself.

WORKED SOLUTION

1. False. Take $\mathbb{F}_1 = \mathbb{R}$, $\mathbb{F}_2 = \mathbb{C}$ and $V = \mathbb{C}$. Over $\mathbb{C}$, the basis (1) gives
$$\dim_{\mathbb{C}} \mathbb{C} = 1.$$
Over $\mathbb{R}$, the basis $(1, i)$ gives
$$\dim_{\mathbb{R}} \mathbb{C} = 2 = 2 \dim_{\mathbb{C}} \mathbb{C}.$$
The underlying vectors are the same, but the dimension changes because the field of admissible coefficients changes.
2. True. Prove both inclusions. If $x \in N(B)$, then $Bx = 0$, and
$$ABx = A(Bx) = A0 = 0.$$
so $x \in N(AB)$. Hence $N(B) \subseteq N(AB)$. Conversely, if $x \in N(AB)$, then $ABx = 0$. Multiplication on the left by A^{-1} gives
$$Bx = (A^{-1}A)Bx = A^{-1}0 = 0.$$
so $x \in N(B)$, proving $N(AB) \subseteq N(B)$. Together the inclusions give the required equality.
3. True. Define
$$A = \frac{k}{n} I_n.$$

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EXERCISE 9.12 EXAM QUESTION CONTINUED

Since $n \geq 1$ and the field is $\mathbb{R}$, the quotient is defined, and

$$\operatorname{tr}(A) = \sum_{j=1}^{n} \frac{k}{n} = n\frac{k}{n} = k.$$

For every invertible $P \in M_n(\mathbb{R})$,

$$P^{-1}AP = P^{-1}\left(\frac{k}{n}I_n\right)P = \frac{k}{n}P^{-1}P = \frac{k}{n}I_n = A.$$

Thus every matrix similar to A equals A. In fact this construction works for every real k, including all the values in the statement.

Additional practice

EXERCISE 9.13 FOUNDATION

Define $T : \mathbb{R}^2 \to \mathbb{R}^2$ by $T(x, y) = (2x + y, x + y)$, and choose the ordered bases

$$B = ((1, 0)^T, (1, 1)^T), \qquad C = ((1, 1)^T, (0, 1)^T).$$

Compute $[T]_C^B$ and $[T^{-1}]_B^C$. For $v = (3, 1)^T$, verify the coordinate identity and recover v using the inverse.

WORKED SOLUTION

For every $(r, s)^T$, its C-coordinates are $(r, s - r)^T$. Since $T((1, 0)^T) = (2, 1)^T$ and $T((1, 1)^T) = (3, 2)^T$,

$$M = [T]_C^B = \begin{pmatrix} 2 & 3 \\ -1 & -1 \end{pmatrix}, \qquad \det M = 1.$$

Thus T is an isomorphism. Reversing the roles of the domain and codomain bases gives

$$[T^{-1}]_B^C = M^{-1} = \begin{pmatrix} -1 & -3 \\ 1 & 2 \end{pmatrix}.$$

Now $v = 2(1, 0)^T + (1, 1)^T$, so $[v]_B = (2, 1)^T$. Direct computation gives $T(v) = (7, 4)^T$, and hence

$$[T(v)]_C = \begin{pmatrix} 7 \\ -3 \end{pmatrix} = M \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

Finally, $M^{-1}(7, -3)^T = (2, 1)^T$, and reconstruction in B gives $(3, 1)^T$. The inverse matrix takes C-coordinates as input and returns B-coordinates.

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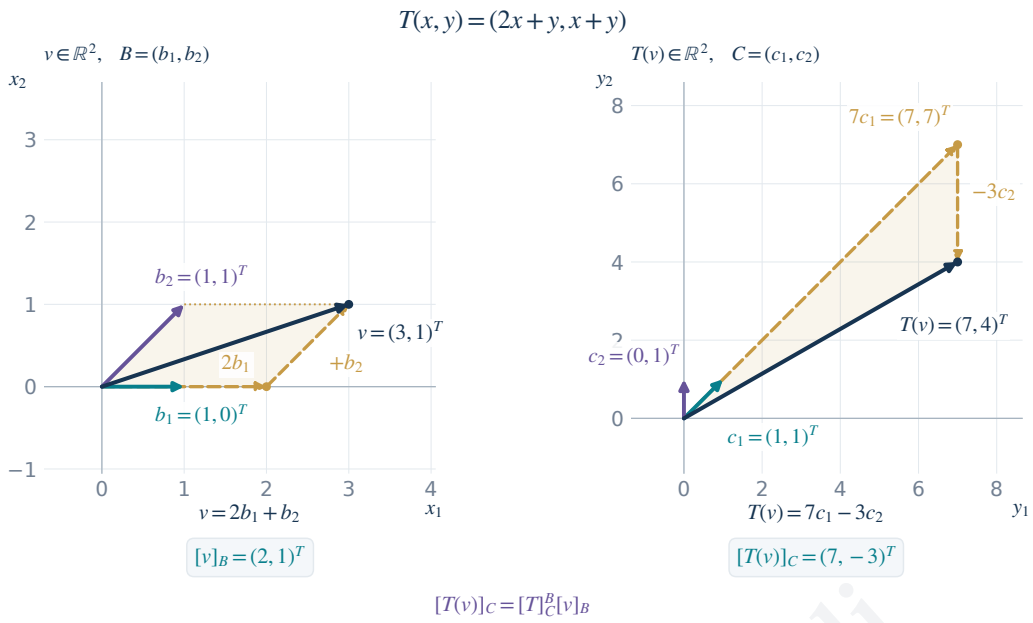


Figure 9.1. The arrows show vectors in standard coordinates; the gold paths show the decompositions $v = 2b_1 + b_2$ and $T(v) = 7c_1 - 3c_2$. Coordinate columns appear separately: $[v]_B = (2,1)^T$ and $[T(v)]_C = (7,-3)^T$. The vector $T(v)$ itself is $(7,4)^T$.

EXERCISE 9.14 CHALLENGE

Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ have standard matrix

$$A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$$

and let $B = ((1,1)^T, (1,-1)^T)$. Compute $D = [T]_B$ and $P = [I]_E^B$. Prove $A^k = PD^kP^{-1}$ for every integer $k \geq 0$, and use it to solve $x_{k+1} = Ax_k$ with $x_0 = (1,0)^T$.

WORKED SOLUTION

The basis images are $T((1,1)^T) = 4(1,1)^T$ and $T((1,-1)^T) = 2(1,-1)^T$. Therefore,

$$D = \begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix}, \qquad P = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad P^{-1} = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

The columns of P reconstruct standard coordinates from B-coefficients. Hence $D = P^{-1}AP$ and $A = PDP^{-1}$. For $k = 0$, the identity is $I = PIP^{-1}$. For $k \geq 1$, cancelling adjacent factors $P^{-1}P$ gives $A^k = PD^kP^{-1}$. Explicitly,

$$A^k = \frac{1}{2}\begin{pmatrix} 4^k + 2^k & 4^k - 2^k \\ 4^k - 2^k & 4^k + 2^k \end{pmatrix}.$$

Induction on the recurrence gives $x_k = A^k x_0$, so

$$x_k = \frac{1}{2}\begin{pmatrix} 4^k + 2^k \\ 4^k - 2^k \end{pmatrix} \qquad (k \geq 0).$$

At $k = 0$ this is $(1,0)^T$; for $k \geq 1$ both entries are integers. In B-coordinates the components

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EXERCISE 9.14 **CHALLENGE CONTINUED**

evolve separately by multiplication by 4 and 2, which explains the simpler calculation.

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TUTORIAL 10

Similarity and Determinants

MOTIVATION AND GUIDING IDEA

A change of basis preserves the linear map but can greatly simplify its matrix. The determinant tests invertibility without computing an inverse, detects dependence among rows, and measures changes in area or volume. Before a long calculation, look for zeros, linear dependence, or block structure.

NOTATION AND CONVENTIONS

- $M_n(\mathbb{F})$ is the space of square matrices of size n over $\mathbb{F}$; I_n is the identity matrix, and A^T is the transpose.
- $A \sim B$ means that $B = P^{-1}AP$ for some invertible P. The columns of P are the new basis vectors expressed in the old basis.
- $\det A$ is the determinant. The alternative notation $|A|$ sometimes means the same scalar; it is not an absolute value of a matrix.
- M_{ij} is obtained by deleting row i and column j; $C_{ij} = (-1)^{i+j} \det(M_{ij})$ is the corresponding cofactor.
- $R_i \leftarrow R_i + cR_j$ means replacing row i by itself plus c times row j. The determinant preservation rule requires $i \neq j$.

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DEFINITION 10.1

For every $n \geq 1$, the determinant is the unique function $\det : M_n(\mathbb{F}) \to \mathbb{F}$ that is linear in each row separately with the other rows fixed, vanishes when two rows are equal, and satisfies $\det I_n = 1$. Row operation rules and cofactor expansions follow. Expansion along row i gives $\det A = \sum_{j=1}^{n} a_{ij}C_{ij}$; the result is independent of the selected row. With the convention $\det(\varnothing) = 1$ for the empty matrix of size 0, the formula also holds for $n = 1$.

NOTE

For square matrices over any field, $\det(AB) = \det A \det B$, $\det(A^T) = \det A$, and $\det(cA) = c^n \det A$. Also, A is invertible exactly when $\det A \neq 0$; then $\det(A^{-1}) = (\det A)^{-1}$. The determinant is not generally additive: $\det(A + B) \neq \det A + \det B$ in general.

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NOTE

Distinguish two transformations. Similarity, $B = P^{-1}AP$, describes an operator after the same basis change in domain and codomain; it preserves the determinant. Congruence, $B = P^T AP$, describes a basis change for a bilinear form; here $\det B = (\det P)^2 \det A$. The formulas cannot be interchanged.

NOTE

The geometric interpretation requires a real matrix. In $\mathbb{R}^2$, $|\det A|$ is the area scale factor; in $\mathbb{R}^3$, it is the volume scale factor. If $\det A = 0$, the image has smaller dimension. If $\det A \neq 0$, the sign indicates whether orientation is preserved or reversed.

MOTIVATION AND GUIDING IDEA

By the end of this tutorial, you will be able to choose an efficient determinant calculation, track the effect of row operations, identify conditions for invertibility, and connect algebraic structure with changes of basis and area.

10.1 Similar Matrices

RECALL AND TOOLS

Matrices $A, B \in \mathbb{F}^{n\times n}$ are called similar if there is an invertible matrix $P \in \mathbb{F}^{n\times n}$ such that

$$A = P^{-1}BP$$

If A, B are similar, then

1. $$\det(A) = \det(B)$$
2. $$\operatorname{trace}(A) = \operatorname{trace}(B)$$
3. $$\operatorname{rank}(A) = \operatorname{rank}(B)$$

EXERCISE 10.1

Let $A, B \in \mathbb{F}^{n\times n}$. Prove or disprove:

1. If A, B are similar, then A^k, B^k are similar for every integer $k \geq 1$.
2. If A^k, B^k are similar for some integer $k \geq 1$, then A, B are similar.

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EXERCISE 10.1 CONTINUED

WORKED SOLUTION

1. If $B = P^{-1}AP$ with P invertible, cancellation using $PP^{-1} = I$ gives

$$B^k = (P^{-1}AP)^k = P^{-1}A^kP.$$

   Thus the powers are similar via the same matrix P.

2. The statement is false. Over any field, take

$$A = 0_{2\times 2}, \qquad B = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

   We have $A^2 = B^2 = 0$, but A, B are not similar: $P^{-1}AP = 0$ for every invertible P, whereas $B \neq 0$. Over $\mathbb{R}$ or $\mathbb{C}$, the original example $A = -I_2$, $B = I_2$ also works: their squares are equal, and a scalar matrix is similar only to itself. This example gives no counterexample in characteristic 2, where $-I_2 = I_2$.

Common pitfall: The equality $A^k = (Q^{-1}BQ)^k$ does not imply $A = Q^{-1}BQ$. Taking a power is not an injective operation on matrices, as these examples show.

10.2 The Determinant

The determinant is defined for square matrices. In the real plane, its absolute value is the area scale factor; its sign indicates whether orientation is preserved or reversed.

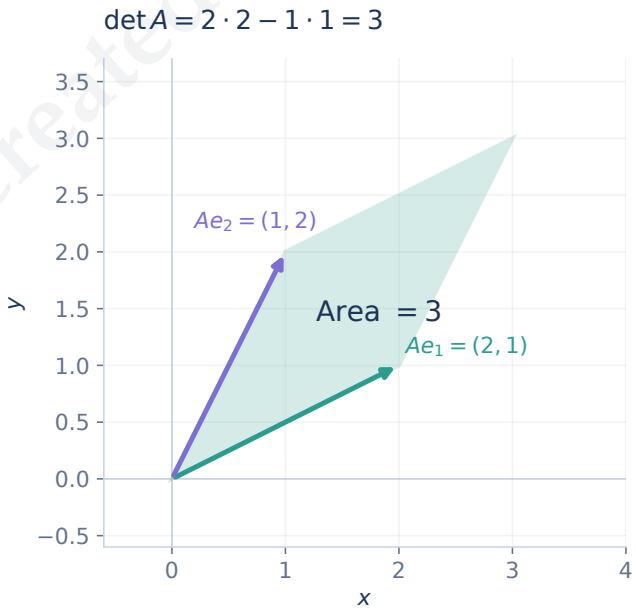


Figure 10.1. A linear map sends the unit square to a parallelogram. The parallelogram's area is the absolute value of the determinant.

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DEFINITION 10.2

Let $n \in \mathbb{N}$. A function

$$\det : M_n(\mathbb{R}) \longrightarrow \mathbb{R}$$

is called the **determinant** if, for every $A \in M_n(\mathbb{R})$, it has the following properties:

1. **Interchanging rows (change of sign):** For $1 \leq i < j \leq n$, if $E^{(1)}_{i,j} \in M_n(\mathbb{R})$ is the elementary matrix interchanging rows i and j, then
$$\det\left(E^{(1)}_{i,j} A\right) = -\det(A).$$
2. **Multiplying a row by a scalar (homogeneity):** For $1 \leq i \leq n$ and $\lambda \in \mathbb{R} \setminus \{0\}$, if $E^{(2)}_{i,\lambda} \in M_n(\mathbb{R})$ is the elementary matrix multiplying row i by λ, then
$$\det\left(E^{(2)}_{i,\lambda} A\right) = \lambda \det(A).$$
3. **Adding a multiple of another row (invariance):** For $1 \leq i \neq j \leq n$ and $\lambda \in \mathbb{R}$, if $E^{(3)}_{i,j,\lambda} \in M_n(\mathbb{R})$ is the elementary matrix performing $R_i \leftarrow R_i + \lambda R_j$, then
$$\det\left(E^{(3)}_{i,j,\lambda} A\right) = \det(A).$$
4. **Normalization:**
$$\det(I_n) = 1.$$

RECALL AND TOOLS

Two useful basic properties:

- If $A = (a_{ij}) \in M_n(\mathbb{R})$ is upper triangular, meaning that $a_{ij} = 0$ for every $i > j$, then
$$\det(A) = \prod_{k=1}^{n} a_{kk}.$$
- If rows i and j are equal for some $i \neq j$, or if A has a zero row, then $\det(A) = 0$. Similarly, two equal columns or a zero column imply $\det(A) = 0$.

EXERCISE 10.2

Compute the determinants using row operations, stating the effect of each operation on the determinant.

1. For $t \in \mathbb{R}$,
$$A(t) = \begin{pmatrix} 1 & 1 & 0 & -1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 \\ 2 & 1 & t & 7 \end{pmatrix}.$$

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EXERCISE 10.2 CONTINUED

2.
$$B = \begin{pmatrix} 2 & 1 & 0 & 3 & -1 \\ 0 & 1 & 2 & -1 & 1 \\ 2 & -1 & -4 & 5 & -3 \\ 1 & 0 & 1 & 0 & 2 \\ 0 & 2 & 0 & 1 & 1 \end{pmatrix}.$$

WORKED SOLUTION

1. Add multiples of other rows to the fourth row; these operations preserve the determinant:
$$A(t) \xrightarrow{R_4 \leftarrow R_4 - 2R_1} \begin{pmatrix} 1 & 1 & 0 & -1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & -1 & t & 9 \end{pmatrix}$$
$$\xrightarrow{R_4 \leftarrow R_4 + R_2} \begin{pmatrix} 1 & 1 & 0 & -1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & t+2 & 10 \end{pmatrix}$$
$$\xrightarrow{R_4 \leftarrow R_4 - (t+2)R_3} \begin{pmatrix} 1 & 1 & 0 & -1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 6-2t \end{pmatrix}.$$
Multiplying the diagonal entries gives $\det A(t) = 1 \cdot 1 \cdot 1 \cdot (6 - 2t) = 6 - 2t$.

2. The rows of B satisfy $r_3 = r_1 - 2r_2$. Thus the two operations $R_3 \leftarrow R_3 - R_1$ and $R_3 \leftarrow R_3 + 2R_2$ produce a zero row without changing the determinant. A matrix with a zero row has determinant zero, so $\det B = 0$.

RECALL AND TOOLS

Basic properties (from the lectures):

- **Multiplicativity:** For every $A, B \in M_n(\mathbb{R})$,
$$\det(AB) = \det(A)\det(B).$$
- **Failure of additivity (in general):** In general, the following equality does not hold:
$$\det(A + B) = \det(A) + \det(B).$$
- **Invariance under transposition:** For every $A \in M_n(\mathbb{R})$,
$$\det(A^T) = \det(A).$$

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- **Scaling the entire matrix:** For every $c \in \mathbb{R}$ and $A \in M_n(\mathbb{R})$,

$$\det(cA) = c^n \det(A),$$

where n is the size of the matrix.

Fundamental theorem:

PROPOSITION 10.1

For every $A \in M_n(\mathbb{R})$, A is invertible if and only if $\det(A) \neq 0$.

EXERCISE 10.3

Let $A, B \in M_n(\mathbb{R})$, with A singular and B invertible. Prove or disprove that the matrix in each part is singular:

1. $A^T B$.
2. $B^{-1}AB$.
3. $A + A^2 + A^3 + A^4$.
4. $(A + I)^2 - (A - I)^2$.
5. $(A + B)^2 + (A - B)^2$.

WORKED SOLUTION

Use $\det A = 0$, $\det B \neq 0$, and multiplicativity of the determinant.

1. $\det(A^T B) = \det(A^T) \det B = \det A \det B = 0$.
2. $\det(B^{-1}AB) = \det(B^{-1}) \det A \det B = 0$.
3. Factor out A on the left:

$$A + A^2 + A^3 + A^4 = A(I + A + A^2 + A^3),$$

Therefore the determinant is zero.

4. Since $AI = IA = A$, we obtain

$$(A + I)^2 - (A - I)^2 = (A^2 + 2A + I) - (A^2 - 2A + I) = 4A.$$

Consequently, $\det(4A) = 4^n \det A = 0$.

5. Expand while preserving the order of multiplication:

$$(A + B)^2 + (A - B)^2 = (A^2 + AB + BA + B^2) + (A^2 - AB - BA + B^2) = 2(A^2 + B^2).$$

The determinant is therefore $2^n \det(A^2 + B^2)$, but it need not vanish. A counterexample is $A = 0$, $B = I_n$: the resulting matrix is $2I_n$, which is invertible and has determinant $2^n \neq 0$.

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10.2.1 The Determinant Formula

RECALL AND TOOLS

Let $A = (a_{ij}) \in M_n(\mathbb{R})$ be a square matrix of size n. For $1 \leq i, j \leq n$, the **minor matrix** corresponding to entry (i, j), denoted M_{ij}, is the square matrix of size $n - 1$ obtained by deleting row i and column j from A.

EXAMPLE 10.1

Let $A = \begin{pmatrix} 9 & 8 & 7 \\ 6 & 5 & 4 \\ 3 & 2 & 1 \end{pmatrix}$. Then $M_{11} = \begin{pmatrix} 5 & 4 \\ 2 & 1 \end{pmatrix}$ and $M_{23} = \begin{pmatrix} 9 & 8 \\ 3 & 2 \end{pmatrix}$.

DEFINITION 10.3

Let $A = (a_{ij}) \in M_n(\mathbb{R})$ be square of size n. Its **determinant**, denoted $\det(A)$, can be computed recursively in n:

1. If $n = 1$, then $A = (a_{11})$ and we define
$$\det(A) = a_{11}.$$
2. If $n = 2$, then
$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}, \qquad \det(A) = a_{11}a_{22} - a_{12}a_{21}.$$
3. If $n \geq 3$, then for every $1 \leq i \leq n$, expansion along row i gives
$$\det(A) = \sum_{j=1}^{n} a_{ij} \, (-1)^{i+j} \, \det(M_{ij}),$$
where M_{ij} is obtained by deleting row i and column j.
Equivalently, for every $1 \leq j \leq n$, expansion along column j gives
$$\det(A) = \sum_{i=1}^{n} a_{ij} \, (-1)^{i+j} \, \det(M_{ij}).$$

EXERCISE 10.4

Compute the determinants again, this time using cofactor expansion and its consequences.

1. For $t \in \mathbb{R}$,
$$A(t) = \begin{pmatrix} 1 & 1 & 0 & -1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 \\ 2 & 1 & t & 7 \end{pmatrix}.$$

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EXERCISE 10.4 CONTINUED

2.
$$B = \begin{pmatrix} 2 & 1 & 0 & 3 & -1 \\ 0 & 1 & 2 & -1 & 1 \\ 2 & -1 & -4 & 5 & -3 \\ 1 & 0 & 1 & 0 & 2 \\ 0 & 2 & 0 & 1 & 1 \end{pmatrix}.$$

WORKED SOLUTION

1. Expand along the first column. Its two nonzero entries are in positions $(1,1)$ and $(4,1)$, so
$$\det A(t) = \det M_{11} - 2\det M_{41}.$$
The minor matrices are
$$M_{11} = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \\ 1 & t & 7 \end{pmatrix}, \qquad M_{41} = \begin{pmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}.$$
Expand the first along its first column and the second along its first row:
$$\det M_{11} = (1\cdot 7 - 2t) + (2\cdot 2 - 1\cdot 1) = 10 - 2t,$$
$$\det M_{41} = 1(2\cdot 2 - 1\cdot 1) + (-1)(1\cdot 1 - 2\cdot 0) = 3 - 1 = 2.$$
Thus $\det A(t) = 10 - 2t - 2\cdot 2 = 6 - 2t$, as obtained by the previous method.

2. Regard the determinant as a function of the rows. Since $r_3 = r_1 - 2r_2$, linearity in the third row gives
$$\begin{aligned}\det(r_1, r_2, r_1 - 2r_2, r_4, r_5) &= \det(r_1, r_2, r_1, r_4, r_5) \\ &\quad - 2\det(r_1, r_2, r_2, r_4, r_5) = 0.\end{aligned}$$
Each term has two equal rows. Linearity in a row and vanishing for equal rows follow from the determinant expansion, so again $\det B = 0$.

EXERCISE 10.5

Using the determinant, find all $k \in \mathbb{R}$ for which
$$A(k) = \begin{pmatrix} 2 & k & 4 \\ 3 & k+1 & 6 \\ -1 & -1 & k+1 \end{pmatrix}$$
is singular.

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EXERCISE 10.5 CONTINUED

WORKED SOLUTION

A square matrix is singular exactly when its determinant is zero. Expand along the first row:

$$\begin{aligned}\det A(k) &= 2\big((k+1)^2+6\big) - k\big(3(k+1)+6\big) \\ &\quad + 4\big(-3+(k+1)\big) \\ &= 2k^2+4k+14-3k^2-9k+4k-8 \\ &= -k^2-k+6 = -(k+3)(k-2).\end{aligned}$$

Hence the required values are $k=-3,2$. For every other value, the determinant is nonzero, so the matrix is invertible.

EXERCISE 10.6

1. Let $n \geq 2$ and let $x_1,\dots,x_n$ be real (or complex) numbers. Define the Vandermonde matrix

$$V(x_1,\dots,x_n) = \big(x_i^{j-1}\big)_{i,j=1}^{n} = \begin{pmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^{n-1} \\ 1 & x_2 & x_2^2 & \cdots & x_2^{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^{n-1} \end{pmatrix}.$$

Prove that

$$\det\big(V(x_1,\dots,x_n)\big) = \prod_{1\leq i<j\leq n} (x_j - x_i).$$

WORKED SOLUTION

1. Write

$$\Delta_n(x_1,\dots,x_n) = \det\big(V(x_1,\dots,x_n)\big).$$

We prove by induction on n that

$$\Delta_n(x_1,\dots,x_n) = \prod_{1\leq i<j\leq n} (x_j - x_i).$$

Base case, $n=2$:

$$\Delta_2(x_1,x_2) = \det\begin{pmatrix} 1 & x_1 \\ 1 & x_2 \end{pmatrix} = x_2 - x_1,$$

as required.

Inductive step: assume the statement holds for $n-1$, and prove it for n. Regard Δ_n as a polynomial in x_n, with $x_1,\dots,x_{n-1}$ fixed. Expanding the determinant along row n gives

$$\Delta_n(x_1,\dots,x_n) = \sum_{j=1}^{n} x_n^{j-1}\,(-1)^{n+j}\,\det(M_{nj}),$$

Thus Δ_n is a polynomial in x_n of degree at most $n-1$. The only term containing x_n^{n-1}

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EXERCISE 10.6 CONTINUED

comes from column n, so its coefficient is

$$(-1)^{n+n}\det(M_{nn}) = \det\big(V(x_1,\dots,x_{n-1})\big) = \Delta_{n-1}(x_1,\dots,x_{n-1}).$$

If two of $x_1,\dots,x_{n-1}$ are equal, the matrix has two equal rows, and both sides of the desired formula are zero. Hence we may assume these numbers are pairwise distinct. For every $k \in \{1,\dots,n-1\}$, substituting $x_n = x_k$ makes rows k and n of V equal, so

$$\Delta_n(x_1,\dots,x_{n-1},x_k) = 0.$$

Therefore $(x_n - x_k)$ divides Δ_n for every $k = 1,\dots,n-1$ (as a polynomial in x_n). Since these roots are distinct and the degree is at most $n-1$, there is a polynomial P such that

$$\Delta_n(x_1,\dots,x_n) = \Big(\prod_{k=1}^{n-1}(x_n - x_k)\Big)\,P(x_1,\dots,x_{n-1}),$$

where P is independent of x_n.

Compare the coefficients of x_n^{n-1} on both sides. On the left the coefficient is $\Delta_{n-1}(x_1,\dots,x_{n-1})$; on the right the coefficient of x_n^{n-1} in $\prod_{k=1}^{n-1}(x_n - x_k)$ is 1. Hence

$$P(x_1,\dots,x_{n-1}) = \Delta_{n-1}(x_1,\dots,x_{n-1}).$$

It follows that

$$\Delta_n(x_1,\dots,x_n) = \Delta_{n-1}(x_1,\dots,x_{n-1})\prod_{k=1}^{n-1}(x_n - x_k).$$

Finally, by the induction hypothesis,

$$\Delta_{n-1}(x_1,\dots,x_{n-1}) = \prod_{1\le i<j\le n-1}(x_j - x_i),$$

and therefore

$$\Delta_n(x_1,\dots,x_n) = \Big(\prod_{1\le i<j\le n-1}(x_j - x_i)\Big)\Big(\prod_{k=1}^{n-1}(x_n - x_k)\Big) = \prod_{1\le i<j\le n}(x_j - x_i).$$

This proves the formula.

EXERCISE 10.7

1. Let $n \ge 2$ and let $w \in \mathbb{C}$ be a primitive nth root of unity: $w^n = 1$ and $w^k \ne 1$ for every $1 \le k \le n-1$. Define the matrix

$$F = \big(w^{(i-1)(j-1)}\big)_{i,j=1}^{n} \in M_n(\mathbb{C}).$$

Compute $\det(F)$ explicitly.

2. Let $n \ge 3$, let $1 \le p \le n-1$, and write $q = n - p$. For $\alpha, \beta \in \mathbb{R}$, define $A(\alpha,\beta) \in M_n(\mathbb{R})$

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EXERCISE 10.7 CONTINUED

by

$$a_{ii} = 1 \quad (1 \le i \le n),$$

and, for every $i \neq j$,

$$a_{ij} = \begin{cases} \alpha, & 1 \le i,j \le p \text{ or } p+1 \le i,j \le n, \\ \beta, & \text{otherwise.} \end{cases}$$

Let $T : \mathbb{R}^n \to \mathbb{R}^n$ be the linear map $T(v) = A(\alpha,\beta)v$. Determine for which α, β the map T is an isomorphism, and compute $\det(A(\alpha,\beta))$.

WORKED SOLUTION

First display the structure of the matrices:

$$F = \left(w^{(i-1)(j-1)}\right)_{i,j=1}^{n} = \begin{pmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & w & w^2 & \cdots & w^{n-1} \\ 1 & w^2 & w^4 & \cdots & w^{2(n-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & w^{n-1} & w^{2(n-1)} & \cdots & w^{(n-1)(n-1)} \end{pmatrix}.$$

$$A(\alpha,\beta) = \left(\begin{array}{c|c} A_{11} & A_{12} \\ \hline A_{21} & A_{22} \end{array}\right),.$$

The block sizes are

$$A_{11} \in M_p(\mathbb{R}), \qquad A_{22} \in M_q(\mathbb{R}),$$
$$A_{12} \in M_{p\times q}(\mathbb{R}), \qquad A_{21} \in M_{q\times p}(\mathbb{R}).$$

$$A_{11} = \begin{pmatrix} 1 & \alpha & \alpha & \cdots & \alpha \\ \alpha & 1 & \alpha & \cdots & \alpha \\ \alpha & \alpha & 1 & \cdots & \alpha \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \alpha & \alpha & \alpha & \cdots & 1 \end{pmatrix}_{p\times p}, \qquad A_{22} = \begin{pmatrix} 1 & \alpha & \alpha & \cdots & \alpha \\ \alpha & 1 & \alpha & \cdots & \alpha \\ \alpha & \alpha & 1 & \cdots & \alpha \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \alpha & \alpha & \alpha & \cdots & 1 \end{pmatrix}_{q\times q}.$$

$$A_{12} = \begin{pmatrix} \beta & \beta & \cdots & \beta \\ \beta & \beta & \cdots & \beta \\ \vdots & \vdots & \ddots & \vdots \\ \beta & \beta & \cdots & \beta \end{pmatrix}_{p\times q}, \qquad A_{21} = \begin{pmatrix} \beta & \beta & \cdots & \beta \\ \beta & \beta & \cdots & \beta \\ \vdots & \vdots & \ddots & \vdots \\ \beta & \beta & \cdots & \beta \end{pmatrix}_{q\times p}.$$

$$P^{-1}A(\alpha,\beta)P = (1-\alpha)I_{n-2} \oplus B = \left(\begin{array}{c|c} (1-\alpha)I_{n-2} & 0 \\ \hline 0 & B \end{array}\right),$$

where the lower-right block has size 2:

$$B = \begin{pmatrix} 1+(p-1)\alpha & q\beta \\ p\beta & 1+(q-1)\alpha \end{pmatrix}.$$

1. **Fact used below (Vandermonde determinant).** For $x_1, \ldots, x_n \in \mathbb{C}$, define

$$V(x_1,\ldots,x_n) = \left(x_i^{j-1}\right)_{i,j=1}^{n}.$$

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EXERCISE 10.7 CONTINUED

Then

$$\det\big(V(x_1,\dots,x_n)\big) = \prod_{1\le i<j\le n}(x_j - x_i).$$

Set

$$x_i = w^{i-1} \qquad (i = 1,\dots,n).$$

For every i, j,

$$F_{ij} = w^{(i-1)(j-1)} = (w^{i-1})^{j-1} = x_i^{j-1}.$$

Therefore

$$F = V(x_1,\dots,x_n).$$

By the preceding exercise on the Vandermonde determinant,

$$\det(F) = \prod_{1\le i<j\le n}(x_j - x_i) = \prod_{0\le i<j\le n-1}\big(w^j - w^i\big).$$

For every $0 \le i < j \le n-1$,

$$w^j - w^i = w^i\big(w^{j-i} - 1\big),$$

so

$$\det(F) = \left(\prod_{0\le i<j\le n-1} w^i\right)\left(\prod_{0\le i<j\le n-1}(w^{j-i} - 1)\right).$$

Write

$$S = \sum_{0\le i<j\le n-1} i.$$

Then

$$\prod_{0\le i<j\le n-1} w^i = w^S.$$

We now compute S: For each $i \in \{0,\dots,n-2\}$, there are exactly $(n-1-i)$ values of j with $i < j \le n-1$, so

$$S = \sum_{i=0}^{n-2} i(n-1-i) = \frac{n(n-1)(n-2)}{6}.$$

Consequently,

$$\prod_{0\le i<j\le n-1} w^i = w^{\frac{n(n-1)(n-2)}{6}}.$$

Also group the factors by $m = j - i \in \{1,\dots,n-1\}$. For each m, there are exactly $(n-m)$ pairs (i,j) with $j - i = m$, so

$$\prod_{0\le i<j\le n-1}(w^{j-i} - 1) = \prod_{m=1}^{n-1}(w^m - 1)^{n-m}.$$

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EXERCISE 10.7 CONTINUED

Therefore

$$\boxed{\det(F) = w^{\frac{n(n-1)(n-2)}{6}} \prod_{m=1}^{n-1} (w^m - 1)^{n-m}}.$$

Because w is a primitive nth root of unity, $w^m \neq 1$ for $1 \leq m \leq n-1$, so $w^m - 1 \neq 0$. Every factor in the product is therefore nonzero, and $\det(F) \neq 0$.

2. **Theorems used below (from class).**

(a) Isomorphisms and representing matrices. Let V be a finite-dimensional real vector space, let $T : V \to V$ be linear, and let $[T]_{\mathcal{B}}$ represent T in a basis $\mathcal{B}$. Then

$$T \text{ is an isomorphism } \iff [T]_{\mathcal{B}} \text{ is invertible.}$$

(b) Invertibility and the determinant. For every $A \in M_n(\mathbb{R})$,

$$A \text{ is invertible } \iff \det(A) \neq 0.$$

(This is the standard determinant criterion for invertibility.)

(c) Invariance of the determinant under change of basis. If P is invertible, then

$$\det(P^{-1}AP) = \det(A),$$

because

$$\det(P^{-1}AP) = \det(P^{-1}) \det(A) \det(P) = \det(A).$$

We can now solve the problem.

Write $A = A(\alpha, \beta)$. Define

$$U = \Big\{ x \in \mathbb{R}^n : x_{p+1} = \cdots = x_n = 0, \quad \sum_{i=1}^{p} x_i = 0 \Big\},$$

$$W = \Big\{ x \in \mathbb{R}^n : x_1 = \cdots = x_p = 0, \quad \sum_{i=p+1}^{n} x_i = 0 \Big\}.$$

Clearly U, W are subspaces, since they are defined by homogeneous linear equations in the coordinates.

We first prove that $Ax = (1-\alpha)x$ for every $x \in U$. Let $x \in U$. Then $x_{p+1} = \cdots = x_n = 0$ and $\sum_{j=1}^{p} x_j = 0$.

If $1 \leq i \leq p$, the structure of row i of A (diagonal entry 1, entries α elsewhere in its block, and entries β between blocks) gives

$$(Ax)_i = 1 \cdot x_i + \sum_{\substack{1 \leq j \leq p \\ j \neq i}} \alpha x_j + \sum_{j=p+1}^{n} \beta x_j.$$

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EXERCISE 10.7 CONTINUED

The last sum vanishes because $x_{p+1} = \cdots = x_n = 0$, so

$$(Ax)_i = x_i + \alpha \sum_{\substack{1 \le j \le p \\ j \ne i}} x_j.$$

Now

$$\sum_{\substack{1 \le j \le p \\ j \ne i}} x_j = \sum_{j=1}^{p} x_j - x_i = 0 - x_i = -x_i,$$

and hence

$$(Ax)_i = x_i + \alpha(-x_i) = (1-\alpha)x_i.$$

If $p+1 \le i \le n$, the coefficients of coordinates $1, \ldots, p$ in this row are all β (the second block of x is zero). Thus

$$(Ax)_i = \sum_{j=1}^{p} \beta x_j = \beta \sum_{j=1}^{p} x_j = 0 = (1-\alpha)x_i.$$

Therefore $Ax = (1-\alpha)x \in U$. In particular, U is A-invariant, and the restriction $A|_U$ is multiplication by $(1-\alpha)$. The same argument gives $Ay = (1-\alpha)y$ for every $y \in W$. Thus W is also A-invariant and $A|_W = (1-\alpha)\mathrm{Id}_W$.

Define

$$u = (\underbrace{1, \ldots, 1}_{p}, \underbrace{0, \ldots, 0}_{q}), \qquad v = (\underbrace{0, \ldots, 0}_{p}, \underbrace{1, \ldots, 1}_{q}), \qquad S = \mathrm{span}\{u, v\}.$$

We next compute Au and Av.

For every $1 \le i \le p$,

$$(Au)_i = 1 \cdot 1 + \sum_{\substack{1 \le j \le p \\ j \ne i}} \alpha \cdot 1 + \sum_{j=p+1}^{n} \beta \cdot 0 = 1 + (p-1)\alpha.$$

and, for every $p+1 \le i \le n$,

$$(Au)_i = \sum_{j=1}^{p} \beta \cdot 1 + \Big(1 \cdot 0 + \sum_{\substack{p+1 \le j \le n \\ j \ne i}} \alpha \cdot 0\Big) = p\beta.$$

Therefore

$$Au = \big(1 + (p-1)\alpha\big)u + p\beta\, v.$$

Similarly,

$$Av = q\beta\, u + \big(1 + (q-1)\alpha\big)v.$$

Thus S is A-invariant, and its representing matrix in the basis (u, v) is

$$B = \begin{pmatrix} 1 + (p-1)\alpha & q\beta \\ p\beta & 1 + (q-1)\alpha \end{pmatrix}.$$

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EXERCISE 10.7 CONTINUED

We claim that

$$\mathbb{R}^n = (U \oplus W) \oplus S.$$

First, $U \cap W = \{0\}$, since a vector in both subspaces must have every coordinate zero. Also, if $au + bv \in U \oplus W$, the sum of its first-block coordinates is $ap = 0$, so $a = 0$; the sum of its second-block coordinates is $bq = 0$, so $b = 0$. Hence $S \cap (U \oplus W) = \{0\}$.

Next, every $x \in \mathbb{R}^n$ admits a decomposition $x = x_{UW} + x_S$ with $x_{UW} \in U \oplus W$ and $x_S \in S$. Write

$$s_1 = \sum_{i=1}^{p} x_i, \qquad s_2 = \sum_{i=p+1}^{n} x_i,$$

and define

$$x_S = \frac{s_1}{p}u + \frac{s_2}{q}v \in S, \qquad x_{UW} = x - x_S.$$

The sum of the first-block coordinates of x_{UW} is $s_1 - s_1 = 0$, and the second-block sum is $s_2 - s_2 = 0$, so $x_{UW} \in U \oplus W$. This proves the direct-sum decomposition.

Choose a basis $\mathcal{B}$ by concatenating bases of U and W and the basis (u, v) of S. In this basis, the matrix of A has the block diagonal form

$$(1-\alpha)I_{n-2} \oplus B.$$

Let P be the change-of-coordinates matrix from $\mathcal{B}$ to the standard basis. Then

$$P^{-1}AP = (1-\alpha)I_{n-2} \oplus B.$$

Combining facts (b) and (c),

$$\det(A) = \det(P^{-1}AP) = \det\big((1-\alpha)I_{n-2} \oplus B\big).$$

The determinant of a block diagonal matrix is the product of the determinants of its blocks (this follows by cofactor expansion, or by a suitable factorization and multiplicativity $\det(AB) = \det(A)\det(B)$). Therefore

$$\det(A) = \det\big((1-\alpha)I_{n-2}\big) \cdot \det(B) = (1-\alpha)^{n-2}\det(B).$$

Compute $\det(B)$:

$$\det(B) = \big(1 + (p-1)\alpha\big)\big(1 + (q-1)\alpha\big) - pq\,\beta^2.$$

and hence

$$\boxed{\det(A) = (1-\alpha)^{n-2}\Big(\big(1 + (p-1)\alpha\big)\big(1 + (q-1)\alpha\big) - pq\,\beta^2\Big)}.$$

By fact (a), T is an isomorphism if and only if its representing matrix A in the standard basis is invertible. By fact (b), this is equivalent to $\det(A) \neq 0$. Thus

$$T \text{ is an isomorphism} \iff \det(A) \neq 0 \iff (1-\alpha)^{n-2} \neq 0 \text{ and } \det(B) \neq 0.$$

Since $\alpha, \beta \in \mathbb{R}$ and $n \geq 3$, $(1-\alpha)^{n-2} \neq 0$ holds if and only if $\alpha \neq 1$. The exact condition

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EXERCISE 10.7 CONTINUED

is therefore

$$\boxed{\alpha \neq 1} \quad \text{and} \quad \boxed{\big(1+(p-1)\alpha\big)\big(1+(q-1)\alpha\big) \neq pq\,\beta^2}.$$

EXERCISE 10.8

For $s \in \mathbb{R}$, define $T_s : \mathbb{R}^2 \to \mathbb{R}^2$ by

$$T_s(v) = A_s v, \qquad A_s = \begin{pmatrix} 2 & 1 \\ s & 2 \end{pmatrix}.$$

The triangle Δ has vertices $(0,0)$, $(1,0)$, $(0,1)$. Find the vertices and area of its image, determine when T_s is invertible, and explain what happens when $s = 4$.

WORKED SOLUTION

The images of the vertices are $(0,0)$, $(2,s)$, $(1,2)$. The original triangle has area $1/2$, and the area scale factor is $|\det A_s| = |2\cdot 2 - 1\cdot s| = |4-s|$. Hence the image has area $|4-s|/2$. Also, T_s is invertible exactly when $4 - s \neq 0$, namely when $s \neq 4$. When $s = 4$, the columns satisfy $(2,4)^T = 2(1,2)^T$. Every image is therefore a multiple of $(1,2)^T$, and the triangle collapses to a line segment on $y = 2x$, with zero area. For $s < 4$, orientation is preserved; for $s > 4$, it is reversed. For example, at $s = 1$ the image has area $3/2$.

EXERCISE 10.9 EXAM QUESTION

1. Let $A \in M_n(\mathbb{R})$ satisfy $\operatorname{rank}(A^2) < \operatorname{rank}(A)$. Prove that there exists $v \in \mathbb{R}^n$ such that $A^2 v = 0$ and $Av \neq 0$.
2. Let $A, B \in M_n(\mathbb{R})$. Prove that
$$\operatorname{rank}(A) + \operatorname{rank}(B) \leq n + \operatorname{rank}(AB).$$
Deduce that if $\operatorname{rank}(A) + \operatorname{rank}(B) > n$, then $AB \neq 0$.
3. Let $N \in M_n(\mathbb{R})$ satisfy $N^2 = 0$ and $\operatorname{rank}(N) = r$. Find $\det(I_n + tN)$ and $\operatorname{rank}(I_n + tN)$ for every $t \in \mathbb{R}$.

WORKED SOLUTION

1. If $Av = 0$, then $A^2 v = A(Av) = 0$. Hence $\ker A \subseteq \ker A^2$. By the rank-nullity theorem and the given inequality,
$$\begin{aligned} \dim\ker A^2 &= n - \operatorname{rank}(A^2) \\ &> n - \operatorname{rank}(A) = \dim\ker A. \end{aligned}$$
Thus the inclusion is strict: $\ker A \subsetneq \ker A^2$. Choose $v \in \ker A^2 \setminus \ker A$. Membership in the first kernel gives $A^2 v = 0$, while exclusion from the second gives $Av \neq 0$. This vector remains nonzero after one application of A and becomes zero after two applications.

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EXERCISE 10.9 **EXAM QUESTION CONTINUED** **10**

2. Consider the restricted map

$$T : \operatorname{Im} B \longrightarrow \mathbb{R}^n, \qquad T(u) = Au.$$

Its image is exactly $\operatorname{Im}(AB)$: every $u \in \operatorname{Im} B$ has the form $u = Bv$, so $T(u) = ABv$; conversely, every vector ABv arises in this way. Its kernel is $\operatorname{Im} B \cap \ker A$. Applying rank-nullity to T gives

$$\operatorname{rank}(B) = \dim(\operatorname{Im} B \cap \ker A) + \operatorname{rank}(AB).$$

However,

$$\dim(\operatorname{Im} B \cap \ker A) \leq \dim \ker A = n - \operatorname{rank}(A).$$

Substitute and rearrange:

$$\begin{aligned} \operatorname{rank}(B) &\leq n - \operatorname{rank}(A) + \operatorname{rank}(AB), \\ \operatorname{rank}(A) + \operatorname{rank}(B) &\leq n + \operatorname{rank}(AB). \end{aligned}$$

If the ranks sum to more than n, then $\operatorname{rank}(AB) > 0$, so AB is not the zero matrix.

3. First compute the two products:

$$\begin{aligned} (I_n + tN)(I_n - tN) &= I_n - tN + tN - t^2N^2 = I_n, \\ (I_n - tN)(I_n + tN) &= I_n + tN - tN - t^2N^2 = I_n. \end{aligned}$$

Thus $I_n + tN$ is invertible for every $t \in \mathbb{R}$, with inverse $I_n - tN$. Immediately,

$$\operatorname{rank}(I_n + tN) = n.$$

To compute the determinant, note that $N^2 = 0$ implies $\operatorname{Im} N \subseteq \ker N$. Hence $r \leq n - r$, or $2r \leq n$. Choose a basis $(u_1, \ldots, u_r)$ of $\operatorname{Im} N$ and extend it to a basis $(u_1, \ldots, u_r, w_1, \ldots, w_{n-2r})$ of $\ker N$. For each j, choose v_j with $Nv_j = u_j$.

The list

$$\mathcal{B} = (u_1, \ldots, u_r, w_1, \ldots, w_{n-2r}, v_1, \ldots, v_r)$$

is a basis. Indeed, applying N to a vanishing linear combination gives a combination of the u_j, so every coefficient of a v_j is zero. The remaining coefficients vanish because they multiply a basis of $\ker N$. Thus the list contains n linearly independent vectors. In this basis,

$$[N]_{\mathcal{B}} = \begin{pmatrix} 0 & 0 & I_r \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad [I_n + tN]_{\mathcal{B}} = \begin{pmatrix} I_r & 0 & tI_r \\ 0 & I_{n-2r} & 0 \\ 0 & 0 & I_r \end{pmatrix}.$$

A block of size zero is simply omitted. This is upper triangular with every diagonal entry equal to 1. Since the determinant is preserved under change of basis,

$$\det(I_n + tN) = 1 \qquad \text{for every } t \in \mathbb{R}.$$

Neither answer depends on r: the determinant is 1 and the rank is n.

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Additional practice

EXERCISE 10.10 **FOUNDATION**

For $t \in \mathbb{R}$, consider

$$A_t = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 3 & 1 \\ 2 & 4 & t \end{pmatrix}.$$

1. Compute $\det A_t$ using one row operation, and determine when A_t is invertible.
2. At $t = 0$, find a basis of $\ker A_0$ and find $\operatorname{rank} A_0$.
3. Construct B_t from A_t by, in this order, interchanging the first two rows, multiplying the third row by 2, and adding 5 times the third row to the first row. Compute $\det B_t$ without writing B_t explicitly.

WORKED SOLUTION

1. The operation $R_3 \leftarrow R_3 - 2R_1$ preserves the determinant and gives

$$\begin{pmatrix} 1 & 2 & 0 \\ 0 & 3 & 1 \\ 0 & 0 & t \end{pmatrix}.$$

   Hence $\det A_t = 1 \cdot 3 \cdot t = 3t$, and A_t is invertible exactly when $t \neq 0$.

2. At $t = 0$, the kernel equations are $x_1 + 2x_2 = 0$ and $3x_2 + x_3 = 0$. Setting $x_2 = s$ gives

$$\ker A_0 = \{s(-2, 1, -3)^T : s \in \mathbb{R}\} = \operatorname{span}\{(-2, 1, -3)^T\}.$$

   The kernel has dimension 1, so rank-nullity gives $\operatorname{rank} A_0 = 3 - 1 = 2$.

3. The interchange multiplies the determinant by -1, scaling the row multiplies it by 2, and the final operation preserves it. Thus

$$\det B_t = (-1) \cdot 2 \cdot \det A_t = -6t.$$

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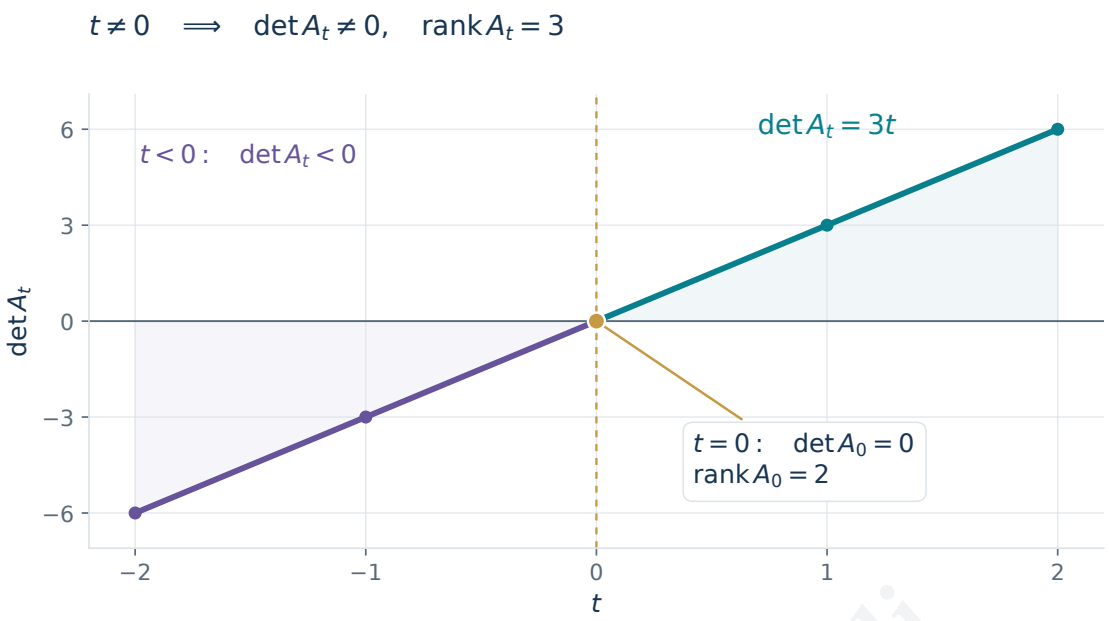


Figure 10.2. For the matrix A_t in the exercise, the graph $\det A_t = 3t$ crosses zero only at $t = 0$. For every $t \neq 0$, the matrix is invertible and has rank 3; at the gold point, $\operatorname{rank} A_0 = 2$, with kernel $\operatorname{span}\{(-2, 1, -3)^T\}$.

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EXERCISE 10.11 CHALLENGE

Let $A \in M_n(\mathbb{F})$ be invertible, $U \in \mathbb{F}^{n\times k}$, and $V \in \mathbb{F}^{k\times n}$, with $n, k \geq 1$.

1. Use determinants of block matrices to prove

$$\det(A + UV) = \det A \, \det(I_k + VA^{-1}U).$$

Do not assume that U or V is square or invertible.

2. For $s \in \mathbb{R}$, use the identity to compute the determinant of

$$C_s = \begin{pmatrix} s+1 & 0 & 1 \\ 0 & s+1 & -1 \\ s & s & 1 \end{pmatrix}$$

and determine when it is invertible. At the singular parameter value, find its kernel and rank.

WORKED SOLUTION

1. First prove $\det(I_n + XY) = \det(I_k + YX)$ for $X \in \mathbb{F}^{n\times k}$ and $Y \in \mathbb{F}^{k\times n}$. Set

$$K = \begin{pmatrix} I_n & X \\ -Y & I_k \end{pmatrix}, \quad L = \begin{pmatrix} I_n & 0 \\ Y & I_k \end{pmatrix}, \quad H = \begin{pmatrix} I_n & -X \\ 0 & I_k \end{pmatrix}.$$

Both L and H are block triangular with determinant 1. We have

$$LK = \begin{pmatrix} I_n & X \\ 0 & I_k + YX \end{pmatrix}, \qquad HK = \begin{pmatrix} I_n + XY & 0 \\ -Y & I_k \end{pmatrix}.$$

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EXERCISE 10.11 **CHALLENGE CONTINUED**

Multiplicativity and the determinant formula for block triangular matrices give

$$\det K = \det(I_k + YX) = \det(I_n + XY).$$

Now $A + UV = A(I_n + A^{-1}UV)$. Substitute $X = A^{-1}U$ and $Y = V$ to obtain the required identity.

2. Take $A = I_3$ and write $C_s = I_3 + UV$, where

$$U = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}, \qquad V = \begin{pmatrix} s & 0 & 1 \\ 0 & s & -1 \end{pmatrix}.$$

Direct multiplication verifies this factorization and gives

$$I_2 + VU = \begin{pmatrix} s+2 & 1 \\ -1 & s \end{pmatrix}.$$

Therefore

$$\det C_s = \det(I_2 + VU) = s(s+2) + 1 = (s+1)^2.$$

Thus C_s is invertible exactly when $s \neq -1$. At $s = -1$, its kernel equations are $x_3 = 0$ and $-x_1 - x_2 + x_3 = 0$, so

$$\ker C_{-1} = \operatorname{span}\{(1, -1, 0)^T\}, \qquad \operatorname{rank} C_{-1} = 3 - 1 = 2.$$

The multiplicity of a root of the determinant as a polynomial in s is not the kernel dimension: this root is double, while the kernel has dimension 1.

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TUTORIAL 11

Eigenvalues and Diagonalization

MOTIVATION AND GUIDING IDEA

Diagonalization replaces coordinates whose components interact by coordinates in which each component is multiplied by its own scalar. This helps compute powers, understand update processes, and identify preserved or decaying components. Finding eigenvalues is only the first step: enough linearly independent eigenvectors must exist over the field being used.

NOTATION AND CONVENTIONS

- $A \in M_n(\mathbb{F})$ is square; $\lambda \in \mathbb{F}$ is a scalar, and $v \in \mathbb{F}^n \setminus \{0\}$ is an eigenvector when $Av = \lambda v$.
- $E_\lambda(A) = \ker(A - \lambda I_n) = \ker(\lambda I_n - A)$ is the eigenspace. The notations V_λ and $N(\lambda I_n - A)$ used in the notes describe the same space.
- $p_A(t) = \det(tI_n - A)$ is the characteristic polynomial; t is neither a vector nor a matrix parameter. The notations $P_A(x)$ and $\chi_A(t)$ express the same convention with a different variable name.
- $r_a(\lambda)$ is the algebraic multiplicity, and $r_g(\lambda) = \dim E_\lambda(A)$ is the geometric multiplicity.
- $P = (v_1 \cdots v_n)$ diagonalizes A when its columns form an eigenbasis. If $Av_i = \lambda_i v_i$, then $AP = PD$, with $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$.
- For $f(t) = \sum_{j=0}^{m} c_j t^j$, define $f(A) = \sum_{j=0}^{m} c_j A^j$, where $A^0 = I_n$.

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DEFINITION 11.1

For $\lambda \in \mathbb{F}$ and $A \in M_n(\mathbb{F})$, λ is an eigenvalue exactly when $\ker(A - \lambda I_n) \neq \{0\}$, equivalently $p_A(\lambda) = 0$. The eigenspace contains zero, but zero itself is not an eigenvector. For an eigenvalue, the algebraic multiplicity is the integer $r \geq 1$ such that $p_A(t) = (t - \lambda)^r q(t)$ and $q(\lambda) \neq 0$; the geometric multiplicity satisfies $1 \leq r_g(\lambda) \leq r_a(\lambda)$.

DEFINITION 11.2

A matrix $A \in M_n(\mathbb{F})$ is diagonalizable over $\mathbb{F}$ if there are an invertible $P \in M_n(\mathbb{F})$ and a diagonal $D \in M_n(\mathbb{F})$ with $P^{-1}AP = D$. Equivalently, $\mathbb{F}^n$ has a basis of eigenvectors. The precise criterion is that p_A split into linear factors over $\mathbb{F}$ and that $r_g(\lambda) = r_a(\lambda)$ for every eigenvalue. Splitting is automatic over $\mathbb{C}$; over $\mathbb{R}$, it must be checked.

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NOTE

If $A = PDP^{-1}$, then $f(A) = Pf(D)P^{-1}$ for every polynomial f. In particular, $A^k = PD^kP^{-1}$ for every integer $k \geq 0$; negative powers require invertibility. If $Av = \lambda v$ and $v \neq 0$, then $f(A)v = f(\lambda)v$, but distinct eigenvalues of A can give the same value of f. Diagonalizability of a power or polynomial of A therefore does not imply diagonalizability of A.

NOTE

Sum eigenspace dimensions over distinct eigenvalues. Eigenspaces for distinct eigenvalues form a direct sum. Their dimensions sum to n exactly when there are enough eigenvectors for a basis; a smaller sum means that an eigenbasis is missing.

MOTIVATION AND GUIDING IDEA

By the end of this tutorial, you will be able to identify directions preserved by a linear map, compute eigenspaces and multiplicities, test diagonalizability, and use an eigenbasis to calculate powers and understand an iterative process. We distinguish statements that hold generally from those requiring a particular field or additional hypotheses.

11.1 Similar Matrices

DEFINITION 11.3

Let $A, B \in M_n(\mathbb{F})$.
We say that A is **similar** to B (and write $A \sim B$) if there is an invertible $P \in M_n(\mathbb{F})$ such that

$$A = P^{-1}BP$$

PROPOSITION 11.1

Let $A, B \in M_n(\mathbb{F})$. They are similar if and only if there exist an n-dimensional vector space V, a linear operator $T : V \to V$, and two ordered bases $\mathcal{E}, \mathcal{F}$ such that

$$A = [T]_{\mathcal{E}}, \qquad B = [T]_{\mathcal{F}}.$$

If P converts coordinates from $\mathcal{E}$ to $\mathcal{F}$, so that $[v]_{\mathcal{F}} = P[v]_{\mathcal{E}}$, then

$$A = P^{-1}BP.$$

Thus similarity describes the same operation in two coordinate systems.

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PROPOSITION 11.2

If $A \sim B$, the following quantities and properties are preserved:

1. **Determinant:** $\det(A) = \det(B)$.
2. **Trace:** $\text{tr}(A) = \text{tr}(B)$.
3. **Characteristic polynomial:** $P_A(x) = P_B(x)$.
4. **Eigenvalues:** A and B have exactly the same eigenvalues, with the same algebraic and geometric multiplicities.
5. **Rank:** $\text{rank}(A) = \text{rank}(B)$.
6. **Diagonalizability:** A is diagonalizable $\iff$ B is diagonalizable.

NOTE

Warning:
Equality of characteristic polynomials (and hence of eigenvalues, determinants, and traces) **does not guarantee** similarity.
A standard counterexample is $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$. Both have characteristic polynomial x^2, but they are not similar: the zero matrix is similar only to itself (prove that the same is true of every scalar matrix).

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11.2 Eigenvalues and Eigenvectors

11.2.1 Motivation and Background

MOTIVATION AND GUIDING IDEA

Eigenvalues and eigenvectors help simplify complex systems. Linear operators (and matrices) generally rotate vectors and mix their coordinates. We seek nonzero vectors on which the action is multiplication by a scalar. Such vectors are **eigenvectors** , and the scalar is an **eigenvalue** . Over $\mathbb{R}$, a positive eigenvalue stretches or contracts along the same direction, a negative one also reverses direction, and a zero eigenvalue sends the vector to zero.

Eigenvalues describe models throughout engineering. The following examples show how one algebraic equation appears in different settings:

- **Google's ranking algorithm (PageRank):** In a random browsing model, a transition matrix M sends a probability distribution to the next one. A stationary ranking distribution $\vec{r}$ satisfies
$$M\vec{r} = \vec{r}, \qquad r_i \geq 0, \qquad \sum_i r_i = 1.$$
This is an eigenvalue equation with $\lambda = 1$. The full PageRank model adds a random jump mechanism to address the link structure; the exercise below studies a simplified model.
- **Quantum mechanics:** In a finite-dimensional quantum model, a pure state is a normalized

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vector and a Hermitian Hamiltonian H represents energy. The eigenvalue equation

$$H\vec{\psi} = E\vec{\psi}$$

identifies states of definite energy and possible energy values. It is an application of the same linear equation; a general description of mixed states requires the additional concept of a density matrix.

- **Systems of differential equations:** If A is diagonalizable and $A = PDP^{-1}$, substituting $\vec{y} = P\vec{z}$ into $\vec{y}' = A\vec{y}$ gives

$$P\vec{z}' = AP\vec{z} = PD\vec{z} \quad \Longrightarrow \quad \vec{z}' = D\vec{z}.$$

Each component satisfies $z_i' = \lambda_i z_i$, so $z_i(t) = z_i(0)e^{\lambda_i t}$. For a continuous linear system, all solutions decay to zero if every eigenvalue has negative real part. If an eigenvalue has positive real part, some solutions grow. A zero real part requires further analysis; its sign alone does not imply decay.

- **Classifying critical points in calculus:** For a twice continuously differentiable real function, consider the symmetric Hessian at a critical point. If all its eigenvalues are positive, the point is a strict local minimum; if all are negative, it is a strict local maximum. Positive and negative eigenvalues imply a saddle point. When zero eigenvalues occur without both signs, the second derivative test may be inconclusive.

DEFINITION 11.4

Let V be a vector space over a field $\mathbb{F}$, and let $T : V \to V$ be a linear map (a linear operator). A scalar $\lambda \in \mathbb{F}$ is an **eigenvalue** of T if there is a **nonzero** vector $\vec{v} \in V$ (that is, $\vec{v} \neq \vec{0}_V$) such that

$$T(\vec{v}) = \lambda\vec{v}$$

This is the **eigenvalue equation** . The vector $\vec{v}$ is an **eigenvector** of T associated with λ.

DEFINITION 11.5

Let $A \in \mathbb{F}^{n\times n}$ be a square matrix of size n.
A scalar $\lambda \in \mathbb{F}$ is an **eigenvalue** of A if there is a **nonzero** column vector $\vec{x} \in \mathbb{F}^n$ (that is, $\vec{x} \neq \vec{0}$) such that

$$A\vec{x} = \lambda\vec{x}$$

This is the **eigenvalue equation** . The vector $\vec{x}$ is an **eigenvector** of A associated with λ.

NOTE

Over $\mathbb{R}$, the line $\text{span}\{\vec{x}\}$ is invariant. If $\lambda < 0$, direction is reversed; if $\lambda = 0$, the image is the zero vector. The algebraic concept also applies over $\mathbb{C}$, but a complex eigenvalue cannot always be interpreted as stretching a real line.

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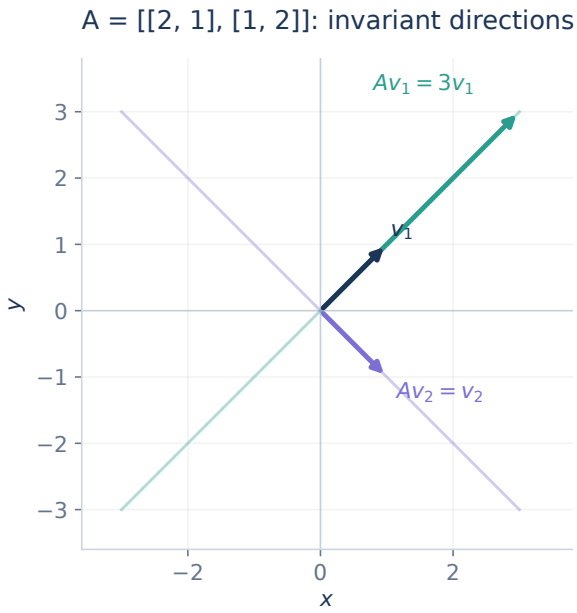


Figure 11.1. Vectors in eigendirections remain on the lines they span. A general vector may change direction because its components are stretched by different factors.

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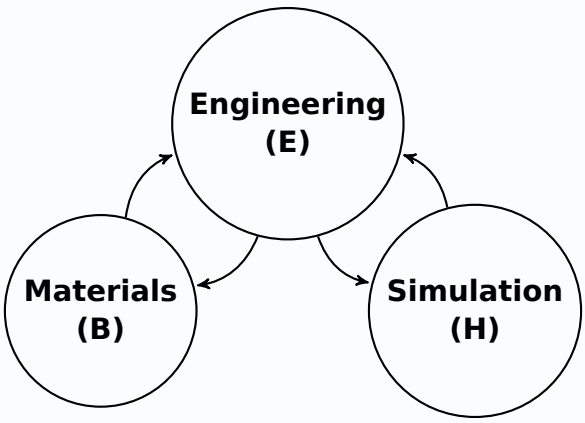
EXERCISE 11.1

Consider a hypothetical internal network at Tel Aviv University consisting of three websites:

- **The central site:** the Faculty of Engineering website (denoted E).
- **Resource site 1:** the materials repository (denoted B).
- **Resource site 2:** the simulation environment (denoted H).

The links are as follows:

- The **materials repository** (B) and **simulation site** (H) link only to the faculty site (E).
- The **faculty site** (E) sends a browser to the materials repository or the simulation environment with equal probability.



Find a stationary ranking distribution $\vec{r}$ whose components sum to 1. Compare the faculty site's ranking with those of the two resource sites.

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EXERCISE 11.1 CONTINUED

WORKED SOLUTION

1. Order the sites as 1=Materials (B), 2=Engineering (E), 3=Simulation (H). Construct the transition matrix M:

   - Column B: links only to E, so $M_{2,1} = 1$.
   - Column E: splits equally between B and H, so $M_{1,2} = 0.5$, $M_{3,2} = 0.5$.
   - Column H: links only to E, so $M_{2,3} = 1$.

   The resulting matrix is

   $$M = \begin{pmatrix} 0 & 0.5 & 0 \\ 1 & 0 & 1 \\ 0 & 0.5 & 0 \end{pmatrix}$$

2. Seek the eigenvector $\vec{r} = (r_B, r_E, r_H)^T$ for $\lambda = 1$ satisfying $M\vec{r} = \vec{r}$:

   $$\begin{cases} 0.5r_E = r_B \\ r_B + r_H = r_E \\ 0.5r_E = r_H \end{cases}$$

   The first and third equations immediately give $r_B = r_H = 0.5r_E$.

   Normalize so that the rankings sum to 1:

   $$r_B + r_E + r_H = 1 \implies 0.5r_E + r_E + 0.5r_E = 1 \implies 2r_E = 1$$

   Hence $r_E = 0.5$, and therefore $r_B = 0.25$ and $r_H = 0.25$.

   The ranking vector is

   $$\vec{r} = \begin{pmatrix} 0.25 \\ 0.5 \\ 0.25 \end{pmatrix}$$

3. **Conclusion:** The Faculty of Engineering site ($r_E = 0.5$) has **twice** the ranking of either the materials repository ($r_B = 0.25$) or the simulation environment ($r_H = 0.25$).

NOTE

We found a stationary distribution; we did not prove convergence of every iteration. In fact, browsing initially only at the faculty site produces the cycle

$$(0, 1, 0)^T \longmapsto (1/2, 0, 1/2)^T \longmapsto (0, 1, 0)^T.$$

This model illustrates the difference between existence of a normalized eigenvector and convergence of a process.

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EXERCISE 11.2

Consider the vector space V of all infinite sequences of real numbers. A vector in this space is an infinite sequence

$$\vec{v} = (x_1, x_2, x_3, x_4, \dots)$$

Define the linear operator $T : V \to V$ by multiplying each entry by its position (its index):

$$T(x_1, x_2, x_3, \dots) = (1 \cdot x_1,\ 2 \cdot x_2,\ 3 \cdot x_3,\ 4 \cdot x_4, \dots)$$

For example, if $\vec{v} = (1, 1, 1, \dots)$, then $T(\vec{v}) = (1, 2, 3, \dots)$.
Find **all** eigenvalues and **all** eigenvectors of T.

WORKED SOLUTION

1. Write $T(\vec{v}) = \lambda\vec{v}$ explicitly for a general vector $\vec{v} = (x_1, x_2, \dots)$:

$$(1x_1,\ 2x_2,\ 3x_3,\ \dots,\ nx_n, \dots) = \lambda \cdot (x_1,\ x_2,\ x_3,\ \dots,\ x_n, \dots)$$

2. Equality of vectors means equality in every coordinate. Compare the nth entries:

$$n \cdot x_n = \lambda \cdot x_n$$

   Rearrange:

$$(n - \lambda) \cdot x_n = 0$$

3. This equation must hold for **every** index $n = 1, 2, 3, \dots$. Suppose λ is an eigenvalue. At every position n with $n \neq \lambda$, the factor in parentheses is nonzero, so necessarily $x_n = 0$.

4. Thus a nonzero vector (with at least one nonzero entry) can occur only if λ is a positive integer.

   Check $\lambda = 1$: The equation is $(n - 1)x_n = 0$. For $n = 1$, it becomes $0 \cdot x_1 = 0$ (every x_1 is allowed). For $n \neq 1$, necessarily $x_n = 0$. One eigenvector is $\vec{e}_1 = (1, 0, 0, 0, \dots)$.

   Check $\lambda = 2$: The equation is $(n - 2)x_n = 0$. For $n = 2$, it becomes $0 \cdot x_2 = 0$ (every x_2 is allowed). For $n \neq 2$, necessarily $x_n = 0$. One eigenvector is $\vec{e}_2 = (0, 1, 0, 0, \dots)$.

5. **Conclusion:** There are infinitely many eigenvalues: all the positive integers,

$$\lambda_n = n \quad (n = 1, 2, 3, \dots)$$

   For each eigenvalue n, all eigenvectors are $c\vec{e}_n$, where $c \in \mathbb{R} \setminus \{0\}$ and $\vec{e}_n$ is the unit sequence in position n:

$$\vec{e}_n = (0, 0, \dots, \underbrace{1}_{n\text{-th place}}, \dots, 0)$$

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NOTE

The exercise concerns all sequences, with no summability or norm assumption. The unit sequences $\{e_n\}$ are not an algebraic basis of this space: finite linear combinations give only finitely supported sequences, not, for example, $(1, 1, 1, \ldots)$. An algebraically similar model can use indices $n = 0, 1, 2, \ldots$ and the rule $Ne_n = ne_n$. This resembles a number operator. The exercise alone does not imply that all physical energies are discrete or that every physical operator acts on the space of all sequences.

EXERCISE 11.3

Let $n \geq 2$, let $V = M_n(\mathbb{C})$ be the space of square matrices, and let $p, q \in \mathbb{C}$ with $q \neq 0$. Define the linear operator $T : V \to V$ by

$$T(A) = pA + qA^T$$

Find the eigenvalues and eigenvectors of T directly from the eigenvalue equation.

WORKED SOLUTION

Let $A \neq 0$ be an eigenvector of T with eigenvalue λ. Then

$$pA + qA^T = \lambda A \tag{11.1}$$

Isolate the transpose term:

$$qA^T = (\lambda - p)A$$

Divide by q (which is nonzero) to express A^T in terms of A:

$$A^T = \frac{\lambda - p}{q} A \tag{11.2}$$

Transpose both sides of (11.2):

$$(A^T)^T = \left(\frac{\lambda - p}{q} A\right)^T$$

Using the properties of transposition,

$$A = \frac{\lambda - p}{q} A^T$$

Substitute the expression for A^T from (11.2) into the last equation:

$$A = \frac{\lambda - p}{q} \cdot \underbrace{\left(\frac{\lambda - p}{q} A\right)}_{A^T}$$

$$A = \left(\frac{\lambda - p}{q}\right)^2 A$$

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EXERCISE 11.3 CONTINUED

Rearrange and factor out A on the right:

$$\left[1 - \left(\frac{\lambda - p}{q}\right)^2\right] A = 0$$

Since $A \neq 0$ (an eigenvector), the scalar in brackets must vanish, so $\left(\frac{\lambda-p}{q}\right)^2 = 1$. There are two cases:

1. **First case:** $\frac{\lambda-p}{q} = 1$. Solving for λ gives the first eigenvalue:

   $$\lambda_1 = p + q$$

   To identify its eigenvectors, substitute $\frac{\lambda-p}{q} = 1$ directly into (11.2):

   $$A^T = 1 \cdot A \implies A^T = A$$

   Thus the eigenvectors for λ_1 are precisely the nonzero symmetric matrices.

2. **Second case:** $\frac{\lambda-p}{q} = -1$. Solving for λ gives the second eigenvalue:

   $$\lambda_2 = p - q$$

   Substitute $\frac{\lambda-p}{q} = -1$ into (11.2):

   $$A^T = -1 \cdot A \implies A^T = -A$$

   Thus the eigenvectors for λ_2 are precisely the nonzero skew-symmetric matrices.

11.3 Eigenspaces, the Characteristic Polynomial, and Multiplicities

DEFINITION 11.6

Definition: the eigenspace V_λ

Let λ be an eigenvalue. The eigenspace V_λ is the set of all eigenvectors for λ together with the zero vector. The zero vector belongs to this subspace but is not an eigenvector. It follows directly from the eigenvalue equation:

1. **For a linear operator T:** Starting from $T(\vec{v}) = \lambda\vec{v}$, rearrange:

   $$T(\vec{v}) - \lambda\vec{v} = \vec{0} \implies T(\vec{v}) - \lambda I_V(\vec{v}) = \vec{0}$$

   Factor:

   $$(T - \lambda I_V)\vec{v} = \vec{0}$$

   Thus the eigenspace is the kernel of the operator

   $$V_\lambda = \ker(T - \lambda I_V)$$

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2. **For a matrix A:** Starting from $A\vec{v} = \lambda\vec{v}$, rearrange:

$$\lambda\vec{v} - A\vec{v} = \vec{0} \implies \lambda I_n\vec{v} - A\vec{v} = \vec{0}$$

Factor out $\vec{v}$ on the right:

$$(\lambda I_n - A)\vec{v} = \vec{0}$$

Thus the eigenspace is the nullspace of the matrix:

$$V_\lambda = N(\lambda I_n - A)$$

DEFINITION 11.7

Definition: characteristic polynomial and multiplicities

1. **Characteristic polynomial $P_A(x)$:** It is defined by

$$P_A(x) = \det(xI_n - A)$$

2. **Algebraic multiplicity $r_a(\lambda)$:** The multiplicity (exponent) of λ as a root of the characteristic polynomial $P_A(x)$.

3. **Geometric multiplicity $r_g(\lambda)$:** The dimension of the eigenspace for λ:

$$r_g(\lambda) = \dim(V_\lambda) = \dim(N(\lambda I_n - A))$$

EXERCISE 11.4

Consider the real matrix

$$A = \begin{pmatrix} -1 & 2 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix}.$$

Find its characteristic polynomial, eigenspaces, and algebraic and geometric multiplicities. Determine whether A is diagonalizable.

WORKED SOLUTION

The matrix is triangular, so

$$p_A(\lambda) = \det(\lambda I - A) = (\lambda + 1)^2(\lambda - 2)^2.$$

Both eigenvalues $-1, 2$ have algebraic multiplicity 2. For -1, solve $(A + I)v = 0$:

$$\begin{pmatrix} 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix} v = 0.$$

The equations are $2v_2 = 0, 3v_3 = 0, 3v_4 = 0$, so $E_{-1} = \text{span}\{e_1\}$ and its geometric multiplicity

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EXERCISE 11.4 CONTINUED

is 1. For 2, the first rows of $A - 2I$ give $-3v_1 + 2v_2 = 0$, $-3v_2 = 0$. Thus $v_1 = v_2 = 0$, and

$$E_2 = \text{span}\{e_3, e_4\}, \qquad \dim E_2 = 2.$$

The sum of the eigenspace dimensions is $1 + 2 = 3 < 4$, so there is no eigenbasis and A is not diagonalizable.

11.4 Diagonalization and Eigenbases

MOTIVATION AND GUIDING IDEA

Main objective: find a basis of eigenvectors

DEFINITION 11.8

Definition: diagonalizable matrix and diagonalizing basis

Let $A \in M_n(\mathbb{F})$.

1. A is **diagonalizable over** $\mathbb{F}$ if it is similar over $\mathbb{F}$ to a diagonal matrix. That is, there are an invertible matrix P and a diagonal matrix D, both over $\mathbb{F}$, such that

   $$D = P^{-1}AP \iff A = PDP^{-1}$$

   In this case, the diagonal entries of D are the eigenvalues of A:

   $$D = \text{diag}(\lambda_1, \ldots, \lambda_n)$$

2. The columns of the diagonalizing matrix P form a basis of **eigenvectors** of A, in the order of the diagonal entries of D. Since P is invertible, its columns form a basis of $\mathbb{F}^n$. This is a **diagonalizing basis** , or **eigenbasis** .

PROPOSITION 11.3

Diagonalization theorem (criterion for diagonalizability)

A matrix $A \in M_n(\mathbb{F})$ is diagonalizable over $\mathbb{F}$ **if and only if** both conditions hold:

1. **Complete splitting:** The characteristic polynomial $P_A(x)$ splits into linear factors over $\mathbb{F}$, meaning that all its roots (counted with multiplicity) lie in $\mathbb{F}$.

2. **Equal multiplicities:** For each eigenvalue λ, the geometric multiplicity equals the algebraic multiplicity:

   $$r_g(\lambda) = r_a(\lambda)$$

Useful consequence: If a matrix A of size n has n **distinct** eigenvalues in $\mathbb{F}$, then $r_a = 1$ for each. Consequently $r_g = 1$ (since $1 \le r_g \le r_a$), and A is diagonalizable over $\mathbb{F}$.

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NOTE

Why does $1 \leq r_g(\lambda) \leq r_a(\lambda)$ always hold? Write $r = \dim V_\lambda$. Since λ is an eigenvalue, a nonzero eigenvector exists, so $r \geq 1$. Extend a basis of V_λ to a basis of the full space. In this basis, the matrix has the form

$$\begin{pmatrix} \lambda I_r & C \\ 0 & D \end{pmatrix}.$$

The lower-left block is zero because each of the first r basis vectors maps to a multiple of itself. Thus the characteristic polynomial is $(x-\lambda)^r \det(xI - D)$, and $r_g(\lambda) = r \leq r_a(\lambda)$.

DEFINITION 11.9

Definition: a diagonalizable linear operator

Let $T : V \to V$ be a linear operator on a finite-dimensional vector space V of dimension n over $\mathbb{F}$. The operator T is **diagonalizable over** $\mathbb{F}$ if there is a basis $\mathcal{B}$ of V in which $[T]_\mathcal{B}$ is diagonal.

Such a basis necessarily consists of eigenvectors of T and is called a **diagonalizing basis** .

PROPOSITION 11.4

Let $v_1, \ldots, v_k$ be eigenvectors of a matrix A (or an operator T), associated with **distinct** eigenvalues $\lambda_1, \ldots, \lambda_k$.

Then $\{v_1, \ldots, v_k\}$ is **linearly independent** .

MOTIVATION AND GUIDING IDEA

An algorithm for diagonalizing a linear operator

To diagonalize T (or find a diagonalizing basis), proceed as follows:

1. Choose a convenient basis $\mathcal{C}$ of V (often the standard basis).
2. Form its representing matrix $A = [T]_\mathcal{C}$.
3. Solve the matrix problem:
   - Compute the characteristic polynomial $P_A(x)$.
   - Find the eigenvalues λ in $\mathbb{F}$.
   - For each λ, find a basis of the eigenspace (the kernel) V_λ.
4. Test diagonalizability: add the geometric multiplicities ($\sum r_g(\lambda)$). If the sum is less than n, the operator is not diagonalizable over $\mathbb{F}$. If it equals n, continue.
5. Recover the diagonalizing basis: The vectors found in step 3 are coordinate vectors (in $\mathbb{F}^n$). Convert them back to vectors of V using the initial basis $\mathcal{C}$. The combined lists form the diagonalizing basis $\mathcal{B}$.
6. The representing matrix in the new basis is

$$[T]_\mathcal{B} = D = \text{diag}(\lambda_1, \ldots, \lambda_n)$$

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11.4.1 Exercises on Diagonalization

EXAMPLE 11.1

A polynomial that splits into linear factors over $\mathbb{R}$: $p(x) = x(x-1)(x-2)$.

EXAMPLE 11.2

A polynomial that **does not** split into linear factors over $\mathbb{R}$: $q(x) = x^2 + 1$.
Observe that $q(x)$ **does** split into linear factors over $\mathbb{C}$: $q(x) = (x-i)(x+i)$

EXERCISE 11.5

For each matrix, determine whether it is diagonalizable over $\mathbb{R}$ and over $\mathbb{C}$. Whenever it is diagonalizable, give suitable P, D over the relevant field:

$$A_1 = \begin{pmatrix} 4 & 2 \\ 0 & 4 \end{pmatrix}, \quad A_2 = \begin{pmatrix} -1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{pmatrix}, \quad A_3 = \begin{pmatrix} 0 & -9 \\ 1 & 0 \end{pmatrix}.$$

WORKED SOLUTION

1. For A_1, the characteristic polynomial is $(\lambda-4)^2$. The equation $(A_1-4I)v = 0$ forces $2v_2 = 0$, so the eigenspace is $\text{span}\{(1,0)^T\}$ and has dimension 1. Its algebraic multiplicity 2 exceeds its geometric multiplicity, so it is not diagonalizable over either field.

2. A_2 has distinct eigenvalues $-1, 1, 3$. For -1, choose e_1. For 1, the equation $-2v_1+2v_2 = 0$ gives $(1,1,0)^T$. For 3, choose e_3. Hence

$$P = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad D = \text{diag}(-1, 1, 3).$$

   The columns are linearly independent, and their images give $A_2P = PD$. This is a diagonalization over $\mathbb{R}$, and hence also over $\mathbb{C}$.

3. The characteristic polynomial of A_3 is $\lambda^2 + 9$. It has no real root, so there is no real diagonalization. Over $\mathbb{C}$, the eigenvalues are $3i, -3i$, with respective eigenvectors $(3i, 1)^T, (-3i, 1)^T$:

$$A_3(\lambda, 1)^T = (-9, \lambda)^T = \lambda(\lambda, 1)^T \quad \text{when } \lambda^2 = -9.$$

   Therefore

$$P_{\mathbb{C}} = \begin{pmatrix} 3i & -3i \\ 1 & 1 \end{pmatrix}, \quad D_{\mathbb{C}} = \text{diag}(3i, -3i).$$

   Since $\det P_{\mathbb{C}} = 6i \neq 0$, this is a valid complex diagonalization.

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EXERCISE 11.6

As in the earlier transpose exercise, assume $n \geq 2$ and consider $T : M_n(\mathbb{R}) \to M_n(\mathbb{R})$ defined by

$$T(A) = pA + qA^T$$

where $p, q \in \mathbb{R}$ and $q \neq 0$.
Find the characteristic polynomial $P_T(x)$ and the algebraic and geometric multiplicities (r_a, r_g) of each eigenvalue found earlier.

WORKED SOLUTION

Previously we decomposed the full space $V = M_n(\mathbb{R})$ into a direct sum of eigenspaces:

$$V = V_{p+q} \oplus V_{p-q} = S \oplus K$$

Here S is the space of symmetric matrices and K is the space of skew-symmetric matrices.

1. **For the eigenvalue $\lambda_1 = p + q$:** The eigenspace consists of the symmetric matrices:

   $$V_{\lambda_1} = S = \{A \in M_n(\mathbb{R}) \mid A^T = A\}$$

   Its geometric multiplicity is its dimension (the number of entries on or above the diagonal):

   $$r_g(p+q) = \dim(S) = \frac{n(n+1)}{2}$$

2. **For the eigenvalue $\lambda_2 = p - q$:** The eigenspace consists of the skew-symmetric matrices:

   $$V_{\lambda_2} = K = \{A \in M_n(\mathbb{R}) \mid A^T = -A\}$$

   Its geometric multiplicity is its dimension (the number of entries strictly above the diagonal, since diagonal entries must be zero):

   $$r_g(p-q) = \dim(K) = \frac{n(n-1)}{2}$$

3. **The characteristic polynomial $P_T(x)$:** The geometric multiplicities sum to the dimension of the full space (n^2), so the operator is diagonalizable. Thus each algebraic multiplicity equals its geometric multiplicity ($r_a = r_g$).

   The characteristic polynomial is the product of the linear factors, each raised to its algebraic multiplicity:

   $$P_T(x) = (x - \lambda_1)^{r_a(\lambda_1)} \cdot (x - \lambda_2)^{r_a(\lambda_2)}$$

   Substitute the values found above:

   $$P_T(x) = \Big(x - (p+q)\Big)^{\frac{n(n+1)}{2}} \cdot \Big(x - (p-q)\Big)^{\frac{n(n-1)}{2}}$$

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NOTE

The dimension of a solution subspace can be found by counting independent homogeneous linear constraints:

$$\dim(\text{Subspace}) = \dim(\text{Ambient space}) - (\text{number of independent constraints})$$

Here the ambient space is $M_n(\mathbb{R})$, of dimension n^2.

1. **For symmetric matrices (S):** The condition is $A_{ij} = A_{ji}$. This is a constraint only for $i < j$ (on the diagonal $A_{ii} = A_{ii}$ is an identity, and the equations below the diagonal repeat those above it). There are $\frac{n(n-1)}{2}$ independent constraints, one per entry strictly above the diagonal. Thus the dimension is

$$n^2 - \frac{n(n-1)}{2} = \frac{2n^2 - n^2 + n}{2} = \frac{n(n+1)}{2}$$

2. **For skew-symmetric matrices (K):** The condition is $A_{ji} = -A_{ij}$. There are two types of constraints:

   - For $i < j$, there are $\frac{n(n-1)}{2}$ constraints (as in the symmetric case).
   - For $i = j$ (on the diagonal), $A_{ii} = -A_{ii}$ implies $A_{ii} = 0$. There are n such constraints.

   The total number of independent constraints is $\frac{n(n-1)}{2} + n$. Thus the dimension is

$$n^2 - \left(\frac{n(n-1)}{2} + n\right) = \frac{n(n-1)}{2}$$

NOTE

A useful check: the sum of the exponents (degrees) in the characteristic polynomial must equal the dimension of the space on which the operator acts.

$$\frac{n(n+1)}{2} + \frac{n(n-1)}{2} = \frac{n^2 + n + n^2 - n}{2} = \frac{2n^2}{2} = n^2$$

This is indeed the dimension of $M_n(\mathbb{R})$.

EXERCISE 11.7

Let $n \in \mathbb{N}$, and consider a general upper triangular matrix $A \in M_n(\mathbb{R})$:

$$A = \begin{pmatrix} \lambda_1 & * & \cdots & * \\ 0 & \lambda_2 & \cdots & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix}$$

Its diagonal entries $\lambda_1, \ldots, \lambda_n$ are **pairwise distinct** (that is, $\lambda_i \neq \lambda_j$ whenever $i \neq j$). Define the polynomial $P(x) = 2026x^{2025} - 58x^{58}$.

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EXERCISE 11.7 CONTINUED

Prove that there **exists** a matrix $X \in M_n(\mathbb{R})$ satisfying

$$P(X) = A$$

WORKED SOLUTION

Solution strategy: Although A has unspecified entries (asterisks) above the diagonal, the essential fact is that its diagonal entries are **distinct** . A matrix of size n with n distinct eigenvalues is diagonalizable. We can therefore work with the simpler diagonal matrix, solve the resulting scalar problem using calculus, and return to the original coordinates.

1. The characteristic polynomial of an upper triangular matrix is the product of its diagonal factors:
$$P_A(x) = (x - \lambda_1)(x - \lambda_2) \cdots (x - \lambda_n)$$
Its eigenvalues are therefore exactly $\lambda_1, \ldots, \lambda_n$. They are distinct, so A has n linearly independent eigenvectors. Thus A is diagonalizable: there are an invertible M and a diagonal matrix $D = \text{diag}(\lambda_1, \ldots, \lambda_n)$ such that
$$A = MDM^{-1}$$

2. Seek a solution of the form $X = MEM^{-1}$, where $E = \text{diag}(c_1, \ldots, c_n)$ is a diagonal matrix to be constructed. The requirement $P(X) = A$ becomes the following equation for diagonal matrices:
$$P(E) = D$$
Since both are diagonal, this is equivalent to n separate scalar equations:
$$P(c_i) = \lambda_i \quad \text{for every } i = 1, \ldots, n$$

3. Consider the real function $f(t) = 2026t^{2025} - 58t^{58}$.
   - It is continuous.
   - Its leading term has odd degree and positive coefficient, so $\lim_{t\to\pm\infty} f(t) = \pm\infty$.

   The intermediate value theorem implies that f maps onto $\mathbb{R}$. Thus for each eigenvalue λ_i, regardless of the unspecified entries in A, there exists $c_i \in \mathbb{R}$ such that $f(c_i) = \lambda_i$.

4. **Construct the solution:** Choose these c_i as the diagonal entries of E. Define
$$X = MEM^{-1}$$
and verify:
$$P(X) = MP(E)M^{-1} = M\,\text{diag}(P(c_1), \ldots, P(c_n))M^{-1}$$
$$= M\,\text{diag}(\lambda_1, \ldots, \lambda_n)M^{-1} = MDM^{-1} = A$$
This proves existence of the required matrix.

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11.5 Diagonalization, Powers, and Parameters

EXERCISE 11.8

Let $A \in M_{4\times 4}(\mathbb{R})$ have 1 on the superdiagonal and 0 elsewhere (that is, $A_{12} = A_{23} = A_{34} = 1$, with all other entries zero).

1. Show that A is nilpotent, meaning that some positive power is zero, and find its characteristic polynomial.
2. Find $\dim\ker(A^2)$ and $\dim\ker(A)$.
3. Prove that A is not diagonalizable.

WORKED SOLUTION

1. The matrix is upper triangular with zero diagonal, so $P_A(x) = x^4$. By the Cayley-Hamilton theorem, $A^4 = 0$, and hence A is nilpotent.
2. For A, there is one free variable (the first column is zero and the remaining columns are linearly independent), so $\dim\ker(A) = 1$. In A^2, the entries equal to 1 lie on the second superdiagonal. There are two zero columns and two linearly independent nonzero columns, giving two free variables: $\dim\ker(A^2) = 2$.
3. The only eigenvalue is $\lambda = 0$, of algebraic multiplicity $r_a = 4$. The previous part gives geometric multiplicity $r_g = \dim\ker(A) = 1$. Since $1 \neq 4$, the matrix is not diagonalizable.

EXERCISE 11.9

Let $A \in M_{n\times n}(\mathbb{F})$, where $\mathbb{F}$ has characteristic different from 2 (so $1_{\mathbb{F}} + 1_{\mathbb{F}} \neq 0_{\mathbb{F}}$, and $2^{-1} = \frac{1}{2}$ exists). Suppose $A^2 = I$.

1. Prove that every eigenvalue of A lies in $\{1, -1\}$.
2. Prove that A is diagonalizable. (Hint: express every vector as a sum of vectors in the two eigenspaces.)
3. In this part, assume additionally that $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. Given $\mathrm{tr}(A) = t$, determine $\dim\ker(A - I)$ and $\dim\ker(A + I)$ in terms of n and t.

WORKED SOLUTION

1. Let $\lambda \in \mathbb{F}$ be an eigenvalue. By definition, there is a nonzero vector $v \in \mathbb{F}^n$, $v \neq 0$, with

$$Av = \lambda v$$

Multiply the equality on the left by A:

$$A(Av) = A(\lambda v)$$

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EXERCISE 11.9 CONTINUED

Use associativity on the left and take the scalar outside on the right (linearity):

$$A^2v = \lambda(Av)$$

Substitute $A^2 = I$ on the left and $Av = \lambda v$ on the right:

$$Iv = \lambda(\lambda v)$$

$$v = \lambda^2 v$$

Move all terms to one side:

$$\lambda^2 v - v = 0 \implies (\lambda^2 - 1)v = 0$$

Since $v \neq 0$ (an eigenvector is nonzero), its scalar coefficient must vanish:

$$\lambda^2 - 1 = 0 \implies \lambda^2 = 1$$

The solutions in the field are $\lambda = 1$ or $\lambda = -1$.

2. To prove diagonalizability, it suffices to find a basis of eigenvectors for the full space. Equivalently, the eigenspaces must span the space: $V = V_1 + V_{-1}$. For any $v \in \mathbb{F}^n$, use the identity below (division by 2 is permitted because the characteristic is not 2):

$$v = \frac{1}{2}(v + Av) + \frac{1}{2}(v - Av)$$

Denote the two summands by

$$u_1 = \frac{1}{2}(v + Av), \quad u_{-1} = \frac{1}{2}(v - Av)$$

Then $v = u_1 + u_{-1}$. Each summand lies in its corresponding eigenspace; a summand may be zero, in which case it is not itself an eigenvector: **Check u_1:** Apply A:

$$A(u_1) = A\left(\frac{1}{2}(v + Av)\right) = \frac{1}{2}(Av + A^2v)$$

Substitute $A^2 = I$:

$$= \frac{1}{2}(Av + v) = u_1$$

Thus $u_1 \in \ker(A - I)$ (and if $u_1 \neq 0$, it is an eigenvector for 1). **Check u_{-1}:** Apply A:

$$A(u_{-1}) = A\left(\frac{1}{2}(v - Av)\right) = \frac{1}{2}(Av - A^2v)$$

Substitute $A^2 = I$:

$$= \frac{1}{2}(Av - v) = -\frac{1}{2}(v - Av) = -u_{-1}$$

Thus $u_{-1} \in \ker(A + I)$ (and if $u_{-1} \neq 0$, it is an eigenvector for -1). **Conclusion:** Every vector lies in $\ker(A-I)+\ker(A+I)$. Their intersection is trivial: if $Az = z$ and $Az = -z$, then $2z = 0$, so $z = 0$. Consequently, this direct sum is the entire space. Concatenating bases of the two kernels gives an eigenbasis, so A is diagonalizable.

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EXERCISE 11.9 CONTINUED

3. The matrix A is diagonalizable, so it is similar to a diagonal matrix D whose diagonal entries are 1 and -1. Write $k = \dim\ker(A - I)$ (the geometric multiplicity of 1, equal to its number of occurrences on the diagonal). $m = \dim\ker(A + I)$ (the geometric multiplicity of -1, equal to its number of occurrences on the diagonal).

   Two equations are available: 1. The matrix has size n, so the multiplicities sum to n:

$$k + m = n$$

   2. Trace is preserved under similarity, so it equals the sum of the diagonal entries of D:

$$k \cdot (1) + m \cdot (-1) = t \implies k - m = t$$

   This is a system of two equations in the two unknowns k, m. Add the equations:

$$(k + m) + (k - m) = n + t \implies 2k = n + t \implies k = \frac{n + t}{2}$$

   Subtract the second equation from the first:

$$(k + m) - (k - m) = n - t \implies 2m = n - t \implies m = \frac{n - t}{2}$$

   Answer:

$$\dim\ker(A - I) = \frac{n + t}{2}, \quad \dim\ker(A + I) = \frac{n - t}{2}.$$

   The field assumption in this part is essential for interpreting the trace as the difference of integer dimensions. In positive characteristic, the trace determines only their difference as a field element, which need not determine the integer dimensions.

EXERCISE 11.10

For $t \in \mathbb{R}$, consider

$$A(t) = \begin{pmatrix} 2 & t & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 5 \end{pmatrix}.$$

Find the eigenvalues and their algebraic multiplicities, compute $\dim\ker(A(t) - 2I)$, and determine when the matrix is diagonalizable. Prove a formula for $A(t)^n$ for every $n \geq 1$.

WORKED SOLUTION

Triangularity gives $p_A(\lambda) = (\lambda - 2)^2(\lambda - 5)$. Thus 2 has algebraic multiplicity 2, while 5 has algebraic multiplicity 1. Compute

$$A(t) - 2I = \begin{pmatrix} 0 & t & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 3 \end{pmatrix}.$$

The kernel satisfies $tv_2 = 0$, $3v_3 = 0$. If $t \neq 0$, only v_1 is free, giving dimension 1. If $t = 0$, v_2 is also free, giving dimension 2. The eigenvalue 5 has algebraic multiplicity 1, so its geometric multiplicity is 1. The polynomial splits over $\mathbb{R}$, so $A(t)$ is diagonalizable exactly when $t = 0$.

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EXERCISE 11.10 CONTINUED

We claim that

$$A(t)^n = \begin{pmatrix} 2^n & n2^{n-1}t & 0 \\ 0 & 2^n & 0 \\ 0 & 0 & 5^n \end{pmatrix} \qquad (n \geq 1).$$

At $n = 1$ this is $A(t)$. Assuming the formula for n, right multiplication by $A(t)$ gives diagonal entries $2^{n+1}, 2^{n+1}, 5^{n+1}$. The $(1, 2)$ entry is the sum of two products:

$$2^n t + (n2^{n-1}t) \cdot 2 = (n + 1)2^n t.$$

All other off-diagonal entries remain zero, giving exactly the formula for $n+1$. This completes the induction. Separately, the zeroth power is $A^0 = I_3$.

EXERCISE 11.11

Let $A \in M_{n\times n}(\mathbb{F})$ be square over $\mathbb{F}$, and let $u, v \in \mathbb{F}^n$ be nonzero column vectors. Define $B = A + uv^T$ (this is a rank-one update).

1. Suppose $x \in \mathbb{F}^n$, $x \neq 0$, satisfies:
   - x is an eigenvector of A with eigenvalue λ (that is, $Ax = \lambda x$).
   - $v^T x = 0$.

   Prove that x is also an eigenvector of B, and identify its eigenvalue.

2. Now assume $A = \operatorname{diag}(\alpha_1, \ldots, \alpha_n)$ and $u = v = (1, 1, \ldots, 1)^T$. Prove that for every scalar λ outside $\{\alpha_1, \ldots, \alpha_n\}$,

$$\lambda \text{ is an eigenvalue of } B \iff 1 + \sum_{i=1}^{n} \frac{1}{\alpha_i - \lambda} = 0$$

3. For $n = 3$ and $A = \operatorname{diag}(0, 1, 3)$, compute the characteristic polynomial of B explicitly (with u, v as in the preceding part).

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WORKED SOLUTION

1. Check directly by multiplying B by x:

$$Bx = (A + uv^T)x$$

Expand:

$$= Ax + u(v^T x)$$

Substitute $Ax = \lambda x$ (because x is an eigenvector of A) and $v^T x = 0$ (by assumption):

$$= \lambda x + u(0) = \lambda x$$

Conclusion: We obtained $Bx = \lambda x$. Since $x \neq 0$, it is an eigenvector of B with the same eigenvalue λ.

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EXERCISE 11.11 CONTINUED

2. We seek the condition $\det(\lambda I - B) = 0$. Substitute B:

$$\det(\lambda I - (A + uv^T)) = \det((\lambda I - A) - uv^T)$$

Write $D = \lambda I - A$. This is diagonal, with entries $\lambda - \alpha_i$. **Why is D invertible?** By assumption, λ differs from every α_i, so all its diagonal entries are nonzero.

Factor out D inside the determinant:

$$\det(D - uv^T) = \det(D(I - D^{-1}uv^T))$$

By multiplicativity ($\det(XY) = \det(X)\det(Y)$),

$$= \det(D) \cdot \det(I - (D^{-1}u)v^T)$$

For λ to be an eigenvalue, the full expression must be zero. Since D is invertible, $\det(D) \neq 0$, so the second factor must vanish:

$$\det(I - (D^{-1}u)v^T) = 0$$

We use $\det(I + xy^T) = 1 + y^T x$, justifying it even when $y^T x = 0$. If $x = 0$, both sides equal 1. Otherwise, extend x to a basis. In this basis, the image of xy^T lies in $\text{span}\{x\}$ and its first column is $(y^T x, 0, \ldots, 0)^T$. Thus $I + xy^T$ has an upper triangular representing matrix with diagonal $(1 + y^T x, 1, \ldots, 1)$, giving determinant $1 + y^T x$. No nonzero eigenvalue of xy^T needs to be assumed. Here $x = -D^{-1}u$ and $y = v$. Hence the determinant is

$$1 - v^T(D^{-1}u) = 0$$

Compute $v^T D^{-1} u$: The diagonal entries of D^{-1} are $\frac{1}{\lambda - \alpha_i}$. Multiplication by the all-ones vectors adds them:

$$\sum_{i=1}^{n} \frac{1}{\lambda - \alpha_i} = 1$$

Rearranging gives

$$1 - \sum_{i=1}^{n} \frac{1}{\lambda - \alpha_i} = 0 \implies 1 + \sum_{i=1}^{n} \frac{1}{\alpha_i - \lambda} = 0$$

3. Substitute $\alpha_1 = 0, \alpha_2 = 1, \alpha_3 = 3$ into the formula. The characteristic polynomial is the product of $\det(D)$ and the scalar factor:

$$P_B(\lambda) = \underbrace{\lambda(\lambda - 1)(\lambda - 3)}_{\det(D)} \cdot \left(1 + \frac{1}{-\lambda} + \frac{1}{1 - \lambda} + \frac{1}{3 - \lambda}\right)$$

Distribute the polynomial factor over the fractions:

$$= \lambda(\lambda - 1)(\lambda - 3) - (\lambda - 1)(\lambda - 3) - \lambda(\lambda - 3) - \lambda(\lambda - 1)$$

Expand all products:

$$= (\lambda^3 - 4\lambda^2 + 3\lambda) - (\lambda^2 - 4\lambda + 3) - (\lambda^2 - 3\lambda) - (\lambda^2 - \lambda)$$

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EXERCISE 11.11 CONTINUED

Collect like terms:

$$= \lambda^3 + (-4-1-1-1)\lambda^2 + (3+4+3+1)\lambda - 3$$

$$= \lambda^3 - 7\lambda^2 + 11\lambda - 3.$$

The rational expression is used only when its denominators are nonzero. After cancellation, it gives a polynomial identity, so the characteristic polynomial formula is valid also at $\lambda = 0, 1, 3$. It can be checked directly from the matrix

$$B = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 4 \end{pmatrix}.$$

EXERCISE 11.12

Two units exchange information at each step according to

$$\begin{pmatrix} x_{k+1} \\ y_{k+1} \end{pmatrix} = A \begin{pmatrix} x_k \\ y_k \end{pmatrix}, \qquad A = \begin{pmatrix} 3/4 & 1/4 \\ 1/4 & 3/4 \end{pmatrix}.$$

Find an eigenbasis, an explicit formula for A^k and for the two readings, and their limits as $k \to \infty$.

WORKED SOLUTION

Compute the action of A on two directions:

$$A \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad A \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 1/2 \\ -1/2 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

These vectors are linearly independent, so

$$P = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad P^{-1} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad D = \operatorname{diag}(1, 1/2).$$

For every integer $k \geq 0$,

$$A^k = PD^kP^{-1} = \frac{1}{2} \begin{pmatrix} 1 + 2^{-k} & 1 - 2^{-k} \\ 1 - 2^{-k} & 1 + 2^{-k} \end{pmatrix}.$$

Consequently,

$$x_k = \frac{x_0 + y_0}{2} + 2^{-k} \frac{x_0 - y_0}{2}, \qquad y_k = \frac{x_0 + y_0}{2} - 2^{-k} \frac{x_0 - y_0}{2}.$$

The mean is preserved, while the difference $x_k - y_k = 2^{-k}(x_0 - y_0)$ is halved at each step. Both readings therefore converge to the initial mean $(x_0 + y_0)/2$. The eigenvalue 1 describes the preserved component, and $1/2$ describes the decaying component.

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EXERCISE 11.13 **EXAM QUESTION**

Let $\mathbb{C}_3[x] = \{p \in \mathbb{C}[x] : \deg p \leq 3\}$, and let $\gamma \in \mathbb{Z}_{\geq 0}$. Define the linear map $T_\gamma : \mathbb{C}_3[x] \longrightarrow \mathbb{C}_3[x]$ by

$$T_\gamma(a_0 + a_1x + a_2x^2 + a_3x^3) = a_1 - a_0x + (\gamma a_2 - a_3)x^2 + 2a_3x^3.$$

1. Find the representing matrix of T_γ in the standard basis $\mathcal{B} = (1, x, x^2, x^3)$.
2. Determine for which γ the map T_γ is diagonalizable over $\mathbb{C}$.
3. For the smallest γ for which T_γ is diagonalizable, give the characteristic polynomial.
4. For this same γ, find a basis in which the representing matrix is diagonal, and give that diagonal matrix.

WORKED SOLUTION

1. Compute the images of the basis vectors:

$$T_\gamma(1) = -x, \quad T_\gamma(x) = 1,$$
$$T_\gamma(x^2) = \gamma x^2, \quad T_\gamma(x^3) = -x^2 + 2x^3.$$

The matrix columns are their coordinate vectors in the same order:

$$A_\gamma = [T_\gamma]_\mathcal{B} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & \gamma & -1 \\ 0 & 0 & 0 & 2 \end{pmatrix}.$$

2. The matrix consists of two diagonal blocks of size 2. Using the convention $\chi_T(t) = \det(tI - [T])$, its characteristic polynomial is

$$\chi_{T_\gamma}(t) = \det \begin{pmatrix} t & -1 \\ 1 & t \end{pmatrix} \det \begin{pmatrix} t-\gamma & 1 \\ 0 & t-2 \end{pmatrix} = (t^2+1)(t-\gamma)(t-2).$$

The eigenvalues are $i, -i, \gamma, 2$. If $\gamma \neq 2$, these are four distinct values, since γ is real. A matrix of size 4 with four distinct eigenvalues is diagonalizable.

If $\gamma = 2$, compute the eigenspace for 2:

$$A_2 - 2I = \begin{pmatrix} -2 & 1 & 0 & 0 \\ -1 & -2 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

The first equations give $a_1 = 2a_0$ and $-a_0 - 2a_1 = 0$. Thus $-5a_0 = 0$, so $a_0 = a_1 = 0$ over $\mathbb{C}$. The third equation gives $a_3 = 0$, while a_2 is free. Hence

$$\ker(A_2 - 2I) = \operatorname{span}\{(0, 0, 1, 0)^T\}.$$

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EXERCISE 11.13 **EXAM QUESTION CONTINUED**

The geometric multiplicity of 2 is 1, whereas its algebraic multiplicity is 2. Therefore T_2 is not diagonalizable. In conclusion,

$$T_\gamma \text{ is diagonalizable over } \mathbb{C} \iff \gamma \neq 2.$$

3. The smallest allowed value in $\mathbb{Z}_{\geq 0}$ is $\gamma = 0$, which satisfies the diagonalizability condition. Therefore
$$\chi_{T_0}(t) = t(t-2)(t^2+1).$$

4. Check four eigenpolynomials:
$$\begin{aligned} T_0(1+ix) &= -x + i = i(1+ix), \\ T_0(1-ix) &= -x - i = -i(1-ix), \\ T_0(x^2) &= 0, \\ T_0(x^2 - 2x^3) &= 2x^2 - 4x^3 = 2(x^2 - 2x^3). \end{aligned}$$
These are nonzero eigenpolynomials for distinct eigenvalues, so they are linearly independent. There are $4 = \dim \mathbb{C}_3[x]$ of them, so
$$\mathcal{E} = (1+ix,\ 1-ix,\ x^2,\ x^2-2x^3)$$
is a basis. In this order, the representing matrix is
$$[T_0]_\mathcal{E} = \operatorname{diag}(i, -i, 0, 2).$$

EXERCISE 11.14 **EXAM QUESTION**

Work in $V = \mathbb{R}^4$ with its standard basis $\mathcal{E} = (e_1, e_2, e_3, e_4)$. Define another basis $\mathcal{F} = (f_1, f_2, f_3, f_4)$ by
$$f_1 = e_1, \qquad f_2 = e_1 + e_2, \qquad f_3 = e_2 + e_3, \qquad f_4 = e_3 + e_4.$$
Define $T : V \to V$ by $T(e_j) = f_j$ for $j = 1, 2, 3, 4$, and let $S : V \to V$ have representing matrix in $\mathcal{F}$ given by
$$[S]_\mathcal{F} = \begin{pmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -5 \\ 0 & 0 & 5 & 0 \end{pmatrix}.$$
Set $U = T^{-1} \circ S \circ T$.

1. Compute $[U]_\mathcal{E}$.
2. Determine whether U is diagonalizable over $\mathbb{R}$ and over $\mathbb{C}$. Whenever it is diagonalizable, find a diagonalizing basis and a diagonal matrix. Over $\mathbb{C}$, consider the extension of U to $\mathbb{C}^4$.
3. Define $W : V \to V$ by $[W]_\mathcal{E} = 2I_4$. Determine whether there is a basis $\mathcal{B}$ with $[U]_\mathcal{B} = [W]_\mathcal{B}$, and whether there is a basis $\widetilde{\mathcal{B}}$ with $[U]_{\widetilde{\mathcal{B}}} = [W]_\mathcal{E}$.

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EXERCISE 11.14 **EXAM QUESTION CONTINUED** **11**

WORKED SOLUTION

1. Set $J = [S]_{\mathcal{F}}$, and let P have columns $f_1, \dots, f_4$ expressed in $\mathcal{E}$. Then

$$P = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad \det P = 1.$$

Thus $[v]_{\mathcal{E}} = P[v]_{\mathcal{F}}$, and $[T]_{\mathcal{E}} = P$ because $T(e_j) = f_j$. The change-of-basis formula gives $[S]_{\mathcal{E}} = PJP^{-1}$, so

$$\begin{aligned} [U]_{\mathcal{E}} &= [T^{-1}]_{\mathcal{E}}[S]_{\mathcal{E}}[T]_{\mathcal{E}} \\ &= P^{-1}(PJP^{-1})P = J. \end{aligned}$$

Therefore

$$[U]_{\mathcal{E}} = \begin{pmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -5 \\ 0 & 0 & 5 & 0 \end{pmatrix}.$$

2. Using $p_U(t) = \det(tI_4 - J)$, the characteristic polynomial is

$$p_U(t) = \det \begin{pmatrix} t & 1 \\ -1 & t \end{pmatrix} \det \begin{pmatrix} t & 5 \\ -5 & t \end{pmatrix} = (t^2+1)(t^2+25).$$

It has no real roots. Thus there is no nonzero real eigenvector, and in particular U is not diagonalizable over $\mathbb{R}$.

Over $\mathbb{C}$, the four eigenvalues $i, -i, 5i, -5i$ are distinct. Explicit eigenvectors are

$$v_1 = \begin{pmatrix} 1 \\ -i \\ 0 \\ 0 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 1 \\ i \\ 0 \\ 0 \end{pmatrix}, \quad v_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \\ -i \end{pmatrix}, \quad v_4 = \begin{pmatrix} 0 \\ 0 \\ 1 \\ i \end{pmatrix}.$$

For example,

$$Jv_1 = \begin{pmatrix} i \\ 1 \\ 0 \\ 0 \end{pmatrix} = iv_1, \qquad Jv_3 = \begin{pmatrix} 0 \\ 0 \\ 5i \\ 5 \end{pmatrix} = 5iv_3,$$

and similarly $Jv_2 = -iv_2$ and $Jv_4 = -5iv_4$. If $Q = (v_1\ v_2\ v_3\ v_4)$, then

$$\det Q = (2i)(2i) = -4 \neq 0.$$

Hence $\mathcal{B}_{\mathbb{C}} = (v_1, v_2, v_3, v_4)$ is a diagonalizing basis, with

$$[U]_{\mathcal{B}_{\mathbb{C}}} = Q^{-1}JQ = \operatorname{diag}(i, -i, 5i, -5i).$$

3. W is the map $v \mapsto 2v$, so in every basis $\mathcal{B}$, $[W]_{\mathcal{B}} = 2I_4$. If $[U]_{\mathcal{B}} = [W]_{\mathcal{B}}$, the two maps

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EXERCISE 11.14 **EXAM QUESTION CONTINUED**

would be equal: for every v, the coordinate vectors of $U(v)$ and $W(v)$ would coincide. But $U(e_1) = e_2$, whereas $W(e_1) = 2e_1$. Thus no such basis exists.

A basis of the second type does not exist either. Indeed, if R is the matrix whose columns are the vectors of $\widetilde{\mathcal{B}}$ expressed in $\mathcal{E}$, then

$$R^{-1}JR = 2I_4 \quad \Longrightarrow \quad J = R(2I_4)R^{-1} = 2I_4,$$

contradicting the matrix found above. Both answers remain negative if complex bases are allowed.

EXERCISE 11.15 **EXAM QUESTION**

Let $n \geq 4$, and let $A \in M_n(\mathbb{R})$ be diagonalizable over $\mathbb{R}$ with exactly two distinct eigenvalues. Suppose

$$\operatorname{rank}(A - I_n) = n - 3, \qquad \operatorname{tr}(A) = 2n - 3, \qquad \det A = 2^{n-3}.$$

1. Find the two eigenvalues and their algebraic multiplicities.
2. Give the characteristic polynomial $p_A(t)$.
3. Compute $\operatorname{tr}(A^k)$ for every $k \in \mathbb{Z}_{\geq 1}$.

WORKED SOLUTION

1. By rank-nullity,

$$\dim \ker(A - I_n) = n - \operatorname{rank}(A - I_n) = n - (n - 3) = 3.$$

Thus 1 is an eigenvalue with a 3-dimensional eigenspace. Since A is diagonalizable, every algebraic multiplicity equals the corresponding geometric multiplicity. Hence the algebraic multiplicity of 1 is exactly 3.

Denote the other eigenvalue by λ. There are exactly two eigenvalues, so λ has algebraic multiplicity $n - 3$. The trace is the sum of the eigenvalues counted with multiplicity, so

$$3 \cdot 1 + (n - 3)\lambda = 2n - 3.$$

Subtracting 3 gives

$$(n - 3)\lambda = 2n - 6 = 2(n - 3).$$

Since $n \geq 4$, $n - 3 \neq 0$, and therefore $\lambda = 2$. Thus the eigenvalues are 1 of multiplicity 3 and 2 of multiplicity $n - 3$. The given determinant agrees with this result:

$$\det A = 1^3\, 2^{n-3} = 2^{n-3}.$$

The rank, diagonalizability, number of distinct eigenvalues, and trace were sufficient; the determinant provides an additional consistency check.

2. With $p_A(t) = \det(tI_n - A)$,

$$p_A(t) = (t - 1)^3(t - 2)^{n-3}.$$

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EXERCISE 11.15 EXAM QUESTION CONTINUED

Under the alternative convention $\det(A - tI_n)$, multiply this expression by $(-1)^n$.

3. There is an invertible matrix P with $A = PDP^{-1}$, where

$$D = \operatorname{diag}(1, 1, 1, \underbrace{2, \ldots, 2}_{n-3}).$$

Repeated multiplication cancels the intervening factors $P^{-1}P$, giving

$$A^k = PD^kP^{-1}, \qquad D^k = \operatorname{diag}(1, 1, 1, \underbrace{2^k, \ldots, 2^k}_{n-3}).$$

Trace is invariant under similarity, so

$$\operatorname{tr}(A^k) = 3 + (n-3)2^k.$$

For example, at $k = 1$ this gives $3 + 2(n-3) = 2n - 3$, as required by the data.

Additional practice

EXERCISE 11.16 FOUNDATION

Consider the real matrix

$$A = \begin{pmatrix} 3 & 1 & 0 \\ 1 & 3 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

1. Find its characteristic polynomial and all eigenspaces.
2. Give real matrices P, D satisfying $A = PDP^{-1}$.
3. Find an explicit formula for A^k for every integer $k \geq 0$.

WORKED SOLUTION

1. The block structure gives

$$p_A(t) = (t-1)\det\begin{pmatrix} t-3 & -1 \\ -1 & t-3 \end{pmatrix} = (t-1)((t-3)^2 - 1) = (t-1)(t-2)(t-4).$$

The eigenvalues are distinct. Solving the kernel equations, or direct verification, gives

$$E_4 = \operatorname{span}\{(1,1,0)^T\}, \quad E_2 = \operatorname{span}\{(1,-1,0)^T\}, \quad E_1 = \operatorname{span}\{(0,0,1)^T\}.$$

Each eigenspace has dimension 1, as also follows from its algebraic multiplicity 1.

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EXERCISE 11.16 FOUNDATION CONTINUED

2. Place the eigenbasis vectors in the columns:

$$P = \begin{pmatrix} 1 & 1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad P^{-1} = \begin{pmatrix} 1/2 & 1/2 & 0 \\ 1/2 & -1/2 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad D = \operatorname{diag}(4, 2, 1).$$

We have $\det P = -2 \neq 0$, and the eigenvalue equations give $AP = PD$. Thus $A = PDP^{-1}$.

3. Computing PD^kP^{-1} gives

$$A^k = \begin{pmatrix} (4^k + 2^k)/2 & (4^k - 2^k)/2 & 0 \\ (4^k - 2^k)/2 & (4^k + 2^k)/2 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

The formula also holds at $k = 0$, giving I_3. For $k \geq 1$, all entries are integers despite the division by 2 in this expression.

EXERCISE 11.17 CHALLENGE

Let $A \in M_n(\mathbb{F})$ have n distinct eigenvalues $\lambda_1, \ldots, \lambda_n$ in $\mathbb{F}$.

1. Prove that $B \in M_n(\mathbb{F})$ satisfies $AB = BA$ if and only if there is a unique polynomial f of degree at most $n - 1$ (including the zero polynomial) with $B = f(A)$.

2. Over $\mathbb{R}$, find such a polynomial for

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \qquad B = \begin{pmatrix} 5 & 3 \\ 3 & 5 \end{pmatrix}.$$

WORKED SOLUTION

1. First, every power of A commutes with A, so every linear combination of its powers commutes with it. Thus $B = f(A)$ implies $AB = BA$. Conversely, assume $AB = BA$. Since A has n distinct eigenvalues in the field, it has an eigenbasis $(v_1, \ldots, v_n)$ with $Av_i = \lambda_i v_i$, and every eigenspace is one-dimensional. We have

$$A(Bv_i) = B(Av_i) = \lambda_i Bv_i.$$

Thus $Bv_i \in E_{\lambda_i}(A) = \operatorname{span}\{v_i\}$, including the possibility $Bv_i = 0$. Hence $Bv_i = \mu_i v_i$ for some $\mu_i \in \mathbb{F}$. Define explicitly

$$f(t) = \sum_{i=1}^{n} \mu_i \prod_{\substack{1 \leq j \leq n \\ j \neq i}} \frac{t - \lambda_j}{\lambda_i - \lambda_j}.$$

All denominators are nonzero, so this polynomial is defined over $\mathbb{F}$. For $n = 1$, an empty product equals 1. Its degree is at most $n-1$, and at $t = \lambda_i$ only the ith summand remains,

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EXERCISE 11.17 **CHALLENGE CONTINUED**

giving $f(\lambda_i) = \mu_i$. Therefore

$$f(A)v_i = f(\lambda_i)v_i = \mu_i v_i = Bv_i.$$

The two maps agree on a basis, so $f(A) = B$. If another polynomial g of degree at most $n-1$ satisfies $g(A) = B$, then $(f-g)(\lambda_i) = 0$ for every i. A nonzero polynomial of degree at most $n-1$ cannot have n distinct roots, so $f = g$. Uniqueness requires the degree bound; without it, a multiple of p_A may be added.

2. The vectors $(1,1)^T, (1,-1)^T$ form a common eigenbasis. Their eigenvalues for A are $3, 1$, and for B they are $8, 2$, respectively. Seek $f(t) = at + b$, requiring

$$3a + b = 8, \qquad a + b = 2.$$

Subtracting gives $2a = 6$, so over $\mathbb{R}$ we obtain $a = 3$, $b = -1$. Indeed,

$$B = 3A - I_2, \qquad f(t) = 3t - 1.$$

This numerical example is over $\mathbb{R}$ (or $\mathbb{C}$); the general result in the first part holds over every field in which the stated eigenvalues belong to the field and are distinct.

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TUTORIAL 12

Inner Products, Orthogonality, and Projections

MOTIVATION AND GUIDING IDEA

An inner product adds geometric measurement to a vector space: we can ask whether two directions are perpendicular, measure size, and decompose a vector relative to a subspace. A different inner product changes the geometry even when the vectors and subspace remain fixed. Begin every computation by identifying the given inner product, and only then apply the formulas.

NOTATION AND CONVENTIONS

- $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$ is the scalar field; $\overline{z}$ is complex conjugation, with $\overline{z} = z$ over the reals.
- A^T is the transpose and $A^* = \overline{A}^T$ is the conjugate transpose. For column vectors, the standard inner product is $\langle x, y \rangle = x^T \overline{y} = y^* x$.
- $\langle v, w \rangle$ denotes the given inner product, and $\|v\| = \sqrt{\langle v, v \rangle}$ is its induced norm. For a weighted inner product, we sometimes write $\langle v, w \rangle_G$ and $\|v\|_G$.
- $v \perp w$ means $\langle v, w \rangle = 0$; $S^\perp$ is the set of all vectors orthogonal to every element of S.
- δ_{ij} is the Kronecker delta: it is 1 when $i = j$ and 0 otherwise. Thus $\langle e_i, e_j \rangle = \delta_{ij}$ expresses orthonormality.
- P_W is the orthogonal projection map onto W; $P_W(v)$ is the projection of v onto W, and $v - P_W(v)$ is the orthogonal residual.
- $V = W \oplus W^\perp$ means that every $v \in V$ has a unique decomposition $v = w + z$ with $w \in W$ and $z \in W^\perp$.
- $[v]_B$ is the coordinate vector of v relative to the **ordered** basis B; the basis order determines the coordinate order.
- $G \succ 0$ and $G \succeq 0$ denote a positive definite Hermitian matrix and a positive semidefinite Hermitian matrix, respectively.

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DEFINITION 12.1

Let V be an inner product space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. We use linearity in the first argument and conjugate linearity in the second. Thus, for $u \neq 0$, the projection coefficient onto $\text{span}\{u\}$ is

$$\frac{\langle v, u\rangle}{\langle u, u\rangle},$$

with this order of the arguments in the numerator. The denominator is strictly positive. If V is finite dimensional and $W \leq V$, then $P_W(v)$ is uniquely characterized by the two conditions together:

$$P_W(v) \in W, \qquad v - P_W(v) \in W^{\perp}.$$

If $(e_1, \ldots, e_k)$ is an orthonormal basis of W, then $P_W(v) = \sum_{j=1}^{k} \langle v, e_j\rangle e_j$. For an orthogonal basis that is not normalized, divide each coefficient by $\|e_j\|^2$. Membership alone is insufficient, and orthogonality alone is also insufficient.

MOTIVATION AND GUIDING IDEA

We study how an inner product measures lengths and angles, how to choose an orthonormal basis, and how orthogonal projection decomposes a vector into a component in a subspace and a component perpendicular to it. We connect algebraic computation with geometry and compare different inner products.

12.1 Inner Product Spaces

RECALL AND TOOLS

Throughout this chapter, let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$ and let V be a vector space over $\mathbb{F}$. Write $\bar{\cdot} : \mathbb{F} \to \mathbb{F}$ for standard conjugation: that is, if $\mathbb{F} = \mathbb{C}$, then $\overline{a + ib} = a - ib$; if $\mathbb{F} = \mathbb{R}$, then $\overline{x} = x$ for every $x \in \mathbb{R}$.

RECALL AND TOOLS

Let V be a vector space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. An *inner product* on V is a map

$$\langle \cdot, \cdot \rangle : V \times V \to \mathbb{F}$$

satisfying the following axioms for all $u, v, w \in V$ and all $\alpha, \beta \in \mathbb{F}$.

1. **Linearity in the first argument:**

$$\langle \alpha u + \beta v,\ w\rangle = \alpha\langle u, w\rangle + \beta\langle v, w\rangle.$$

2. **Conjugate symmetry (Hermitian symmetry):**

$$\langle u, v\rangle = \overline{\langle v, u\rangle}.$$

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3. **Positive definiteness:**

$$\langle v, v\rangle \in \mathbb{R}, \qquad \langle v, v\rangle \geq 0, \qquad \langle v, v\rangle = 0 \iff v = 0.$$

Formal observation: Linearity in the first argument and conjugate symmetry imply *conjugate linearity in the second argument*: for all $\alpha, \beta \in \mathbb{F}$ and all $u, v, w \in V$,

$$\langle u,\, \alpha v + \beta w\rangle = \overline{\alpha}\,\langle u, v\rangle + \overline{\beta}\,\langle u, w\rangle.$$

When $\mathbb{F} = \mathbb{R}$, this gives ordinary linearity in the second argument as well.

EXAMPLE 12.1

Standard examples of inner products:

- On $\mathbb{C}^n$ (column vectors $u, v \in \mathbb{C}^n$):

$$\langle u, v\rangle := u^T \overline{v}.$$

  On $\mathbb{R}^n$:

$$\langle u, v\rangle := u^T v.$$

- On $M_{m\times n}(\mathbb{C})$:

$$\langle A, B\rangle := \operatorname{tr}\!\left(A^T \overline{B}\right).$$

- On $\mathbb{R}_n[x]$ (or on $C([a, b], \mathbb{R})$), where $a < b$:

$$\langle p, q\rangle := \int_a^b p(x)\, q(x)\, dx.$$

DEFINITION 12.2

Let $(V, \langle\cdot,\cdot\rangle)$ be a vector space equipped with an inner product. The **norm** induced by the inner product is defined, for each $v \in V$, by

$$\|v\| := \sqrt{\langle v, v\rangle}.$$

Thus $\|v\| \in [0, \infty)$; in particular, $\|v\| = 0$ if and only if $v = 0$.

DEFINITION 12.3

Inner product space: If V is a vector space over $\mathbb{F}$ and $\langle\cdot,\cdot\rangle$ is an inner product on V, then the pair

$$\left(V, \langle\cdot,\cdot\rangle\right)$$

is called an **inner product space**.

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PROPOSITION 12.1

Let $(V, \langle\cdot,\cdot\rangle)$ be an inner product space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. For all $n, m \in \mathbb{N}$, all $\alpha_1, \ldots, \alpha_n, \beta_1, \ldots, \beta_m \in \mathbb{F}$, and all $v_1, \ldots, v_n, u_1, \ldots, u_m \in V$,

$$\left\langle \sum_{i=1}^{n} \alpha_i v_i,\ \sum_{j=1}^{m} \beta_j u_j \right\rangle = \sum_{i=1}^{n} \sum_{j=1}^{m} \alpha_i\, \overline{\beta_j}\, \langle v_i, u_j \rangle.$$

When $\mathbb{F} = \mathbb{R}$, we have $\overline{\beta_j} = \beta_j$, so the formula reduces to

$$\left\langle \sum_{i=1}^{n} \alpha_i v_i,\ \sum_{j=1}^{m} \beta_j u_j \right\rangle = \sum_{i=1}^{n} \sum_{j=1}^{m} \alpha_i\, \beta_j\, \langle v_i, u_j \rangle.$$

12.2 Basic Exercises and Results

EXERCISE 12.1

Let $n \geq 1$. On $V = M_n(\mathbb{R})$, define

$$f(A, B) := \operatorname{tr}(AB).$$

Determine for which values of n this is an inner product, and justify your answer.

WORKED SOLUTION

If $n = 1$, identify matrices with real numbers. Then $f(a, b) = ab$ is the standard inner product on $\mathbb{R}$. If $n \geq 2$, choose $A = E_{12}$: that is, $a_{12} = 1$ and all other entries are zero. Then $A \neq 0$, but $A^2 = 0$, so

$$f(A, A) = \operatorname{tr}(A^2) = 0.$$

Positive definiteness therefore fails, and the function is not an inner product when $n \geq 2$.

EXERCISE 12.2

Let $n \in \mathbb{N}$, and let $M \in M_n(\mathbb{R})$ be symmetric and positive definite; that is,

$$M^T = M, \qquad x^T M x > 0 \ \ \forall x \in \mathbb{R}^n \setminus \{0\}.$$

Define $\langle\cdot,\cdot\rangle_M : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ by

$$\langle x, y\rangle_M := x^T M y.$$

1. Prove that $\langle\cdot,\cdot\rangle_M$ is an inner product on $\mathbb{R}^n$.
2. Show that the induced norm is $\|x\|_M = \sqrt{x^T M x}$.
3. Physical example: suppose a linear mechanical system is described by a mass matrix M (in some system of generalized coordinates). Show that its kinetic energy

$$T(v) = \frac{1}{2}\, v^T M v$$

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EXERCISE 12.2 CONTINUED

equals $\frac{1}{2}\|v\|_M^2$.

WORKED SOLUTION

1. We verify the inner product axioms.

   Linearity in the first argument: Let $x_1, x_2, y \in \mathbb{R}^n$ and $\alpha, \beta \in \mathbb{R}$. Then

   $$\langle \alpha x_1 + \beta x_2,\, y\rangle_M = (\alpha x_1 + \beta x_2)^T M y = \alpha x_1^T M y + \beta x_2^T M y = \alpha\langle x_1, y\rangle_M + \beta\langle x_2, y\rangle_M.$$

   Symmetry: For every $x, y \in \mathbb{R}^n$,

   $$\langle x, y\rangle_M = x^T M y.$$

   Since this is a real scalar, $(x^T My)^T = x^T My$, so

   $$\langle x, y\rangle_M = (x^T My)^T = y^T M^T x = y^T Mx = \langle y, x\rangle_M,$$

   where we used $M^T = M$.

   Positive definiteness: For every $x \in \mathbb{R}^n$,

   $$\langle x, x\rangle_M = x^T Mx \geq 0,$$

   and $\langle x, x\rangle_M = 0$ if and only if $x = 0$, because M is positive definite.

   Hence $\langle \cdot, \cdot\rangle_M$ is an inner product on $\mathbb{R}^n$.

2. By the definition of the norm induced by an inner product, for every $x \in \mathbb{R}^n$,

   $$\|x\|_M := \sqrt{\langle x, x\rangle_M} = \sqrt{x^T Mx}.$$

3. For every $v \in \mathbb{R}^n$,

   $$T(v) = \frac{1}{2} v^T Mv = \frac{1}{2}\langle v, v\rangle_M = \frac{1}{2}\|v\|_M^2,$$

   so the kinetic energy is half the square of the norm induced by $\langle \cdot, \cdot\rangle_M$.

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EXERCISE 12.3

Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$ and $A \in M_n(\mathbb{F})$. For any matrix C, write $C^* := \overline{C}^T$; over $\mathbb{R}$, $C^* = C^T$. Following the convention that the inner product is linear in its first argument, define

$$\langle x, y\rangle := y^* x = x^T \overline{y}.$$

1. Prove that $\langle Av, w\rangle = \langle v, A^* w\rangle$ for all $v, w \in \mathbb{F}^n$.
2. Let $G \in M_n(\mathbb{F})$ be Hermitian and positive definite. Define

   $$\langle x, y\rangle_G := y^* Gx.$$

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EXERCISE 12.3 CONTINUED

Prove that this is an inner product, and show that there is a unique matrix A^{*_G} satisfying

$$\langle Av, w\rangle_G = \langle v, A^{*_G} w\rangle_G \quad \text{for all } v, w \in \mathbb{F}^n,$$

and that $A^{*_G} = G^{-1}A^*G$.

WORKED SOLUTION

1. The identity $(A^*w)^* = w^*A$ gives

$$\langle Av, w\rangle = w^*Av = (A^*w)^*v = \langle v, A^*w\rangle.$$

2. Linearity in the first argument follows from the linearity of matrix multiplication. Since $G^* = G$,

$$\overline{\langle y, x\rangle_G} = \overline{x^*Gy} = (x^*Gy)^* = y^*G^*x = y^*Gx = \langle x, y\rangle_G.$$

Furthermore, $\langle x, x\rangle_G = x^*Gx > 0$ for every $x \neq 0$, and $\langle 0, 0\rangle_G = 0$. The matrix G is invertible: otherwise $Gx = 0$ for some $x \neq 0$ would imply $x^*Gx = 0$. Set $B := G^{-1}A^*G$. Then

$$B^* = (G^{-1}A^*G)^* = GAG^{-1},$$

and hence

$$\langle v, Bw\rangle_G = (Bw)^*Gv = w^*B^*Gv = w^*GAv = \langle Av, w\rangle_G.$$

For uniqueness, if C also satisfies the identity, then $w^*(C^*G - GA)v = 0$ for all v, w. Choosing $v = e_j$ and $w = e_i$ shows that every entry of $C^*G - GA$ is zero. Thus

$$C^*G = GA \implies C^* = GAG^{-1} \implies C = G^{-1}A^*G = B.$$

EXERCISE 12.4

Let $(V, \langle\cdot,\cdot\rangle)$ be an inner product space over $\mathbb{C}$, and let $u, v \in V$ satisfy $u \neq v$. Show that there is $w \in V$ such that $\langle w, u\rangle \neq \langle w, v\rangle$.

WORKED SOLUTION

Choose $w = u - v$. Since $u \neq v$, the vector w is nonzero. Conjugate linearity in the second argument implies additivity, so

$$\langle w, u\rangle - \langle w, v\rangle = \langle w, u - v\rangle = \langle u - v, u - v\rangle = \|u - v\|^2 > 0.$$

Therefore $\langle w, u\rangle \neq \langle w, v\rangle$, as required.

12.3 Orthogonality

DEFINITION 12.4

Let $(V, \langle\cdot,\cdot\rangle)$ be an inner product space. Vectors $v, w \in V$ are **orthogonal** if $\langle v, w\rangle = 0$; we write $v \perp w$.

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PROPOSITION 12.2

If $v \perp b$ for every $b \in B$, then $v \perp \text{span}(B)$. Indeed, for any linear combination $s = \sum_{j=1}^{k} \alpha_j b_j$,

$$\langle v, s \rangle = \sum_{j=1}^{k} \overline{\alpha_j} \langle v, b_j \rangle = 0.$$

DEFINITION 12.5

A set $B = \{v_1, \ldots, v_n\} \subseteq V$ is **orthogonal** if

$$\langle v_i, v_j \rangle = 0 \qquad \text{for all } i \neq j.$$

If, in addition, $\|v_i\| = 1$ for every i, it is **orthonormal**. In this case,

$$\langle v_i, v_j \rangle = \delta_{ij} = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

If B is also a basis of V, we call it an orthogonal basis or an orthonormal basis, respectively.

RECALL AND TOOLS

Normalization: an orthogonal set can be converted into an orthonormal set:
if $\{v_1, \ldots, v_n\}$ is orthogonal and contains no zero vector, then $\left\{ \frac{v_1}{\|v_1\|}, \ldots, \frac{v_n}{\|v_n\|} \right\}$ is orthonormal.
Indeed, for every $1 \leq i \leq n$, $\left\langle \frac{v_i}{\|v_i\|}, \frac{v_i}{\|v_i\|} \right\rangle = \frac{1}{\|v_i\|^2} \langle v_i, v_i \rangle = 1$.

PROPOSITION 12.3

- An **orthogonal** set containing no zero vector is linearly independent.
- An **orthonormal** set is linearly independent. (Since each of its elements has norm 1, it contains no zero vector.)
- The converse fails: for example, $\left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\}$ with the standard inner product is linearly independent but is neither orthogonal nor orthonormal.

EXERCISE 12.5

Let $u, v \in \mathbb{R}^n$ be such that $a = 3v - 2u$ and $b = u - v$ are orthonormal for the standard inner product. Compute $\|u\|$, $\|v\|$, and $\langle u, v \rangle$, and prove that u, v are linearly independent.

WORKED SOLUTION

First solve the two vector identities: from $b = u - v$, we obtain $u = v + b$. Substitution into $a = 3v - 2u$ gives $a = v - 2b$, so

$$v = a + 2b, \qquad u = a + 3b.$$

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EXERCISE 12.5 CONTINUED

Since $\langle a,a\rangle = \langle b,b\rangle = 1$ and $\langle a,b\rangle = 0$,

$$\|v\|^2 = \langle a+2b, a+2b\rangle = 1+2\cdot 0+2\cdot 0+4 = 5,$$

$$\|u\|^2 = \langle a+3b, a+3b\rangle = 1+3\cdot 0+3\cdot 0+9 = 10,$$

$$\langle u,v\rangle = \langle a+3b, a+2b\rangle = 1+2\cdot 0+3\cdot 0+6 = 7.$$

Thus $\|v\| = \sqrt{5}$, $\|u\| = \sqrt{10}$, and $\langle u,v\rangle = 7$. If $\alpha u + \beta v = 0$, then

$$(\alpha+\beta)a + (3\alpha+2\beta)b = 0.$$

Since a, b are orthonormal, they are linearly independent. Hence $\alpha+\beta = 0$ and $3\alpha+2\beta = 0$. Subtracting twice the first equation from the second gives $\alpha = 0$, and then $\beta = 0$. The given information therefore determines the norms and inner product and also establishes the linear independence of u, v.

12.4 Coordinates in an Orthogonal or Orthonormal Basis

DEFINITION 12.6

Let $(V, \langle\cdot,\cdot\rangle)$ be an inner product space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. Let $B = (b_1, \ldots, b_n)$ be an ordered basis of V.
Every $v \in V$ has a unique representation as a linear combination of the basis vectors: $v = \alpha_1 b_1 + \cdots + \alpha_n b_n$.
If B is an **orthogonal** basis, its coefficients are given by $\alpha_i = \frac{\langle v, b_i\rangle}{\|b_i\|^2}$.
If B is an **orthonormal** basis, then $\|b_i\|^2 = 1$, so $\alpha_i = \langle v, b_i\rangle$. Thus the coordinate vector relative to B is

$$[v]_B = \begin{pmatrix} \langle v, b_1\rangle \\ \vdots \\ \langle v, b_n\rangle \end{pmatrix}$$

EXERCISE 12.6

On $\mathbb{C}^3$, use the inner product $\langle x, y\rangle = x^T\overline{y}$. Let

$$b_1 = \tfrac{1}{5}(3, 4i, 0)^T, \quad b_2 = \tfrac{1}{5}(4, -3i, 0)^T, \quad b_3 = (0,0,1)^T.$$

Show that $B = (b_1, b_2, b_3)$ is an orthonormal basis and find $[v]_B$ for $v = (1, 3i, 2)^T$.

WORKED SOLUTION

Compute the norms:

$$\|b_1\|^2 = \frac{3^2 + |4i|^2}{25} = \frac{9+16}{25} = 1, \quad \|b_2\|^2 = \frac{4^2 + |-3i|^2}{25} = 1, \quad \|b_3\|^2 = 1.$$

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EXERCISE 12.6 CONTINUED

The only inner product requiring a calculation is

$$\langle b_1, b_2\rangle = \frac{3\cdot 4 + (4i)\overline{-3i}}{25} = \frac{12 + (4i)(3i)}{25} = \frac{12-12}{25} = 0.$$

The two inner products with b_3 vanish because of the third coordinate. Thus B is orthonormal; since it has three vectors in $\mathbb{C}^3$, it is a basis. With our convention of linearity in the first argument, the coefficients are $\langle v, b_j\rangle$:

$$\langle v, b_1\rangle = \frac{1\cdot 3 + (3i)(-4i)}{5} = \frac{3+12}{5} = 3,$$

$$\langle v, b_2\rangle = \frac{1\cdot 4 + (3i)(3i)}{5} = \frac{4-9}{5} = -1, \qquad \langle v, b_3\rangle = 2.$$

Hence $[v]_B = (3, -1, 2)^T$. Check by reconstruction:

$$3b_1 - b_2 + 2b_3 = \left(\frac{9-4}{5}, \frac{12i+3i}{5}, 2\right)^T = (1, 3i, 2)^T = v.$$

12.5 Norms and the Cauchy--Schwarz Inequality

DEFINITION 12.7

Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$ and let V be a vector space over $\mathbb{F}$. A **norm** on V is a function

$$\|\cdot\| : V \to [0, \infty)$$

satisfying the following axioms for all $u, v \in V$ and all $\alpha \in \mathbb{F}$:

1. **Positive definiteness:**
$$\|v\| \geq 0, \qquad \|v\| = 0 \iff v = 0.$$
2. **Homogeneity:**
$$\|\alpha v\| = |\alpha|\,\|v\|.$$
3. **Triangle inequality:**
$$\|u + v\| \leq \|u\| + \|v\|.$$

Here $|\alpha|$ denotes absolute value over $\mathbb{R}$ or modulus over $\mathbb{C}$. A vector space equipped with a norm is a **normed space**, denoted $(V, \|\cdot\|)$.

PROPOSITION 12.4

Cauchy--Schwarz inequality. For all $u, v \in V$,

$$|\langle u, v\rangle| \leq \|u\|\,\|v\|.$$

Equality holds if and only if u, v are linearly dependent: that is, there is $\alpha \in \mathbb{F}$ such that $u = \alpha v$ or $v = \alpha u$.

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PROPOSITION 12.5

Angle in a real space. If $\mathbb{F} = \mathbb{R}$ and $u, v \neq 0$, the number

$$\cos\zeta := \frac{\langle u, v\rangle}{\|u\|\|v\|}$$

satisfies $\cos\zeta \in [-1, 1]$. In particular, there is $\zeta \in [0, \pi]$ for which this equality holds.

EXERCISE 12.7

Let $\omega, \zeta, \xi > 0$. Prove the inequality

$$\frac{\omega}{\zeta+\xi} + \frac{\zeta}{\xi+\omega} + \frac{\xi}{\omega+\zeta} \geq \frac{(\omega+\zeta+\xi)^2}{2(\omega\zeta+\zeta\xi+\xi\omega)}.$$

WORKED SOLUTION

Use the standard inner product on $\mathbb{R}^3$, namely $\langle x, y\rangle = x^T y$.
Define the vectors

$$x = \left(\sqrt{\frac{\omega}{\zeta+\xi}}, \sqrt{\frac{\zeta}{\xi+\omega}}, \sqrt{\frac{\xi}{\omega+\zeta}}\right), \qquad y = \left(\sqrt{\omega(\zeta+\xi)}, \sqrt{\zeta(\xi+\omega)}, \sqrt{\xi(\omega+\zeta)}\right).$$

Then

$$\langle x, y\rangle = \sqrt{\frac{\omega}{\zeta+\xi}}\sqrt{\omega(\zeta+\xi)} + \sqrt{\frac{\zeta}{\xi+\omega}}\sqrt{\zeta(\xi+\omega)} + \sqrt{\frac{\xi}{\omega+\zeta}}\sqrt{\xi(\omega+\zeta)} = \omega+\zeta+\xi.$$

Also,

$$\|x\|^2 = \langle x, x\rangle = \left(\sqrt{\frac{\omega}{\zeta+\xi}}\right)^2 + \left(\sqrt{\frac{\zeta}{\xi+\omega}}\right)^2 + \left(\sqrt{\frac{\xi}{\omega+\zeta}}\right)^2 = \frac{\omega}{\zeta+\xi} + \frac{\zeta}{\xi+\omega} + \frac{\xi}{\omega+\zeta},$$

and

$$\begin{aligned}\|y\|^2 &= \langle y, y\rangle \\ &= \left(\sqrt{\omega(\zeta+\xi)}\right)^2 + \left(\sqrt{\zeta(\xi+\omega)}\right)^2 + \left(\sqrt{\xi(\omega+\zeta)}\right)^2 \\ &= \omega(\zeta+\xi) + \zeta(\xi+\omega) + \xi(\omega+\zeta).\end{aligned}$$

Expand the last sum:

$$\omega(\zeta+\xi) + \zeta(\xi+\omega) + \xi(\omega+\zeta) = (\omega\zeta+\omega\xi) + (\zeta\xi+\zeta\omega) + (\xi\omega+\xi\zeta) = 2(\omega\zeta+\zeta\xi+\xi\omega).$$

The Cauchy--Schwarz inequality gives

$$\langle x, y\rangle^2 \leq \|x\|^2\,\|y\|^2,$$

and hence

$$(\omega+\zeta+\xi)^2 \leq \left(\frac{\omega}{\zeta+\xi} + \frac{\zeta}{\xi+\omega} + \frac{\xi}{\omega+\zeta}\right)\cdot 2(\omega\zeta+\zeta\xi+\xi\omega).$$

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EXERCISE 12.7 CONTINUED

Since $2(\omega\zeta + \zeta\xi + \xi\omega) > 0$, division yields

$$\frac{\omega}{\zeta+\xi} + \frac{\zeta}{\xi+\omega} + \frac{\xi}{\omega+\zeta} \geq \frac{(\omega+\zeta+\xi)^2}{2(\omega\zeta+\zeta\xi+\xi\omega)},$$

as required.

12.6 Orthogonal Complements and Orthogonal Projections

DEFINITION 12.8

Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$ and let $(V, \langle\cdot,\cdot\rangle)$ be an inner product space. For $S \subseteq V$, define the **orthogonal complement** of S by

$$S^\perp := \{\, v \in V : \forall s \in S,\ \langle v, s\rangle = 0 \,\}.$$

In particular, if $W \subseteq V$ is a subspace, then $W^\perp$ is the **orthogonal complement** of W.

DEFINITION 12.9

Let $U, W \subseteq V$ be subspaces. We say that U and W are **orthogonal** (written $U \perp W$) if

$$\forall u \in U\ \forall w \in W \qquad \langle u, w\rangle = 0.$$

PROPOSITION 12.6

Let $S \subseteq V$, and let $U, W \subseteq V$ be subspaces. The following statements hold:

1. $S^\perp$ is a subspace of V.
2. If $S \subseteq T \subseteq V$, then $T^\perp \subseteq S^\perp$.
3. $(\operatorname{span} S)^\perp = S^\perp$.
4. $W \cap W^\perp = \{0\}$.
5. $U \perp W \iff U \subseteq W^\perp \iff W \subseteq U^\perp$.
6. $(U+W)^\perp = U^\perp \cap W^\perp$.
7. If V is finite dimensional, then $(U \cap W)^\perp = U^\perp + W^\perp$.
8. If $U \perp W$, then $U + W = U \oplus W$.
9. $W \subseteq (W^\perp)^\perp$. If V is finite dimensional, then $(W^\perp)^\perp = W$.

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RECALL AND TOOLS

For all projection results below, assume that V is finite dimensional. In infinite dimensions, closedness of W alone is insufficient in a general inner product space: the projection theorem holds when V is complete in the induced norm, that is, a Hilbert space, and W is closed. Alternatively, it suffices that W be finite dimensional, without assuming completeness of V.

DEFINITION 12.10

Suppose V is finite dimensional and $W \subseteq V$ is a subspace. The **orthogonal projection onto** W is the map $P_W : V \to W$ defined by requiring that, for every $v \in V$, $P_W(v) \in W$ be the unique vector satisfying

$$v - P_W(v) \in W^{\perp}.$$

PROPOSITION 12.7

Suppose V is finite dimensional and $W \subseteq V$ is a subspace. The following hold:

1. For every $v \in V$, there are unique $w \in W$ and $z \in W^{\perp}$ such that

   $$v = w + z, \qquad P_W(v) = w.$$

   In particular, $V = W \oplus W^{\perp}$.

2. P_W is linear.

3. $$P_W^2 = P_W, \qquad \operatorname{Im}(P_W) = W, \qquad \ker(P_W) = W^{\perp}.$$

4. For every $v \in V$,

   $$P_W(v) \perp \big(v - P_W(v)\big), \qquad \|v\|^2 = \|P_W(v)\|^2 + \|v - P_W(v)\|^2.$$

5. For all $u, v \in V$,

   $$\langle P_W(u), v\rangle = \langle u, P_W(v)\rangle.$$

6. For every $v \in V$, $\|P_W(v)\| \le \|v\|$.

RECALL AND TOOLS

If $\dim V < \infty$ and $\{e_1, \dots, e_k\}$ is an orthonormal basis of W, then for every $v \in V$,

$$P_W(v) = \sum_{i=1}^{k} \langle v, e_i\rangle\, e_i,$$

and, in particular,

$$W^{\perp} = \{\, v \in V : \ \langle v, e_i\rangle = 0 \ \forall i = 1, \dots, k \,\}.$$

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PROPOSITION 12.8

If $\dim V < \infty$ and $W \subseteq V$ is a subspace, then

$$\dim W + \dim W^{\perp} = \dim V.$$

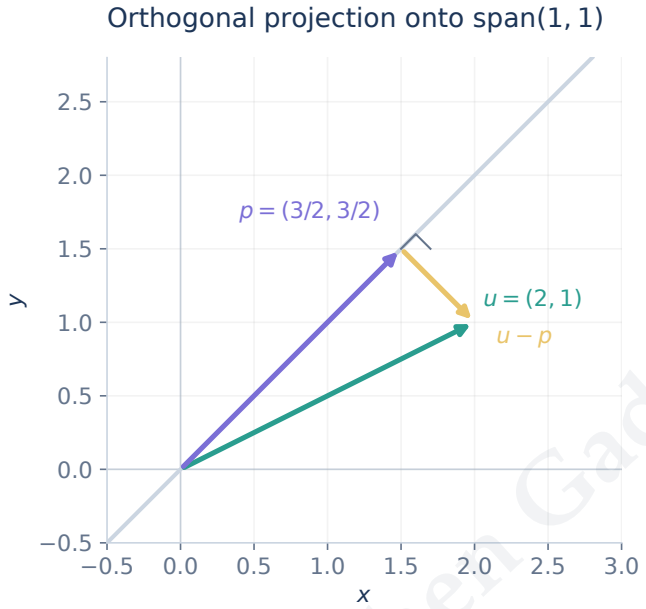


Figure 12.1. Orthogonal projection decomposes a vector into a component in the subspace and a perpendicular residual. The two components satisfy the Pythagorean identity.

EXERCISE 12.8

Let $y = (-2, 9)^T$, $u = (2, 1)^T$, and $W = \operatorname{span}\{u\}$. For the weighted inner product $\langle x, z\rangle_M = x^T M z$, where $M = \operatorname{diag}(1, 4)$, compute $P_W^M(y)$ and check that the residual is orthogonal to W. Compare with the projection $P_W^I(y)$ for the standard inner product.

WORKED SOLUTION

M is symmetric and positive definite, since $x^T M x = x_1^2 + 4x_2^2 > 0$ when $x \neq 0$. The coefficient of the weighted projection is

$$t_M = \frac{y^T M u}{u^T M u} = \frac{(-2)\cdot 2 + 4 \cdot 9 \cdot 1}{2^2 + 4 \cdot 1^2} = \frac{-4 + 36}{8} = 4.$$

Thus $P_W^M(y) = 4u = (8, 4)^T$, and the residual is $r_M = (-10, 5)^T$. Check:

$$r_M^T M u = (-10)\cdot 2 + 4 \cdot 5 \cdot 1 = -20 + 20 = 0.$$

For the standard inner product,

$$t_I = \frac{y^T u}{u^T u} = \frac{-4 + 9}{4 + 1} = 1, \qquad P_W^I(y) = u = (2, 1)^T.$$

The residual is $r_I = (-4, 8)^T$; indeed, $r_I^T u = -4\cdot 2 + 8 \cdot 1 = 0$. In both cases, the projection lies in W and the residual is orthogonal to it for the relevant inner product. Changing the inner

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EXERCISE 12.8 CONTINUED

product changes the projection; here both projections have integer coordinates.

12.7 Linear Maps on Inner Product Spaces

EXERCISE 12.9

Let $(V, \langle\cdot,\cdot\rangle)$ be a real inner product space of dimension $n \geq 3$. Let $v, w \in V$ be orthonormal: that is, $\langle v, v\rangle = \langle w, w\rangle = 1$ and $\langle v, w\rangle = 0$. Define the linear map $T : V \to V$ by

$$T(u) = \langle u, v\rangle\, w + \langle u, w\rangle\, v.$$

1. Find all eigenvalues of T, all eigenvectors, and the characteristic polynomial of T.
2. Prove that $\|T(u)\| \leq \|u\|$ for every $u \in V$, and find all u for which equality holds.

WORKED SOLUTION

1. Start with the eigenvalue equation and compare components using an orthogonal decomposition:

   $$T(u) = \lambda u.$$

   Decompose each $u \in V$ using the orthogonal direct sum

   $$V = \text{span}\{v, w\} \oplus (\text{span}\{v, w\})^{\perp}.$$

   Here $\{v, w\}^{\perp}$ denotes the orthogonal complement of $\text{span}\{v, w\}$. Thus there are unique $\alpha, \beta \in \mathbb{R}$ and $y \in \{v, w\}^{\perp}$ such that

   $$u = \alpha v + \beta w + y.$$

   Now compute $T(u)$: since $y \perp v, w$, we have $\langle y, v\rangle = \langle y, w\rangle = 0$, so

   $$\langle u, v\rangle = \langle \alpha v + \beta w + y,\ v\rangle = \alpha, \qquad \langle u, w\rangle = \langle \alpha v + \beta w + y,\ w\rangle = \beta.$$

   Hence

   $$T(u) = \alpha w + \beta v.$$

   Substitute into the eigenvalue equation $T(u) = \lambda u$ to obtain

   $$\alpha w + \beta v = \lambda(\alpha v + \beta w + y).$$

   Compare components in the direct sum $\text{span}\{v, w\} \oplus \{v, w\}^{\perp}$. The component in $\{v, w\}^{\perp}$ on the left is zero, so necessarily

   $$\lambda y = 0.$$

   Consequently:

   - If $\lambda \neq 0$, then $y = 0$, so $u \in \text{span}\{v, w\}$.
   - If $\lambda = 0$, then $T(u) = 0$, and $\alpha w + \beta v = 0$ gives $\alpha = \beta = 0$, so $u = y \in \{v, w\}^{\perp}$.

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EXERCISE 12.9 CONTINUED

Now consider $\lambda \neq 0$. Then $u = \alpha v + \beta w$ and $y = 0$. The eigenvalue equation becomes

$$\alpha w + \beta v = \lambda(\alpha v + \beta w).$$

Comparing coefficients of v and w gives the system

$$\beta = \lambda\alpha, \qquad \alpha = \lambda\beta.$$

Substitution gives $\alpha = \lambda(\lambda\alpha) = \lambda^2\alpha$, so

$$(\lambda^2 - 1)\alpha = 0.$$

If $\alpha = 0$, then $\beta = \lambda\alpha = 0$ as well, giving $u = 0$, contrary to the definition of an eigenvector. Thus $\alpha \neq 0$, and

$$\lambda^2 - 1 = 0 \quad \Longleftrightarrow \quad \lambda \in \{1, -1\}.$$

Now determine the eigenvectors:

- For $\lambda = 1$, $\beta = \alpha$, so $u = \alpha(v + w)$; hence
$$V_1 = \text{span}\{v + w\}.$$
- For $\lambda = -1$, $\beta = -\alpha$, so $u = \alpha(v - w)$; hence
$$V_{-1} = \text{span}\{v - w\}.$$
- For $\lambda = 0$, we obtained $u \in \{v, w\}^\perp$, so
$$V_0 = \{v, w\}^\perp.$$

In particular, there are no other eigenvalues, because the eigenvalue equation forced $\lambda \in \{0, 1, -1\}$.

To compute the characteristic polynomial, choose an orthonormal basis of V of the form

$$B = (v, w, e_3, \ldots, e_n),$$

where $e_3, \ldots, e_n$ is an orthonormal basis of $\{v, w\}^\perp$. Then

$$T(v) = w, \qquad T(w) = v, \qquad T(e_i) = 0 \ \ (i = 3, \ldots, n).$$

The matrix representing T relative to B is therefore

$$[T]_B = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 1 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 0 \end{pmatrix}.$$

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EXERCISE 12.9 CONTINUED

Hence its characteristic polynomial is

$$p_T(\lambda) = \det(\lambda I - [T]_B) = \det\begin{pmatrix} \lambda & -1 & 0 & \cdots & 0 \\ -1 & \lambda & 0 & \cdots & 0 \\ 0 & 0 & \lambda & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & \lambda \end{pmatrix} = (\lambda^2 - 1)\lambda^{n-2}.$$

2. Set $\alpha := \langle u, v\rangle$ and $\beta := \langle u, w\rangle$, and define

$$s := \alpha v + \beta w \in \operatorname{span}\{v, w\}.$$

Since v, w are orthonormal,

$$\|s\|^2 = \langle s, s\rangle = \alpha^2 + \beta^2 = \langle u, v\rangle^2 + \langle u, w\rangle^2 = \|T(u)\|^2.$$

Also,

$$\langle u, s\rangle = \alpha\langle u, v\rangle + \beta\langle u, w\rangle = \alpha^2 + \beta^2 = \|s\|^2.$$

By the Cauchy--Schwarz inequality,

$$|\langle u, s\rangle| \le \|u\|\,\|s\|.$$

Since $|\langle u, s\rangle| = \|s\|^2$, we obtain

$$\|s\|^2 \le \|u\|\,\|s\|.$$

If $s = 0$, then $\|T(u)\| = \|s\| = 0 \le \|u\|$. If $s \neq 0$, divide by $\|s\|$ to obtain

$$\|T(u)\| = \|s\| \le \|u\|.$$

To check equality, write $u = s + y$ with $y \in \{v, w\}^{\perp}$. By the Pythagorean identity,

$$\|u\|^2 = \|s\|^2 + \|y\|^2 = \|T(u)\|^2 + \|y\|^2.$$

Thus $\|T(u)\| = \|u\|$ if and only if $y = 0$, or equivalently $u \in \operatorname{span}\{v, w\}$. This argument also includes $s = 0$, in which case equality requires $u = 0$.

EXERCISE 12.10

Let $n \ge 3$ be a natural number and let $a \in \mathbb{R}$. Define $A_n(a) \in M_n(\mathbb{R})$ by

$$(A_n(a))_{ij} = \begin{cases} 1, & i = j, \\ a, & i \neq j. \end{cases}$$

Recall the definition: a Hermitian matrix $H \in M_n(\mathbb{C})$ is positive semidefinite if $x^*Hx \ge 0$ for every $x \in \mathbb{C}^n$, and positive definite if $x^*Hx > 0$ for every $x \neq 0$.

1. For every a, find all eigenvalues of $A_n(a)$ and their multiplicities.
2. Determine for which values of a the matrix $A_n(a)$ is positive definite, and for which

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EXERCISE 12.10 CONTINUED

values it is positive semidefinite.

3. Find $\operatorname{rank}(A_n(a))$ and $\ker(A_n(a))$ in the two boundary cases where $A_n(a)$ is positive semidefinite but not positive definite.

In addition, let $B \in M_n(\mathbb{C})$ and write $B^* = \overline{B}^T$. Define

$$H := B^*B + I.$$

(4) Prove that H is positive definite and hence invertible.

(5) Prove that, for every $x \in \mathbb{C}^n$,

$$x^*H^{-1}x \le \|x\|^2,$$

and also prove that

$$\det(H) \ge 1,$$

with equality if and only if $B = 0$.

WORKED SOLUTION

Write $e = (1, 1, \dots, 1)^T \in \mathbb{R}^n$. We have the decomposition

$$A_n(a) = (1-a)I + a\,ee^T,$$

because every entry of ee^T equals 1.

1. Note that $ee^Te = (e^Te)\,e = n\,e$, so

$$A_n(a)e = (1-a)e + a\,ee^Te = (1-a)e + an\,e = \big(1 + (n-1)a\big)e,$$

Thus e is an eigenvector with eigenvalue $\lambda_1 = 1 + (n-1)a$.

Now let $x \in \mathbb{R}^n$ satisfy $e^Tx = 0$. Then $ee^Tx = e(e^Tx) = 0$, so

$$A_n(a)x = (1-a)x + a \cdot 0 = (1-a)x.$$

Consequently, every $x \in e^\perp \setminus \{0\}$ is an eigenvector with eigenvalue $\lambda_2 = 1 - a$. Since $\mathbb{R}^n = \operatorname{span}\{e\} \oplus e^\perp$, these are all eigenvalues. If $a \neq 0$, then $\lambda_1 \neq \lambda_2$, so

$$\begin{aligned} &\lambda_1 = 1 + (n-1)a \quad \text{with algebraic and geometric multiplicity } 1, \\ &\lambda_2 = 1 - a \qquad\quad\ \ \text{with algebraic and geometric multiplicity } n-1. \end{aligned}$$

If $a = 0$, then $A_n(0) = I$, so its only eigenvalue is 1, with algebraic and geometric multiplicity n.

2. The matrix $A_n(a)$ is real symmetric and hence Hermitian. It is therefore positive definite exactly when all its eigenvalues are strictly positive, and positive semidefinite exactly when all its eigenvalues are nonnegative. Thus:

$$A_n(a) \succ 0 \iff \begin{cases} 1-a > 0, \\ 1 + (n-1)a > 0, \end{cases} \iff -\frac{1}{n-1} < a < 1,$$

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EXERCISE 12.10 CONTINUED

and

$$A_n(a) \succeq 0 \iff \begin{cases} 1-a \geq 0, \\ 1+(n-1)a \geq 0, \end{cases} \iff -\frac{1}{n-1} \leq a \leq 1.$$

3. If $a = 1$, then $\lambda_2 = 0$ has multiplicity $n-1$, and $\lambda_1 = n$. Hence

$$\operatorname{rank}(A_n(1)) = 1, \qquad \ker(A_n(1)) = e^{\perp} = \{x \in \mathbb{R}^n : \sum_{i=1}^{n} x_i = 0\}.$$

If $a = -\frac{1}{n-1}$, then $\lambda_1 = 0$ has multiplicity 1, and $\lambda_2 = 1 - a = \frac{n}{n-1} > 0$ has multiplicity $n-1$. Hence

$$\operatorname{rank}\Big(A_n\big(-\tfrac{1}{n-1}\big)\Big) = n-1, \qquad \ker\Big(A_n\big(-\tfrac{1}{n-1}\big)\Big) = \operatorname{span}\{e\}.$$

Now turn to $H = B^*B + I$.

(4) Let $x \in \mathbb{C}^n$. Then

$$x^*Hx = x^*(B^*B)x + x^*Ix = \|Bx\|^2 + \|x\|^2.$$

Hence $x^*Hx \geq 0$ for every x, and $x^*Hx > 0$ for every $x \neq 0$, since $\|x\|^2 > 0$. Thus $H \succ 0$, so H is invertible.

(5) From $H = I + B^*B$, we obtain $H - I = B^*B \succeq 0$; thus $x^*Hx \geq x^*Ix = \|x\|^2$ for every x. Let $\lambda_{\min}$ be the smallest eigenvalue of H. The preceding inequality gives $\lambda_{\min} \geq 1$. Every eigenvalue of H^{-1} is therefore at most 1, so for every x,

$$x^*H^{-1}x \leq \|x\|^2.$$

For the determinant inequality, write the eigenvalues of H as $\lambda_1, \ldots, \lambda_n$. Since $H \succ 0$, we have $\lambda_i > 0$; since $\lambda_{\min} \geq 1$, in fact $\lambda_i \geq 1$ for every i. Hence

$$\det(H) = \prod_{i=1}^{n} \lambda_i \geq 1.$$

Equality holds if and only if $\lambda_i = 1$ for every i: equivalently, $H = I$, or $B^*B = 0$. Finally, $B^*B = 0$ implies $\|Bx\|^2 = x^*B^*Bx = 0$ for every x. Thus $Bx = 0$ for all x, so $B = 0$.

EXERCISE 12.11 EXAM QUESTION

1. Work in $M_{3\times4}(\mathbb{R})$. Define

$$C = \operatorname{diag}(1,0,1), \qquad D = \operatorname{diag}(1,0,0,1),$$

and

$$W = \{X \in M_{3\times4}(\mathbb{R}) : CXD = 0\}.$$

Prove that W is a subspace of $M_{3\times4}(\mathbb{R})$.

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EXERCISE 12.11 **EXAM QUESTION CONTINUED** **12**

2. On $M_{3\times4}(\mathbb{R})$, define

$$\langle X, Y\rangle = \operatorname{tr}(X^T QY), \qquad Q = \operatorname{diag}(1,2,3).$$

Prove that this is an inner product. In particular, verify linearity in the second argument.

3. Now work in $\mathbb{R}^4$ with the standard inner product. Let

$$M = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}, \qquad W = N(M), \qquad v = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix}.$$

Compute the orthogonal projection $P_W(v)$.

WORKED SOLUTION

1. Define $L : M_{3\times4}(\mathbb{R}) \longrightarrow M_{3\times4}(\mathbb{R})$ by $L(X) = CXD$. For all $\alpha, \beta \in \mathbb{R}$ and all X_1, X_2,

$$\begin{aligned} L(\alpha X_1 + \beta X_2) &= C(\alpha X_1 + \beta X_2)D \\ &= \alpha C X_1 D + \beta C X_2 D \\ &= \alpha L(X_1) + \beta L(X_2). \end{aligned}$$

Thus L is linear and $W = \ker L$, so W is a subspace. The condition can also be read directly: multiplication by C zeros the second row, and multiplication by D zeros the second and third columns. Hence

$$CXD = 0 \iff x_{11} = x_{14} = x_{31} = x_{34} = 0.$$

These are homogeneous linear conditions on the entries of X.

2. Write $Q = \operatorname{diag}(q_1, q_2, q_3)$, where $(q_1, q_2, q_3) = (1, 2, 3)$. If $X = (x_{ij})$ and $Y = (y_{ij})$, computation of the diagonal entries gives

$$(X^T QY)_{jj} = \sum_{i=1}^{3} x_{ij} q_i y_{ij},$$

$$\langle X, Y\rangle = \sum_{j=1}^{4} \sum_{i=1}^{3} q_i x_{ij} y_{ij}.$$

For all $\alpha, \beta \in \mathbb{R}$,

$$\begin{aligned} \langle X, \alpha Y + \beta Z\rangle &= \sum_{j=1}^{4} \sum_{i=1}^{3} q_i x_{ij} (\alpha y_{ij} + \beta z_{ij}) \\ &= \alpha\langle X, Y\rangle + \beta\langle X, Z\rangle. \end{aligned}$$

This proves linearity in the second argument. Since all entries are real, the same calcu-

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EXERCISE 12.11 **EXAM QUESTION CONTINUED** **12**

lation also proves linearity in the first argument. In addition,

$$\langle X, Y\rangle = \sum_{j=1}^{4}\sum_{i=1}^{3} q_i y_{ij} x_{ij} = \langle Y, X\rangle,$$

so symmetry holds. Finally,

$$\langle X, X\rangle = \sum_{j=1}^{4}(x_{1j}^2 + 2x_{2j}^2 + 3x_{3j}^2) \geq 0.$$

Equality holds if and only if all entries x_{ij} vanish, or equivalently $X = 0$. We have proved linearity, symmetry, and positive definiteness; thus this is a real inner product.

3. Solve $Mx = 0$:

$$x_1 + x_3 = 0, \qquad x_2 + x_4 = 0.$$

Thus

$$W = \text{span}\{w_1, w_2\}, \qquad w_1 = \begin{pmatrix} 1 \\ 0 \\ -1 \\ 0 \end{pmatrix}, \quad w_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ -1 \end{pmatrix}.$$

The two vectors are linearly independent and orthogonal, and

$$\begin{aligned} \langle w_1, w_2\rangle &= 0, & \|w_1\|^2 &= \|w_2\|^2 = 2, \\ \langle v, w_1\rangle &= 1 - 3 = -2, & \langle v, w_2\rangle &= 2 - 4 = -2. \end{aligned}$$

The projection formula for an orthogonal basis therefore gives

$$\begin{aligned} P_W(v) &= \frac{\langle v, w_1\rangle}{\|w_1\|^2} w_1 + \frac{\langle v, w_2\rangle}{\|w_2\|^2} w_2 \\ &= -w_1 - w_2 = \begin{pmatrix} -1 \\ -1 \\ 1 \\ 1 \end{pmatrix}. \end{aligned}$$

As a check, $MP_W(v) = 0$, whereas

$$v - P_W(v) = \begin{pmatrix} 2 \\ 3 \\ 2 \\ 3 \end{pmatrix}$$

is orthogonal to both w_1 and w_2, since $2 - 2 = 0$ and $3 - 3 = 0$. The projection indeed belongs to W, and the residual is orthogonal to all of W.

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Additional practice

EXERCISE 12.12 **FOUNDATION**

On $\mathbb{R}^2$, consider the inner product $\langle x, y\rangle_G = 2x_1y_1 + x_2y_2$. Let $u = (1,1)^T$, $v = (2,0)^T$, and $W = \text{span}\{u\}$.

1. Compute $p = P_W(v)$ and the residual $r = v - p$.
2. Verify both conditions characterizing the projection and the identity $\|v\|_G^2 = \|p\|_G^2 + \|r\|_G^2$.

WORKED SOLUTION

1. The matrix $G = \text{diag}(2,1)$ is symmetric and positive definite, since $2x_1^2 + x_2^2 > 0$ for every $x \neq 0$. We have

$$\langle v, u\rangle_G = 4, \qquad \langle u, u\rangle_G = 3,$$

so

$$p = \frac{4}{3}u = \begin{pmatrix} 4/3 \\ 4/3 \end{pmatrix}, \qquad r = \begin{pmatrix} 2/3 \\ -4/3 \end{pmatrix}.$$

2. The expression $p = \frac{4}{3}u$ shows that $p \in W$. Also,

$$\langle r, u\rangle_G = 2 \cdot \frac{2}{3} - \frac{4}{3} = 0,$$

so $r \perp W$. These two conditions guarantee that p is the required projection. Finally,

$$\|v\|_G^2 = 8, \qquad \|p\|_G^2 = 2 \cdot \frac{16}{9} + \frac{16}{9} = \frac{16}{3}, \qquad \|r\|_G^2 = 2 \cdot \frac{4}{9} + \frac{16}{9} = \frac{8}{3}.$$

Indeed, $8 = \frac{16}{3} + \frac{8}{3}$.

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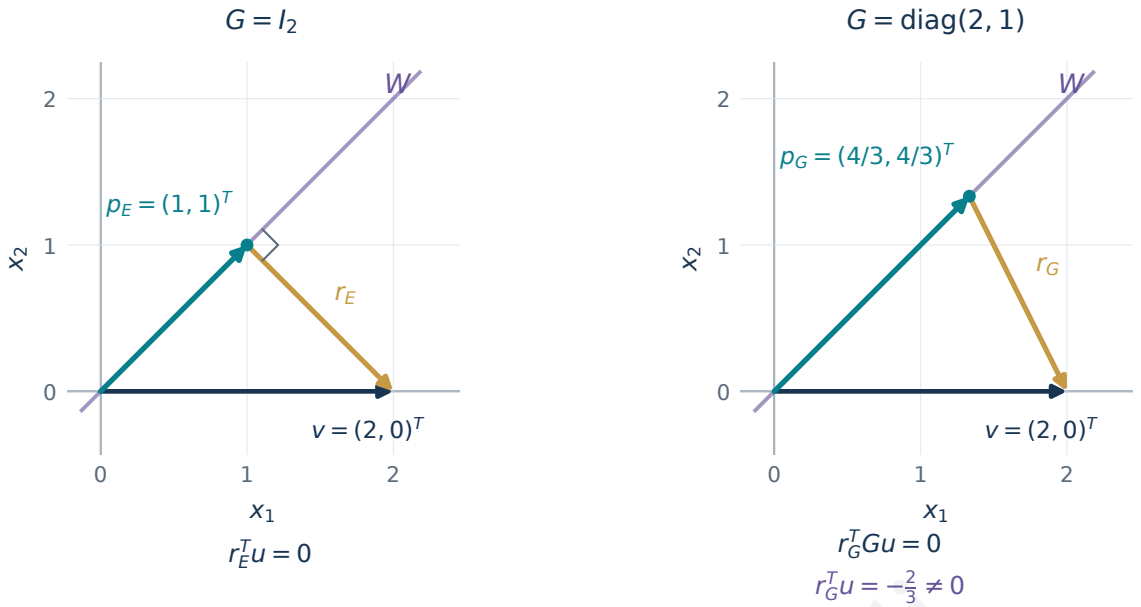


Figure 12.2. Both panels use $v = (2,0)^T$ and $W = \mathrm{span}\{u\}$, where $u = (1,1)^T$. On the left, the standard inner product gives $p_E = (1,1)^T$; on the right, $\langle x, y\rangle_G = x^T G y$ with $G = \mathrm{diag}(2,1)$ gives $p_G = (4/3, 4/3)^T$. Gold arrows are the residuals. On the right, $r_G^T G u = 0$, whereas $r_G^T u = -2/3 \neq 0$: orthogonality is with respect to the weighted inner product, so no Euclidean right-angle marker appears there.

EXERCISE 12.13 **CHALLENGE**

Let V be a finite-dimensional inner product space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, and let $U, W \leq V$. Write $P = P_U$ and $Q = P_W$, viewed as maps from V to itself.

1. Prove that PQ is the orthogonal projection onto $U \cap W$ if and only if $PQ = QP$.
2. In $\mathbb{R}^2$ with its standard inner product, take $U = \mathrm{span}\{(1,0)^T\}$ and $W = \mathrm{span}\{(1,1)^T\}$. Compute PQ and explain why a product of two orthogonal projections need not be an orthogonal projection.

WORKED SOLUTION

1. Use $\langle Px, y\rangle = \langle x, Py\rangle$, valid for every orthogonal projection, and the corresponding identity for Q. First suppose $PQ = QP$, and set $R = PQ$. For each $x \in V$, the vector Rx belongs to U; since $Rx = QPx$, it also belongs to W. For every $z \in U \cap W$, we have $Pz = Qz = z$, so $Rz = z$. Furthermore,

$$\langle Rx, z\rangle = \langle PQx, z\rangle = \langle Qx, Pz\rangle = \langle x, QPz\rangle = \langle x, z\rangle.$$

Thus $x - Rx \perp U \cap W$. The two defining conditions for projection show that $R = P_{U\cap W}$. Conversely, suppose $R = PQ = P_{U\cap W}$. For all $x, y \in V$,

$$\langle x, QPy\rangle = \langle PQx, y\rangle = \langle Rx, y\rangle = \langle x, Ry\rangle = \langle x, PQy\rangle.$$

Hence $\langle x, (QP - PQ)y\rangle = 0$ for all x, y. For fixed y, choose $x = (QP - PQ)y$; positive definiteness gives $(QP - PQ)y = 0$. Therefore $QP = PQ$.

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EXERCISE 12.13 **CHALLENGE CONTINUED**

2. Here

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad Q = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \qquad PQ = \begin{pmatrix} 1/2 & 1/2 \\ 0 & 0 \end{pmatrix}.$$

However,

$$(PQ)^2 = \begin{pmatrix} 1/4 & 1/4 \\ 0 & 0 \end{pmatrix} \neq PQ.$$

Every orthogonal projection satisfies $R^2 = R$, so PQ is not an orthogonal projection. Indeed, $U \cap W = \{0\}$, whose projection is the zero map, whereas $PQ \neq 0$.

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TUTORIAL 13

Gram--Schmidt and Orthogonal Diagonalization

MOTIVATION AND GUIDING IDEA

Gram--Schmidt replaces a list of directions with an orthonormal list spanning the same space, simplifying coordinate and projection computations. Orthogonal diagonalization imposes an additional requirement: the basis vectors must also be eigenvectors. For diagonalization, therefore, begin with the eigenspaces and apply the algorithm within each one; applying it to an arbitrary basis does not generally produce eigenvectors.

NOTATION AND CONVENTIONS

- v_j is an input vector, w_j is the residual after subtracting projections, and e_i is a unit vector already obtained. Never divide by $\|w_j\|$ when $w_j = 0$.
- $\text{span}\{v_1, \ldots, v_j\}$ is the span of the input prefix; it equals the span of the output vectors obtained up to that step.
- $\mathbb{R}_n[x]$ and $\mathbb{C}_n[x]$ denote polynomials of degree at most n, including the zero polynomial. Evaluation values and integrals are computed according to the inner product specified in the question.
- $E_\lambda(T) = \ker(T - \lambda I)$ is the eigenspace belonging to λ; an eigenvector is a **nonzero** element of this space.
- T^* is the adjoint map; $A^* = \overline{A}^T$ is the conjugate transpose. The identity $[T^*]_B = [T]_B^*$ requires an orthonormal basis B.
- $T = T^*$ means self-adjointness; $TT^* = T^*T$ means normality.
- $Q^TQ = QQ^T = I$ describes a real orthogonal matrix; $U^*U = UU^* = I$ describes a complex unitary matrix. In particular, $Q^{-1} = Q^T$ and $U^{-1} = U^*$.
- In $Q^TAQ = D$ or $U^*AU = D$, column j of the change-of-basis matrix is an eigenvector belonging to d_{jj}. The column order must match the order of the eigenvalues on the diagonal.

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DEFINITION 13.1

Conditions for the tools. In an inner product space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, Gram--Schmidt applies to any finite list, including a dependent list: after obtaining $e_1, \ldots, e_r$, compute

$$w_j = v_j - \sum_{i=1}^{r} \langle v_j, e_i \rangle e_i.$$

If $w_j = 0$, skip that input vector; otherwise append $e_{r+1} = w_j / \|w_j\|$. The resulting list is an orthonormal basis of the input span. For a linearly independent input, no vector is skipped. All projection coefficients use the convention of linearity in the first argument.
If V is finite dimensional and $T : V \to V$ is linear, there is a unique linear map $T^* : V \to V$ satisfying $\langle Tv, w \rangle = \langle v, T^* w \rangle$ for all v, w. Over $\mathbb{C}$, a map is normal if and only if it has an orthonormal eigenbasis. If it is self-adjoint, all its eigenvalues are real. Over $\mathbb{R}$, every symmetric matrix admits an orthogonal diagonalization; normality alone does not suffice for a real orthogonal diagonalization. For example,

$$R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad RR^T = R^T R = I,$$

but $R^2 = -I$, so R has no real eigenvector. When constructing an orthonormal eigenbasis, apply Gram--Schmidt **within each eigenspace** so that the linear combinations remain in that eigenspace.

MOTIVATION AND GUIDING IDEA

We construct orthonormal bases by systematically subtracting projections and compute projections in polynomial spaces. We study the conditions for orthogonal diagonalization over the reals and unitary diagonalization over the complex numbers, and use an orthonormal eigenbasis to prove normality.

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13.1 Review and Notation

RECALL AND TOOLS

Throughout this tutorial, the inner product space is over $\mathbb{R}$ or $\mathbb{C}$; in the complex case, our convention is:

1. $\langle x, x \rangle \geq 0$, with $\langle x, x \rangle = 0 \iff x = 0$.
2. $\langle x, y \rangle = \overline{\langle y, x \rangle}$.
3. Linearity in the first argument: $\langle ax + by, z \rangle = a\langle x, z \rangle + b\langle y, z \rangle$.

It follows that the second argument is conjugate linear: $\langle x, ay + bz \rangle = \overline{a}\langle x, y \rangle + \overline{b}\langle x, z \rangle$. This is the same convention as in the preceding tutorial; over $\mathbb{R}$, the two linearity conventions coincide.

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DEFINITION 13.2

Let $(V, \langle\cdot,\cdot\rangle)$ be an inner product space.

1. Vectors $u, v \in V$ are orthogonal if $\langle u, v\rangle = 0$.
2. A list $(e_1, \ldots, e_k)$ is orthonormal if $\langle e_i, e_j\rangle = \delta_{ij}$ for all i, j.
3. For each $x \in V$, define the norm $\|x\| := \sqrt{\langle x, x\rangle}$.

RECALL AND TOOLS

Orthogonal projection onto a line: if $u \neq 0$, the vector

$$\text{proj}_{\text{span}\{u\}}(v) := \frac{\langle v, u\rangle}{\langle u, u\rangle}\, u$$

is the unique vector of the form αu for which $v - \alpha u \perp u$.

13.2 The Gram--Schmidt Algorithm

We present Gram--Schmidt with immediate normalization at each step, constructing an orthonormal list directly.

Let $v_1, \ldots, v_k \in V$ be linearly independent. Define recursively:

1. Set $w_1 := v_1$, and then $e_1 := \dfrac{w_1}{\|w_1\|}$.
2. For $j = 2, \ldots, k$, first define
$$w_j := v_j - \sum_{i=1}^{j-1} \langle v_j, e_i\rangle\, e_i,$$
and then normalize:
$$e_j := \frac{w_j}{\|w_j\|}.$$

The main conclusions are:

1. $(e_1, \ldots, e_k)$ is orthonormal.
2. For every m, $\text{span}\{v_1, \ldots, v_m\} = \text{span}\{e_1, \ldots, e_m\}$.

Formal justification: assume inductively that $e_1, \ldots, e_{j-1}$ are orthonormal. For every $r < j$,

$$\begin{aligned}\langle w_j, e_r\rangle &= \left\langle v_j - \sum_{i=1}^{j-1} \langle v_j, e_i\rangle e_i,\ e_r \right\rangle \\ &= \langle v_j, e_r\rangle - \sum_{i=1}^{j-1} \langle v_j, e_i\rangle \langle e_i, e_r\rangle \\ &= \langle v_j, e_r\rangle - \langle v_j, e_r\rangle = 0.\end{aligned}$$

If $w_j = 0$, then v_j belongs to the span of $e_1, \ldots, e_{j-1}$, which by induction equals the span of $v_1, \ldots, v_{j-1}$. This contradicts the linear independence of the input. Thus $w_j \neq 0$, normalization is permitted, and $\|e_j\| = 1$. Also, $w_j \in \text{span}\{v_1, \ldots, v_j\}$, and the subtraction formula implies $v_j \in \text{span}\{e_1, \ldots, e_{j-1}, w_j\}$. Hence the corresponding prefix spans agree at every step.

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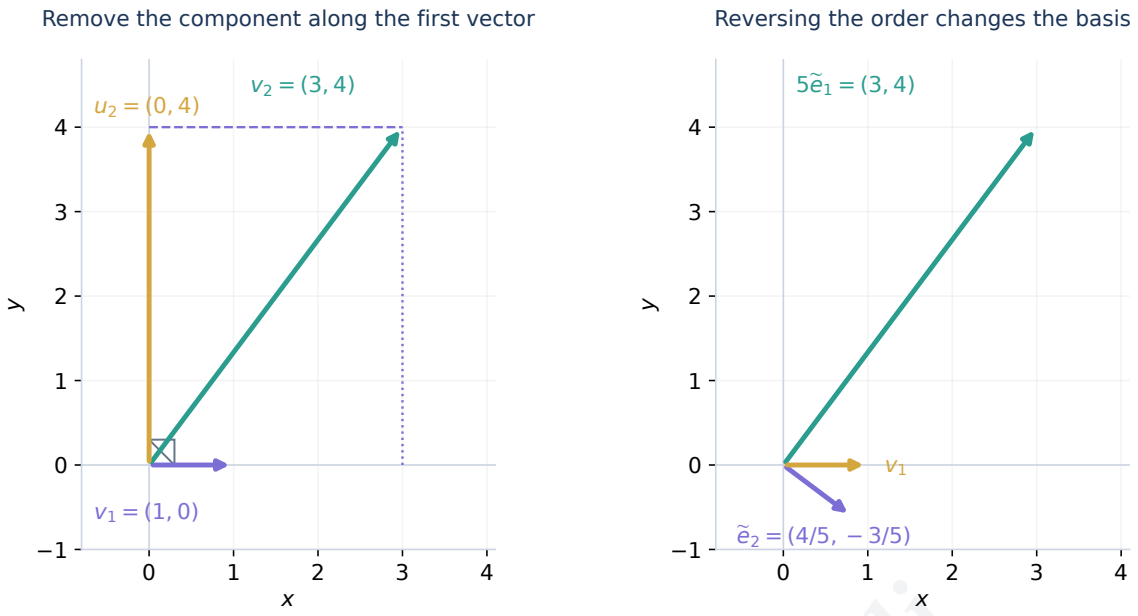


Figure 13.1. At each step, subtract from the next vector its components in the directions already constructed. The residual is orthogonal to the preceding directions; normalization produces the next unit vector.

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EXERCISE 13.1

Apply Gram--Schmidt, normalizing at every step, to the vectors

$$v_1 = (3, 4, 0)^T, \quad v_2 = (1, 3, 0)^T, \quad v_3 = (2, 1, 5)^T$$

in $\mathbb{R}^3$ with the standard inner product. Explicitly verify that the output is orthonormal.

WORKED SOLUTION

The first norm is $\|v_1\| = \sqrt{9 + 16} = 5$, so $e_1 = (3/5, 4/5, 0)^T$. Compute the projection of v_2:

$$\langle v_2, e_1 \rangle = \frac{3}{5} + \frac{12}{5} = 3,$$

$$w_2 = v_2 - 3e_1 = (1 - 9/5, 3 - 12/5, 0)^T = (-4/5, 3/5, 0)^T.$$

Since $\|w_2\|^2 = 16/25 + 9/25 = 1$, we have $e_2 = w_2$. The two coefficients for v_3 are

$$\langle v_3, e_1 \rangle = 6/5 + 4/5 = 2, \qquad \langle v_3, e_2 \rangle = -8/5 + 3/5 = -1.$$

Thus

$$w_3 = v_3 - 2e_1 - (-1)e_2 = (2 - 6/5 - 4/5, 1 - 8/5 + 3/5, 5)^T = (0, 0, 5)^T.$$

Its norm is 5, so $e_3 = (0, 0, 1)^T$. The output is

$$e_1 = \tfrac{1}{5}(3, 4, 0)^T, \qquad e_2 = \tfrac{1}{5}(-4, 3, 0)^T, \qquad e_3 = (0, 0, 1)^T.$$

The first two norms are $\sqrt{25/25} = 1$, and the third is 1. The inner product $\langle e_1, e_2 \rangle = (-12 + 12)/25 = 0$, and both inner products with e_3 vanish. We have obtained an orthonormal basis with rational coordinates only.

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EXERCISE 13.2

In $\mathbb{C}^2$ with $\langle x, y\rangle = x_1\overline{y_1} + x_2\overline{y_2}$, let $v_1 = (3, 4i)^T$ and $v_2 = (1, i)^T$. Apply Gram--Schmidt with immediate normalization, taking care to conjugate the correct argument in the projection coefficient.

WORKED SOLUTION

First, $\|v_1\|^2 = 9 + |4i|^2 = 25$, so $e_1 = (3, 4i)^T/5$. With the convention of linearity in the first argument,

$$\langle v_2, e_1\rangle = 1 \cdot \frac{3}{5} + i \cdot \frac{\overline{4i}}{5} = \frac{3}{5} + i\frac{-4i}{5} = \frac{7}{5}.$$

Subtract the projection:

$$w_2 = v_2 - \frac{7}{5}e_1 = \left(1 - \frac{21}{25}, i - \frac{28i}{25}\right)^T = \frac{1}{25}(4, -3i)^T.$$

Thus $\|w_2\|^2 = (16 + 9)/625 = 1/25$, so $\|w_2\| = 1/5$, and $e_2 = (4, -3i)^T/5$. Check:

$$\|e_2\|^2 = \frac{16 + 9}{25} = 1, \qquad \langle e_1, e_2\rangle = \frac{3 \cdot 4 + (4i)(3i)}{25} = 0.$$

The output (e_1, e_2) is orthonormal. The product $i\overline{4i} = -4i^2 = 4$ illustrates why conjugation is essential.

EXERCISE 13.3

On $\mathbb{R}[t]_{\le 2}$, define the inner product

$$\langle p, q\rangle := \int_0^1 p(t)\, q(t)\, dt.$$

Apply Gram--Schmidt to $(v_1, v_2, v_3) = (1, t, t^2)$ to obtain an orthonormal list (e_1, e_2, e_3), normalizing immediately at every step.

WORKED SOLUTION

1. $$\|1\|^2 = \int_0^1 1\, dt = 1 \quad \Rightarrow \quad e_1 = 1.$$

2. Compute
$$\langle t, e_1\rangle = \int_0^1 1 \cdot t\, dt = \frac{1}{2}, \qquad w_2 = t - \langle t, e_1\rangle e_1 = t - \frac{1}{2}.$$
Norm:
$$\|w_2\|^2 = \int_0^1 \left(t - \frac{1}{2}\right)^2 dt = \int_0^1 \left(t^2 - t + \frac{1}{4}\right) dt = \frac{1}{3} - \frac{1}{2} + \frac{1}{4} = \frac{1}{12},$$
and hence
$$e_2 = \frac{w_2}{\|w_2\|} = \frac{t - \frac{1}{2}}{\sqrt{1/12}} = 2\sqrt{3}\left(t - \frac{1}{2}\right).$$

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EXERCISE 13.3 CONTINUED

3. First compute

$$\langle t^2, e_1\rangle = \int_0^1 t^2 dt = \frac{1}{3}.$$

and

$$\langle t^2, e_2\rangle = \int_0^1 2\sqrt{3}\left(t-\frac{1}{2}\right)t^2 dt = 2\sqrt{3}\int_0^1 \left(t^3-\frac{1}{2}t^2\right)dt = 2\sqrt{3}\left(\frac{1}{4}-\frac{1}{6}\right) = \frac{\sqrt{3}}{6}.$$

Thus

$$\begin{aligned} w_3 &= t^2 - \langle t^2, e_1\rangle e_1 - \langle t^2, e_2\rangle e_2 \\ &= t^2 - \frac{1}{3} - \frac{\sqrt{3}}{6}\cdot 2\sqrt{3}\left(t-\frac{1}{2}\right) \\ &= t^2 - \frac{1}{3} - \left(t-\frac{1}{2}\right) \\ &= t^2 - t + \frac{1}{6}. \end{aligned}$$

Norm:

$$\begin{aligned} \|w_3\|^2 &= \int_0^1 \left(t^2 - t + \frac{1}{6}\right)^2 dt \\ &= \int_0^1 \left(t^4 - 2t^3 + \frac{4}{3}t^2 - \frac{1}{3}t + \frac{1}{36}\right) dt \\ &= \frac{1}{5} - \frac{1}{2} + \frac{4}{9} - \frac{1}{6} + \frac{1}{36} = \frac{1}{180}, \end{aligned}$$

Therefore

$$\|w_3\| = \sqrt{\frac{1}{180}} = \frac{1}{6\sqrt{5}}.$$

and hence

$$e_3 = \frac{w_3}{\|w_3\|} = 6\sqrt{5}\left(t^2 - t + \frac{1}{6}\right).$$

The resulting basis is:

$$e_1 = 1, \qquad e_2 = 2\sqrt{3}\left(t-\frac{1}{2}\right), \qquad e_3 = 6\sqrt{5}\left(t^2 - t + \frac{1}{6}\right).$$

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EXERCISE 13.4

In $\mathbb{R}^2$, compare the Gram--Schmidt outputs for the two orders (v_1, v_2) and (v_2, v_1), where $v_1 = (1,0)^T$ and $v_2 = (3,4)^T$. Explain what remains the same despite the different outputs.

WORKED SOLUTION

For the first order, $e_1 = (1,0)^T$. The projection coefficient is $\langle v_2, e_1\rangle = 3$, so

$$w_2 = (3,4)^T - 3(1,0)^T = (0,4)^T, \qquad e_2 = (0,1)^T.$$

For the reversed order, start with $\widetilde{e}_1 = (3,4)^T/5$. The projection coefficient of v_1 onto it is $3/5$,

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EXERCISE 13.4 CONTINUED

so

$$\widetilde{w}_2 = (1,0)^T - \frac{3}{5}\frac{1}{5}(3,4)^T = \left(\frac{16}{25}, -\frac{12}{25}\right)^T.$$

Its norm is $\sqrt{256+144}/25 = 20/25 = 4/5$. Normalization gives

$$\widetilde{e}_2 = (4/5, -3/5)^T.$$

Thus the first order gives the standard basis, whereas the second gives $((3/5,4/5)^T, (4/5,-3/5)^T)$. Both are orthonormal and span the same plane. At every step, the span of the input prefix is preserved, but changing the input order changes the first direction and can therefore change the entire output.

EXERCISE 13.5

Let $V = \mathbb{C}_3[x]$ be the space of complex polynomials of degree at most 3. For $f, g \in V$, define

$$\langle f, g\rangle := f(-1)\,\overline{g(-1)} + f(0)\,\overline{g(0)} + f(1)\,\overline{g(1)} + f(2)\,\overline{g(2)}.$$

1. Prove that this is an inner product on V.
2. Let $W = \mathbb{C}_2[x] \subseteq V$. Find an orthonormal basis of W.
3. Find $W^{\perp}$.

WORKED SOLUTION

1. Verify the axioms.

   (a) Positive definiteness: for every $f \in V$,

   $$\langle f, f\rangle = \sum_{t\in\{-1,0,1,2\}} f(t)\,\overline{f(t)} = \sum_{t\in\{-1,0,1,2\}} |f(t)|^2 \geq 0.$$

   If $\langle f, f\rangle = 0$, then $f(-1) = f(0) = f(1) = f(2) = 0$. A nonzero polynomial of degree at most 3 has at most 3 distinct roots, so necessarily $f \equiv 0$.

   (b) Conjugate symmetry: for every $f, g \in V$,

   $$\overline{\langle g, f\rangle} = \overline{\sum_t g(t)\,\overline{f(t)}} = \sum_t f(t)\,\overline{g(t)} = \langle f, g\rangle.$$

   (c) Linearity in the first argument: for all $f, g, h \in V$ and all $\alpha, \beta \in \mathbb{C}$,

   $$\begin{aligned}\langle \alpha f + \beta g, h\rangle &= \sum_t \big(\alpha f(t) + \beta g(t)\big)\overline{h(t)} \\ &= \alpha \sum_t f(t)\overline{h(t)} + \beta \sum_t g(t)\overline{h(t)} \\ &= \alpha\langle f, h\rangle + \beta\langle g, h\rangle.\end{aligned}$$

   Thus this is an inner product on V.

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EXERCISE 13.5 CONTINUED

2. Apply Gram--Schmidt to $(1, x, x^2)$ and normalize at every step. Write $v_1 = 1$, $v_2 = x$, $v_3 = x^2$.

   First,

$$\|v_1\|^2 = \langle 1, 1\rangle = 1 + 1 + 1 + 1 = 4, \qquad e_1 = \frac{v_1}{\|v_1\|} = \frac{1}{2}.$$

   Next,

$$\langle v_2, e_1\rangle = \left\langle x, \frac{1}{2}\right\rangle = \frac{1}{2}\big((-1) + 0 + 1 + 2\big) = 1, \qquad w_2 = v_2 - \langle v_2, e_1\rangle e_1 = x - \frac{1}{2}.$$

   Compute the norm:

$$\begin{aligned}\|w_2\|^2 &= \left(-\frac{3}{2}\right)^2 + \left(-\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2 + \left(\frac{3}{2}\right)^2 \\ &= \frac{9}{4} + \frac{1}{4} + \frac{1}{4} + \frac{9}{4} = 5.\end{aligned}$$

   Therefore

$$e_2 = \frac{w_2}{\|w_2\|} = \frac{x - \frac{1}{2}}{\sqrt{5}}.$$

   Next,

$$\langle v_3, e_1\rangle = \left\langle x^2, \frac{1}{2}\right\rangle = \frac{1}{2}(1 + 0 + 1 + 4) = 3,$$

   and

$$\begin{aligned}\langle v_3, e_2\rangle &= \left\langle x^2, \frac{x - \frac{1}{2}}{\sqrt{5}}\right\rangle \\ &= \frac{1}{\sqrt{5}}\left(\left(-\frac{3}{2}\right)\cdot 1 + \left(-\frac{1}{2}\right)\cdot 0 + \left(\frac{1}{2}\right)\cdot 1 + \left(\frac{3}{2}\right)\cdot 4\right) \\ &= \frac{5}{\sqrt{5}} = \sqrt{5}.\end{aligned}$$

   Thus

$$w_3 = v_3 - \langle v_3, e_1\rangle e_1 - \langle v_3, e_2\rangle e_2 = x^2 - 3\cdot\frac{1}{2} - \sqrt{5}\cdot\frac{x - \frac{1}{2}}{\sqrt{5}} = x^2 - x - 1.$$

   Compute the norm:

$$\|w_3\|^2 = 1^2 + (-1)^2 + (-1)^2 + 1^2 = 4, \qquad e_3 = \frac{w_3}{\|w_3\|} = \frac{x^2 - x - 1}{2}.$$

   We have obtained the following orthonormal basis of W:

$$e_1 = \frac{1}{2}, \qquad e_2 = \frac{x - \frac{1}{2}}{\sqrt{5}}, \qquad e_3 = \frac{x^2 - x - 1}{2}.$$

3. Since $\dim V = 4$ and $\dim W = 3$, we have $\dim(W^\perp) = 1$. For $p \in W^\perp$, write its values

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EXERCISE 13.5 CONTINUED

at the four points as

$$A = p(-1), \quad B = p(0), \quad C = p(1), \quad D = p(2).$$

The condition $p \perp W$ is equivalent to $\langle p, q \rangle = 0$ for each $q \in \{1, x, x^2\}$; that is,

$$A+B+C+D=0, \qquad (-1)A+0\cdot B+1\cdot C+2D=0, \qquad 1\cdot A+0\cdot B+1\cdot C+4D=0.$$

Solving the system gives a vector of values proportional to $(-1, 3, -3, 1)$, so we may choose

$$p(-1) = -1, \quad p(0) = 3, \quad p(1) = -3, \quad p(2) = 1.$$

Find a polynomial of degree at most 3 with these values: write $p(x) = ax^3+bx^2+cx+d$. From $p(0) = 3$, we obtain $d = 3$, and also

$$\begin{aligned} p(1) = -3 &\implies a+b+c=-6, \\ p(-1) = -1 &\implies -a+b-c=-4, \\ p(2) = 1 &\implies 8a+4b+2c=-2. \end{aligned}$$

The first two equations give $b = -5$, hence $a+c = -1$. The third gives $4a+2b+c = -1$, so $4a - 10 + c = -1$, or $4a + c = 9$. Subtracting yields $3a = 10$, so

$$a = \frac{10}{3}, \qquad c = -\frac{13}{3}, \qquad d = 3, \qquad b = -5.$$

Multiply by 3 to avoid fractions and obtain a spanning vector:

$$u(x) := 10x^3 - 15x^2 - 13x + 9.$$

Hence

$$W^{\perp} = \operatorname{span}\{u\}.$$

For a unit vector, note that $\|u\|^2 = |-3|^2 + |9|^2 + |-9|^2 + |3|^2 = 180$, so

$$\frac{u}{6\sqrt{5}}$$

is an orthonormal basis of $W^{\perp}$.

EXERCISE 13.6

Continue with $V = \mathbb{C}_3[x]$ and the inner product

$$\langle f, g \rangle = f(-1)\,\overline{g(-1)} + f(0)\,\overline{g(0)} + f(1)\,\overline{g(1)} + f(2)\,\overline{g(2)},$$

and take $W = \mathbb{C}_2[x]$.
Write $P_W : V \to W$ for the orthogonal projection onto W.

1. Prove that P_W is linear and that $P_W^2 = P_W$.
2. Compute $P_W(x^3)$ explicitly using the orthonormal basis of W found in the preceding exercise.

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EXERCISE 13.6 CONTINUED

3. Define $T : V \to V$ by
$$(Tf)(x) := f(x+1) - f(x).$$
Find all $q \in W$ such that $T(q) \in W^{\perp}$. Is the solution unique? Also find the unique solution satisfying $\langle q, 1\rangle = 1$, and determine whether $\|q\| = 1$ would guarantee uniqueness.

WORKED SOLUTION

Use the orthonormal basis of W:
$$e_1 = \frac{1}{2}, \qquad e_2 = \frac{x - \frac{1}{2}}{\sqrt{5}}, \qquad e_3 = \frac{x^2 - x - 1}{2}.$$

1. Every $f \in V$ has a unique orthogonal decomposition $f = w + u$, where $w \in W$ and $u \in W^{\perp}$. Define $P_W(f) := w$. Linearity: if $f = w + u$ and $g = w' + u'$ with $w, w' \in W$ and $u, u' \in W^{\perp}$, then
$$\alpha f + \beta g = (\alpha w + \beta w') + (\alpha u + \beta u')$$
where $\alpha w + \beta w' \in W$ and $\alpha u + \beta u' \in W^{\perp}$, so
$$P_W(\alpha f + \beta g) = \alpha w + \beta w' = \alpha P_W(f) + \beta P_W(g).$$
Furthermore, if $f = w + u$, then $P_W(f) = w \in W$, so $P_W(P_W(f)) = P_W(w) = w$; hence $P_W^2 = P_W$.

2. Since e_1, e_2, e_3 are orthonormal, for every $f \in V$,
$$P_W(f) = \sum_{i=1}^{3} \langle f, e_i\rangle e_i.$$
Compute this for $f(x) = x^3$.
First,
$$\langle x^3, e_1\rangle = \left\langle x^3, \frac{1}{2}\right\rangle = \frac{1}{2}\left((-1)^3 + 0^3 + 1^3 + 2^3\right) = \frac{1}{2}(-1 + 0 + 1 + 8) = 4.$$
Second,
$$\langle x^3, e_2\rangle = \left\langle x^3, \frac{x - \frac{1}{2}}{\sqrt{5}}\right\rangle = \frac{1}{\sqrt{5}} \sum_{t \in \{-1,0,1,2\}} \left(t - \tfrac{1}{2}\right) t^3.$$
Compute the sum:
$$\begin{aligned} t = -1: &\quad (-\tfrac{3}{2}) \cdot (-1) = \tfrac{3}{2}, \\ t = 0: &\quad (-\tfrac{1}{2}) \cdot 0 = 0, \\ t = 1: &\quad (\tfrac{1}{2}) \cdot 1 = \tfrac{1}{2}, \\ t = 2: &\quad (\tfrac{3}{2}) \cdot 8 = 12. \end{aligned}$$
Thus the sum is 14, giving
$$\langle x^3, e_2\rangle = \frac{14}{\sqrt{5}}.$$

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EXERCISE 13.6 CONTINUED

Third,

$$\langle x^3, e_3\rangle = \left\langle x^3, \frac{x^2-x-1}{2}\right\rangle = \frac{1}{2}\sum_t (t^2-t-1)\,t^3.$$

Compute:

$$\begin{aligned} t=-1: &\quad (1+1-1)\cdot(-1) = -1,\\ t=0: &\quad (-1)\cdot 0 = 0,\\ t=1: &\quad (1-1-1)\cdot 1 = -1,\\ t=2: &\quad (4-2-1)\cdot 8 = 8. \end{aligned}$$

Thus the sum is 6, giving

$$\langle x^3, e_3\rangle = 3.$$

Consequently,

$$P_W(x^3) = 4e_1 + \frac{14}{\sqrt{5}}e_2 + 3e_3 = 4\cdot\frac{1}{2} + \frac{14}{\sqrt{5}}\cdot\frac{x-\frac{1}{2}}{\sqrt{5}} + 3\cdot\frac{x^2-x-1}{2}.$$

that is,

$$P_W(x^3) = 2 + \frac{14}{5}\left(x-\frac{1}{2}\right) + \frac{3}{2}(x^2-x-1) = \frac{3}{2}x^2 + \frac{13}{10}x - \frac{9}{10}.$$

3. We found $W^\perp = \text{span}\{u\}$, where

$$u(x) = 10x^3 - 15x^2 - 13x + 9.$$

Seek $q \in W$ with $T(q) \in W^\perp$: that is, there is $\lambda \in \mathbb{C}$ such that

$$T(q) = \lambda u.$$

Write $q(x) = ax^2 + bx + c$.

Compute:

$$q(x+1) = a(x+1)^2 + b(x+1) + c = ax^2 + (2a+b)x + (a+b+c),$$

Therefore

$$(Tq)(x) = q(x+1) - q(x) = 2ax + (a+b).$$

Thus $T(q)$ is a polynomial of degree at most 1.

On the other hand, if $T(q) \in W^\perp$, then $T(q) \in W^\perp \cap W$ (because its degree is at most 1, so $T(q) \in W$). But $W \cap W^\perp = \{0\}$, so necessarily

$$T(q) = 0.$$

Hence

$$2ax + (a+b) \equiv 0 \quad \Rightarrow \quad 2a = 0,\ a+b = 0 \quad \Rightarrow \quad a = 0,\ b = 0.$$

and therefore $q(x) = c$ is constant.

Conversely, $T(q) = 0 \in W^\perp$ for every constant, so every $q = c$ is a solution. Without an additional condition, the solution is therefore not unique. Under the condition $\langle q, 1\rangle = 1$, we obtain $4c = 1$, so the unique solution is $q = \frac{1}{4}$. In contrast, the condition $\|q\| = 1$

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EXERCISE 13.6 CONTINUED

does not guarantee uniqueness. For $q = c$,

$$\|q\|^2 = \langle c, c\rangle = |c|^2 + |c|^2 + |c|^2 + |c|^2 = 4|c|^2,$$

so $\|q\| = 1 \Rightarrow |c| = \frac{1}{2}$. Over the reals there are two possibilities, $c = \pm\frac{1}{2}$; over the complex numbers there are infinitely many possibilities of modulus $\frac{1}{2}$.

EXERCISE 13.7 EXAM QUESTION

Let $V = \mathbb{R}_4[x]$, the space of real polynomials of degree at most 4, including the zero polynomial. Define $L : V \to \mathbb{R}^3$ by

$$L(p) = (p(0), p(1), p'(1)).$$

Equip V with the inner product

$$\langle p, q\rangle = \int_0^1 p(t)q(t)\,dt + p(0)q(0) + p(1)q(1).$$

Write $W = \ker L$ and $s(x) = 1 + x + x^2 + x^3 + x^4$.

1. Prove that L is linear.
2. Compute $\ker L$ and $\operatorname{Im} L$, and find a basis of each.
3. Compute the orthogonal projection $P_W(s)$ for the given inner product.

WORKED SOLUTION

1. Let $p, q \in V$ and $a, b \in \mathbb{R}$. By the linearity of evaluation and differentiation,

$$\begin{aligned} L(ap + bq) &= ((ap + bq)(0), (ap + bq)(1), (ap + bq)'(1)) \\ &= (ap(0) + bq(0), ap(1) + bq(1), ap'(1) + bq'(1)) \\ &= aL(p) + bL(q). \end{aligned}$$

   so L is linear.

2. The condition $p \in \ker L$ is equivalent to the three conditions

$$p(0) = 0, \qquad p(1) = 0, \qquad p'(1) = 0.$$

   The first says that x divides p. The last two say that 1 is a root of multiplicity at least 2: if $p(x) = (x-1)q(x)$, then $p'(x) = q(x) + (x-1)q'(x)$, so $p'(1) = q(1)$. Thus $(x-1)^2$ divides p. Since x and $(x-1)^2$ have no common root, and p has degree at most 4,

$$W = \{x(x-1)^2(a + bx) : a, b \in \mathbb{R}\}.$$

   Define

$$w_1 = x(x-1)^2 = x^3 - 2x^2 + x, \qquad w_2 = x^2(x-1)^2 = x^4 - 2x^3 + x^2.$$

   The polynomials w_1, w_2 span W and are linearly independent, for example because they

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EXERCISE 13.7 **EXAM QUESTION CONTINUED** **13**

have different degrees. Hence (w_1, w_2) is a basis of W, and $\dim W = 2$.

To determine the image, exhibit polynomials mapped to the three standard basis vectors:

$$h_1 = (x-1)^2, \qquad h_2 = 2x - x^2, \qquad h_3 = x^2 - x.$$

A direct calculation gives

$$L(h_1) = (1,0,0), \qquad L(h_2) = (0,1,0), \qquad L(h_3) = (0,0,1).$$

Thus $\operatorname{Im} L = \mathbb{R}^3$, with the standard basis as a basis of the image. As a dimension check, $\dim \ker L + \dim \operatorname{Im} L = 2 + 3 = 5 = \dim V$.

3. Seek $p = P_W(s)$ in the form $p = aw_1 + bw_2$. The definition of orthogonal projection requires $p \in W$ and $s - p \in W^\perp$. Since w_1, w_2 form a basis of W, it suffices to require

$$\langle s - p, w_1 \rangle = 0, \qquad \langle s - p, w_2 \rangle = 0.$$

Every $w \in W$ satisfies $w(0) = w(1) = 0$, so the endpoint terms in the inner product vanish whenever one factor is w_1 or w_2. First compute the inner products between the basis vectors:

$$\begin{aligned}
\langle w_1, w_1 \rangle &= \int_0^1 (t^2 - 4t^3 + 6t^4 - 4t^5 + t^6)\, dt \\
&= \frac{1}{3} - 1 + \frac{6}{5} - \frac{2}{3} + \frac{1}{7} = \frac{1}{105}, \\
\langle w_1, w_2 \rangle &= \int_0^1 (t^3 - 4t^4 + 6t^5 - 4t^6 + t^7)\, dt \\
&= \frac{1}{4} - \frac{4}{5} + 1 - \frac{4}{7} + \frac{1}{8} = \frac{1}{280}, \\
\langle w_2, w_2 \rangle &= \int_0^1 (t^4 - 4t^5 + 6t^6 - 4t^7 + t^8)\, dt \\
&= \frac{1}{5} - \frac{2}{3} + \frac{6}{7} - \frac{1}{2} + \frac{1}{9} = \frac{1}{630}.
\end{aligned}$$

Multiplying out the polynomials also gives

$$\begin{aligned}
\langle s, w_1 \rangle &= \int_0^1 (t^7 - t^6 - t^2 + t)\, dt = \frac{1}{8} - \frac{1}{7} - \frac{1}{3} + \frac{1}{2} = \frac{25}{168}, \\
\langle s, w_2 \rangle &= \int_0^1 (t^8 - t^7 - t^3 + t^2)\, dt = \frac{1}{9} - \frac{1}{8} - \frac{1}{4} + \frac{1}{3} = \frac{5}{72}.
\end{aligned}$$

The orthogonality conditions are therefore

$$\begin{pmatrix} 1/105 & 1/280 \\ 1/280 & 1/630 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 25/168 \\ 5/72 \end{pmatrix}.$$

Multiplication by 2520 gives the system

$$24a + 9b = 375, \qquad 9a + 4b = 175.$$

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EXERCISE 13.7 **EXAM QUESTION CONTINUED**

Multiply the first equation by 4 and the second by 9, and subtract:

$$(96a + 36b) - (81a + 36b) = 1500 - 1575 \quad \Longrightarrow \quad 15a = -75.$$

Hence $a = -5$; the second equation gives $-45 + 4b = 175$, so $b = 55$. We conclude that

$$\begin{aligned} P_W(s) &= -5x(x-1)^2 + 55x^2(x-1)^2 \\ &= 55x^4 - 115x^3 + 65x^2 - 5x. \end{aligned}$$

The polynomial belongs to W by the factorization in the first line, and the two inner product equations ensure that its difference from s is orthogonal to all of W. These are exactly the two conditions that uniquely characterize the orthogonal projection.

13.3 Adjoints, Normality, and Diagonalization in an Orthonormal Basis

DEFINITION 13.3

Let V be a finite-dimensional inner product space over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, and let $T : V \to V$ be linear. The **adjoint** T^* is the unique linear map satisfying

$$\langle Tv, w \rangle = \langle v, T^* w \rangle \qquad \text{for all } v, w \in V.$$

In an orthonormal basis, if A represents T, then $A^* = \overline{A}^{\,T}$ represents T^*. We call T **self-adjoint** if $T^* = T$, and **normal** if $TT^* = T^*T$. A matrix $U \in M_n(\mathbb{C})$ is **unitary** if $U^*U = UU^* = I$; its columns form an orthonormal basis of $\mathbb{C}^n$. Over the reals, a square matrix Q is **orthogonal** if $Q^TQ = QQ^T = I$.

PROPOSITION 13.1

On a finite-dimensional complex inner product space, every self-adjoint map has an orthonormal eigenbasis, and all its eigenvalues are real. This is the spectral theorem for self-adjoint maps. Thus for every Hermitian matrix $A = A^*$, there are a unitary matrix U and a real diagonal matrix D such that

$$U^*AU = D, \qquad A = UDU^*.$$

Column j of U is an eigenvector of A belonging to d_{jj}. Over $\mathbb{R}$, every real symmetric matrix $A = A^T$ has a real orthonormal eigenbasis; hence there is an orthogonal matrix Q such that

$$Q^TAQ = D, \qquad A = QDQ^T.$$

Diagonalization by an orthogonal matrix is called **orthogonal diagonalization**; over the complex numbers, we use the term **unitary diagonalization**.

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RECALL AND TOOLS

To find an orthogonal diagonalization of a real symmetric matrix, find a basis of each eigenspace, apply Gram--Schmidt within each eigenspace as needed, and normalize. Eigenvectors of a symmetric matrix belonging to distinct eigenvalues are already orthogonal. Place the unit vectors in the columns of Q, and place the corresponding eigenvalues in the same order on the diagonal of D. Over the complex numbers, the same construction applies to a Hermitian matrix, using U^* in place of Q^T.

EXERCISE 13.8

Let V be a finite-dimensional complex inner product space, and let $T : V \to V$ be linear. Suppose every eigenvector of $T + T^*$ is also an eigenvector of $T - T^*$. Prove that T is normal.

WORKED SOLUTION

Write

$$H := T + T^*, \qquad K := T - T^*.$$

We aim to find an orthonormal basis in which both maps act by scalar multiplication on every basis vector. First compute the adjoints:

$$H^* = (T + T^*)^* = T^* + (T^*)^* = T^* + T = H,$$

and

$$K^* = (T - T^*)^* = T^* - (T^*)^* = T^* - T = -K.$$

Thus H is self-adjoint. If $V = \{0\}$, the result is immediate. Otherwise, the spectral theorem provides an orthonormal basis $B = (e_1, \dots, e_n)$ and real numbers $h_1, \dots, h_n$ such that

$$He_j = h_j e_j \qquad (1 \le j \le n).$$

Every e_j is an eigenvector of H, so the assumption provides $k_j \in \mathbb{C}$ such that

$$Ke_j = k_j e_j.$$

The number k_j may depend on j; we need not assume that the eigenvalues of H are distinct. In the basis B, both matrices are diagonal:

$$[H]_B = D_H := \operatorname{diag}(h_1, \dots, h_n), \qquad [K]_B = D_K := \operatorname{diag}(k_1, \dots, k_n).$$

For every j,

$$HKe_j = H(k_j e_j) = k_j h_j e_j, \qquad KHe_j = K(h_j e_j) = h_j k_j e_j.$$

Scalars commute, so $HKe_j = KHe_j$ for every basis vector. Hence $HK = KH$ on all of V. Now recover T and T^*:

$$H + K = 2T, \qquad H - K = 2T^*, \qquad T = \frac{H + K}{2}, \quad T^* = \frac{H - K}{2}.$$

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EXERCISE 13.8 CONTINUED

Compute the two products separately, without assuming in advance that the factors commute:

$$\begin{aligned} TT^* &= \frac{1}{4}(H+K)(H-K) \\ &= \frac{1}{4}\big(HH - HK + KH - KK\big) \\ &= \frac{1}{4}\big(H^2 - HK + KH - K^2\big) \\ &= \frac{1}{4}(H^2 - K^2), \end{aligned}$$

where the last line uses $HK = KH$. Similarly,

$$\begin{aligned} T^*T &= \frac{1}{4}(H-K)(H+K) \\ &= \frac{1}{4}\big(HH + HK - KH - KK\big) \\ &= \frac{1}{4}\big(H^2 + HK - KH - K^2\big) \\ &= \frac{1}{4}(H^2 - K^2). \end{aligned}$$

Thus $TT^* = T^*T$, so T is normal.
We also make explicit the connection with unitary diagonalization and the order of the matrices. First choose a fixed orthonormal basis of V, write $A = [T]$ in that basis, and form U from the coordinate columns of $e_1, \dots, e_n$ in the same basis. Then U is unitary, and

$$U^*(A + A^*)U = D_H, \qquad U^*(A - A^*)U = D_K.$$

Adding the identities and dividing by 2 gives

$$U^*AU = D := \frac{1}{2}(D_H + D_K), \qquad A = UDU^*.$$

Thus $[T]_B = D$ and $[T^*]_B = D^*$. For diagonal matrices,

$$(DD^*)_{ij} = \sum_{\ell=1}^{n} d_{i\ell}\overline{d_{j\ell}} = \begin{cases} |d_{ii}|^2, & i = j, \\ 0, & i \neq j, \end{cases}$$

and

$$(D^*D)_{ij} = \sum_{\ell=1}^{n} \overline{d_{\ell i}}\, d_{\ell j} = \begin{cases} |d_{ii}|^2, & i = j, \\ 0, & i \neq j. \end{cases}$$

We can therefore check the equality again in matrix form:

$$\begin{aligned} AA^* &= (UDU^*)(UD^*U^*) = UD(U^*U)D^*U^* = U(DD^*)U^*, \\ A^*A &= (UD^*U^*)(UDU^*) = UD^*(U^*U)DU^* = U(D^*D)U^*. \end{aligned}$$

The two results agree. The basis that diagonalizes the self-adjoint part also diagonalizes the other part by the hypothesis, and consequently diagonalizes T itself.

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Additional practice

EXERCISE 13.9 **FOUNDATION**

In $\mathbb{R}^3$ with the standard inner product, consider the list

$$v_1 = (1, 1, 0)^T, \qquad v_2 = (2, 2, 0)^T, \qquad v_3 = (1, 0, 1)^T.$$

Apply Gram--Schmidt, explicitly handling any zero residual. Find and check an orthonormal basis of the input span, and explain why the output does not contain three unit vectors.

WORKED SOLUTION

First, $\|v_1\| = \sqrt{2}$, so

$$e_1 = \frac{1}{\sqrt{2}}(1, 1, 0)^T.$$

For the second input,

$$\langle v_2, e_1\rangle = 2\sqrt{2}, \qquad w_2 = v_2 - 2\sqrt{2}\, e_1 = 0.$$

We cannot normalize w_2; we skip it because $v_2 = 2v_1$. For the third input,

$$\langle v_3, e_1\rangle = \frac{1}{\sqrt{2}}, \qquad w_3 = v_3 - \frac{1}{\sqrt{2}}e_1 = (1/2, -1/2, 1)^T.$$

We have $\|w_3\|^2 = 1/4 + 1/4 + 1 = 3/2$, so the second output vector is

$$e_2 = \frac{1}{\sqrt{6}}(1, -1, 2)^T.$$

A direct check gives

$$\|e_1\|^2 = \frac{1+1}{2} = 1, \qquad \|e_2\|^2 = \frac{1+1+4}{6} = 1, \qquad \langle e_1, e_2\rangle = \frac{1-1}{\sqrt{12}} = 0.$$

The vectors v_1, v_3 are linearly independent: in $av_1 + bv_3 = 0$, the third coordinate gives $b = 0$, and then $a = 0$. Thus the input span has dimension 2, and (e_1, e_2) is its orthonormal basis. It cannot contain three orthonormal vectors, since those vectors would be linearly independent.

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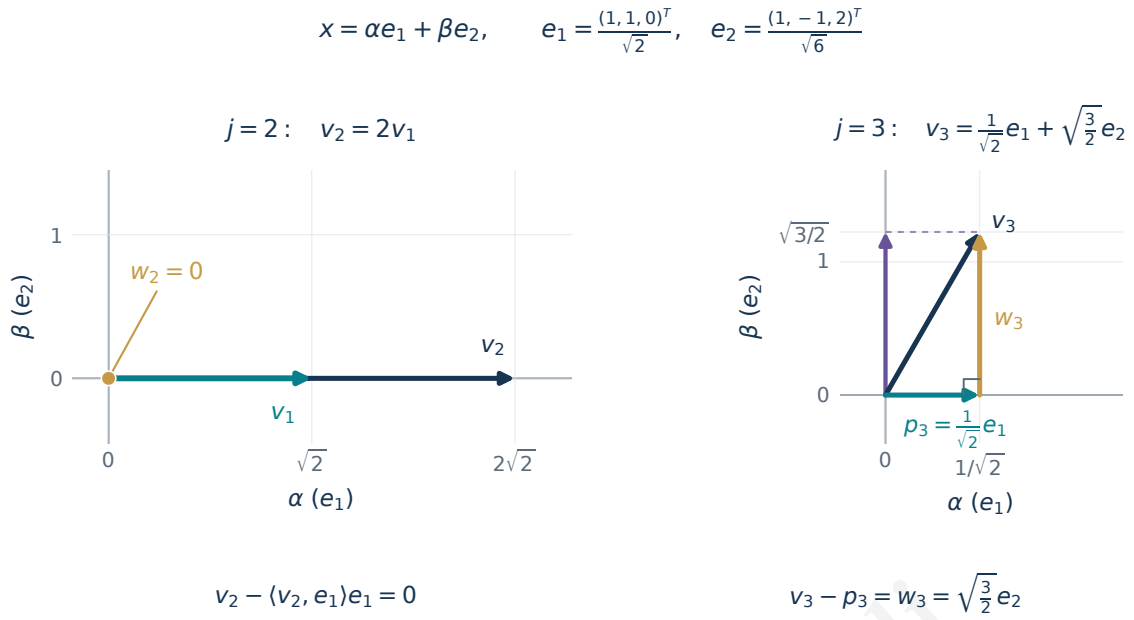


Figure 13.2. The axes α, β are coordinates in the orthonormal basis (e_1, e_2) of the plane $\mathrm{span}\{v_1, v_3\}$: each point represents $x = \alpha e_1 + \beta e_2$. The input coordinates in this chart are $(\sqrt{2}, 0)^T$, $(2\sqrt{2}, 0)^T$, and $(1/\sqrt{2}, \sqrt{3/2})^T$, respectively. On the left, $v_2 = 2v_1$, so the residual $w_2 = 0$ must be skipped. On the right, subtract $p_3 = \langle v_3, e_1 \rangle e_1$ from v_3; the residual $w_3 = (1/2, -1/2, 1)^T = \sqrt{3/2}\, e_2$ is orthogonal to e_1.

EXERCISE 13.10 CHALLENGE

Let $A, B \in M_n(\mathbb{R})$ be symmetric matrices.

1. Prove that there is an orthogonal matrix Q for which both $Q^T AQ$ and $Q^T BQ$ are diagonal if and only if $AB = BA$.
2. Find such a matrix for

$$A = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 5 \end{pmatrix}, \qquad B = \begin{pmatrix} 3 & 1 & 0 \\ 1 & 3 & 0 \\ 0 & 0 & 7 \end{pmatrix},$$

and also give the diagonal matrix representing AB in the same basis.

WORKED SOLUTION

1. If $Q^T AQ = D_A$ and $Q^T BQ = D_B$ are diagonal, then $A = QD_A Q^T$ and $B = QD_B Q^T$. Since diagonal matrices commute,

$$AB = QD_A D_B Q^T = QD_B D_A Q^T = BA.$$

Conversely, suppose $AB = BA$. By the spectral theorem, $\mathbb{R}^n$ is the orthogonal direct sum of the eigenspaces of A. If $x \in E_\lambda(A)$, then

$$A(Bx) = B(Ax) = B(\lambda x) = \lambda Bx,$$

so $Bx \in E_\lambda(A)$. Thus B preserves every eigenspace of A. On each such space, the

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EXERCISE 13.10 **CHALLENGE CONTINUED**

restriction of B is self-adjoint for the induced inner product: for all x, y in that space,

$$\langle Bx, y\rangle = \langle x, By\rangle,$$

because $B^T = B$. By the spectral theorem on that space, choose an orthonormal basis of eigenvectors of B. Each vector is also an eigenvector of A belonging to λ. Combining these bases over all eigenvalues of A gives an orthonormal basis of $\mathbb{R}^n$ common to both matrices. Placing it in the columns of Q gives the required diagonalizations, including the case of repeated eigenvalues.

2. Choose

$$q_1 = \frac{1}{\sqrt{2}}(1,1,0)^T, \qquad q_2 = \frac{1}{\sqrt{2}}(1,-1,0)^T, \qquad q_3 = (0,0,1)^T.$$

Their norms are 1 and their pairwise inner products are 0. Furthermore,

$$Aq_1 = 2q_1, \quad Aq_2 = 2q_2, \quad Aq_3 = 5q_3, \qquad Bq_1 = 4q_1, \quad Bq_2 = 2q_2, \quad Bq_3 = 7q_3.$$

Thus, for

$$Q = \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

we obtain

$$Q^T AQ = \operatorname{diag}(2,2,5), \qquad Q^T BQ = \operatorname{diag}(4,2,7), \qquad Q^T ABQ = \operatorname{diag}(8,4,35).$$

The choice within the two-dimensional eigenspace of A allows the basis to be adapted to B as well.

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APPENDIX A

Logic and Proof Methods

A.1 Logic

This appendix introduces some foundations of mathematical logic. We will study propositions, quantifiers, and logical connectives, and see how they are used to construct proofs and assess mathematical claims.

A.1.1 Propositions

A proposition is a statement to which exactly one of the truth values ``true'' and ``false'' can be assigned unambiguously. Examples of propositions include:

- Paris is the capital of France. (A true proposition.)
- $0 = 1$. (A false proposition.)

Consider the statement that the number of grains of sand on Earth exceeds 10^{25}. We may not know its truth value, and checking it would be difficult. Nevertheless, under an agreed meaning of the statement, it is either true or false; our lack of knowledge does not itself prevent it from being a proposition.

Examples of expressions that are not propositions include:

- How are you? (A question is not a proposition.)
- Go to sleep! (An instruction is not a proposition.)
- $x^2 > x$. (No truth value is determined until the meaning or quantification of x has been specified.)

The sentence ``The Mona Lisa is a beautiful painting'' expresses a subjective judgement. To treat it as a formal proposition, one would first need to specify a precise meaning of ``beautiful.'' In mathematical logic, we work with statements whose meanings and domains of discourse have been specified.

A.1.2 Quantifiers

- Consider ``He is the prime minister.'' Without specifying the referent, this does not determine a proposition. Similarly, $n > 2$ does not determine a truth value when n is unspecified. To bind variables in statements, we use two logical symbols called **quantifiers**.

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RECALL AND TOOLS

- **For every**, denoted by $\forall$. For example, ``Every natural number is greater than 2'' is written as
$$\forall n \in \mathbb{N}\ (n > 2).$$
- **There exists**, denoted by $\exists$. For example, ``There exists a natural number greater than 2'' is written as
$$\exists n \in \mathbb{N}\ (n > 2).$$

Specify the domain of each quantifier, either explicitly or through a clearly established context. The following example explains why this matters.

- The expression $\forall x\ x^2 \geq x$ is incomplete without an agreed domain. The operations and comparison may not even be defined for the objects under consideration. In contrast, the following are propositions because the domain is stated:
  - $\forall x \in \mathbb{N}\ x^2 \geq x$ is true, since $x^2 \geq x$ for every natural number x.
  - $\forall x \in \mathbb{Q}\ x^2 \geq x$ is false. Choosing $x = \frac{1}{2}$ gives a counterexample.

EXERCISE A.1

Write the following statement in mathematical notation: a set is contained in another set if and only if every element of the first set also belongs to the second set.

WORKED SOLUTION

$$A \subseteq B \iff \forall a \in A,\ a \in B.$$

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Logical Connectives: Conjunction, Disjunction, and Negation

Conjunction ($\wedge$): both conditions hold.

- Saying that ``our car is large and green'' means that ``our car is large'' *and* ``our car is green.'' The conjunction is true exactly when both statements are true. If either statement is false, the conjunction is false.

Truth table: The table gives the truth value of $A \wedge B$ for every possible pair of truth values of A, B.

A	B	$A \wedge B$
T	T	T
T	F	F
F	T	F
F	F	F

Disjunction ($\vee$): logical ``or.''

- $A \vee B$ is true when at least one of A, B is true, and false when both are false. This is **inclusive** ``or'': it permits both statements to be true.

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Truth table:

A	B	$A \vee B$
T	T	T
T	F	T
F	T	T
F	F	F

Negation ($\neg$): reversing a truth value.

- For example, the negation of ``Every rectangle is a parallelogram'' is ``Not every rectangle is a parallelogram,'' or equivalently, ``There exists a rectangle that is not a parallelogram.'' A proposition and its negation have opposite truth values: one is true exactly when the other is false.

Truth table:

A	$\neg A$
T	F
F	T

EXERCISE A.2

Use logical quantifiers to write the following proposition and its negation: there exists a natural number M such that $n \leq M$ for every natural number n.

WORKED SOLUTION

The proposition is

$$\exists M \in \mathbb{N}\, \forall n \in \mathbb{N},\ n \leq M.$$

Its negation is

$$\forall M \in \mathbb{N}\, \exists n \in \mathbb{N},\ n > M.$$

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Implication ($\Rightarrow$) and Equivalence ($\Leftrightarrow$)

**Implication ($\Rightarrow$): ``if … then …."** Consider the statement ``If Yossi received a final grade of at least 70, then Yossi passed the course.'' Let A denote the assumption and B the conclusion.

- If A is true and B is true, the implication is true.
- If A is true and B is false, the implication is false.
- If A is false and B is true, the implication is true.
- If A is false and B is false, the implication is true.

This example illustrates the meaning of implication; it does not state the grading policy of any particular course.

The implication is **false** only when A, the assumption, is true and B, the conclusion, is false. In every other case it is true.

Truth table:

A	B	$A \Rightarrow B$
T	T	T
T	F	F
F	T	T
F	F	T

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Example: The proposition

$$0 = 1 \Rightarrow 2 = 3$$

is **true**: its assumption is false, even though its conclusion is also false.

Important remarks. For a statement of the form ``If $P$, then $Q$,'' we ask whether $Q$ must hold whenever $P$ holds. Everyday language often uses ``if … then …'' causally (for example, ``If you eat, you will feel full''). Formal implication has the truth-table meaning given above; it need not express a causal connection.

The negation of

$$P \Rightarrow Q$$

is not

$$\neg P \Rightarrow \neg Q.$$

(Confusing these statements is a common error.) The implication fails exactly when P holds and Q does not. Thus

$$\neg(P \Rightarrow Q) \equiv P \wedge \neg Q.$$

In words: The statement $P \Rightarrow Q$ says that whenever P is true, Q is true as well. To negate it, identify the case in which it fails: P is true but Q is false.

Equivalence ($\Leftrightarrow$): ``if and only if.''

- $P \Leftrightarrow Q$ is true exactly when P, Q have the same truth value: either both are true or both are false.

Truth table:

A	B	$A \Leftrightarrow B$
T	T	T
T	F	F
F	T	F
F	F	T

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Proof Techniques

A.1.3 Proof Templates

NOTE

The following templates summarize proof methods that have already appeared and will be used throughout the course.

Universal Statements

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RECALL AND TOOLS

A universal statement asserts that every object satisfying a property p also satisfies a property q. It can be written as an implication: if an object satisfies p, then it satisfies q.
To prove the statement, take an arbitrary object and assume only that it satisfies p. Show that q follows. Be careful not to impose additional assumptions on the object.
To disprove the statement, find an object satisfying p but not q. One such counterexample is enough to disprove a universal statement.

EXERCISE A.3

Prove that

$$\forall x \in \mathbb{R},\ (x > 2) \Rightarrow (x^2 > 4).$$

WORKED SOLUTION

Let $x \in \mathbb{R}$ satisfy $x > 2$. Since x is positive, multiply both sides of the inequality by x without reversing its direction:

$$x \cdot x > 2 \cdot x.$$

Hence

$$x^2 > 2x.$$

Also, $x > 2$ implies $2x > 4$. By transitivity,

$$x^2 > 4.$$

We have therefore shown that

$$(x > 2) \Rightarrow (x^2 > 4).$$

Because x was arbitrary, the conclusion holds universally:

$$\forall x \in \mathbb{R},\ (x > 2) \Rightarrow (x^2 > 4).$$

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EXERCISE A.4

Prove by contradiction: if $A \setminus B = \varnothing$, then $A \subseteq B$.

WORKED SOLUTION

Assume that $A \setminus B = \varnothing$ and, for a contradiction, that $A \not\subseteq B$. Then some $x \in A$ satisfies $x \notin B$, so $x \in A \setminus B$. This contradicts the assumption that the difference is empty. Therefore $A \subseteq B$.

EXERCISE A.5

Disprove the following statement by a counterexample: $x^2 > x$ for every $x \in \mathbb{R}$.

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EXERCISE A.5 CONTINUED

WORKED SOLUTION

The statement is false. Take $x = \frac{1}{2}$. Then

$$\left(\frac{1}{2}\right)^2 = \frac{1}{4} < \frac{1}{2},$$

so the proposed strict inequality fails.

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